Copernicus Global Land Operations

"Vegetation and Energy"

"CGLOPS-1"

Framework Service Contract N° 945120-IPR-2023

VALIDATION REPORT

SENTINEL-3 SURFACE REFLECTANCE

VERSION 2.3

Issue 1.00

Organization name of lead contractor for this deliverable: EOLAB

Book Captain: Jorge Sánchez-Zapero (EOLAB)

Contributing Authors: Enrique Martínez-Sánchez (EOLAB)
Fernando Camacho (EOLAB)
Sarah Gebruers (VITO)
Carolien Toté (VITO)

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

DOCUMENT RELEASE SHEET

Book captain:	Jorge Sánchez-Zapero	Sign Jorge Sanchez-Zapéro	Date 30.04.2025
Approval:	Roselyne Lacaze	Sign Roselyne Lacaze	Date 13.05.2025
Endorsement:	Nadine Gobron	Sign	Date
Distribution:	Public		

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CHANGE RECORD

Date	Page(s)	Description of Change	Release
30.04.2025	All	First version	I1.00

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LIST OF ACRONYMS

6S	Second Simulation of a Satellite Signal in the Solar Spectrum code
AATSR	Advanced Along-Track Scanning Radiometer
AC	Atmospheric Correction
ACIX	Atmospheric Correction Inter-Comparison Exercise
AD	Applicable Document
AERONET	Aerosol Robotic Network
AIR	Aerospace Information Research Institute
AOI	Area Of Interest
AOT	Aerosol Optical Thickness
APR	Annual Performance Report
APU	Accuracy, Precision and Uncertainty
ATBD	Algorithm Theoretical Basis Document
B	Mean Bias
BA	Bare Areas
BRDF	Bidirectional Reflectance Distribution Function
BSCN	Baoutou Sand ChiNa
C3S	Copernicus Climate Change Service
CAMS	Copernicus Atmosphere Monitoring Service
CCI	Climate Change Initiative
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land OPerationS
CLMS	Copernicus Land Monitoring Service
CNES	Centre National D'Etudes Spatiales (French Space Agency)
CUL	CULtivated
DBF	Deciduous Broadleaved Forest
DMP	Dry Matter Productivity
EBF	Evergreen Broadleaf Forest
EO	Earth Observation
ESA	European Space Agency
EU	European Union
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FCOVER	Fraction of green vegetation Cover
FRM	Fiducial Reference Measurements
FRM4Veg	Fiducial Reference Measurements for Vegetation
FWHM	Full Width at Half Maximum
GLC-100	Global Land Cover 100m

GONA	GObabeb NAmibia
HBK	HandBooK
HER	HERbaceous
HYPERNETS	HYPERspectral radiometer integrated in automated NETworkS
IdePix	Pixel Identification tool
JCGM-GUM	Joint Committee for Guides in Metrology - Guides to the expression of Uncertainty in Measurement
JRC	Joint Research Centre
LAI	Leaf Area Index
LANDVAL	LAND VALidation network
LCFR	La Crau FRance
LPV	Land Product Validation subgroup of CEOS
L2A	Level 2A
MAD	Median Absolute Deviation
MAR	Major Axis Regression
MD	Median Deviation
MERIS	Medium Resolution Imaging Spectrometer
MERRA	Modern Era Retrospective-Analysis for Research and Applications
MODIS	Moderate Resolution Imaging Spectroradiometer
MPC	Mission Performance Cloud
N	Number of samples
NDVI	Normalized Difference Vegetation Index
NIR	Near InfraRed
NLF	Needle-Leaf Forest
NRT	Near Real Time
OF	Other Forests
OLCI	Ocean and Land Colour Instrument
OLS	Ordinary Least Squares
OP	Operations
OPT	OPTical
PN	Product Notice
PROBA-V	Project for On Board Autonomy – Vegetation
PUM	Product User Manual
QA4EO	Quality Assurance Framework for Earth Observation
QAR	Quality Assessment Report
R	Correlation coefficient
RadCalNet	Radiometric Calibration Network
R_{CV}	Relative Coefficient of Variation

RMSD	Root Mean Square Deviation
R_{SE}	Scale Requirement Index
R_{ST}	Relative STrength of the spatial correlation
R_{SV}	Relative proportion of the Structural Variation
RVUS	Railroad Valley USA
SAA	Solar Azimuth Angle
SDR	Surface Directional Reflectance
SHR	SHRublands
SI	Système International (d'unités)
SLSTR	Sea and Land Surface Temperature Radiometer
SMAC	Simplified Model for Atmospheric Corrections
SNAP	SentiNel Application Platform
SRF	Spectral Response Function
STD	Standard deviation
SWIR	Short-wave infrared
SYN	SYNergy
SZA	Solar Zenith Angle
TG	Technical Guide
TOA	Top-of-Atmosphere
TOC	Top-of-Canopy
UTC	Universal Time Coordinated
VAA	Viewing Azimuth Angle
VIIRS	Visible Infrared Imaging Radiometer Suite
VNIR	Visible and Near InfraRed
VZA	Viewing Zenith Angle
WGCV	Working Group on Calibration and Validation

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

CLMS portfolio contains a number of global "Vegetation" biophysical products based upon the data acquired by the Ocean and Land Colour Instrument (OLCI) and the Sea and Land Surface Temperature Radiometer (SLSTR) onboard the Sentinel-3 platform. All various biophysical algorithms ingest the same surface Reflectances derived from the Level 1B Top-Of-Atmosphere (TOA) data by a harmonized pre-processing chain.

This document presents the quality assessment and validation of the CLMS OLCI and SLSTR surface reflectances 300m Version 2.3 (V2.3), derived from Collection 4 Level-1B data for the period from July 2018 to December 2023. The methodology focuses on the evaluation of the product content (surface reflectances, associated uncertainties and quality layers), completeness, spatial and temporal consistency and error evaluation (accuracy, precision and uncertainty) of the products. The product intercomparison is performed using the equivalent European Space Agency (ESA) Sentinel-3 Synergy Level-2 product (ESA L2 SYN). Moreover, the consistency of surface reflectances V2.3 is evaluated for different satellites (Sentinel-3A versus Sentinel-3B during the tandem phase) and sensors (OLCI versus SLSTR similar spectral channels). For the direct validation, surface reflectances V2.3 are compared with ground reference values from the Committee on Earth Observation Satellites (CEOS) Radiometric Calibration Network (RadCalNet) database and other initiatives, such as initial test samples from EU HYPERNETS project and the ESA Fiducial Reference Measurements (FRMs) programme.

CLMS Sentinel-3 surface reflectance V2.3 products provide reliable values at global scale and similar spatial patterns than ESA L2 SYN at local scale over different areas covering the main biome types (different forests, cultivated areas, herbaceous, shrublands and savannas, desert and snow). Good temporal consistency is found between different OLCI and SLSTR channels and with ground measurements over four RadCalNet sites. Observations from different satellites (Sentinel-3A versus -3B) during the tandem phase show very good agreement over desert calibration sites: bias within ±0.9% for OLCI and within ±1.5% for SLSTR, precision and uncertainty up to 1.5% for OLCI and up to 2.5% for SLSTR. Observations for equivalent OLCI and SLSTR channels are also consistent, with positive bias lower than 4% for green channels and low bias within ±0.3% and ±0.8% for red and Near-InfraRed (NIR) bands over desert calibration sites. CLMS TOC reflectances V2.3 also improve the consistency between OLCI and SLSTR than previous Version 1. As compared to ESA L2 SYN for OLCI channels, CLMS surface reflectances V2.3 show typically lower values for visible (except for low blue reflectances) and NIR for most biome cases. The exception is the snow case,

with positive differences over large reflectance values. Visible and NIR channels, SLSTR surface reflectances V2.3 provide lower values than ESA L2 SYN (except for snow areas). On the contrary, SLSTR surface reflectances V2.3 provides higher values than ESA L2 SYN for SWIR channels.

The direct validation of CLMS surface reflectances V2.3 is performed against CEOS RadCalNet multi-temporal data from 4 sites (desert and sparse vegetation). Good accuracy is obtained, within ±2% for most OLCI channels and within ±1.5% for SLSTR, showing in overall better precision and uncertainty in case of OLCI. CLMS surface reflectance V2.3 shows improved accuracy than surface reflectance V1 and ESA L2 SYN. The CLMS surface reflectances V2.3 agree also well with in-situ data from other initiatives: median deviation within -1% to 6% versus HYPERNETS multi-angular measurements and similar spectral signatures than ESA FRMs.

In conclusion, the CLMS Sentinel-3 surface reflectance V2.3 products demonstrate good quality. It can therefore be released to users and deployed in near real time operations to produce biophysical products. It should be noted that some cloud contamination is not identified by the quality layers. Consequently, it is recommended to apply OLCI quality layers to SLSTR observations to have more restrictive cloud screening. Furthermore, the snow identification of IdePix has limitations.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document presents the results of the quality assessment of the Copernicus Land Monitoring Service (CLMS) surface (i.e. Top-Of-Canopy (TOC)) reflectance Version 2.3 (hereafter TOC V2.3) based on Sentinel-3 OLCI and SLSTR sensors Collection 4 Level-1B data. The objective is to evaluate the quality of the TOC reflectances with the aim of releasing time series to users, as well as for the deployment of V2.3 in NRT operations for the production of the biophysical products.

The quality assessment and validation are performed for the outputs of the Copernicus Global Land Operations (CGLOPS-1) processing line v2.3.4 from July 2018 to December 2023.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements
- Chapter 3 describes the methodology for quality assessment, the metrics and the criteria of evaluation.
- Chapter 4 presents the results of the analysis.
- Chapter 5 summarizes the main conclusions of the study.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PUM_TOC-reflectance300m-V2.3	Product User Manual of Sentinel-3 OLCI and SLSTR TOC Reflectance 300m Version 2.3

CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document of the Atmospheric Correction Version 1.1 applied on Sentinel-3 OLCI and SLSTR data.
CGLOPS1_QAR_S3-AC	Quality assessment and validation report of the Atmospheric Correction Version 1.1 applied on Sentinel-3 OLCI and SLSTR data.
CGLOPS-1_QAR_S3-CloudMask	Report presenting the evaluation of performances of cloud, snow and shadow detection performed on the Sentinel-3 sensors data for the specific needs of global biophysical production
CGLOPS-1_QAR_S3-TOCr300m-V1	Quality Assessment Report of Sentinel-3 TOC reflectance 300m Version 1

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements
Available at https://gcos.wmo.int/site/global-climate-observing-system-gcos/publications	
S3-OLCI TG	Sentinel-3 OLCI instrument Technical Guide.
Available at https://sentiwiki.copernicus.eu/web/s3-olci-instrument	
S3-SLSTR TG	Sentinel-3 SLSTR instrument Technical Guide.
Available at https://sentiwiki.copernicus.eu/web/s3-slstr-instrument	
S3.PN-OLCI-L1.12	Product Notice introducing the Sentinel-3A and -3B OLCI Level-1B Collection 4
Available at https://sentiwiki.copernicus.eu/__attachments/1672112/S3.PN-OLCI-L1.12 - Sentinel 3 OLCI Product Notice Level 1B 2024 - 1.0.pdf	
S3.PN-SLSTR-L1.11	Product Notice of the Sentinel-3A and -3B SLSTR Level-1B Collection 4
Available at https://sentiwiki.copernicus.eu/__attachments/1672112/S3.PN.SLSTR-L1.11 - Sentinel 3 SLSTR Product Notice L1 PB SL__L1_.004.07.00 2024 - 1.0.pdf	

S3MPC.RAL.TN.020	Sentinel-3 SLSTR VIS and SWIR Channel Vicarious Calibration Adjustments, I3.0, 22 May 2021

Available at https://sentinel.esa.int/documents/247904/3347387/Sentinel-3-SLSTR-VIS-and-SWIR-Channel-Vicarious-Calibration-Adjustments.pdf

ESA-S3-SLSTR Anomalies and Events	Sentinel-3 SLSTR Anomalies and Events report

Available at https://sentinels.copernicus.eu/documents/d/sentinel/slstr_anomaly_events

ESA-S3-OLCI Anomalies and Events	Sentinel-3 OLCI Anomalies and Events report

Available at https://sentinels.copernicus.eu/documents/d/sentinel/olci_anomaly_events_2024

L2-SYN-SDR-processing	ESA L2 SYN Surface Directional Reflectance processing

Available at https://sentiwiki.copernicus.eu/web/synergy-products#S3-Synergy-Products-L2-SYN-SDR-processing

S3-L2-SD-03-S02-ATBD	SYN Algorithm Theoretical Basis Document

Available at https://sentiwiki.copernicus.eu/__attachments/1672112/S3-L2-SD-03-S02-ATBD - Sentinel-3 Synergy Level-2 and Level-3 ATBD 2017 - 2.5.pdf

L2-SYN-processing-baseline	Sentinel-3 L2 SYN processing baseline

Available at https://sentiwiki.copernicus.eu/web/synergy-processing#Processing-Baseline

OMPC.ACR.HBK.003	Copernicus Sentinel-3 SYN Land User Handbook

Available at https://sentinel.esa.int/documents/247904/4598110/Sentinel-3-Synergy-Land-Handbook.pdf

OMPC.ACR.APR.004	Sentinel-3 Optical Annual Performance Report – Year 2023

Available at https://sentiwiki.copernicus.eu/__attachments/1681931/OMPC.ACR.APR.004 - S3 Optical Annual Performance Report 2023 - 1.1.pdf

S3_SYN_DataHub	Sentinel-3 SYNERGY data hub availability

Available at https://sentiwiki.copernicus.eu/__attachments/1672112/Sentinel-3 SYNERGY data hub availability.pdf

S3.PN-SYN-L2.10	Sentinel-3 Product Notice- SYNergy. Issue/Rev Date 29/02/2024.

Available at https://sentinel.esa.int/documents/d/sentinel/s3-pn-syn-l2-10-sentinel-3-synergy-product-notice-level-2-2024-1-0

JCGM-GUM Joint Committee for Guides in Metrology (JCGM) – Guides to the expression of Uncertainty in Measurement (GUM). ISO/IEC Guide 98 - Part 3, 2008]. (2008).

Available at https://www.iso.org/sites/JCGM/GUM-introduction.htm

2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for the TOC Reflectance products are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300m
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global

TOP-OF-CANOPY REFLECTANCES	
Top-Of-Canopy (TOC) spectral reflectance of the Sentinel-3 OLCI and SLSTR sensor bands	
Uncertainty (2-sigma)	**Threshold: 5%** Goal: 3%
Stability (per decade, i.e. 10 years)	**Threshold: 3%** Goal: 1.5%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NetCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

2.2.5 Input data

Copernicus Sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, PROBA-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 QUALITY ASSESSMENT METHOD

3.1 OVERVALL PROCEDURE

The general premise behind the current state-of-art surface reflectance validation approach is to compare the surface reflectance derived from atmospheric correction using the image derived Aerosol Optical Thickness (AOT) against the equivalent using the station derived AOT. This assumes that the surface reflectance derived from the station AOT is the "truth" (Vermote et al., 2016). This is the approach performed on the quality assessment of the Atmospheric Correction (AC) of the Top-Of-Atmosphere (TOA) to TOC pre-processing chain [CGLOPS1_QAR_S3-AC]. However, some common themes begin to emerge when analyzing the studies using these networks (Origo et al., 2020). Firstly, the combined impact of the sensor spatial resolution (300m in this case) and the site heterogeneity can cause significant disagreement. Secondly, there is little mention or quantitative assessment of the uncertainties associated with the in-situ data, nor any endeavour to perform the validation in a metrologically robust way. Thirdly, many of the published validation activities (for example: Sentinel-2 (Gascon et al., 2017), Moderate Resolution Imaging Spectroradiometer (MODIS) (Vermote et al., 2016), and Advanced Along-Track Scanning Radiometer (AATSR) (Sogacheva et al., 2015)) utilize reference data which cannot be considered independent (e.g. those utilizing other satellite products). For this reason, the main assumption is that the aerosol correction is the dominant error. This may be the case, but only comparing the impact of the aerosol retrieval, the impact of other errors in the processing chain (radiometric, geolocation, etc.) is lost and the validation is only testing the aerosol retrieval. Validation of satellite surface reflectance data against independent in situ reflectance estimates was more common for assessing new retrieval algorithms (e.g., Li et al., 2010; Liang et al., 2002) rather than for assessing operational products. Nevertheless, in the recent years more independent in situ surface reflectance reference measurements are available allowing the direct validation of satellite products over a variety of conditions (Origo et al., 2020).

Validation of surface reflectance should be considered in the context of the validation of the related products that are Bidirectional Reflectance Distribution Function (BRDF) parameters and surface albedo. All three fall within the same category in the CEOS Land Product Validation Subgroup (LPV). For that, the validation methodology follows, as much as possible, the CEOS LPV good practice protocol for the validation of satellite-derived albedo products (Wang et al., 2019). Two main validation approaches are defined:

- The direct point-to-pixel validation, which involves comparisons of satellite products with TOC reflectance measured from in situ tower-based instruments or dedicated field campaigns (i.e. direct validation). In this quality assessment, the CEOS RadCalNet data, as well as other in situ datasets (HYPERNETS, ESA FRM campaign) were used as independent in situ references (see section 3.3).

- Product-to-product validation approach, referred to satellite product intercomparison (i.e., indirect validation). The intercomparison offers a means of assessing discrepancies between products and provides relative uncertainties. It is very helpful due to limited availability of ground data, and several Areas Of Interest (AOI) are selected over different biomes and latitudes (e.g., desert, forest, agriculture areas). In this quality assessment, the ESA SYN product is used as reference (see section 3.2.2).

The following criteria of performance and metrics were assessed: completeness, spatial consistency, temporal consistency and error evaluation, which involves Accuracy, Precision and Uncertainty (APU) metrics.

1) Product Completeness

Completeness corresponds to the absence of spatial and temporal gaps in the data. Missing data are mainly due to cloud or snow contamination, poor atmospheric conditions or technical problems during the acquisition of the images and is generally considered by users as a severe limitation of a given product. It is therefore mandatory to document the completeness of the product (i. e. the distribution in space and time of missing data). Monthly global maps of valid observations are generated.

2) Spatial consistency

Spatial consistency refers to the realism and repeatability of the spatial distribution of retrievals over the globe.

- A first qualitative check of the realism and repeatability of spatial distribution of retrievals and the absence of strange patterns or artifacts (e.g., missing values, stripes, unrealistic values, etc.) can be achieved through systematic visual analysis of global maps based on the expert knowledge of the scientist. The methodology includes the visualization of global maps at reduced resolution (1/48 pixels) and the visualization of 4 tiles at full resolution (X17Y04 [35ºN – 45ºN; 10ºW – 0º], X18Y03 [45ºN – 55ºN; 0º - 10º E], X20Y02 [55ºN – 65ºN; 20ºE – 30ºE and X05Y01 [65ºN - 75ºN; 130ºW - 120ºW]). X17Y04 is selected because it covers most of Spain with a vegetation gradient from South (arid regions) to North (humid regions with dense vegetation). X18Y03 is displayed with the focus on the evaluation of TOC reflectance values and flags over the Alps (snow covered area). X20Y02 is chosen because is located at northern latitudes in Norway and Finland where the latitude/longitude grid is stretched, and the viewing angles could be larger. X05Y01 located in northern Canada is selected because it is an area which typically mixes snow and cloud events, useful to evaluate the quality of flag layers.
- Maps of associated errors to TOC reflectance, observation angles and flags are also checked.

- Maps and difference maps versus equivalent ESA Sentinel-3 SYN products over selected AOIs (see section 3.4) are computed to evaluate the spatial distribution of differences.

3) Temporal consistency

The realism of the temporal variations of the products is assessed over sites with existing ground data (see 3.3) and 48 selected LANDVAL sites (Fuster et al., 2020; Sánchez-Zapero et al., 2020) representatives of the different biomes. The selected 48 LANDVAL sites include 20 desert pseudo-invariant calibration sites (Lacherade et al., 2013) and 4 sites for each of the rest of main biome types: Evergreen Broadleaved Forest (EBF), Needle-Leaf Forest (NLF), Deciduous Broadleaved Forest (DBF), Other Forests (OF), Cultivated (CUL), Herbaceous (HER) and Shrublands (SHR). Qualitative inspection of the temporal variations of the products is conducted.

4) Error evaluation (APU)

Accuracy, precision and uncertainty are quantified by several metrics reporting the goodness of fit between the products and the corresponding reference dataset. Validation metrics are summarized in Table 1.

Table 1: Validation metrics.

Statistics	Comment
N	Number of samples. Indicative of the power of the validation
B	Mean Bias. Difference between average values of x and y. Indicative of accuracy and offset. Bias (%) is the relative mean bias between the average of x and y.
MD	Median deviation between x and y. CEOS LPV best practice reporting the accuracy. MD (%) is the relative MD between the average of x and y.
STD	Standard deviation of the pair differences. Indicates precision. STD (%) is the relative STD between the average of x and y.
MAD	Median absolute deviation between x and y. CEOS LPV Best practice reporting the precision. MAD (%) is the relative MAD between the average of x and y.
RMSD	Root Mean Square Deviation. RMSD is the square root of the average of squared errors between x and y. CEOS LPV best practice reporting uncertainty. RMSD (%) is the relative RMSD between the average of x and y.
R	Correlation coefficient. Indicates descriptive power of the linear accuracy test. Pearson coefficient is used.
MAR	Slope and offset of the Major Axis Regression linear fit. Indicates some possible bias

- Scatter-plots and validation metrics versus reference datasets (ESA SYN and ground data) are analyzed. Percentages of cases within goal and threshold requirement levels (see section 2.1) are provided. The analysis of bias over time versus ground data is also evaluated.
- Additionally, the consistency between Sentinel-3A and Sentinel-3B over desert calibration sites is assessed, as well as the consistency between OLCI and SLSTR equivalent channels over calibration sites and AOIs.

Accuracy is the degree of the "*closeness of the agreement between the result of a measurement and a true value of the measurand*" [JCGM-GUM]. Commonly, accuracy represents systematic errors and often is computed as the statistical mean bias (B). Recently, for albedo products, the median deviation (MD) has been suggested as best practice for reporting accuracy (Wang et al., 2019).

Uncertainty is a "*parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand*" [JCGM-GUM]. Uncertainty includes systematic and random errors and can be estimated by the Root Mean Square Deviation (RMSD). Median Absolute Deviation (MAD) is also calculated to quantify the precision, according to the CEOS LPV best practices (Wang et al., 2019).

Precision or repeatability is the "*closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement*" [JCGM-GUM]. Commonly, precision represents the dispersion of product retrievals around their expected value and can be estimated by the standard deviation of the difference (STD) between retrieved satellite product and the corresponding reference estimates.

Additionally, linear model fits are used to quantify the goodness of fit. For this purpose, Major Axis Regression (MAR) are calculated instead of Ordinary Least Squares (OLS) because it is specifically formulated to handle error in both of the x and y variables (Harper, 2014). Other metrics are used, such as number of samples (N), indicative of the power of the validation, or the correlation coefficient (R, estimated as Pearson coefficient), which indicates descriptive power of the linear accuracy test.

- **Summary of the Quality Assessment Procedure**

The summary of the quality criteria, datasets involved, and different results generated for each criterion is showed in Table 2.

Table 2: Summary of the quality assessment of CLMS Sentinel-3 TOC reflectance V2.3.

Quality Criteria	Product Evaluated	Reference	Coverage
Product completeness	CLMS S3 TOC V2.3	*none*	Global
	-Monthly global maps of valid good quality observations.		
Spatial Consistency	CLMS S3 TOC V2.3	*"Expert knowledge"* ESA S3 SYN	Global AOIs
	-Visual inspection of global maps and selected AOIs. Check of ancillary layers. -Difference maps over selected AOIs.		
Temporal Consistency	CLMS S3 TOC V2.3	*"Expert knowledge"* RadCalNet ground data	48 LANDVAL sites 4 RadCalNet sites
	-Qualitative inspection of temporal profiles.		
APU (product intercomparison)	CLMS S3 TOC V2.3	Sentinel-3 (A vs B) OLCI vs SLSTR ESA S3 SYN	Calibration sites Calibration sites + AOIs AOIs
	-Scatter-plots and validation metrics.		
APU (direct validation)	CLMS S3 TOC V2.3	RadCalNet, HYPERNETS ESA FRM ground data	RadCalNet, HYPERNETS & ESA FRM sites
	-Scatter-plots and validation metrics. Analysis of bias over time.		

3.2 SATELLITE PRODUCTS

3.2.1 Evaluated dataset: CLMS Sentinel-3 TOC reflectances V2.3 (CLMS TOC V2.3)

The CLMS TOC reflectances V2.3 are derived from Sentinel-3 OLCI and SLSTR sensors acquisitions through the CGLOPS-1 pre-processing chain (Figure 1), using as input the L1B latest collection 4 for OLCI [S3.PN-OLCI-L1.12] and SLSTR [S3.PN-SLSTR-L1.11]. CGLOPS-1 pre-processing chain is made of various modules which perform the pixel classification (Idepix), the co-registration between OLCI and SLSTR acquisitions, the reprojection and resampling in the CGLOPS-1 regular grid and the atmospheric correction. Idepix is developed in the framework of the Sentinel-3 Optical Mission Performance Cluster (OPT-MPC) under ESA contracts and imbedded in the Sentinel Application Platform (SNAP). The co-registration, reprojection and resampling module and the AC module are developed by CGLOPS-1.

More information about pixel classification, co-registration, reprojection and resampling module can be found in the Product User Manual (PUM) [CGLOPS1_PUM_TOC-reflectance300m-V2.3].

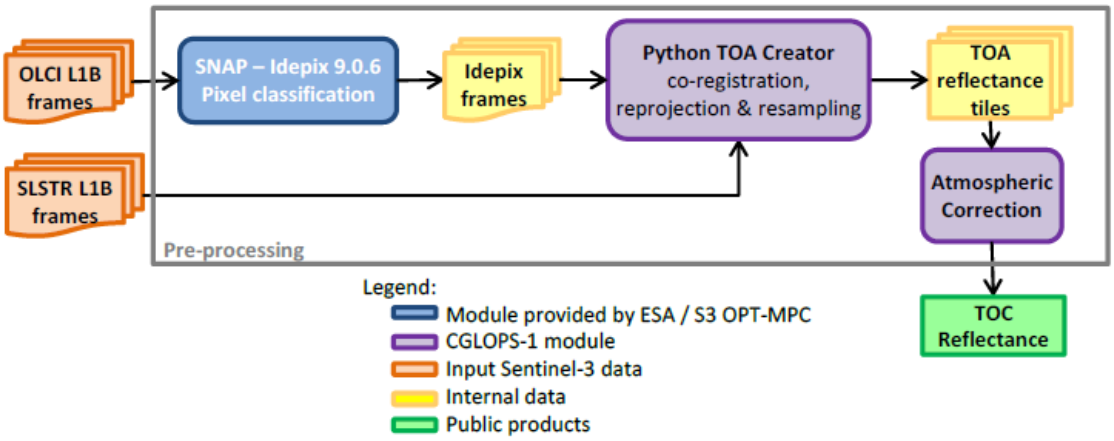


Figure 1: Pre-processing chain of the CGLOPS-1 Sentinel-3 based biophysical chains (source: CGLOPS1_PUM_TOC-reflectance300m-V2.3).

At the input of the AC module, some adjustment factors are applied on the TOA reflectances:

- A factor of 0.98 is applied on all OLCI-A spectral bands, except band 0a21, to reduce the brightness of OLCI-A acquisitions by 2% and make them more consistent with the OLCI-B data. This calibration adjustment is recommended by the S3 OPT-MPC and results from analysis of OLCI data acquired during the tandem phase (Clerc et al., 2020).
- Calibration adjustment coefficients are applied on SLSTR visible and Short Wavelength InfraRed (SWIR) bands. They are based upon the analysis of vicarious calibration results by several expert groups and are recommended by ESA [S3MPC.RAL.TN.020].

The atmospheric correction [CGLOPS1_ATBD_S3-AC-V1.1] is finally performed by the Simplified Model for Atmospheric Corrections (SMAC) algorithm (Rahman and Dedieu, 1994), based on the 6S radiative transfer formulation of the satellite signal (Vermote et al., 1997). Table 21 and Table 22 in Annex I summarize the spectral bands of OLCI and SLSTR, respectively, for which TOC reflectances are calculated. Errors are also propagated from TOA to TOC levels, using as inputs the TOA reflectances, the cloud information, the solar and view angles and ancillary data characterizing the surface and atmosphere (surface elevation, water vapor, pressure, aerosol, ozone). Necessary ancillary data like the Aerosol Optical Thickness (AOT), the surface pressure, the total column of ozone, the water vapour column and the surface temperature, are from different sources: AOT and ozone total column are from the Copernicus Atmosphere Monitoring Service (CAMS) NRT products; water vapour column and surface pressure are from Sentinel-3 inputs files while the surface temperature is from a climatology. The aerosol model is chosen to be a continental model. It should

be noted that TOC reflectance V2.3 incorporates improved TOC reflectances uncertainty assessment compared to previous V1, as the uncertainty due to the aerosol model is now taken into account [CGLOPS1_ATBD_S3-AC-V1.1].

The OLCI and SLSTR TOC reflectances are the input of CGLOPS biophysical modules that generate different CLMS products: BRDF-corrected Normalized Difference Vegetation Index (NDVI), the Leaf Area Index (LAI), the Fraction of Absorbed Photosynthetically Active Radiation (FAPAR), the Fraction of green vegetation Cover, the Dry Matter Productivity (DMP) and the burnt areas (BA). They are also used in the Copernicus Climate Change Service[3] (C3S) for calculating the albedo (Sánchez-Zapero et al., 2023a) and other biophysical variables (LAI, FAPAR), and in the Vegetation Parameters project[4] of the ESA Climate Change Initiative (CCI).

The performance of the AC was assessed following the approach applied in the Atmospheric Correction Inter-Comparison Exercise "ACIX" (Doxani et al., 2018). The TOC reflectances were compared with a reference product retrieved using an accurate radiative transfer code and inversion products of the in-situ Aerosol Robotic Network (AERONET) for aerosols optical thickness and model [CGLOPS1_QAR_S3-AC]. A spectral dependency of the accuracy (from a few 10^{-3} to 1.2 10^{-2}) was obtained. Precision and uncertainty were between 0.012 and 0.033, concluding that the uncertainty was mainly driven by the dispersion not the bias. Additionally, the QAR of the AC recommends to exclude pixels with an AOT greater than 1.0 and zenith angles higher than 65 degrees, (i) to avoid conditions of the limit of the SMAC method based on fit of radiative transfer simulations and (ii) to remove bad conditions as the validity of the plane-parallel assumption in radiative transfer code and to remove cases with high atmospheric contribution com-pared to the surface.

The Sentinel-3 TOC reflectance V2.3 products contain 4 quality layers (see Annex II):

- **cloud_an**: cloud flags from SLSTR Level 1B stored as 16-bit integer values.
- **quality_flags**: quality flags from OLCI Level 1B stored as 32-bit unsigned integer values.
- **pixel_classif_flags**: output of OLCI-IdePix stored as 16-bit integer values.
- **AC_process_flag**: output of atmospheric corrections [CGLOPS1_ATBD_S3-AC-V1.1] module stored as one-byte values.

We filter suboptimal quality pixels according to Table 3 based on the recommendations raised in the Product User Manual [CGLOPS1_PUM_TOC-reflectance300m-V2.3].

[3] https://climate.copernicus.eu/ (accessed on 2024.11.21)
[4] https://climate.esa.int/en/projects/vegetation-parameters/ (accessed on 2024.11.21)

Table 3: Flags used to filter suboptimal quality pixels.

TOC reflectance bands	Quality layer	Flags used to discard pixels
OLCI	**quality_flags**	'saturated_Oa*' depending on which band is used, exclude when raised fresh_inland_water: exclude when raised
	pixel_classif_flags	IdePix_INVALID: exclude when raised IdePix_LAND: include when raised IdePix_CLOUD: exclude when raised IdePix_CLOUD_AMBIGUOUS: exclude when raised IdePix_CLOUD_BUFFER: exclude when raised IdePix_CLOUD_SHADOW: exclude when raised IdePix_SNOW_ICE: exclude when raised Idepix_MOUNTAIN_SHADOW: depending on the application, exclude when raised
SLSTR	**cloud_an**	Exclude when not zero
For both OLCI and SLSTR	**AC_process_flag**	Exclude when AOT > 1 (Bit 2 set) Exclude when SZA > 65° (Bit 3 set)

Known limitations:

- The pixel classification by IdePix, which is the OLCI cloud detection software, suffers some well-known limitations which are confirmed by the CGLOPS-1 specific analysis [CGLOPS-1_QAR_S3-CloudMask]. Idepix tends to omit thin or semi-transparent clouds and cloud edges. In some cases, IdePix over-detects clouds over bright surfaces, such as bare areas and, less frequently, water shorelines. Furthermore, the snow mask of IdePix is not fully adequate: clear sky snow pixels are sometimes incorrectly masked as clouds.
- The SLSTR cloud flag performs well to identify clouds but over-detects the thin cirrus.
- The reprojection and resampling method creates some gaps at high latitudes. Indeed, for every pixel in the target 1/336° grid, the tool identifies the nearest pixel in the frames, up to a distance of 2/336° (~=0.006°). At the edge of the swath (where viewing angles are the largest) and at higher latitudes (where the regular lat/lon grid is more "stretched"), this causes gaps in the resulting image, because the actual ground sampling distance of OLCI is larger than 0.006°. At high latitudes, the effect is slightly more pronounced for SLSTR bands, related to the larger ground sampling distance.

- The SMAC algorithm is a parameterization of the radiative transfer model 6S (Vermote et al., 1997). 6S commonly agreed accuracy is 1%. SMAC approximation to 6S is also claimed to be within 1% for most situations under the assumption of lambertian surface (e. g. isotropic). However, for anisotropic surfaces, the SMAC approximation could be quite inaccurate to several %, especially for high atmospheric turbidity. Neither adjacency effects nor terrain slope correction is applied. These effects are, however, secondary at the spatial scale of 300 meters. Approximations in radiative transfer lead essentially to biases, depending on surface, angles and atmospheric content, and are certainly not a source of random noise. This source of error is thus not included in the error propagation model yet.

3.2.2 Reference product: ESA Sentinel-3 SYN Level-2 (ESA L2 SYN)

The ESA Synergy Level-2 Surface Directional Reflectance (SDR) product [L2-SYN-SDR-processing] is used for benchmarking in direct validation and for the spatial consistency analysis of CLMS TOC reflectance V2.3. It provides the atmospherically corrected surface bi-directional reflectance for the sun-reflective SLSTR channels (S1 to S6 for both nadir and oblique views except S4) and for all OLCI channels (except for the oxygen absorption bands Oa13, Oa14, Oa15, and the water vapor bands Oa19 and Oa20) and the corresponding information on aerosol properties.

The ESA L2 SYN SDR products have a resolution of 300 m and are mapped on the same regular grid as OLCI. They are created from a combination of OLCI and SLSTR Level-1B products. The synergy swath is then corresponding to the common area between OLCI and SLSTR nadir view swaths, restricted to the daylight part of the Sentinel-3 orbit (i.e., where the sun zenith angle at satellite ground track is lower than 80°).

The algorithm of L2 SYN processor [S3-L2-SD-03-S02-ATBD] is implemented to estimate surface reflectance from OLCI and SLSTR TOA measurements and is based on four main components: (i) radiative transfer modelling of TOA reflectance to generate look-up tables for a range of aerosol optical depths and aerosol model types, (ii) definition of parameterized models of land surface spectral and angular reflectance, (iii) an inversion algorithm to estimate the optimal atmospheric profile, and (iv) atmospheric correction to estimate surface reflectance for SLSTR and OLCI channels. The core algorithm of this Level-2 SYN processor is an aerosol retrieval based on an approach combining dark object and multiple view-angle methods derived from an existing work for Medium Resolution Imaging Spectrometer (MERIS) and AATSR. Sentinel-3 L2 SYN data is produced since October 2018 for Sentinel-3A and February 2019 for Sentinel-3B. The track record and evolutions of the processing baseline are available at [L2-SYN-processing-baseline].

NOTE: The version used in this study corresponds to the processing baseline version 2 [L2-SYN-processing-baseline] (data access from 1st to 30th November, 2024). The study period where ESA L2 SYN product is used for benchmarking covers 5 years of data (2019-2023). It should be noted that the recommended ESA calibration factors for OLCI-A and SLSTR are only applied from July 2023 onwards [S3.PN-SYN-L2.10].

The quality assessment of Sentinel-3 L2 SYN SDR products [OMPC.ACR.HBK.003] was performed by indirect validation against the normalized surface reflectance measurements observed from MODIS (MCD43A4.061, a combined Terra and Aqua Product) at 500 m spatial resolution and Visible Infrared Imaging Radiometer Suite (VIIRS) (VNP43MA4.001) at 1 km resolution. Several regional inter-comparisons were performed. A good agreement is observed between both Sentinel 3 A/B derived SYN SDR and MODIS (VIIRS) surface reflectance products, in particular after the correction of discrepancies with respect to acquisition geometry. The observed statistical scores are slightly better with S3B than compared to S3A. There is also a clear influence of surface biome on the intercomparisons results, with better agreement for needle and broad evergreen classes than to cultivated and shrub classes.

3.3 In-situ Reference Products

3.3.1 RadCalNet

RadCalNet (Bouvet et al., 2019) is an initiative of the CEOS Working Group on Calibration and Validation (WGCV). RadCalNet is a network of sites that can be used to compare different satellite sensors to each other and to a common reference. For each site, measurements of nadir-viewing TOC reflectance along with atmospheric parameters (such as surface pressure, columnar water vapor, columnar ozone, aerosol optical thickness, and the Angstrom coefficient to define aerosol sizes) are provided. Simulation to TOA reflectance is performed through the central RadCalNet Processing for all sites, ensuring that the in-situ data from those sites are SI-traceable (SI – from Système International (d'unités)). TOA reflectance and TOC reflectance data are provided for a nadir view at 30 min intervals from 9 am to 3 pm local standard time for a given site, along with their associated uncertainties, to users through the RadCalNet portal[5].

[5] https://www.radcalnet.org (RadCalNet data access on 14th January 2025)

NOTE: RadCalNet detected an instrument issue for GONA site from 13th August 2023 to May 2024, resulting in data gap or higher RadCalNet uncertainty than usual (see RadCalNet forum: https://forum.radcalnet.org/).

As we detected several outliers in the GONA data for the period from 1st July 2023 onwards (period close to the reported instrument issue), the GONA data from 1st July 2023 onwards was removed from our study.

Two assumptions are considered to deal with differences in data acquisition between Sentinel-3 TOC reflectances and RadCalNet ground reference measurements:

- Similar illumination and viewing configurations are considered. For that, Sentinel-3 observations with viewing zenith angle higher than 10º are discarded, as RadCalNet are measured at nadir viewing angle. Same dates and acquisition times were compared.
- The spectral range of RadCalNet measurements is from 380 nm to 2500 nm (except for BSCN site, with maximum wavelength of 1000 nm), with 10 nm of spectral sampling interval. These narrow-band measurements are made comparable with Sentinel-3 channels using the Spectral Response Function (SRF) coefficients of Sentinel-3 [S3-OLCI TG, S3-SLSTR TG].

Table 4 summarizes the main information of the 4 RadCalNet sites used in this study. For each site, the spatial representativeness of the measurement was evaluated. Note that RadCalNet also provides data for a fifth site (Baotou - BTCN), but was discarded as it relies on a very small artificial surface.

Table 4: Characteristics of the four RadCalNet sites considered in this study.

ID Site	Name	Country	Latitude	Longitude	Land Cover	Institution
GONA	Gobabeb	Namibia	-23.59999°N	15.119215° E	desert	ESA/CNES
LCFR	La Crau	France	43.56° N	4.86° E	sparse vegetation	CNES
RVUS	Railroad Valley Playa	USA	38.497° N	-115.69° E	desert	University of Arizona
BSCN	Baotou Sand	China	40.8658° N	109.6155° E	sand desert	AIR

The reference reflectance measured from towers cover a circular footprint that varies with the tower height. It is unlikely that the footprint of the ground measurements exactly matches the satellite pixel sizes. Then, the spatial representativeness of the tower-based measurements should be evaluated to

minimize the issues associated with spatial representativeness in the point-to-pixel comparison (Román et al., 2010, 2009).

We adopted CEOS LPV recommendations based of albedo validation protocols (Wang et al., 2019) for the analysis of the spatial representativeness of tower based measurements. The semi-variogram (Carroll and Cressie, 1996; Hohn, 1991) is one of the most efficient tools for describing spatial representativeness. Semi-variograms are computed for the stations under evaluation using Sentinel-2 surface reflectances at a spatial resolution of 10 m near-nadir. The spatial attributes (e.g., the range, sill and nugget) are computed by fitting the variogram estimator to an isotropic spherical model (Matheron, 1963), and they can reveal the spatial variability of land surfaces and the scaling effects associated with remotely sensed data (Román et al., 2010, 2009; Wang et al., 2017, 2014, 2012; Woodcock et al., 1988a, 1988b).

The methodology adopted in this study for the estimation of the geostatistical indexes is based on the comparison of the variogram model parameters retrieved at different spatial resolutions (i.e., from 300 m to 500 m squared subsets, to allow for errors in geolocation for 300 m resolution). Four different geostatistical attributes are used (Cescatti et al., 2012; Román et al., 2010): the relative coefficient of variation (R_{CV}), the scale requirement index (R_{SE}), the relative strength of the spatial correlation (R_{ST}) and the relative proportion of the structural variation (R_{SV}). The four geostatistical attributes can be combined in a compact metric (ST_{score} (Román et al., 2010, 2009), see Eq. 1), which represents a spatial representativeness score using R_{SE} as the primary marker and the others like secondary weights. When the spherical variogram model does not provide a good fit to the variogram estimator, another indicator (RAW_{score} (Román et al., 2010, 2009), see Eq. 2) could be used to provide a spatial representativeness score, which is only based on the R_{CV}.

$$ST_{score} = \left(\frac{|R_{CV}|+|R_{ST}|+|R_{SV}|}{3} + R_{SE}\right)^{-1} \quad \text{Eq. 1}$$

$$RAW_{score} = |2\,R_{CV}|^{-1} \quad \text{Eq. 2}$$

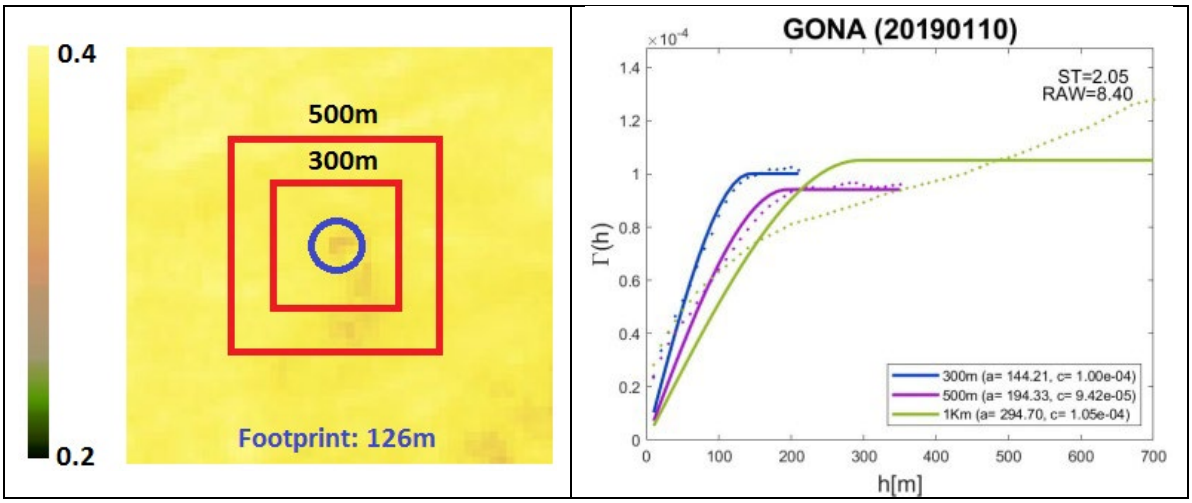
Both scores are directly proportional to the site representativeness. In case of desert sites (GONA, RVUS and BSCN), only one season is considered as stable reflectance measurements values are provided. Two different periods are considered for LCFR site (leaf-off (October-March) and leaf-on (April-September)) as seasonal dependency is expected in this site. Table 5 summarizes the geospatial statistics for the 4 evaluated RadCalNet sites.

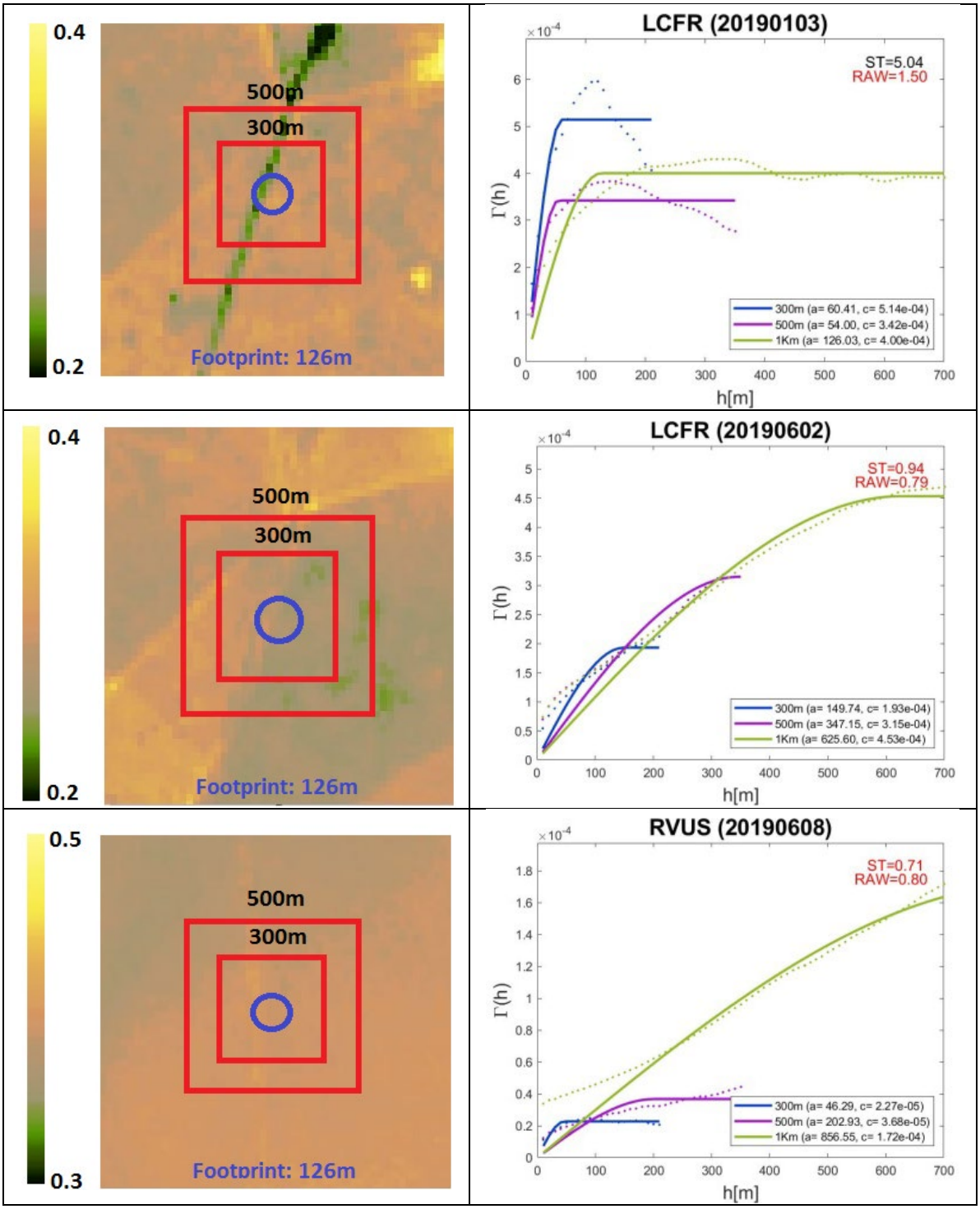
Better agreement between satellite pixels and tower measurements are expected when scores are higher than 2.0 (Cescatti et al., 2012; Sánchez-Zapero et al., 2023b, 2020). Based on our analysis of spatial representativeness, better results are expected for GONA. However, it should be noted that

the analysis is based on the near infrared and slightly different results could be found for visible channels mainly in sites with some vegetation (LCFR). Figure 2 displays the maps of Sentinel-2 L2A NIR Band8A around the 4 RadCalNet sites, and the results of the semi-variogram functions, and the relevant spatial attributes (range and still which are, respectively, the parameters a and c in Figure 2) using these images.

Table 5: Geospatial statistics of the four RadCalNet sites for direct validation. Scores are computed for 300 m versus 500 m.

Site	Seasonal Period	Footprint (m)	Tower Height (m)	300m range	500m range	Rcv (%)	Rse (%)	Rst (%)	Rsv (%)	ST_{score}	RAW_{score}
GONA	Single season	126	10	144.2	194.3	-6	33.6	4.7	34.8	2.05	8.40
LCFR	Leaf-off	126	10	60.4	54	-33.4	4.3	-2.5	-10.6	5.04	1.50
	Leaf-on			149.7	347.2	63.1	39.9	-5.4	131.8	0.94	0.79
RVUS	Single season	126	10	46.3	202.9	62.2	6.1	-1.4	338.4	0.71	0.80
BSCN	Single season	19	2	66.2	84	183.8	8.8	-11	26.9	1.21	0.27





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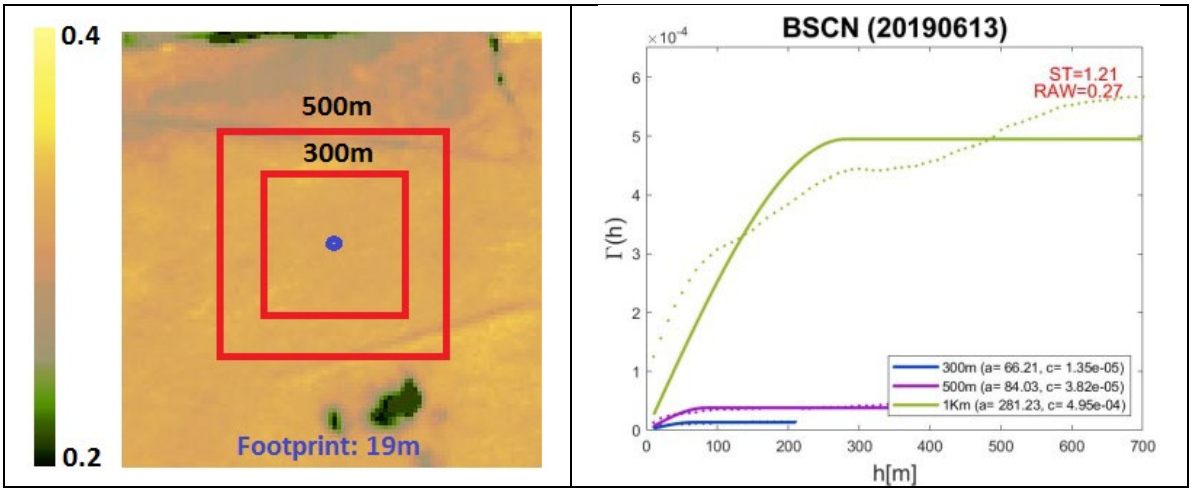


Figure 2: Left side: Sentinel-2 surface reflectance (Band 8A) images (1 km × 1 km) at a spatial resolution of 10 m centered over the 4 RadCalNet sites. From Top to Bottom: GONA, LCFR leaf-off, LCFR leaf-on, RVUS and BSCN. Right side: variogram estimator (points) and isotropic spherical variogram model (lines) using 300 m, 500 m and 1 km subsets, respectively.

3.3.2 HYPERNETS

The HYPERNETS project[6] (Ruddick et al., 2024) aims to ensure that high-quality hyperspectral surface reflectance measurements are available to support the validation of optical Visible and Near InfraRed (VNIR) and SWIR Copernicus products. Therefore, it established a new autonomous hyperspectral spectroradiometer (HYPSTAR® - www.hypstar.eu) dedicated to land and water surface reflectance validation with instrument-pointing capabilities. In the prototype phase, the instrument is being deployed at 24 sites covering a range of water and land types and a range of climatic and logistic conditions.

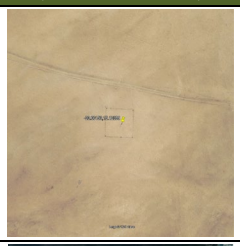
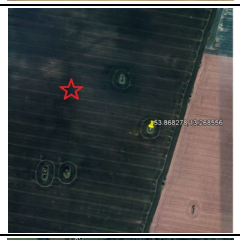
HYPERNETS dataset corresponds to the reflectance L2A products measured with HYPSTAR®-XR sensors every 30 minutes between 9:00 and 17:00 (Universal Time Coordinated - UTC) from different set of geometries (full zenith/azimuth coverage, up to 60° from nadir). The HYPSTAR®-XR (eXtended Range) instruments deployed at each land HYPERNETS site consist of a VNIR and a SWIR sensor and autonomously collect data between 380-1700 nm at various viewing geometries and send it to a central server for quality control and processing. The HYPERNETS processor (Goyens et al., 2021) automatically processes all this data into various products, including the L2A surface reflectance product used here.

The data used in this QAR corresponds to initial test samples published data (De Vis et al., 2023; Morris et al., 2023; Saberioon et al., 2023) over three selected sites (Table 6) over homogeneous land

[6] www.hypernets.eu (access on 14th January 2025)

cover types. Other HYPERNETS sites are not used as they are installed over heterogeneous areas or fields of size less than 100 m x 100 m, which are not suitable for validation of Sentinel-3 products with nominal spatial resolution of 300 m. At the time of this analysis, these datasets are limited in terms of temporal coverage and angular configuration availability as they are a subset of the complete data record expected as outcome of the project.

Table 6: Characteristics of the three HYPERNETS sites considered in this study.

ID Site	Name (Country)	Latitude; Longitude	Temporal coverage	Land Cover	Reference	Google Earth view (1 km x 1 km)
GHNA	Gobabeb (Namibia)	-23.60153°N; 15.12589° E	May 2022 – April 2023	desert	(De Vis et al., 2023)	
DEGE	Demmin (Germany)	53.868278°N; 13.268556°E(*)	July 2021 – April 2023	agricultural	(Saberioon et al., 2023)	
WWUK	Wytham Woods (UK)	51.777206ºN; 1.338494ºW	April 2022 – April 2023	DBF	(Morris et al., 2023)	

(*) For DEGE, the Sentinel-3 extraction coordinate was moved towards the center of the field in order to cover the same field measured by the HYPERNETS (see red star in the Google Earth view).

In summary, HYPERNETS complements the direct validation of Sentinel-3 TOC reflectance V2.3 using the RadCalNet network, which does not cover all surface types and viewing angles (RadCalNet provides only nadir-viewing data). Two assumptions are considered to deal with differences in data acquisition between Sentinel-3 TOC reflectances and HYPERNETS in situ data:

- Similar illumination and viewing configurations are considered. For that, Sentinel-3 and HYPERNETS observations with differences in viewing zenith angles larger than 5º are discarded. Same dates and acquisition times are compared.

- The spectral range of HYPERNETS VNIR sensors is from 380 to 1000 nm (3 nm spectral interval), and the spectral range of HYPERNETS SWIR sensor is from 1000 and 1700 nm (10 nm spectral interval). These narrow-band measurements are made comparable with Sentinel-3 channels using the SRF coefficients of Sentinel-3 [S3-OLCI TG, S3-SLSTR TG]. Note that HYPERNETS measurements do not cover the spectral resolution of SLSTR S6 channel.

3.3.3 ESA Fiducial Reference Measurements (FRM)

The Quality Assurance Framework for Earth Observation (QA4EO) guidelines state that Earth Observation (EO) data and derived products should be provided with an indicator of quality traceable to reference standards (preferably SI) so that users can assess suitability for their application (i.e., their fitness for purpose). Key element of QA4EO relates to validation through independent reference data. Consequently, the reference data must also be characterised following QA4EO principles. The need to evidence these principles for Cal/Val was the basis to create the term "Fiducial Reference Measurement" (Goryl et al., 2023) and the ESA FRM programme. This aims to provide fully characterized and independent in situ data with uncertainties where the measurements are SI-traceable, or community-agreed standards, to underpin the calibration and validation of satellite products.

In this study, surface reflectance FRMs from Las Tiesas, Barrax, Spain (Figure 3) collected the 2nd August 2018 in the framework of the FRM4Veg project[7] are used (Origo et al., 2020). Measurements were made over alfalfa and bare soil surface types. The procedure utilized several key apparatuses in order to collect the appropriate data for the validation: a spectrometer, spectralon panel, and sunphotometer. The spectrometer has a 400 nm – 2500 nm wavelength range with a variable bandwidth of 3 nm at 700 nm and 10 nm at 1400/2100 nm. These measurements are made comparable with Sentinel-3 data using the SRF coefficients of Sentinel-3 [S3-OLCI TG, S3-SLSTR TG]. The measurement window was selected with the aim of conducting measurements no more than 30 min either side of the Sentinel-2A overpass, and these measurements could be useful for validating Sentinel-3 derived products as well. This was done to minimize the change in solar zenith angle (θ_s) during the measurement window and thereby minimize any BRDF effects. For each of the cover types (alfalfa and soil), 10 sample points were selected over a 200 × 200 m^2 area, where it was ensured that every point was at least 50 m from the cover type.

[7] https://frm4veg.org/ (accessed on 2025.02.05)

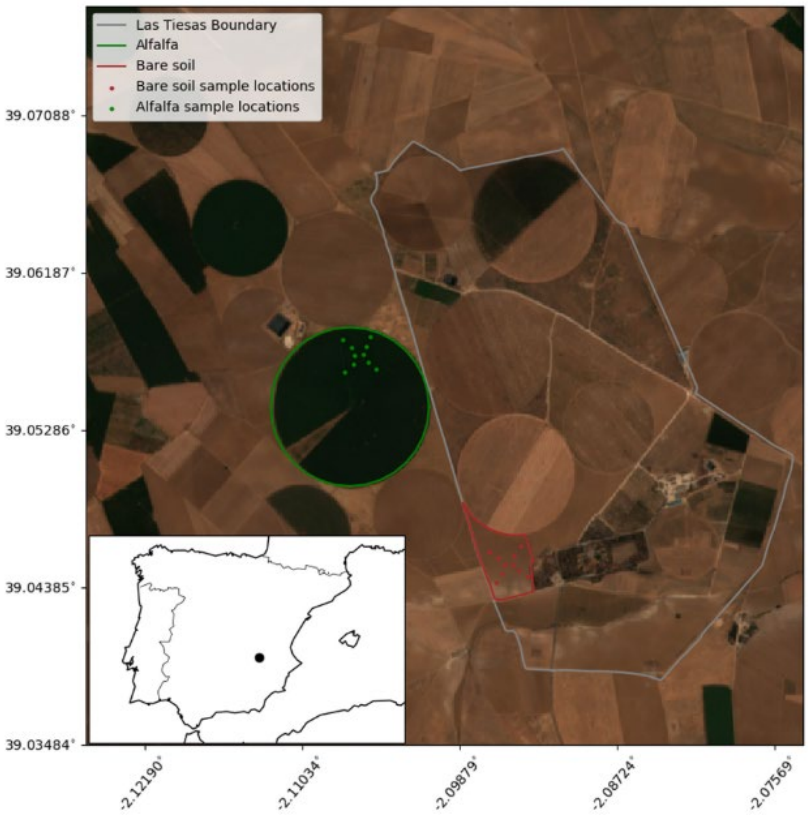


Figure 3: The location of the field campaign with sample points shown. Las Tiesas farm boundary shown in grey, with boundaries and sample points of the Alfalfa and bare soil shown in green and red, respectively (source: Origo et al., 2020).

Figure 4 displays the spatial distribution of the 20 FRMs (10 points of alfalfa + 10 point of bare soil) over the Sentinel-3 TOC reflectance V2.3 (OLCI Oa17 band) image. To compare the FRMs against Sentinel-3 pixels, the average value of the 10 points of each cover type was performed and compared to the value of the Sentinel-3 pixel that contains most of these points. In case of the Alfalfa, the measurements are distributed along two Sentinel-3 different pixels, and the mean value of both pixels is calculated. In case of bare soil, most of the FRM measurements are located over the same pixel, and the TOC reflectance value of this pixel was used.

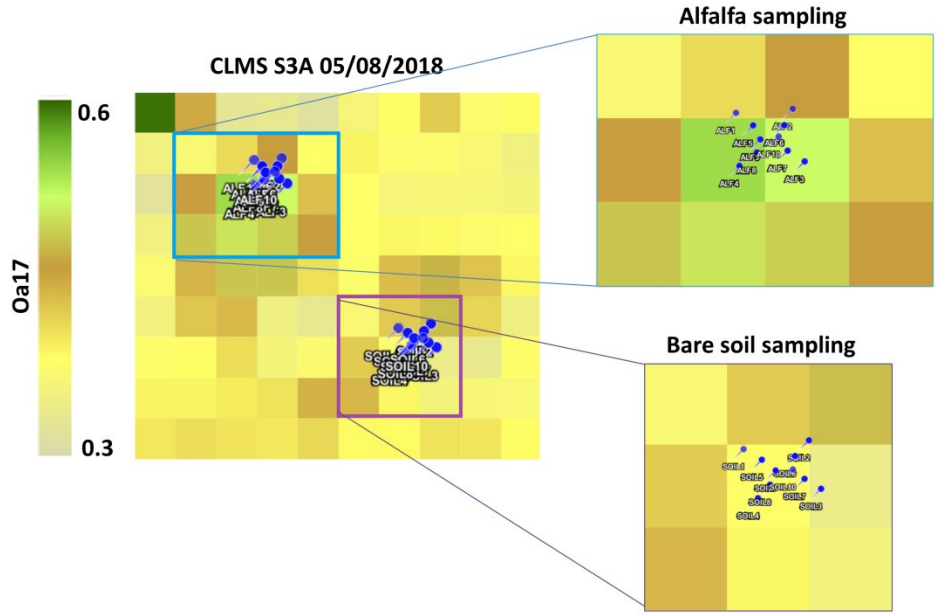
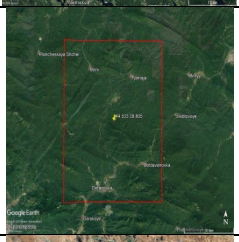
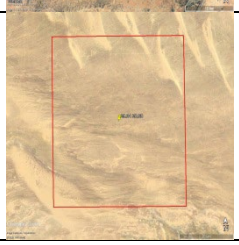


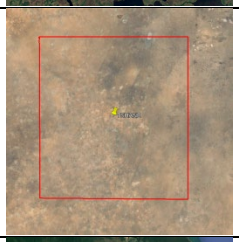
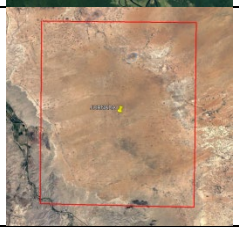
Figure 4: The location of the 10 sample points of each cover type (Alfalfa A1 – A10, bare soil S1 – S10) over Sentinel-3 TOC reflectance V2.3 Oa17 band.

3.4 AREAS OF INTEREST (AOIS)

Ten AOIs (approximately 25km x 25km) representatives of the main biome type have been selected for the spatial consistency analysis and error evaluation in the product intercomparison between Sentinel-3 CLMS and ESA products (see Table 7). The majority class of each AOI is defined according to the Global Land Cover 100m (GLC-100m) epoch 2015 classification (Buchhorn et al., 2020).

Table 7: Characteristics of the selected 25 km x 25 km AOIs.

AOI	Biome	Latitude / Longitude	Google Earth View
#1 Vieremä (Finland)	NLF	63.801°N / 26.687°E	
#2 Tjamaja (Russia)	DBF	44.533°N / 38.805°E	
#3 Barrax (Spain)	Cultivated	39.0544°N / -2.1007°E	
#4 Azuel (Spain)	Herbaceous	38.3°N / -4.45°E	
#5 Ouled Zouyad (Tunisia)	Shrublands	34.7228°N / 9.4836°E	
#6 Calibration Site (Egypt)	Bare Areas	29.34°N / 28.58°E	

#7 Oblast de Murmansh (Russia)	Snow & Ice (in winter)	66.494°N / 37.986°E	
#8 Tshane (Botswana)	Savannah	24.2872°S / 21.9572°E	
#9 Daintree Rainforest (Australia)	EBF	16.0976°S / 145.2948°E	
#10 Jornada (New Mexico)	Shrublands	32.6332°N / 106.8504°W	

4 RESULTS

4.1 PRODUCT CONTENT

Figure 5 shows the time series of available TOC reflectances counts per day over the full analysis period of 2019-2023. The count includes all TOC tiles with at least one land pixel, so sea-only tiles are filtered out. A seasonal variability is visible with more observations available during the March-October period (summer period in the northern hemisphere) and less from October to February (winter period in the northern hemisphere). Apart from this low frequency variation, there are also small, daily fluctuations present. These are inherent to the slightly different orbit coverage from day to day. The timeseries also displays a few anomalies. The large dips can be identified in the ESA OLCI and SLSTR anomaly reports [ESA-S3-OLCI Anomalies and Events, ESA-S3-OLCI Anomalies and EventsESA-S3-SLSTR Anomalies and Events] as periods of instrument maintenance and instrument anomalies. To identify other possible problems, we analyse one year (2019) in more detail in the sections below.

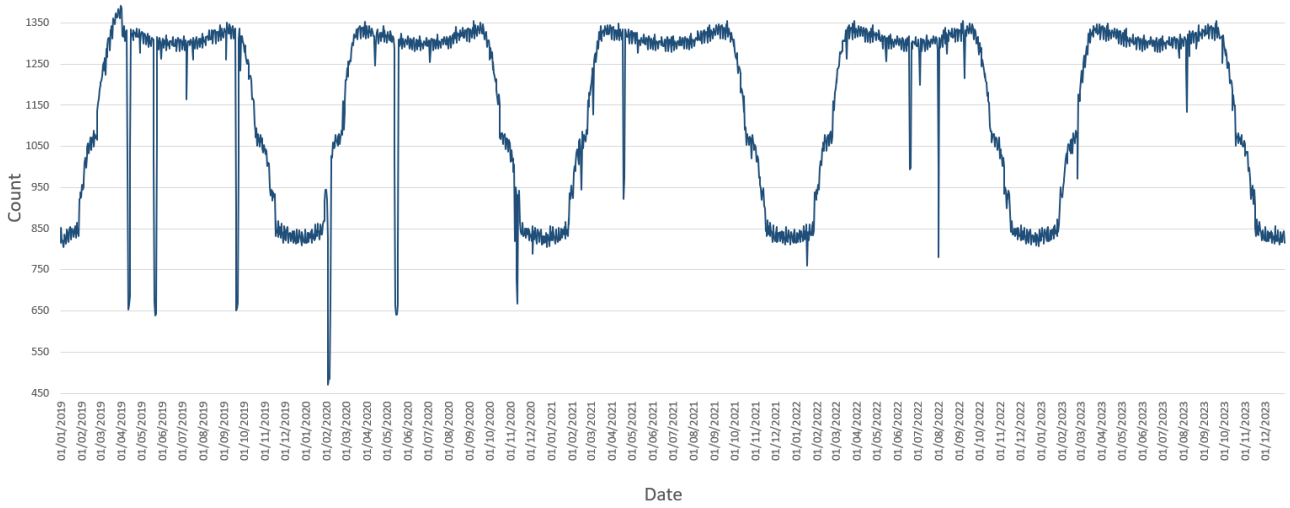


Figure 5: Number of TOC reflectance frames per day for the period 2019-2023.

4.1.1 TOC Reflectance bands

Sentinel-3 TOC reflectance values for 15 OLCI bands and 5 SLSTR bands (see Annex I) are given in the corresponding **Oa[xx]_toc** and **S[x]_an_toc** layers. We generated global maps of Sentinel-3 CLMS TOC V2.3 good quality pixels (Table 3) at reduced resolution (1/84) for the selected period of January 2019 – December 2019. To check the coverage and possible artefacts, reflectances from one OLCI (Oa08) and one SLSTR (S2) channel are displayed for this whole period. Additionally, all OLCI and SLSTR bands are checked for two whole months of data (April 2019 and August 2019).

Figure 6 and Figure 7 show daily global mosaics for the first day of each month in 2019 of Sentinel-3 OLCI Oa08 and SLSTR S2 TOC reflectance values, respectively.

The main findings of the visual inspection of these global maps are:

- The distribution of CLMS TOC V2.3 values is consistent for both OLCI and SLSTR (see Figure 6 and Figure 7) without suspicious pattern for almost all the dates.
- Due to the use of different quality flags, SLSTR tends to provide a higher number of 'good quality' TOC reflectances than OLCI. This is at the cost of a large fraction of unrealistically high TOC reflectances (i.e., undetected clouds) values over equatorial areas and other areas around the whole globe.
- At full resolution maps show data gaps at high latitudes and in the edge of the swath (Figure 8). This is a known issue, related to the reprojection to CLMS grid, as described in the PUM [CGLOPS1_PUM_TOC-reflectance300m-V2.3].
- Some patterns of missing data were identified in the global inspection of the maps. Missing data patterns were divided into four groups: 3' frame missing data (see Figure 9), full orbits missing data (see Figure 10), full images missing data, and vertical bands (see Figure 11). Table 8 shows the calendar of missing data from this analysis of the visual inspection of the global maps. This calendar was compared to the ESA anomalies and events calendar [ESA-S3-OLCI Anomalies and Events].
    - 3' frame missing data occur because some frames are not processed. All occurrences can be linked to an issue in the ESA anomalies report. Explanations are sequencing errors and uncorrectable frames due to a drop in the signal, a gap due to instrument calibration, missing frames due to radio frequency interference, a backend problem, and pointing errors.
    - Full orbits missing data is defined as images for which at least one orbit is missing. This could affect any orbit. According to the ESA OLCI anomaly report, these missing orbits are caused by lost signals due to antenna issues, or payload data handling unit memory scrubbing activities.
    - Full image missing data means that there is no data available for a certain date. It is the least frequent event. For none of the occurrences, an explanation can be found in the ESA OLCI anomaly report. However, two of them are mentioned in the ESA SLSTR anomalies report [ESA-S3-SLSTR Anomalies and Events]. The TOC v2.3 data is only available when there are both OLCI and SLSTR data, therefore the full missing image data gaps are also detected in the OLCI data. The full image missing data sequence of 19-23/09/2019 is not mentioned in the anomalies report, but after some investigation it was found that this gap is caused by the absence of SLSTR data for this period due to a scheduled decontamination process.

- Vertical bands are missing segments of the orbit along the movement direction of the satellites. This only occurred from 29-31/05/2019 and an instrument error causing processing errors due to the data content was recorded for this period in the ESA anomalies report. The vertical bands can be explained by one or more missing OLCI camera inputs.

In summary, all the missing TOC data patterns can be identified in the ESA anomaly reports or explained due to missing L1 input data.

Table 8: Overview of missing data patterns in the Sentinel-3 Oa08 TOC V2.3 reflectance data for year 2019, in pink for S3A and in blue for S3B. There are five missing data patterns detected: 3' frames missing (T), full orbits missing (FO), full image missing (FI), gaps (G), and vertical bands (B).

Month	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Jan				T		FO																									
Feb																															
Mar					T				FO						T																
Apr					T						FO	FI	FI	FI	FO	T	T												FO		
May																				FO	FI	FI	FI	FO					B	B	B
Jun									T	T											T										
Jul	T							FO																							
Aug																															
Sep					FO														FO	FI	FI	FI	FO			FO					
Oct							T										T													FO	
Nov												T																			
Dec																															

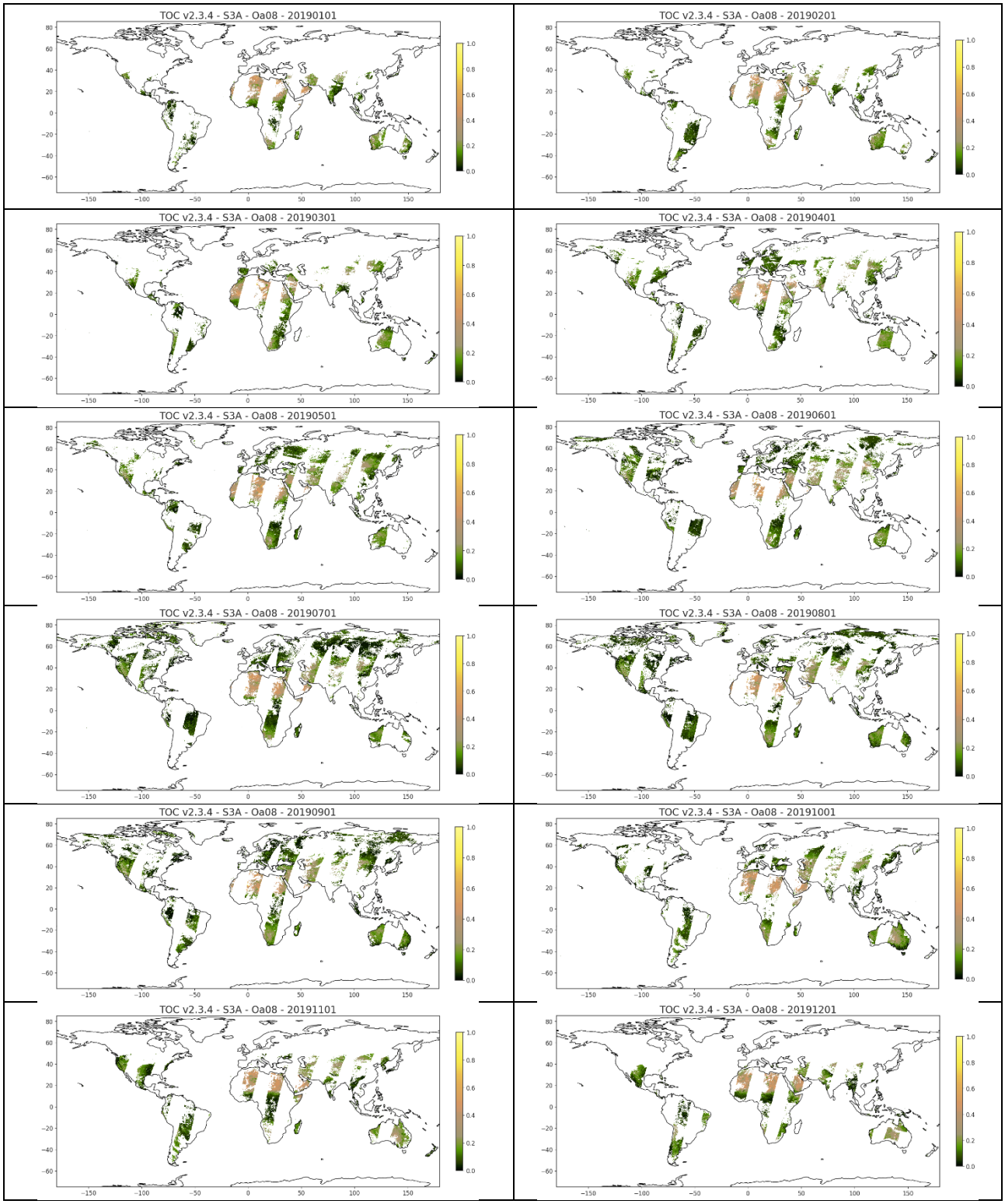


Figure 6: Global maps of good quality Sentinel-3A OLCI (Oa08) CLMS TOC V2.3 observations for year 2019 (1st day of each month). White land pixels correspond to no data or suboptimal quality pixels according to the annotation flags.

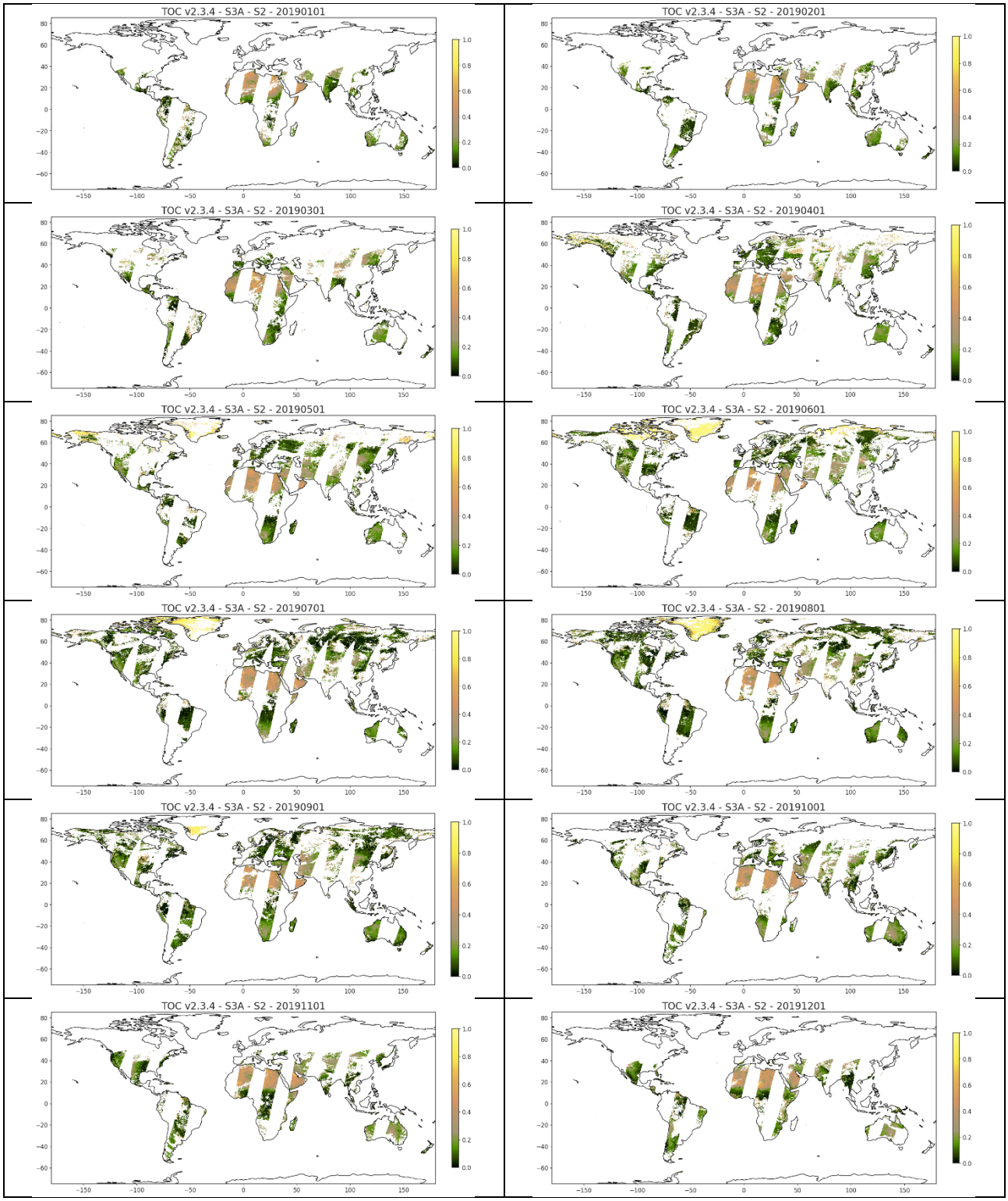


Figure 7: As in Figure 6, but for Sentinel-3A SLSTR (S2) CLMS TOC V2.3 reflectance.

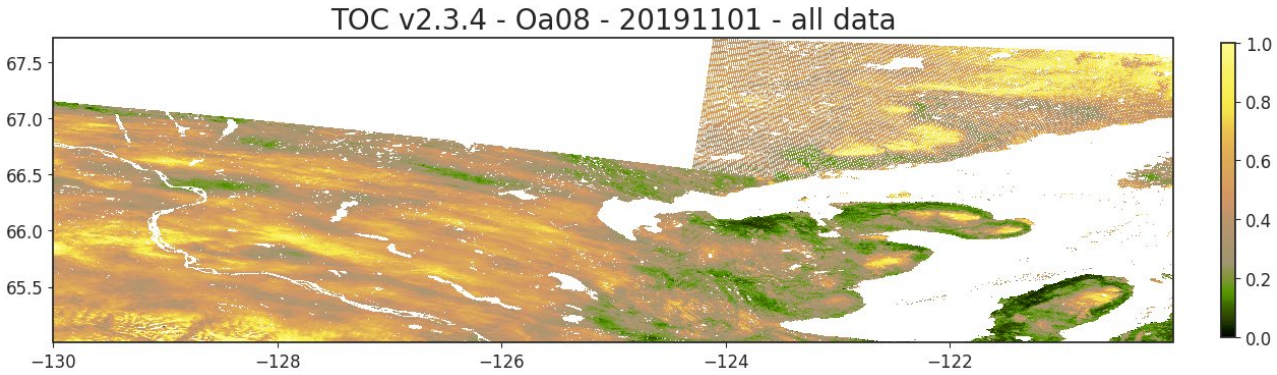


Figure 8: Sentinel-3 Oa08 CLMS TOC V2.3 reflectances for tile X05Y01 on 01/11/2019 at full resolution. Data gaps are visible in the upper right part of the image.

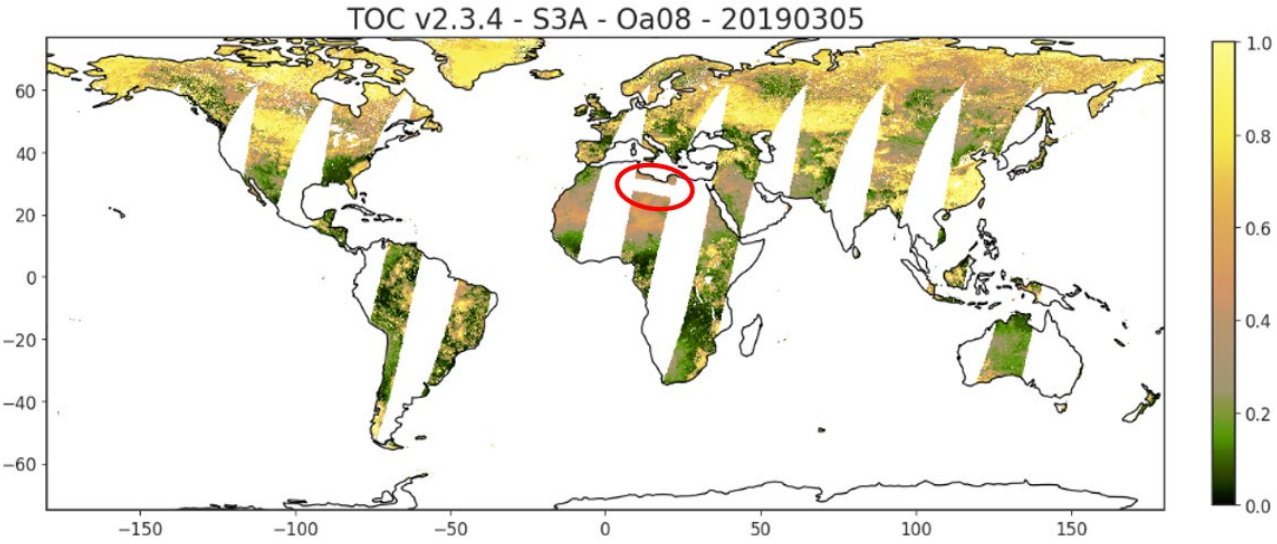


Figure 9: Global map of Sentinel-3A OLCI (Oa08) CLMS TOC V2.3 observations for 05/03/2019, containing 3' frame missing data (red circle).

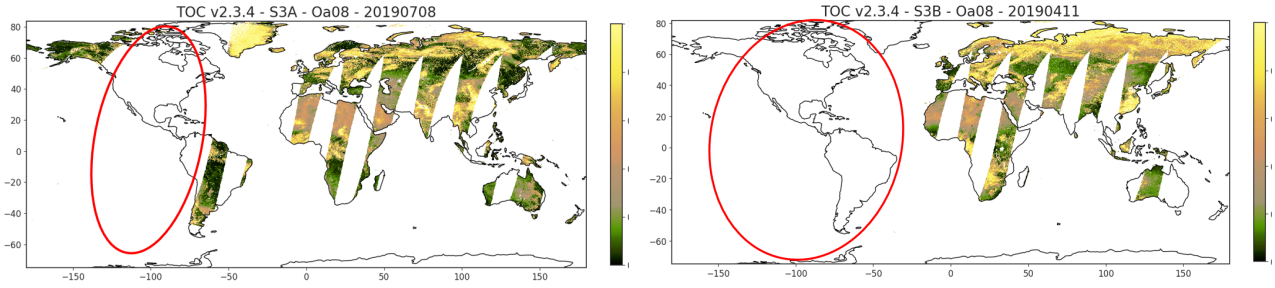


Figure 10: Global maps of Sentinel-3 OLCI (Oa08) CLMS TOC V2.3 observations for S3A 08/07/2019 (left) and S3B 11/04/2019 (right), showing whole orbits with missing data.

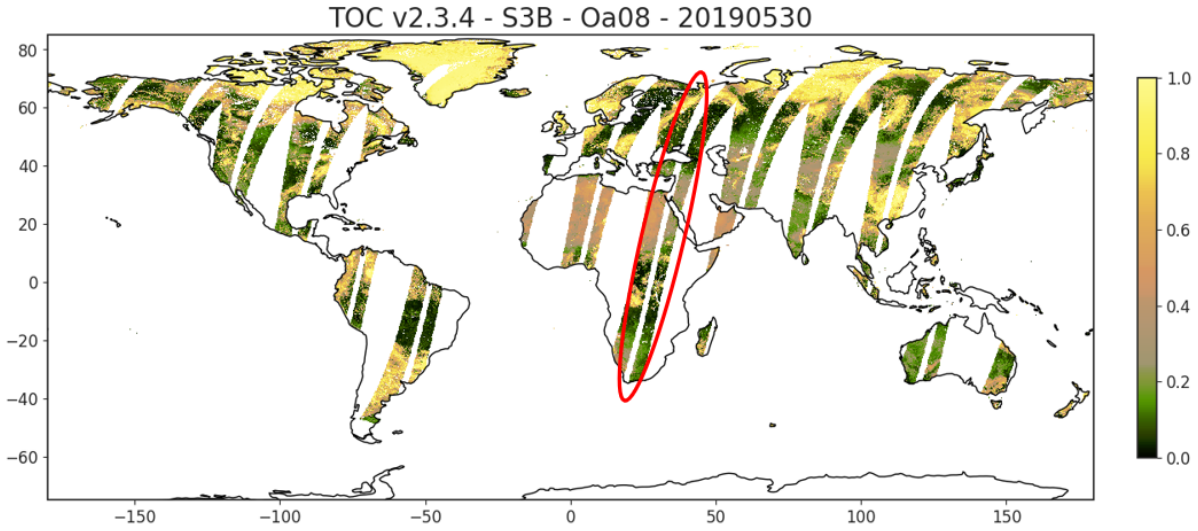


Figure 11: Global map of Sentinel-3B OLCI (Oa08) CLMS TOC V2.3 observations for 30/05/2019, containing vertical bands with missing data in the center of the orbits.

4.1.2 Associated TOC reflectance errors bands

The TOC reflectance physical values have associated errors which are given in **the Oa[xx]_toc_error** bands for OLCI and in **S[x]_an_toc_error** bands for SLSTR. It should be noted that TOC reflectance V2.3 incorporates improved TOC reflectances error assessment compared to previous V1, as the uncertainty due to the aerosol model is now taken into account [CGLOPS1_ATBD_S3-AC-V1.1].

Figure 12 and Figure 13 show the associated errors for some bands (Oa02_toc_err, Oa06_toc_err, Oa08_toc_err, Oa12_toc_err, and Oa17_toc_err for OLCI and S1_an_toc_err, S2_an_toc_err, S3_an_toc_err, and S6_an_toc_err for SLSTR), and the relative errors (in percentage) with respect to the associated TOC reflectance physical values bands. Visual inspection of the TOC reflectance error layers global maps shows:

- OLCI TOC errors are typically between 0.02 and 0.03 except in some non-vegetated areas where the errors could be higher due to the impact of aerosols. Relative errors are typically between 2% and 6%, except in the blue region (Oa02) where the TOC values are lower, and the relative errors are typically above 8%.
- SLSTR TOC errors are, like the OLCI TOC errors, lower than 0.03 over almost the whole globe. Relative errors are typically between 2% and 6%.
- The errors and relative errors have similar global patterns when comparing the five OLCI TOC channels with each other, and for the four SLSTR channels.
- OLCI and SLSTR show similar errors and relative errors for equivalent TOC channels.

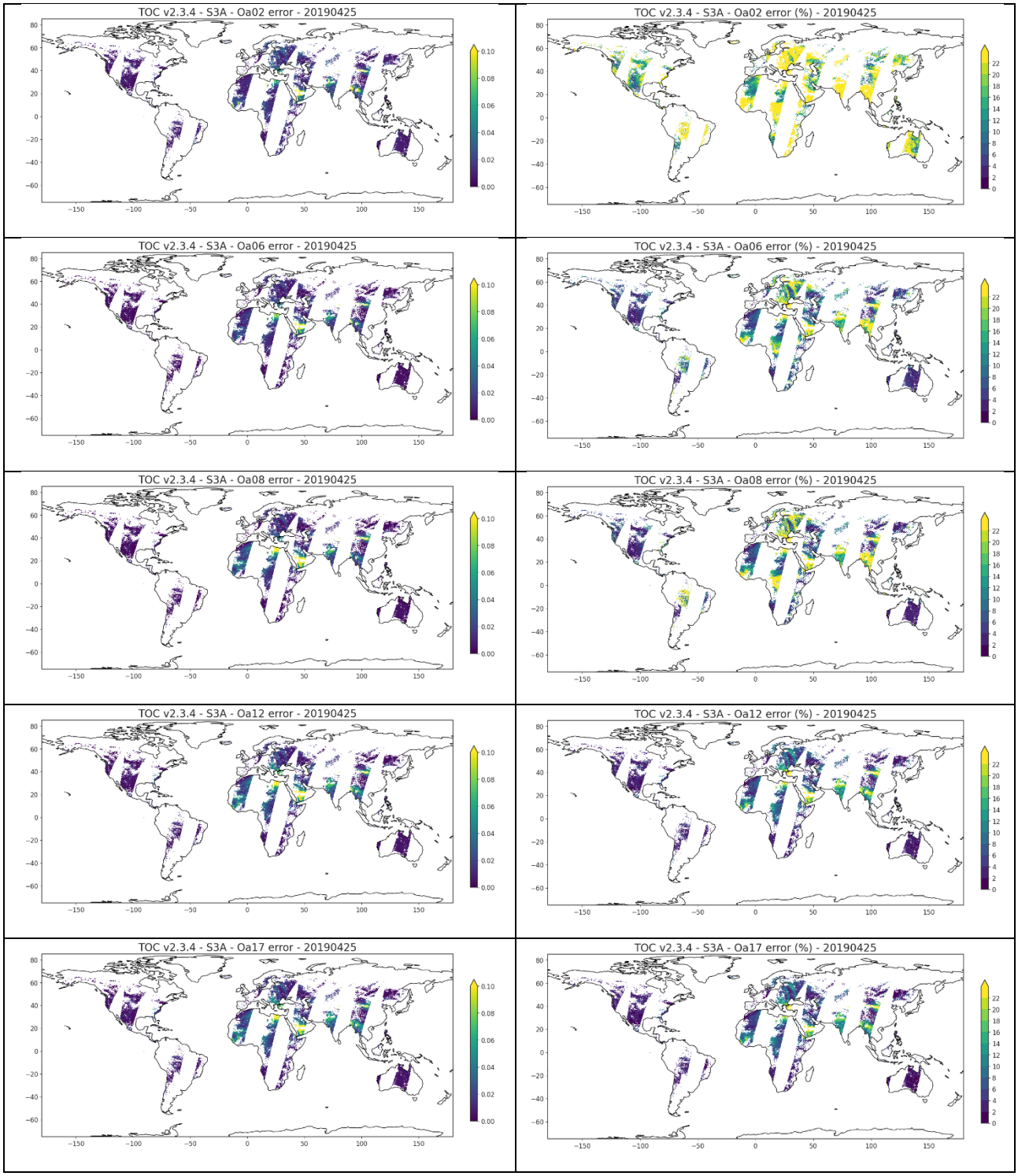


Figure 12: Global maps of Sentinel-3A OLCI absolute (*left*) and relative (*right*) CLMS TOC V2.3 reflectance errors for all good quality observations on 25/04/2019. White land pixels correspond to no data or suboptimal quality pixels. From *Top* to *Bottom*: Oa02, Oa06, Oa08, Oa12, and Oa17.

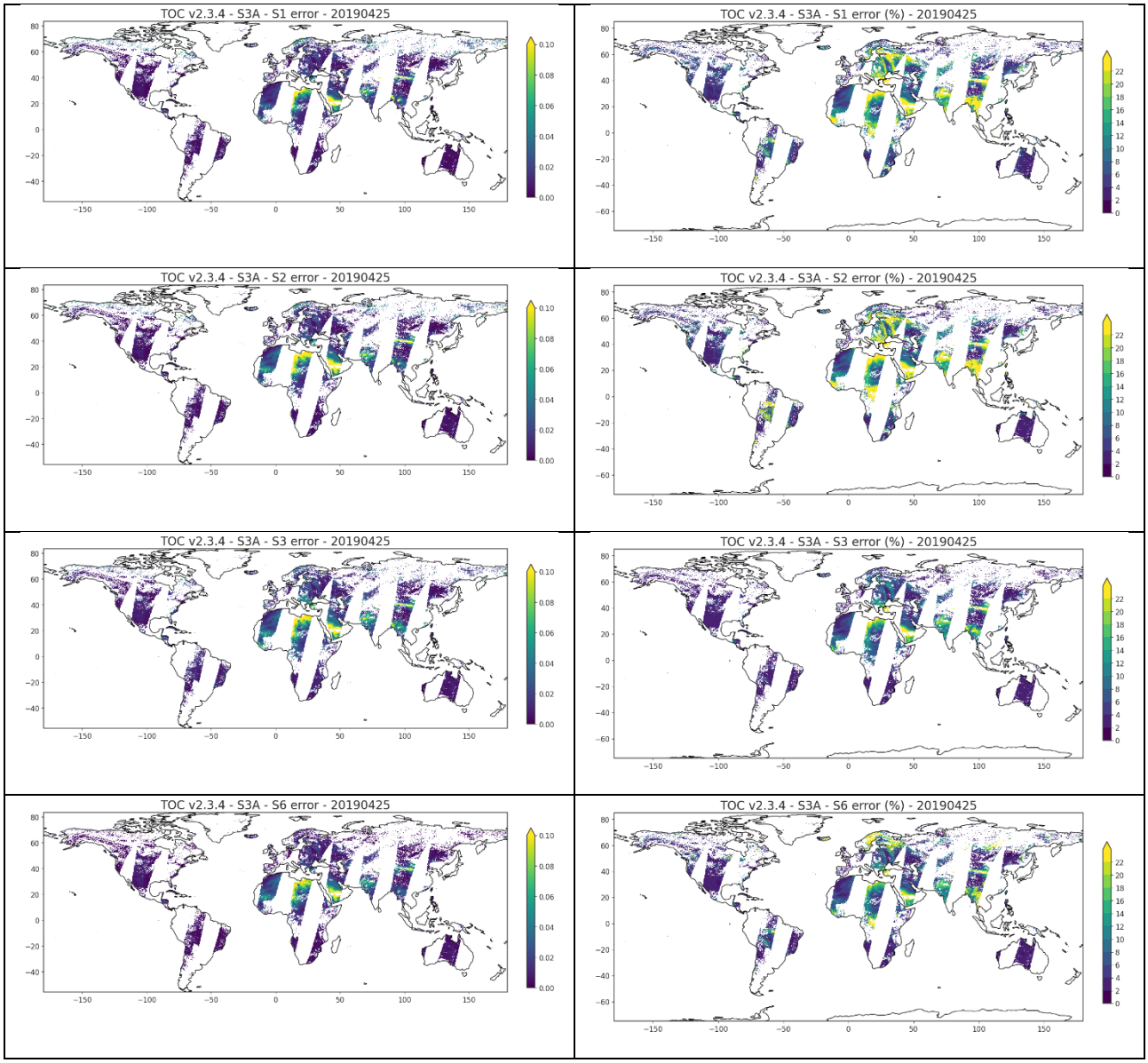


Figure 13: Global maps of Sentinel-3A SLSTR absolute (*left*) and relative (*right*) CLMS TOC V2.3 reflectance errors for all good quality observations on 25/04/2019. White land pixels correspond to no data or suboptimal quality pixels. From *Top* to *Bottom*: S1, S2, S3, and S6.

4.1.3 Angles bands

The Sentinel-3 TOC reflectance includes the angles of the OLCI and SLSTR observations. The provided angles are Solar Zenith Angle (SZA), Viewing Zenith Angle (VZA), Solar Azimuth Angle (SAA) and Viewing Azimuth Angle (VAA). The physical value range of zenith angles is [0º, 180º].

The physical value range of OLCI azimuth angles is [-180º, 180º] while it is [0º, 360º] for SLSTR azimuth angles.

Figure 14 and Figure 15 show the SZA and VZA maps for two selected dates (15/04/2019 and 15/08/2019) for OLCI (top figures) and SLSTR (middle). The maps of differences are also presented (bottom figures). The difference in VZA between OLCI and SLSTR is typically between 0º and 2º (and up to 5º in the extreme of the swath) while the differences in SZA is typically lower than 0.2º (very low, as expected).

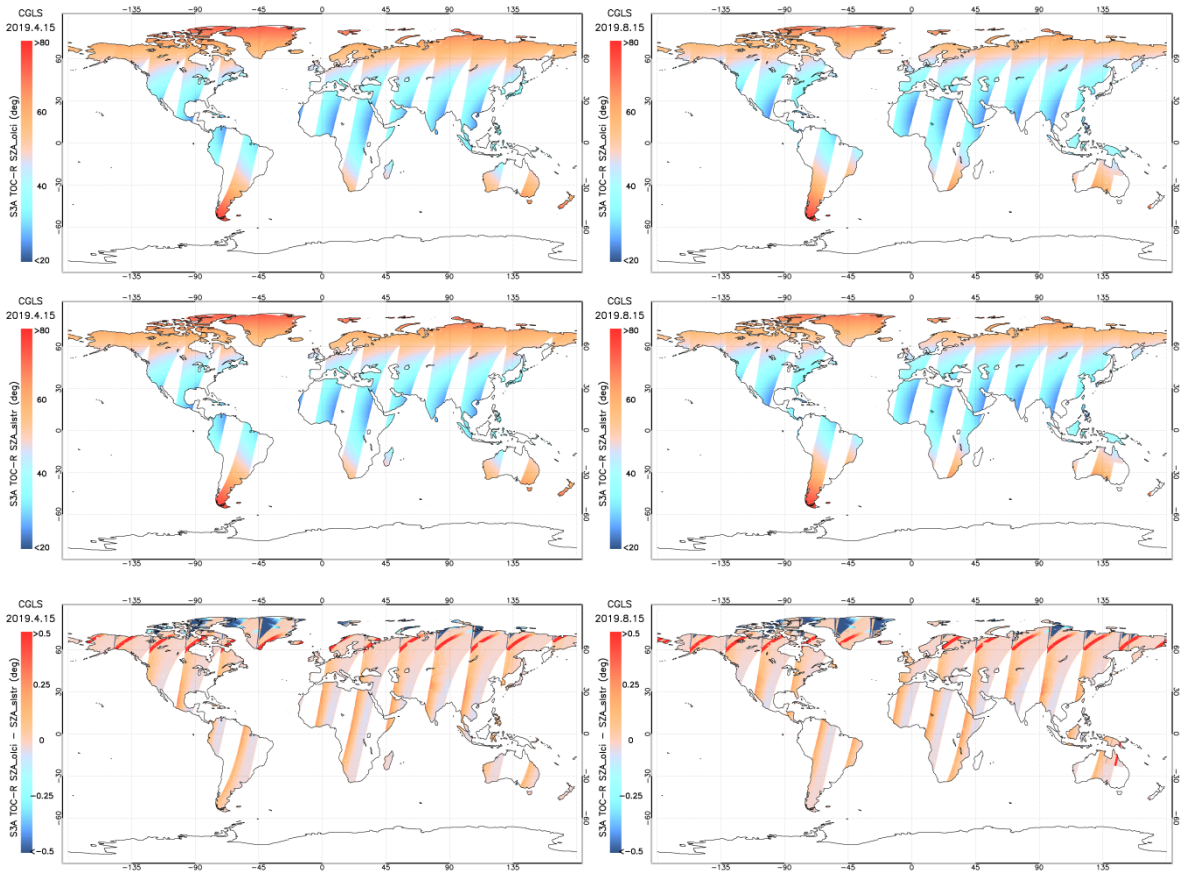


Figure 14: Global maps of Sentinel-3 CLMS TOC V2.3 OLCI (top) and SLSTR (middle) SZA and SZA difference maps between both (OLCI – SLSTR, bottom) for two dates: 15/04/2019 (left) and 15/08/2019 (right).

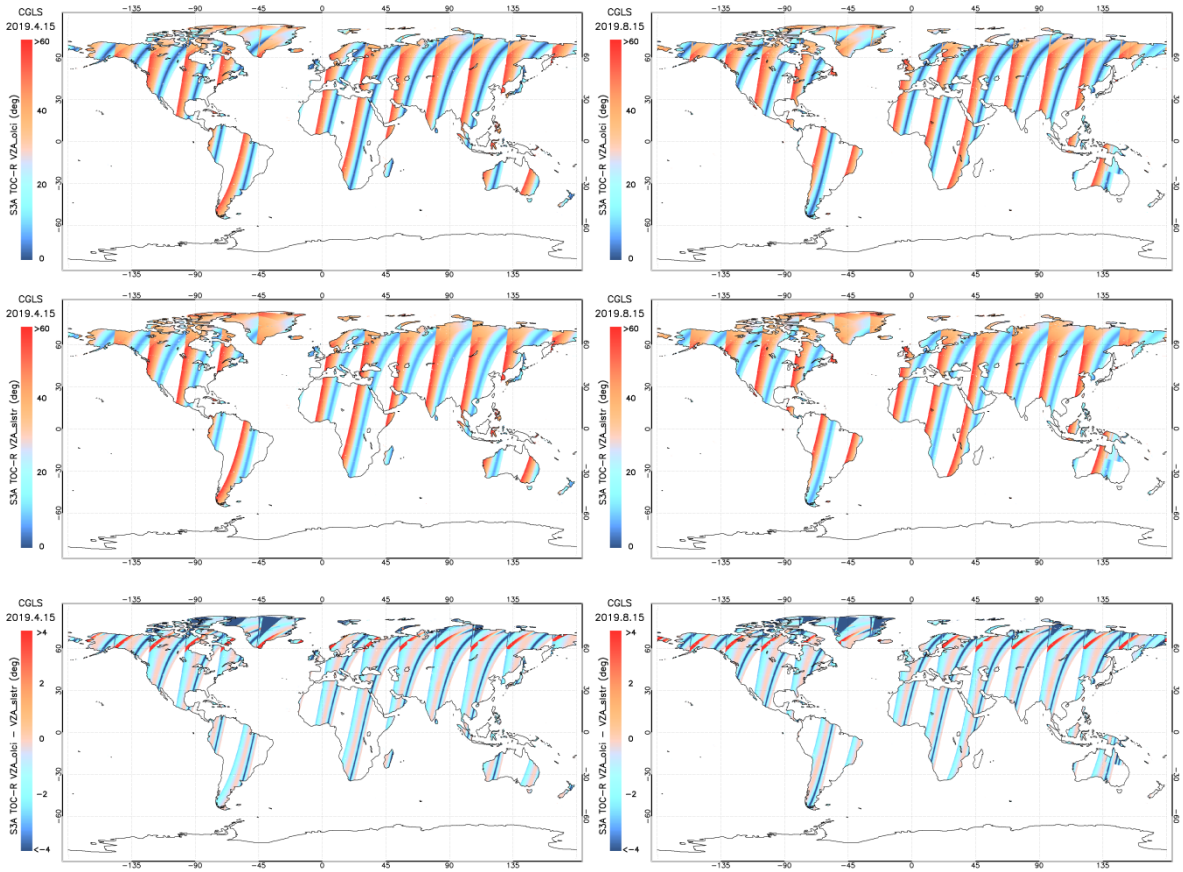


Figure 15: Global maps of Sentinel-3 CLMS TOC V2.3 OLCI (top) and SLSTR (middle) VZA and VZA difference maps between both (OLCI – SLSTR, bottom) for two dates: 15/04/2019 (left) and 15/08/2019 (right).

Figure 16 and Figure 17 show the SAA and VAA maps for the same dates (15/04/2019 and 15/08/2019) for OLCI (top figures) and SLSTR (middle). OLCI and SLSTR VAA are defined differently:

- Between -180° and 180° for OLCI, with negative angles for east-looking and positive angles for west-looking.
- Between 0° and 360° for SLSTR, higher than 180° for east-looking and lower than 180° for west-looking.

Consequently, VAA maps are different. OLCI and SLSTR SAA maps display similar values despite of the angles are differently defined.

Note that SLSTR VAA and SAA need special attention when reading in (see guidelines in section 5.3 of PUM [CGLOPS1_PUM_TOC-reflectance300m-V2.3]).

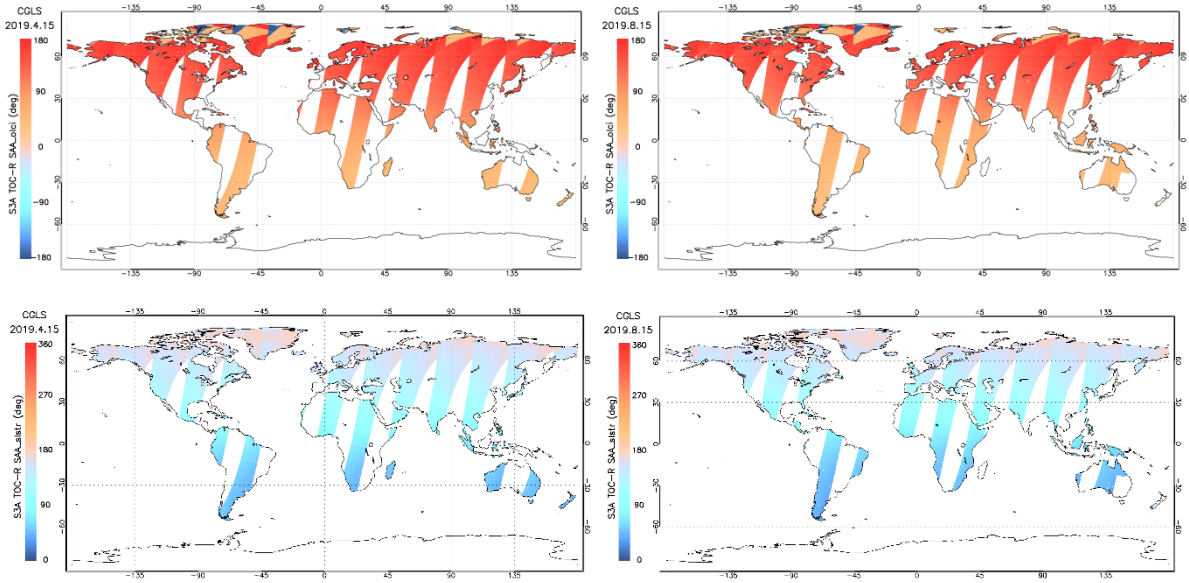


Figure 16: Global maps of Sentinel-3 CLMS TOC V2.3 OLCI (top) and SLSTR (bottom) SAA for two dates: 15/04/2019 (left) and 15/08/2019 (right).

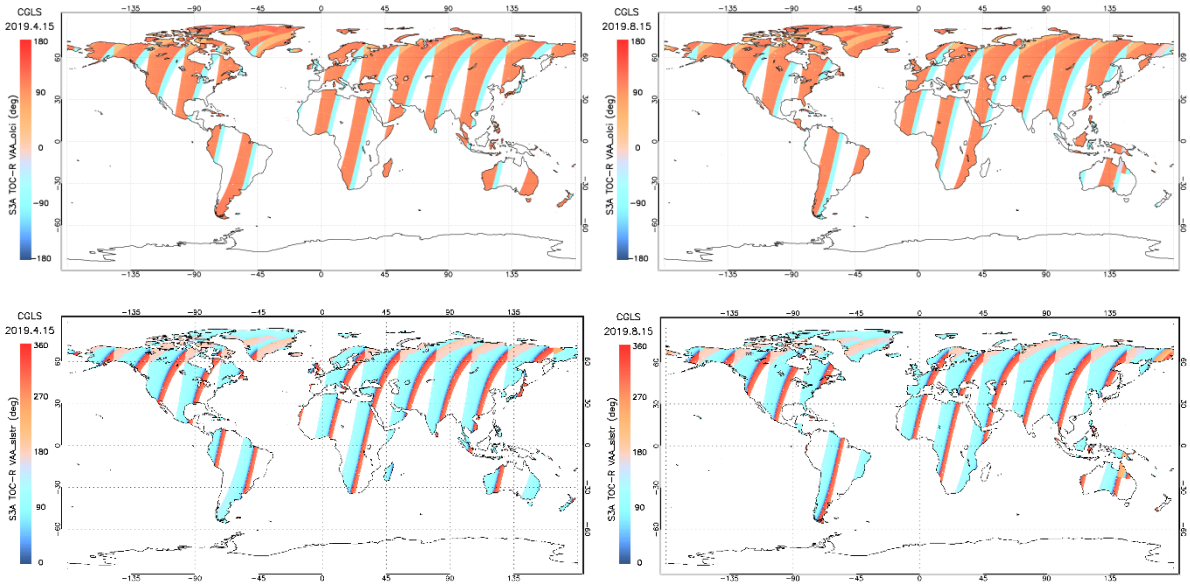


Figure 17: Global maps of Sentinel-3 CLMS TOC V2.3 OLCI (top) and SLSTR (bottom) VAA for two dates: 15/04/2019 (left) and 15/08/2019 (right).

4.1.4 Annotation Flags

The Sentinel-3 TOC reflectance products provide four quality information layers, which are **cloud_an**, **quality_flags**, **pixel_classif_flag** and **AC_process_flag** (Annex II). Although all the layers are provided in the same product, **quality_flags** and **pixel_classif_flag** refer to OLCI, **cloud_an** refers to SLSTR, and **AC_process flag** is common to both sensors.

The impact of the use of each quality information layer at global scale is analysed for OLCI (Oa08_toc channel) and SLSTR (S3_an_toc channel) at reduced resolution (1/84) in Figure 18 and Figure 19 for a date in June (15/06/2019) and in Figure 20 and Figure 21 for a date in December (15/12/2019). The top of the figure shows the global mosaics without the use of quality layers, only the IdePix land classification is applied to filter out sea pixels. The middle figure displays the same global maps when each quality layer is used. The resulting map with the combination of all recommended quality layers can be found at the bottom of the figure.

The main findings in the use of quality layers are:

- For OLCI (Figure 18 and Figure 20):
  - **Pixel_classif_flags** is the output of IdePix (the OLCI cloud and snow detection module) and the most restrictive quality flag layer. Most of unrealistic TOC reflectance retrievals (i.e. cloud/snow contaminated) are filtered when this layer is used. However, other realistic retrievals are wrongly filtered when it is used.
  - **Quality_flags**, indicative of band saturation (from OLCI Level 1B), filters a low number of pixels.
- For SLSTR (Figure 19 and Figure 21):
  - **Cloud_an** is the cloud detection module from SLSTR Level 1B. It filters a large fraction of contaminated pixels, especially in northern regions.
- For both OLCI and SLSTR:
  - **AC_process_flag** is the output of atmospheric correction module. It filters a large area of pixels at northern regions in wintertime (December) because it excludes pixels when SZA is greater than 65º. For the same reason, a lot of pixels are filtered away in southern regions in June (the local wintertime).

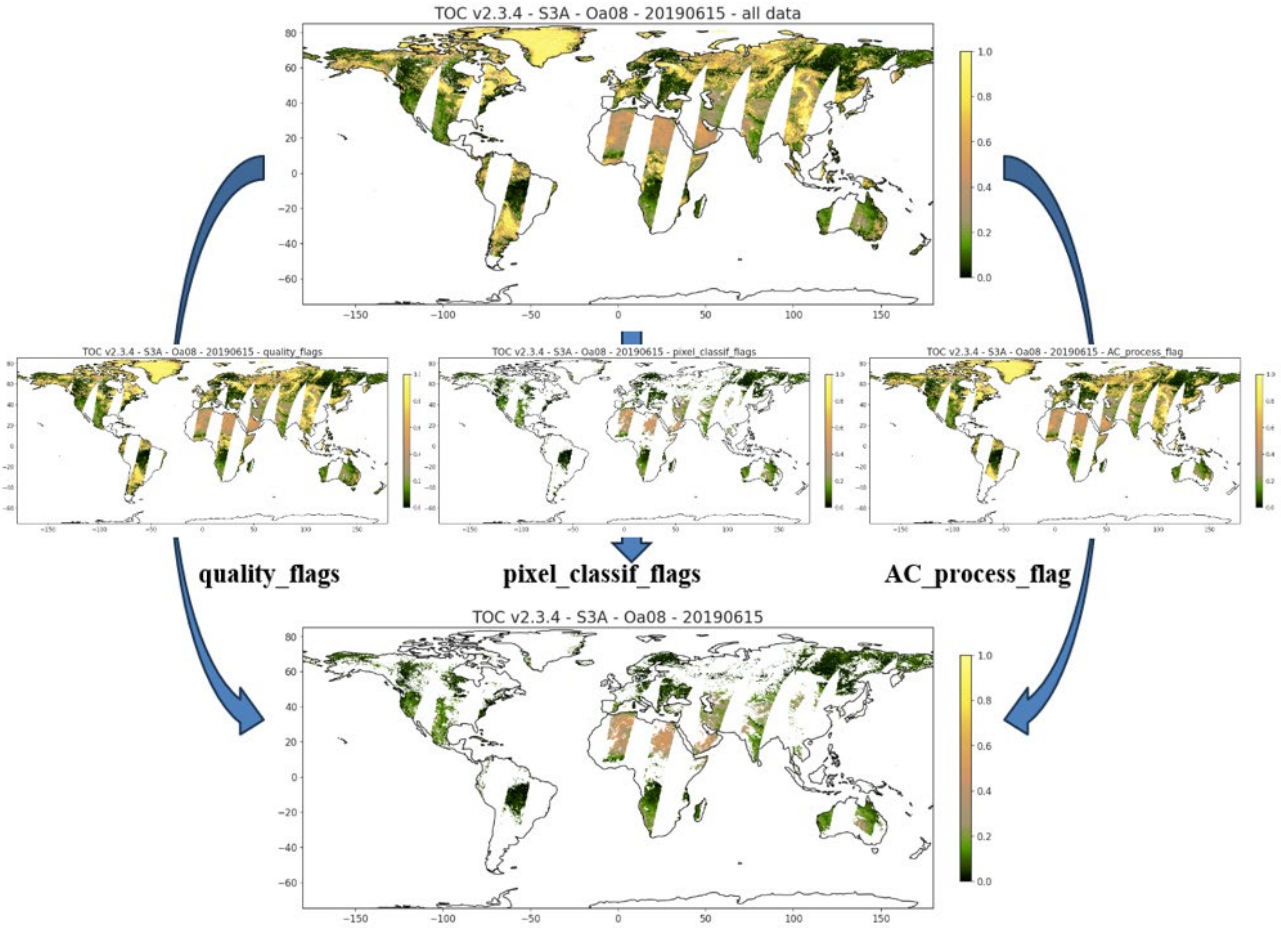


Figure 18: ***Top:*** **Global mosaic (all pixels) of Sentinel-3A OLCI TOC reflectance (Oa08) for 15/06/2019 at reduced spatial resolution (1/84).** ***Middle:*** **global mosaic when applying recommendations for each quality layer: quality_flags, pixel_classif_flags and AC_process_flag.** ***Bottom:*** **global mosaic when applying all quality annotation flags recommendations.**

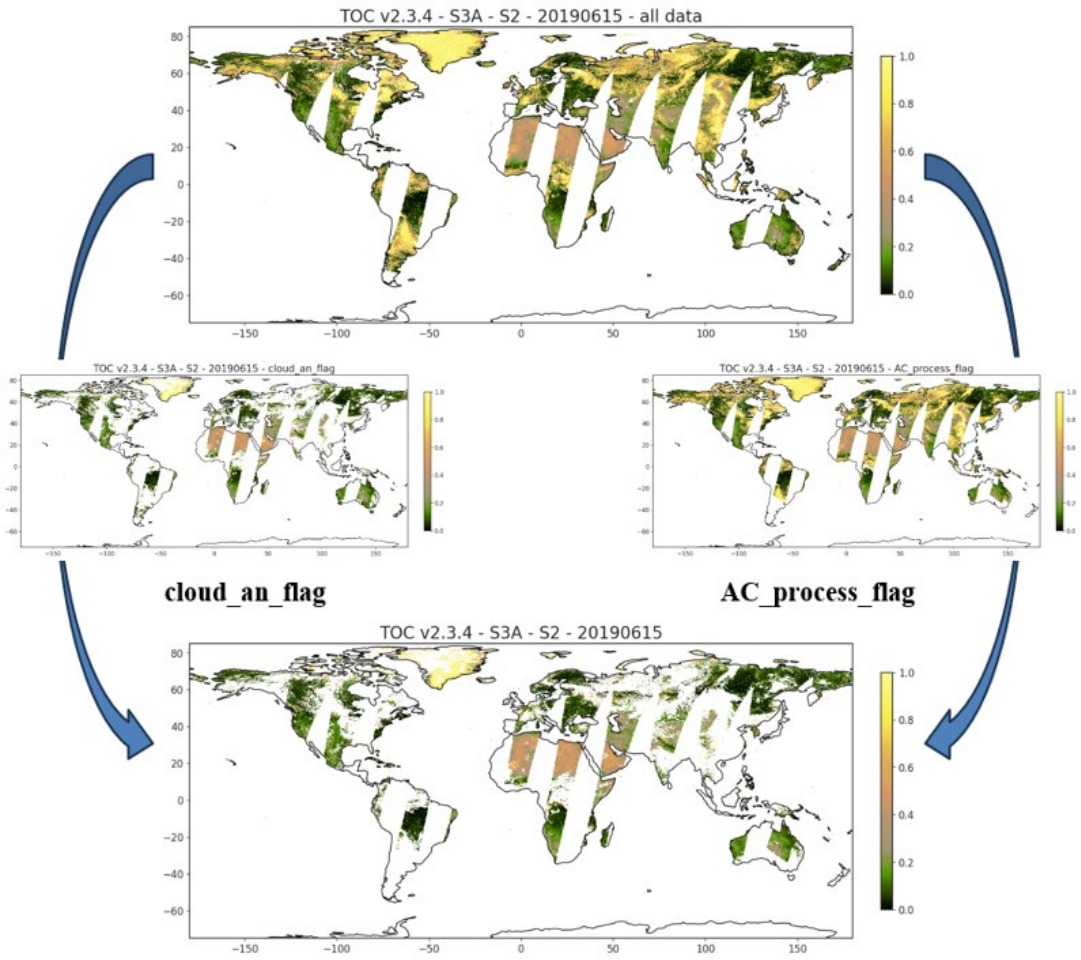


Figure 19: ***Top:*** **Global mosaic (all pixels) of Sentinel-3A SLSTR TOC reflectance (S2) for 15/06/2019 at reduced spatial resolution (1/84).** ***Middle:*** **global mosaic when applying recommendations for each quality layer: cloud_an_flag and AC_process_flag.** ***Bottom:*** **global mosaic when applying all quality annotation flags recommendations.**

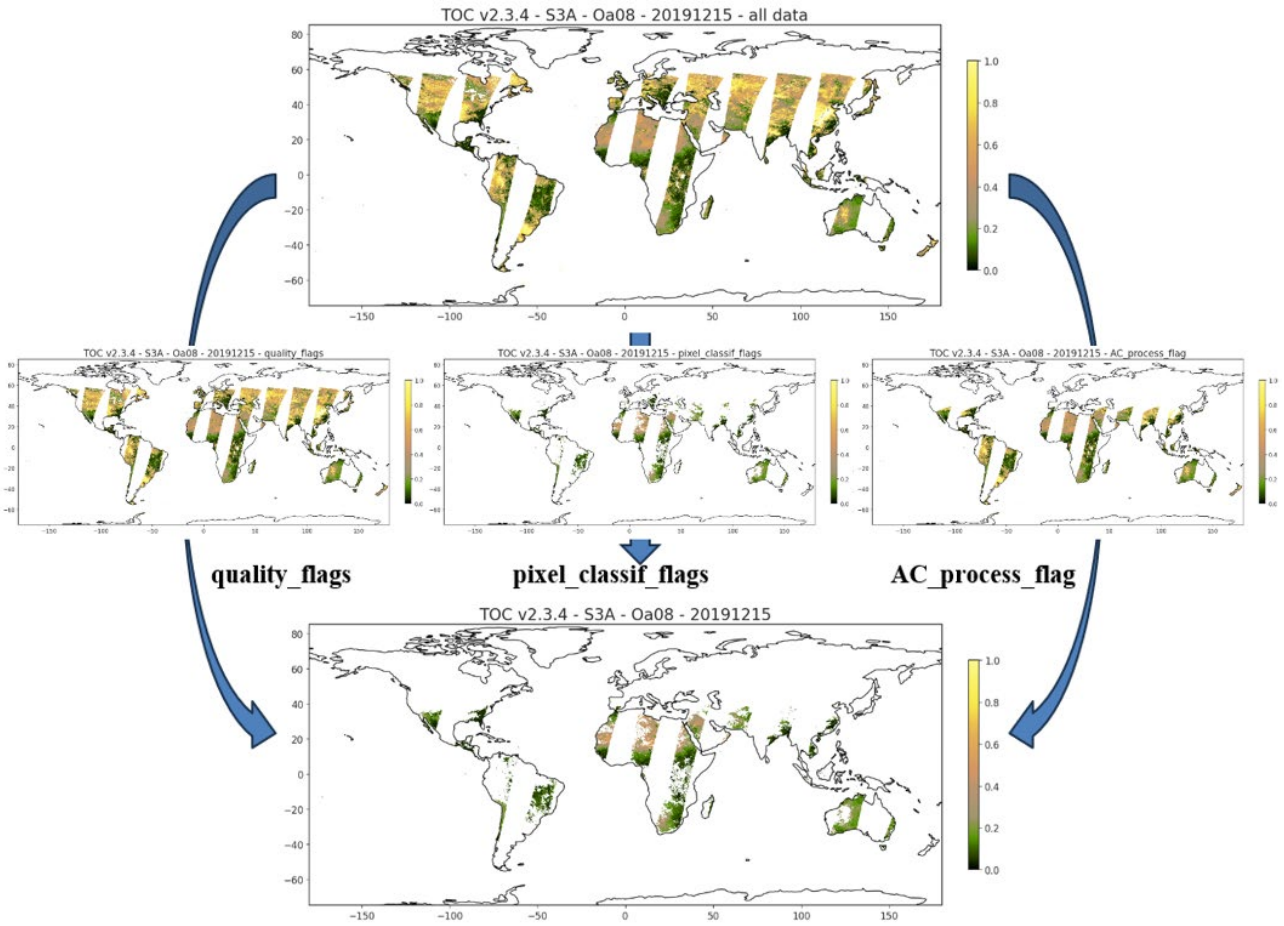


Figure 20: ***Top:*** **Global mosaic (all pixels) of Sentinel-3A OLCI TOC reflectance (Oa08) for 15/12/2019 at reduced spatial resolution (1/84).** ***Middle:*** **global mosaic when applying recommendations for each quality layer: quality_flags, pixel_classif_flags and AC_process_flag.** ***Bottom:*** **global mosaic when applying all quality annotation flags recommendations.**

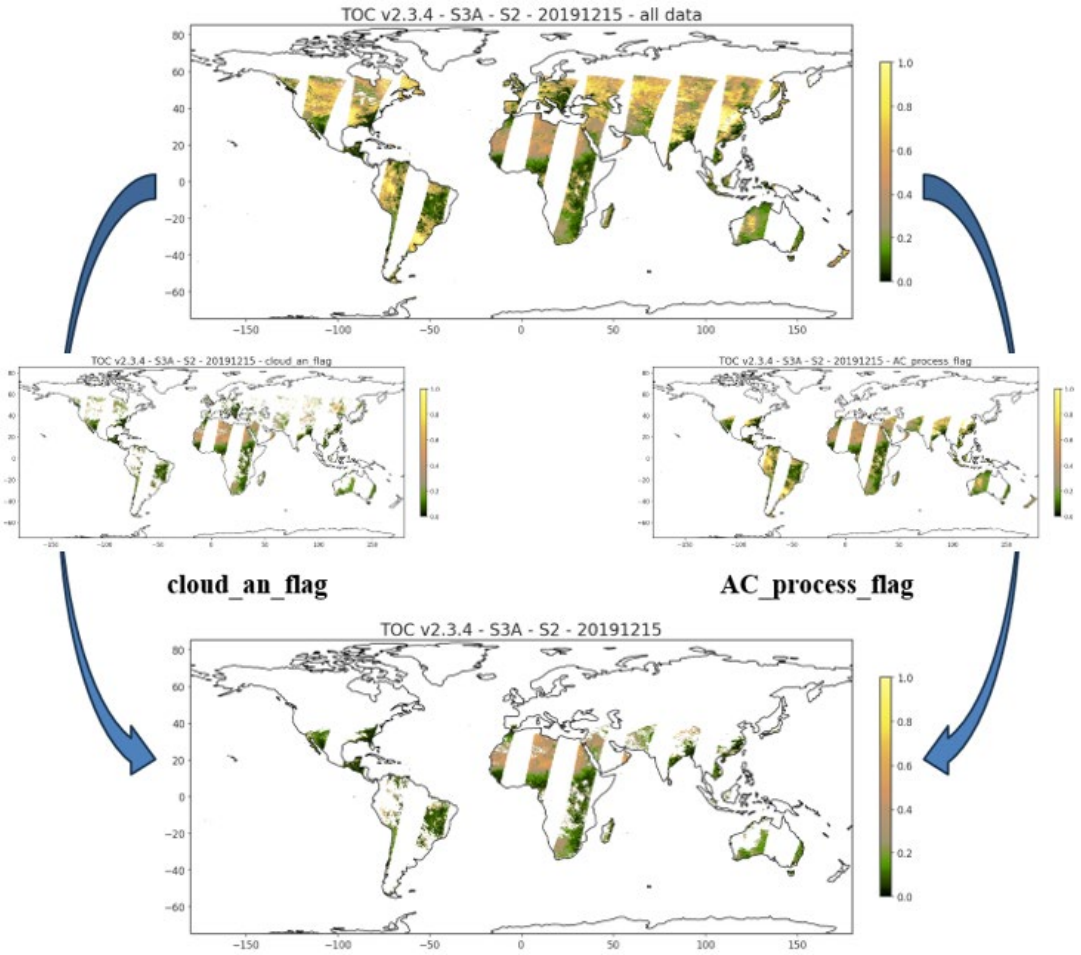


Figure 21: ***Top:*** **Global mosaic (all pixels) of Sentinel-3A SLSTR TOC reflectance (S2) for 15/12/2019 at reduced spatial resolution (1/84).** ***Middle:*** **global mosaic when applying recommendations for each quality layer: cloud_an_flag and AC_process_flag.** ***Bottom:*** **global mosaic when applying all quality annotation flags recommendations.**

To evaluate each recommendation for the quality layer used in more detail, we created maps at full resolution over some selected tiles where each flag (bit per bit) is applied separately. A total of four tiles have been selected for the analysis: X17Y04 (Iberian Peninsula), X18Y03 (Benelux), X20Y02 (Finland/Estonia), and X05Y01 (North of Canada). Maps are plotted for OLCI Oa08 and SLSTR S2 channels. Annex III gathers the figures: the map of all pixels Sentinel-3 TOC reflectance pixels is displayed at the top while the rest of images in the figures show the map when different quality layer recommendations are used bit per bit, as indicated on top of each subfigure.

The main conclusions in the inspection of the full resolution maps distinguishing each recommended flag of each quality layer are:

- **Pixel_classif_flags**:
  - IDEPIX_INVALID identifies pixels where there are invalid OLCI observations. This flag does not filter out many pixels compared to the original image (where the IDEPIX_LAND flag is applied).
  - IDEPIX_CLOUD and IDEPIX_CLOUD_AMBIGUOUS remove most of the pixels affected by cloud contamination. These are the two most restrictive flags.
  - IDEPIX_CLOUD_BUFFER and IDEPIX_CLOUD_SHADOW work properly complementing the cloud mask, as they remove pixels at the boundary of clouds.
  - IDEPIX_SNOW_ICE identifies most of the snow pixels. This is very clear in the Alps (Figure 66) and is a potential improvement compared to TOC V1 [CGLOPS-1_QAR_S3-TOCr300m-V1] due to the version 9.0.6 of Idepix processor used for pixel classification (Figure 1).
  - IDEPIX_MOUNTAIN_SHADOW seems to properly identify the shadow of the mountains.
- **Quality_flags**:
  - Most of the inland water, lakes, rivers, etc. are already identified by the IDEPIX_LAND quality flag. Therefore, the use of the fresh_inland_water flag has no large effect.
  - The Quality_flags saturated_Oa[x] bands almost do not filter any pixels.
- **Cloud_an**:
  - This quality layer removes a large fraction of the pixels affected by cloud contamination, with the most restrictive bits being bit0, bit1, bit7, and bit8. However, the combination of all Cloud_an flags is not enough to filter all the contaminated pixels at the borders of the clouds. Especially for tile X20Y02, it is clear that a large number of pixels is not correctly identified as cloud (see Figure 69).
  - The cloud flags do not correctly identify snow pixels.
- **AC_process_flag**:
  - The AOT flag removes pixels with high AOT values (due to aerosols, …). This is mostly visible in the global maps.
  - The SZA filter properly detects pixels with SZA > 65 º. This can cause large portions of the image to be removed in northern areas during the local winter and similarly in southern areas when it is winter there.

4.2 PRODUCT COMPLETENESS OF SNOW-FREE BEST QUALITY PIXELS

The Sentinel-3 TOC reflectance V2.3 product completeness is analysed for both the OLCI and SLSTR sensors. Maps of monthly valid good quality retrievals, according to Table 3, during the year

2019 are displayed in Figure 22 and Figure 23. For the sake of brevity, only results for the OLCI Oa08 band and SLSTR S3 band are displayed, but similar results were found for the other channels of each sensor. Note that both S3A and S3B observations were ingested in the creation of the monthly completeness maps.

The temporal evolution of the percentage of good quality retrievals over the whole validation period (2019-2023) is shown in Figure 24. The percentages are shown for one OLCI (Oa08) and one SLSTR (S2) band, including both S3A and S3B observations, at every dekad (1st, 11th, and 21st day of the month) in the validation period. In Figure 25, the temporal evolution per quality flag is shown in a similar way.

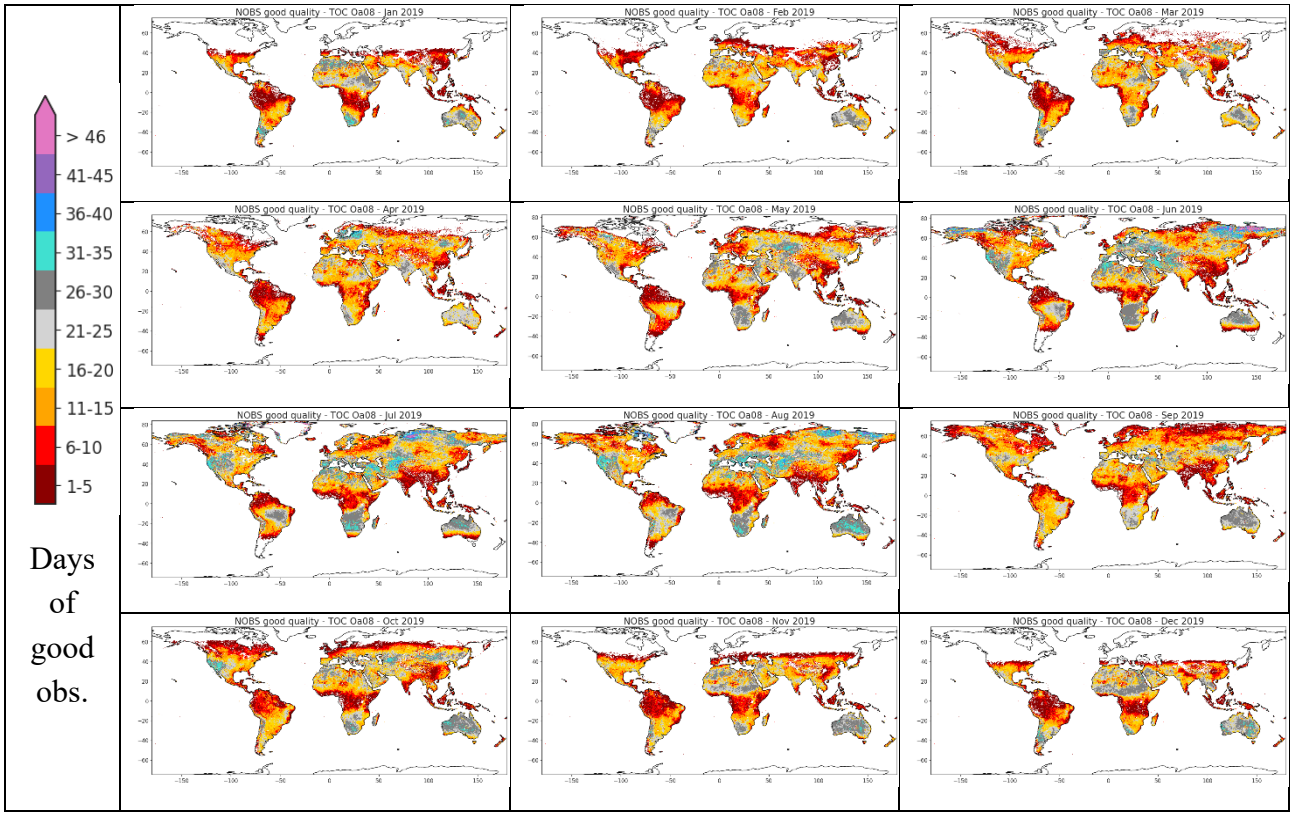


Figure 22: Global monthly composites of Sentinel-3 OLCI TOC reflectance (Oa08) valid good quality observations for year 2019. White land pixels correspond to no data

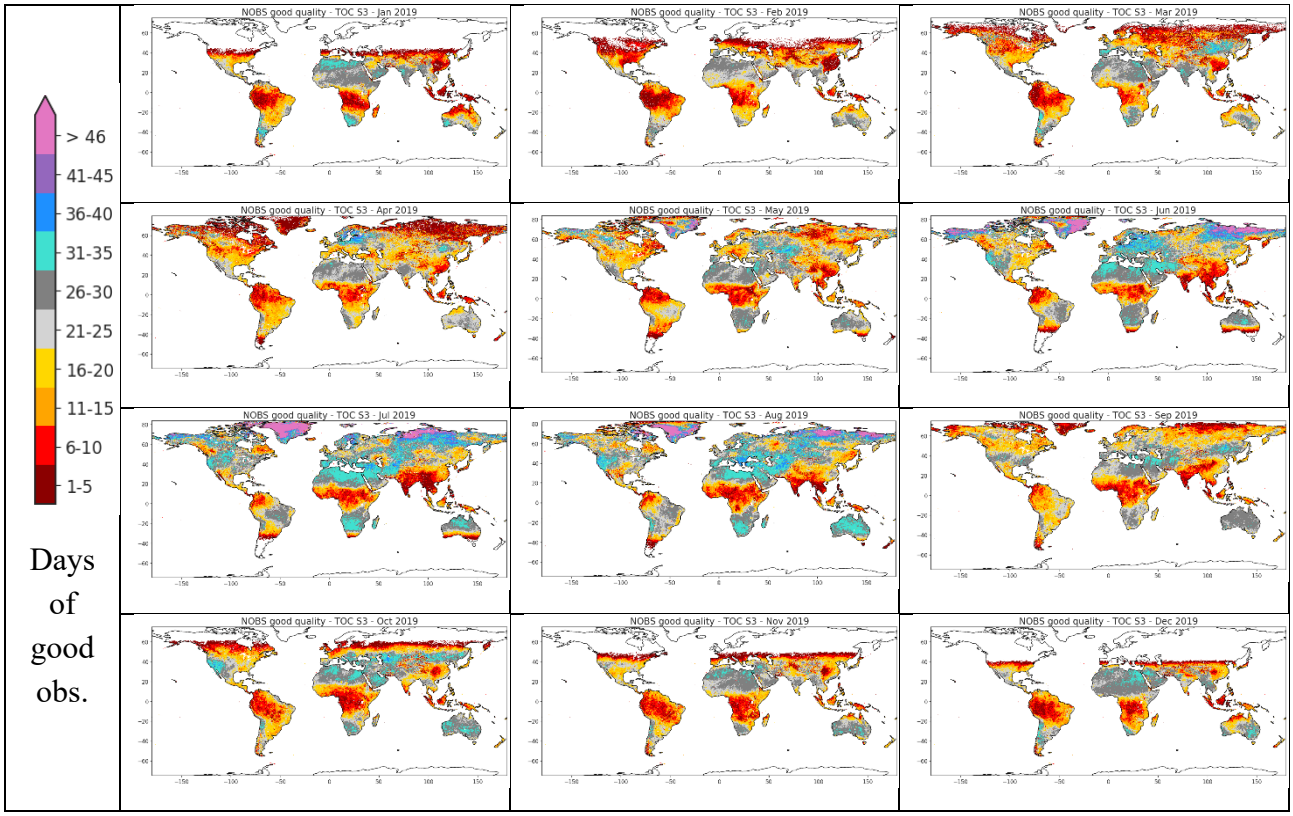


Figure 23: Global monthly composites of Sentinel-3 SLSTR TOC reflectance (S3) valid good quality observations for year 2019. White land pixels correspond to no data.

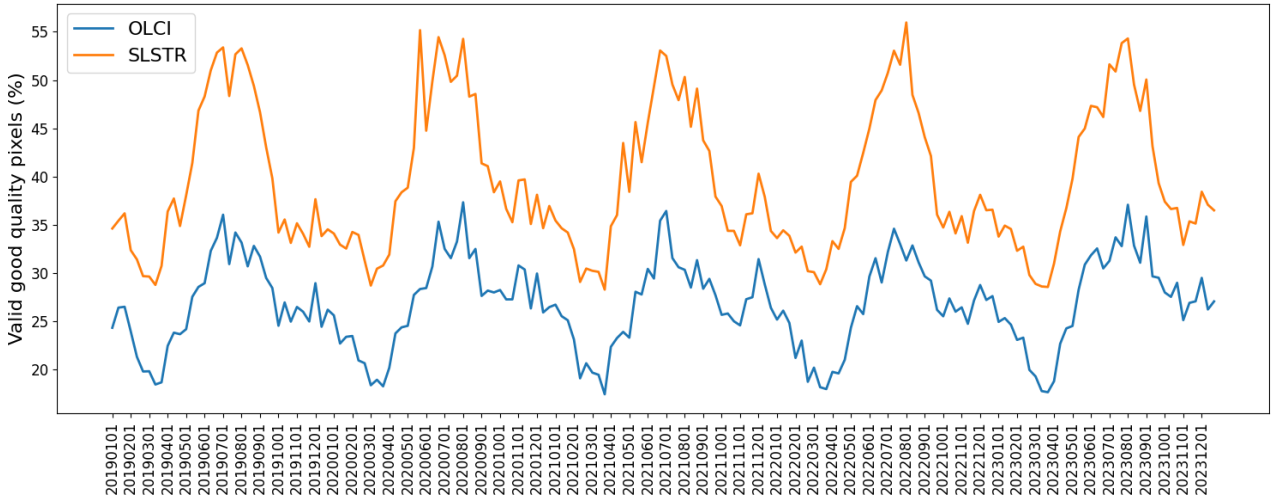


Figure 24: Temporal evolution of the percentage of valid good quality Sentinel-3 OLCI (OA08) and SLSTR (S2) TOC reflectance pixels for the validation period 2019-2023.

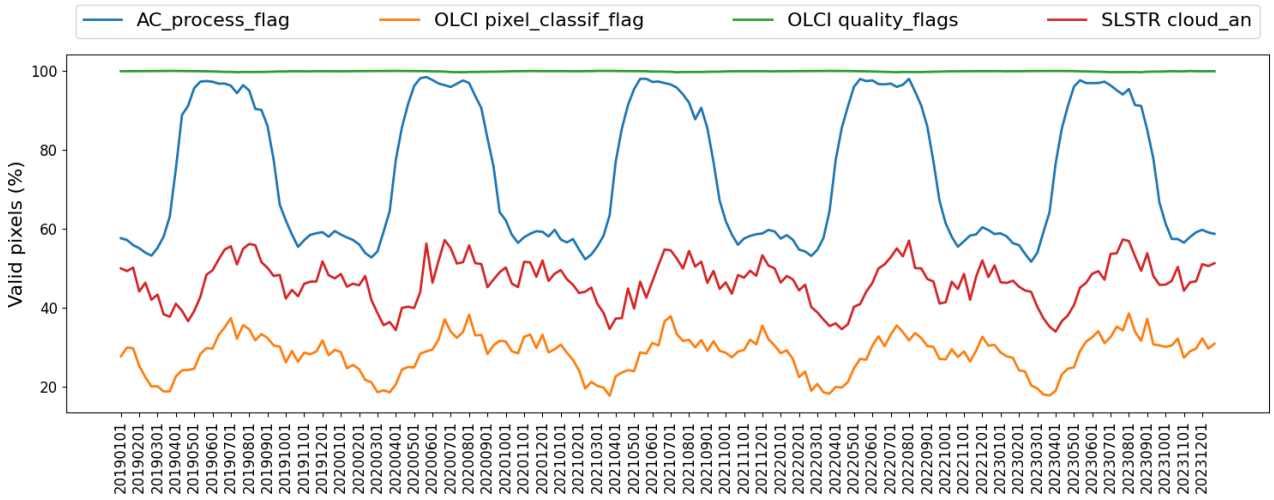


Figure 25: Temporal evolution (2019 - 2023) of the percentage of valid pixels with good quality according to the AC_process_flag (blue), OLCI pixel_classif_flag (orange), OLCI quality_flags (green) and SLSTR cloud_an flag (red). Band OA08 is used for the OLCI quality flags and band S2 fort he SLSTR quality flags.

The product completeness analysis shows:

- The number of days with valid data is typically around 20 observations, being higher over desert areas and at mid-latitudes (typically up to 35 valid observations). High number of observations (up to 45) can be reached at higher latitudes.
- There is no data available over northern latitudes from January to March and from October to December, and from June to August at southern regions. This is because of the SZA < 65° filter that is applied according to the recommendations.
- In general, OLCI provides lower number of good quality observations than SLSTR. This is caused by the more restrictive cloud detection approach (IdePix) applied to OLCI.
- The product completeness is stable over time, with higher completeness during Northern hemispherical summer and lower completeness during its winter. The lowest completeness is found in March, when observations at high latitudes are available but most of them are cloudy.
- The pixel_classif_flag and the cloud_an flag are the most restrictive quality flags, while the quality_flags have almost no effect. The AC_process_flag displays a clear seasonality driven by the flag on SZA (Table 3).

4.3 TEMPORAL CONSISTENCY

4.3.1 Temporal realism

In order to evaluate the temporal realism of Sentinel-3 TOC reflectance products, temporal trajectories of OLCI and SLSTR CLMS TOC V2.3 and ESA L2 SYN are displayed and qualitatively compared with surface reflectance measurements over the four RadCalNet sites. Figure 26 and Figure 27 show the time series plots for OLCI Oa08 and SLSTR S2 channels. The time series for all channels are displayed in Annex IV for OLCI and Annex V for SLSTR.

The analysis of the temporal trajectories of Sentinel-3 products compared with RadCalNet measurements shows:

- CLMS TOC V2.3 reflectances show similar temporal trends than RadCalNet measurements for OLCI (Annex IV) and SLSTR (Annex V), capturing properly the seasonality observed over LCFR, the low variability over GONA pseudo-invariant site and the slight temporal variability in RVUS and BSCN.
    - It should be noted that SLSTR (Figure 27) time series show more outliers than ESA L2 SYN and equivalent CLMS TOC V2.3 OLCI (Figure 26) channels. This is expected since OLCI annotation flags recommendations are more conservative than SLSTR recommendations, as showed in the product completeness analysis (see section 4.2).
- CLMS TOC V2.3 shows lower differences in magnitude and typically smoother profiles compared with RadCalNet than ESA L2 SYN.
    - It should be noted that, for OLCI Oa06, Oa07, Oa08 and Oa09 (see Figure 26 or Figure 73 in Annex IV) and SLSTR S1 and S2 (see Figure 27 or Figure 77 in Annex V) channels, ESA L2 SYN exhibits a significant positive bias from January 2020 onwards in GONA site compared with previous data (January 2019-December 2019). This could be explained in changes in configuration parameters of ESA SYN Level 1 from January 2020 onwards [S3_SYN_DataHub].

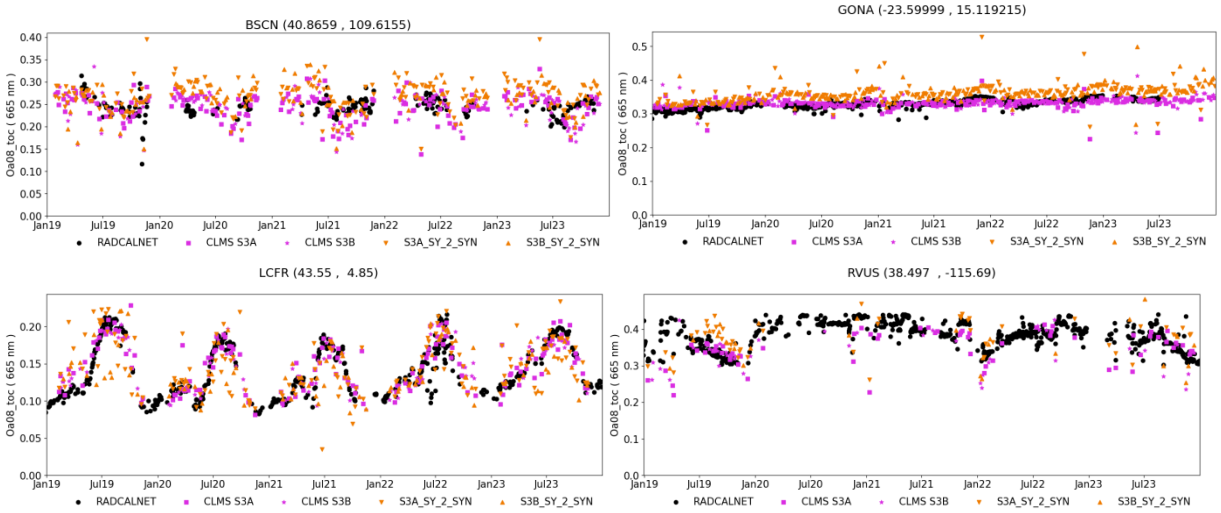


Figure 26: Temporal profiles (2019-2023) of RadCalNet measurements (black dots) and Sentinel-3 OLCI products (VZA<10°) for OLCI Oa08 channel: CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and -3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and -3B).

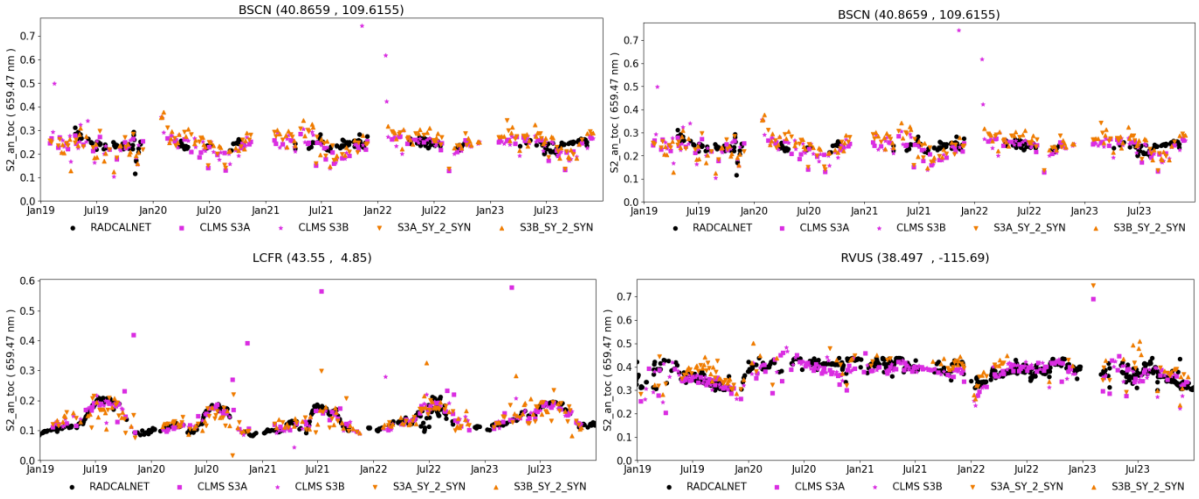


Figure 27: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) and Sentinel-3 SLSTR products (VZA<10°) for S2 channel: CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and -3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and -3B).

4.3.2 Temporal consistency between channels

Additionally, temporal profiles for CLMS Sentinel-3 TOC V2.3 over 48 representative sites (20 desert calibration sites + 4 sites for the rest of main biomes – EBF, DBF, NLF, OF, CUL, SHR, HER) are displayed for all available bands of both OLCI and SLSTR sensors. Note that only good quality observations according to Table 3 are displayed. Figure 28 and Figure 29 show some examples (one for each main biome type + one desert calibration site) for all OLCI and SLSTR bands.

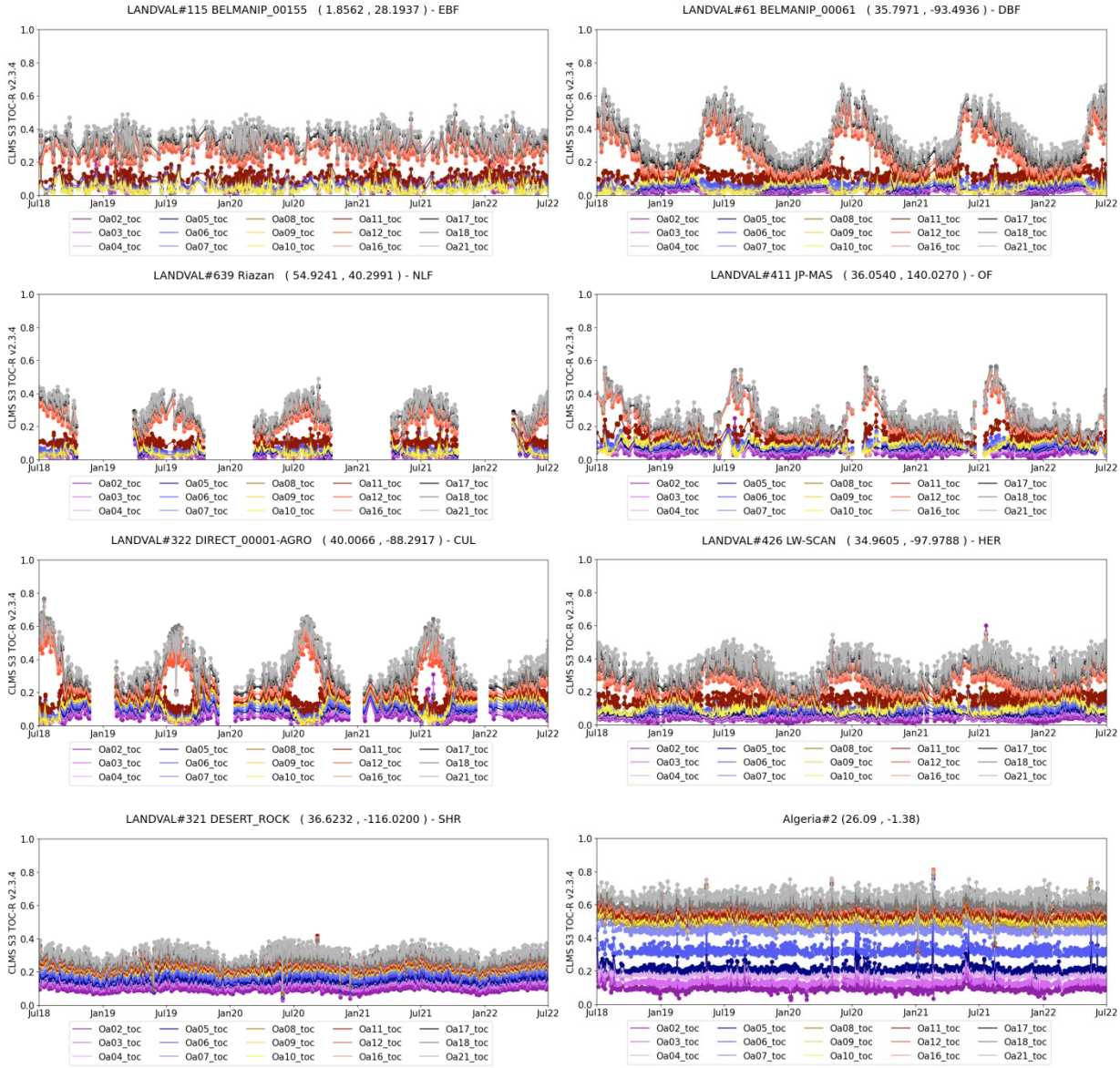


Figure 28: Temporal profiles of CLMS Sentinel-3 OLCI TOC V2.3 over 8 selected LANDVAL sites for each main biome type.

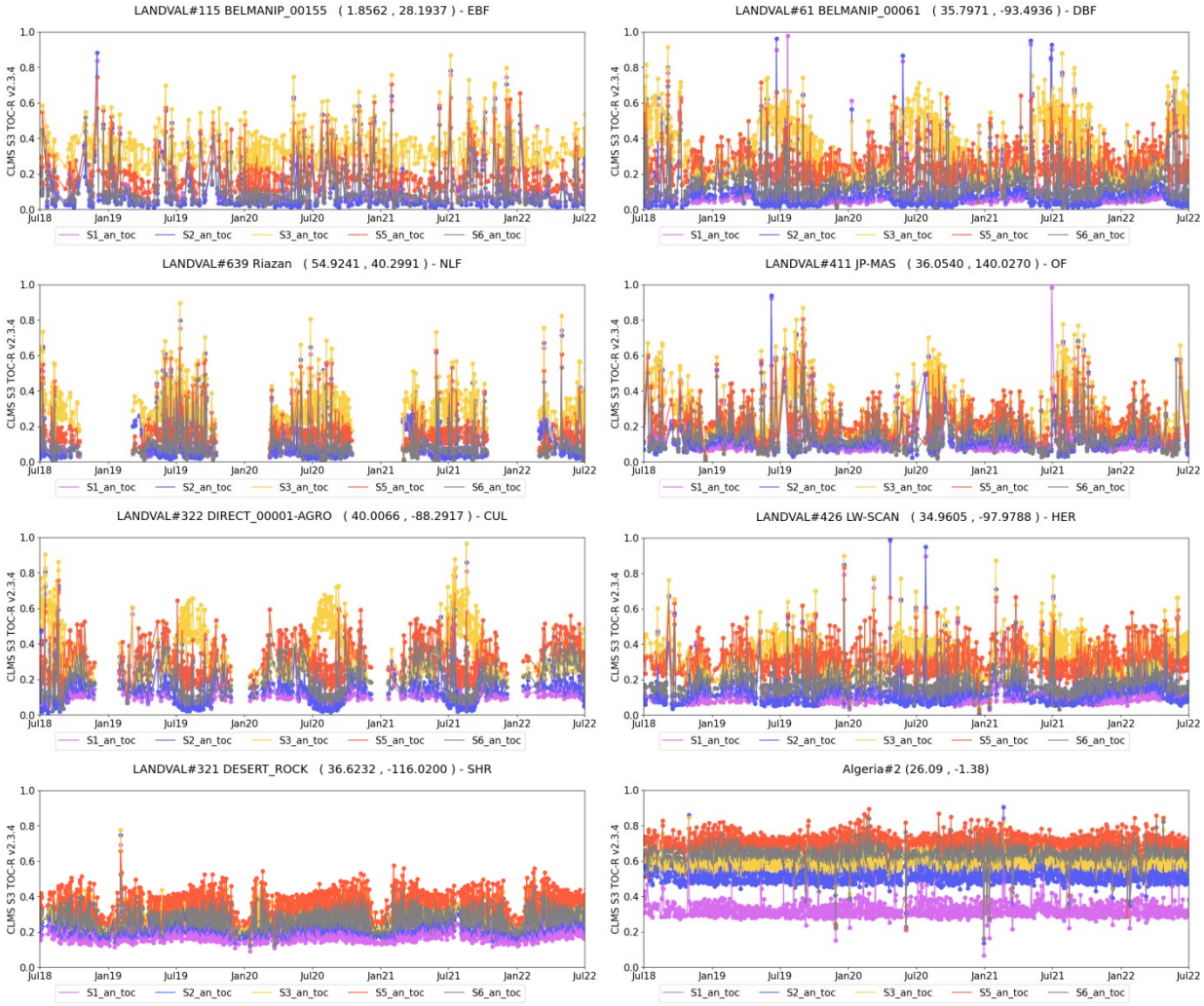


Figure 29: Same temporal profiles displayed in Figure 28 for CLMS Sentinel-3 SLSTR TOC V2.3.

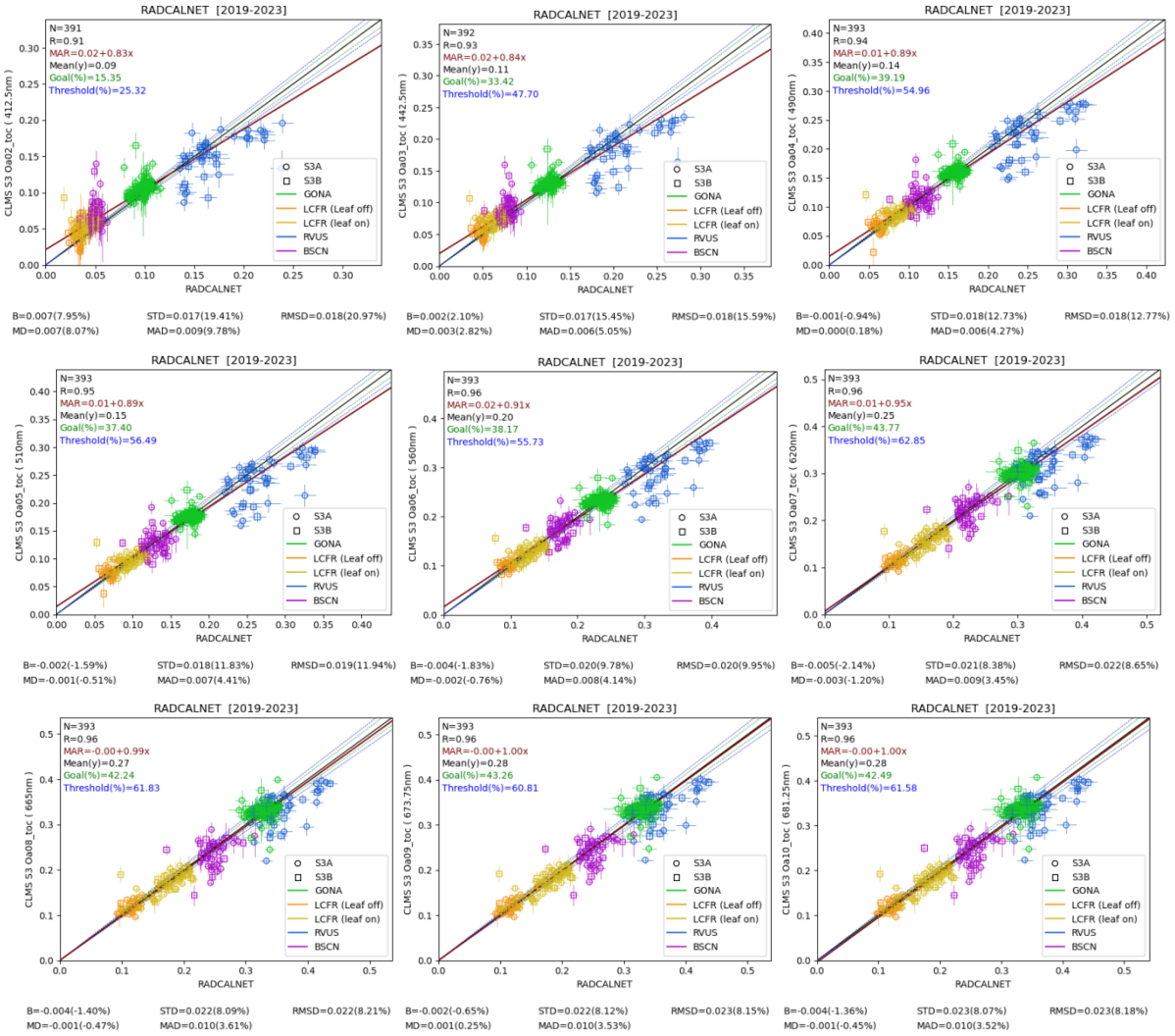
Main conclusions are:

- Good temporal consistency is found between all spectral channels, with very stable temporal trends for bare areas, and similar spectral properties for vegetated targets.
- Sentinel-3 temporal variations show short-time scale variability, which is introduced by differences in the viewing geometry between consecutive observations. It should be noted that SLSTR tends to provide higher variability which is mainly explained by the less restrictive cloud masking, which introduces additional outliers.

4.4 DIRECT VALIDATION

4.4.1 Comparison with RadCalNet measurements

To estimate the error (APU) of the CLMS Sentinel-3 TOC V2.3, scatter-plots and validation metrics versus RadCalNet measurements are produced for OLCI (Figure 30, Table 9) and SLSTR (Figure 32, Table 11) bands. The period of the analysis covers five years (2019-2023). Only Sentinel-3 TOC reflectance retrievals with VZA<10° were used in the comparison, as RadCalNet measurements were acquired at nadir view. Note that only Sentinel-3 pixels showing good quality according to Table 3 are considered. For benchmarking, the equivalent results for ESA L2 SYN products are presented in Figure 31 (Table 10) for OLCI channels and in Figure 33 (Table 12) for SLSTR. Concomitant samples between CLMS and ESA products are used. The percentage of cases within goal and threshold requirements are summarized in Figure 34.



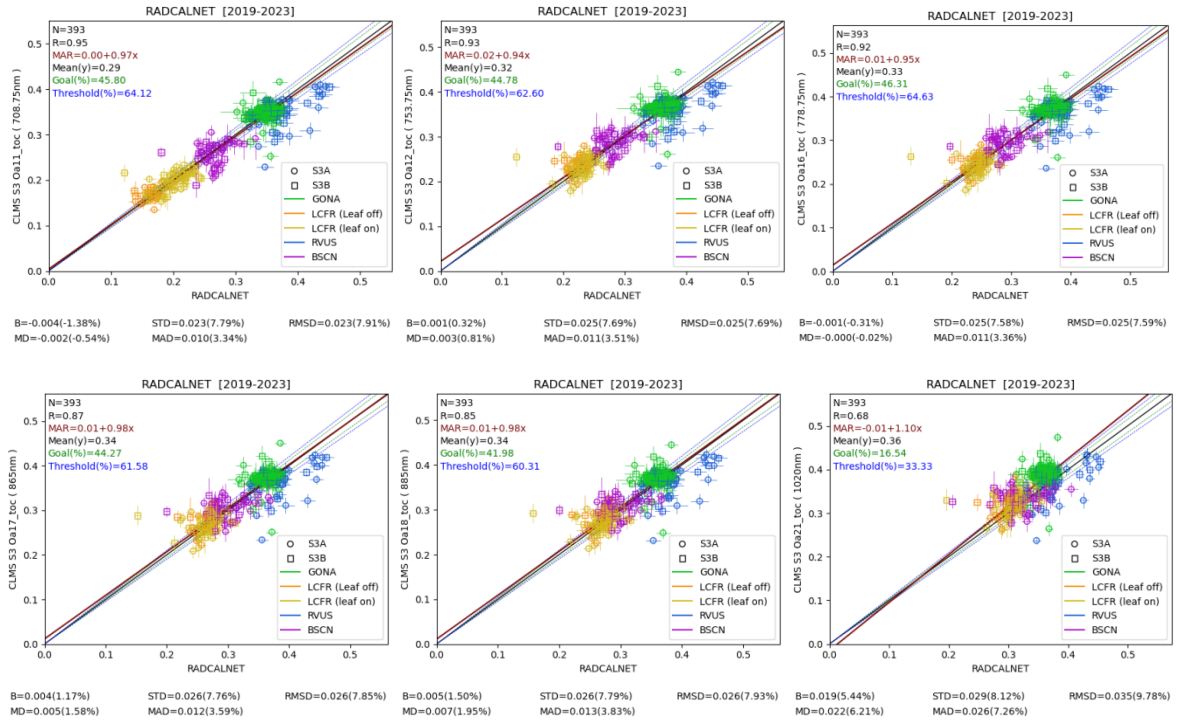
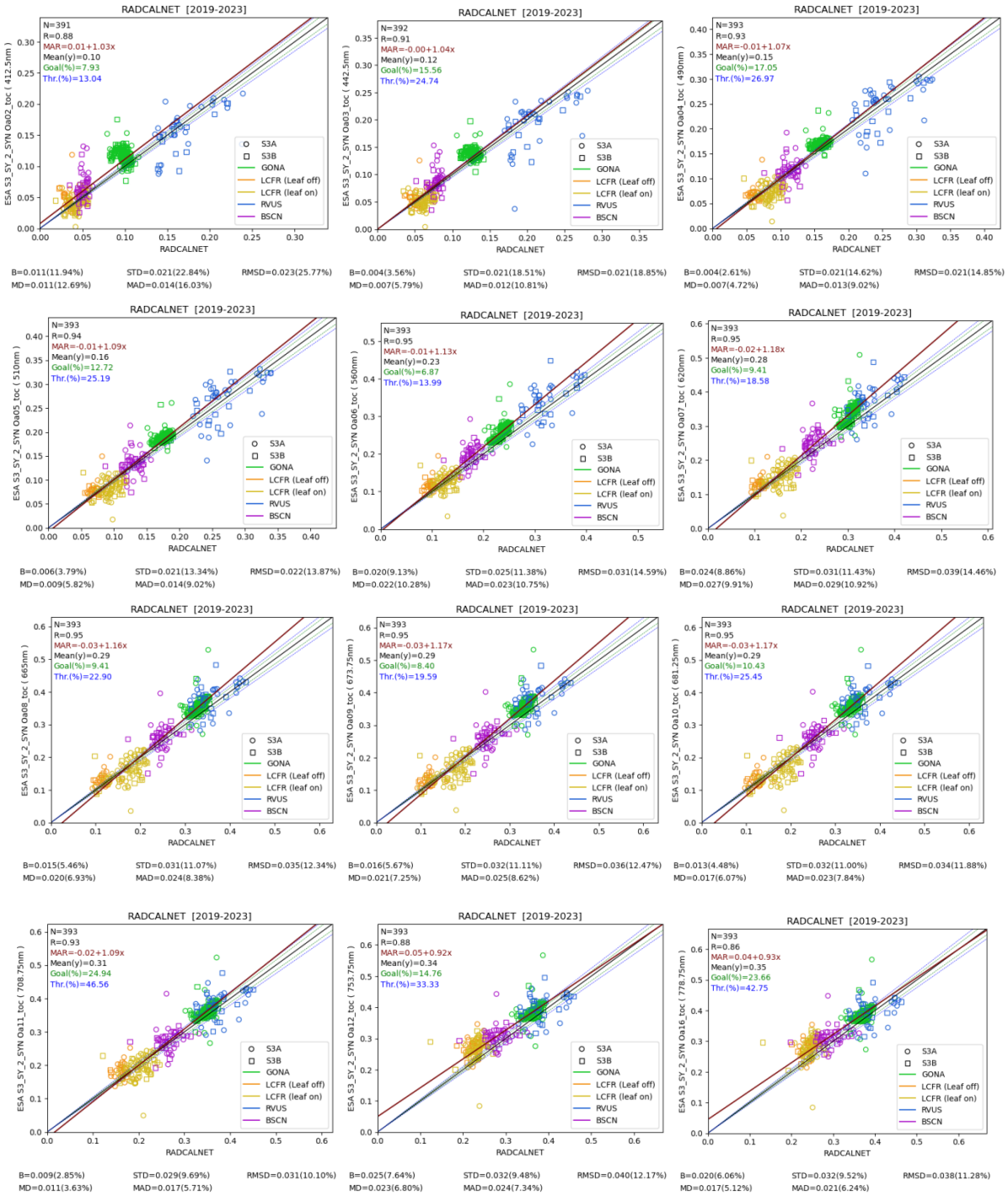


Figure 30: Scatter-plots of CLMS Sentinel-3 TOC V2.3 (VZA<10°) versus RadCalNet measurements (2019-2023) for OLCI channels. Vertical and horizontal bars represent CLMS TOC V2.3 and RadCalNet associated uncertainties. Green and blue lines delimit goal and threshold requirements. Error bars represent the associated uncertainties of CLMS TOC V2.3 and RadCalNet measurements.

Table 9: Performance statistics of CLMS Sentinel-3 TOC V2.3 (VZA<10°) versus RadCalNet measurements (2019-2023) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	391	0.91	0.007 (7.9%)	0.007 (8.1%)	0.017 (19.4%)	0.009 (9.8%)	0.018 (21.0%)	0.02 /0.83	15.4 /25.3
Oa03	392	0.93	0.002 (2.1%)	0.003 (2.8%)	0.017 (15.4%)	0.006 (5.1%)	0.018 (15.6%)	0.02 /0.84	33.4 /47.7
Oa04	393	0.94	-0.001 (-0.9%)	<0.001 (0.2%)	0.018 (12.7%)	0.006 (4.3%)	0.018 (12.8%)	0.01 /0.89	39.2 /55.0
Oa05	393	0.95	-0.002 (-1.6%)	-0.001 (-0.5%)	0.018 (11.8%)	0.007 (4.4%)	0.019 (11.9%)	0.01 /0.89	37.4 /56.5
Oa06	393	0.96	-0.004 (-1.8%)	-0.002 (-0.8%)	0.020 (9.8%)	0.008 (4.1%)	0.020 (10.0%)	0.02 /0.91	38.2 /55.7
Oa07	393	0.96	-0.005 (-2.1%)	-0.003 (-1.2%)	0.021 (8.4%)	0.009 (3.5%)	0.022 (8.7%)	0.01 /0.95	43.8 /62.9
Oa08	393	0.96	-0.004 (-1.4%)	-0.001 (-0.5%)	0.022 (8.1%)	0.010 (3.6%)	0.022 (8.2%)	0.00 /0.99	42.2 /61.8
Oa09	393	0.96	-0.002 (-0.7%)	0.001 (0.3%)	0.022 (8.1%)	0.010 (3.5%)	0.023 (8.2%)	0.00 /1.00	43.3 /60.8
Oa10	393	0.96	-0.004 (-1.4%)	-0.001 (-0.5%)	0.023 (8.1%)	0.010 (3.5%)	0.023 (8.2%)	0.00 /1.00	42.5 /61.6
Oa11	393	0.95	-0.004 (-1.4%)	-0.002 (-0.5%)	0.023 (7.8%)	0.010 (3.4%)	0.023 (7.9%)	0.00 /0.97	45.8 /64.1
Oa12	393	0.93	0.001 (0.3%)	0.003 (0.8%)	0.025 (7.7%)	0.011 (3.5%)	0.025 (7.7%)	0.02 /0.94	44.8 /62.6
Oa16	393	0.92	-0.001 (-0.3%)	>-0.001 (0.0%)	0.025 (7.6%)	0.011 (3.4%)	0.025 (7.6%)	0.01 /0.95	46.3 /64.6
Oa17	393	0.87	0.004 (1.2%)	0.005 (1.6%)	0.026 (7.8%)	0.012 (3.6%)	0.026 (7.9%)	0.01 /0.98	44.3 /61.6
Oa18	393	0.85	0.005 (1.5%)	0.007 (2.0%)	0.026 (7.8%)	0.013 (3.8%)	0.026 (7.9%)	0.01 /0.98	42.0 /60.3
Oa21	393	0.68	0.019 (5.4%)	0.022 (6.2%)	0.029 (8.1%)	0.026 (7.3%)	0.035 (9.8%)	-0.01 /1.10	16.5 /33.3


RADCALNET [2019-2023]
N=391
R=0.88
MAR=0.01+1.03x
Mean(y)=0.10
Goal(%)=7.93
Thr.(%)=13.04
ESA S3_SY_2_SYN Oa02_toc (412.5nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.011(11.94%) STD=0.021(22.84%) RMSD=0.023(25.77%)
MD=0.011(12.69%) MAD=0.014(16.03%)
RADCALNET [2019-2023]
N=392
R=0.91
MAR=-0.00+1.04x
Mean(y)=0.12
Goal(%)=15.56
Thr.(%)=24.74
ESA S3_SY_2_SYN Oa03_toc (442.5nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.004(3.56%) STD=0.021(18.51%) RMSD=0.021(18.85%)
MD=0.007(5.79%) MAD=0.012(10.81%)
RADCALNET [2019-2023]
N=393
R=0.93
MAR=-0.01+1.07x
Mean(y)=0.15
Goal(%)=17.05
Thr.(%)=26.97
ESA S3_SY_2_SYN Oa04_toc (490nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.004(2.61%) STD=0.021(14.62%) RMSD=0.021(14.85%)
MD=0.007(4.72%) MAD=0.013(9.02%)
RADCALNET [2019-2023]
N=393
R=0.94
MAR=-0.01+1.09x
Mean(y)=0.16
Goal(%)=12.72
Thr.(%)=25.19
ESA S3_SY_2_SYN Oa05_toc (510nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.006(3.79%) STD=0.021(13.34%) RMSD=0.022(13.87%)
MD=0.009(5.82%) MAD=0.014(9.02%)
RADCALNET [2019-2023]
N=393
R=0.95
MAR=-0.01+1.13x
Mean(y)=0.23
Goal(%)=6.87
Thr.(%)=13.99
ESA S3_SY_2_SYN Oa06_toc (560nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.020(9.13%) STD=0.025(11.38%) RMSD=0.031(14.59%)
MD=0.022(10.28%) MAD=0.023(10.75%)
RADCALNET [2019-2023]
N=393
R=0.95
MAR=-0.02+1.18x
Mean(y)=0.28
Goal(%)=9.41
Thr.(%)=18.58
ESA S3_SY_2_SYN Oa07_toc (620nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.024(8.86%) STD=0.031(11.43%) RMSD=0.039(14.46%)
MD=0.027(9.91%) MAD=0.029(10.92%)
RADCALNET [2019-2023]
N=393
R=0.95
MAR=-0.03+1.16x
Mean(y)=0.29
Goal(%)=9.41
Thr.(%)=22.90
ESA S3_SY_2_SYN Oa08_toc (665nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.015(5.46%) STD=0.031(11.07%) RMSD=0.035(12.34%)
MD=0.020(6.93%) MAD=0.024(8.38%)
RADCALNET [2019-2023]
N=393
R=0.95
MAR=-0.03+1.17x
Mean(y)=0.29
Goal(%)=8.40
Thr.(%)=19.59
ESA S3_SY_2_SYN Oa09_toc (673.75nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.016(5.67%) STD=0.032(11.11%) RMSD=0.036(12.47%)
MD=0.021(7.25%) MAD=0.025(8.62%)
RADCALNET [2019-2023]
N=393
R=0.95
MAR=-0.03+1.17x
Mean(y)=0.29
Goal(%)=10.43
Thr.(%)=25.45
ESA S3_SY_2_SYN Oa10_toc (681.25nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.013(4.48%) STD=0.032(11.00%) RMSD=0.034(11.88%)
MD=0.017(6.07%) MAD=0.023(7.84%)
RADCALNET [2019-2023]
N=393
R=0.93
MAR=-0.02+1.09x
Mean(y)=0.31
Goal(%)=24.94
Thr.(%)=46.56
ESA S3_SY_2_SYN Oa11_toc (708.75nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.009(2.85%) STD=0.029(9.69%) RMSD=0.031(10.10%)
MD=0.011(3.63%) MAD=0.017(5.71%)
RADCALNET [2019-2023]
N=393
R=0.88
MAR=0.05+0.92x
Mean(y)=0.34
Goal(%)=14.76
Thr.(%)=33.33
ESA S3_SY_2_SYN Oa12_toc (753.75nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.025(7.64%) STD=0.032(9.48%) RMSD=0.040(12.17%)
MD=0.023(6.80%) MAD=0.024(7.34%)
RADCALNET [2019-2023]
N=393
R=0.86
MAR=0.04+0.93x
Mean(y)=0.35
Goal(%)=23.66
Thr.(%)=42.75
ESA S3_SY_2_SYN Oa16_toc (778.75nm)
RADCALNET
S3A
S3B
GONA
LCFR (Leaf off)
LCFR (leaf on)
RVUS
BSCN
B=0.020(6.06%) STD=0.032(9.52%) RMSD=0.038(11.28%)
MD=0.017(5.12%) MAD=0.021(6.24%)

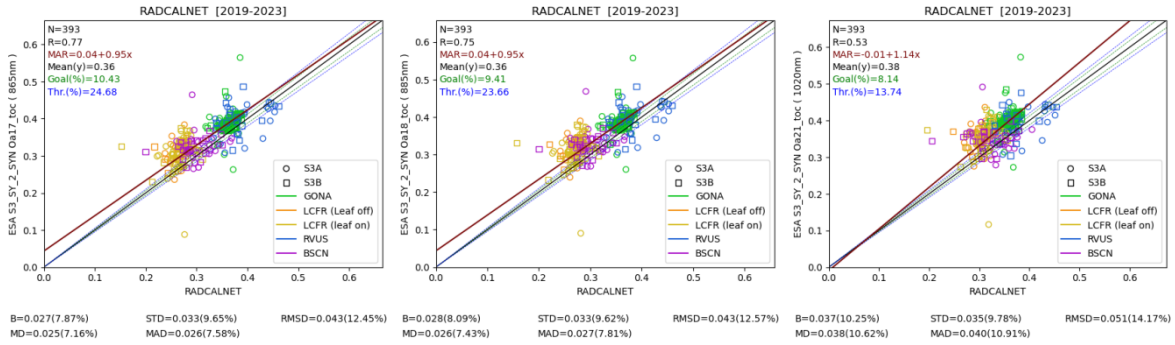


Figure 31: Scatter-plots of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (2019-2023) for OLCI channels. Green and blue lines delimit goal and threshold requirements.

Table 10: Performance statistics of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (2019-2023) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	391	0.88	0.011 (11.9%)	0.011 (12.7%)	0.021 (22.8%)	0.014 (16.0%)	0.023 (25.8%)	0.01 /1.03	7.9 /13.0
Oa03	392	0.91	0.004 (3.5%)	0.007 (5.8%)	0.021 (18.5%)	0.012 (10.8%)	0.021 (18.9%)	0.00 /1.04	15.6 /24.7
Oa04	393	0.93	0.004 (2.6%)	0.007 (4.7%)	0.021 (14.6%)	0.013 (9.0%)	0.021 (14.9%)	-0.01 /1.07	17.1 /27.0
Oa05	393	0.94	0.006 (3.8%)	0.009 (5.8%)	0.021 (13.3%)	0.014 (9.0%)	0.022 (13.9%)	-0.01 /1.09	12.7 /25.2
Oa06	393	0.95	0.020 (9.1%)	0.022 (10.3%)	0.025 (11.4%)	0.023 (10.8%)	0.031 (14.6%)	-0.01 /1.13	6.9 /14.0
Oa07	393	0.95	0.024 (8.9%)	0.027 (9.9%)	0.031 (11.4%)	0.029 (10.9%)	0.039 (14.5%)	-0.02 /1.18	9.4 /18.6
Oa08	393	0.95	0.015 (5.5%)	0.020 (6.9%)	0.031 (11.1%)	0.024 (8.4%)	0.035 (12.3%)	-0.03 /1.16	9.4 /22.9
Oa09	393	0.95	0.016 (5.7%)	0.021 (7.3%)	0.032 (11.1%)	0.025 (8.6%)	0.036 (12.5%)	-0.03 /1.17	8.4 /19.6
Oa10	393	0.95	0.013 (4.5%)	0.017 (6.1%)	0.032 (11.0%)	0.023 (7.8%)	0.034 (11.9%)	-0.03 /1.17	10.4 /25.5
Oa11	393	0.93	0.009 (2.9%)	0.011 (3.6%)	0.029 (9.7%)	0.017 (5.7%)	0.031 (10.1%)	-0.02 /1.09	24.9 /46.6
Oa12	393	0.88	0.025 (7.6%)	0.023 (6.8%)	0.032 (9.5%)	0.024 (7.3%)	0.040 (12.2%)	0.05 /0.92	14.8 /33.3
Oa16	393	0.86	0.020 (6.1%)	0.017 (5.1%)	0.032 (9.5%)	0.021 (6.2%)	0.038 (11.3%)	0.04 /0.93	23.7 /42.8
Oa17	393	0.77	0.027 (7.9%)	0.025 (7.2%)	0.033 (9.7%)	0.026 (7.6%)	0.043 (12.5%)	0.04 /0.95	10.4 /24.7
Oa18	393	0.75	0.028 (8.1%)	0.026 (7.4%)	0.033 (9.6%)	0.027 (7.8%)	0.043 (12.6%)	0.04 /0.95	9.4 /23.7
Oa21	393	0.53	0.037 (10.3%)	0.038 (10.6%)	0.035 (9.8%)	0.040 (10.9%)	0.051 (14.2%)	-0.01 /1.14	8.2 /13.7

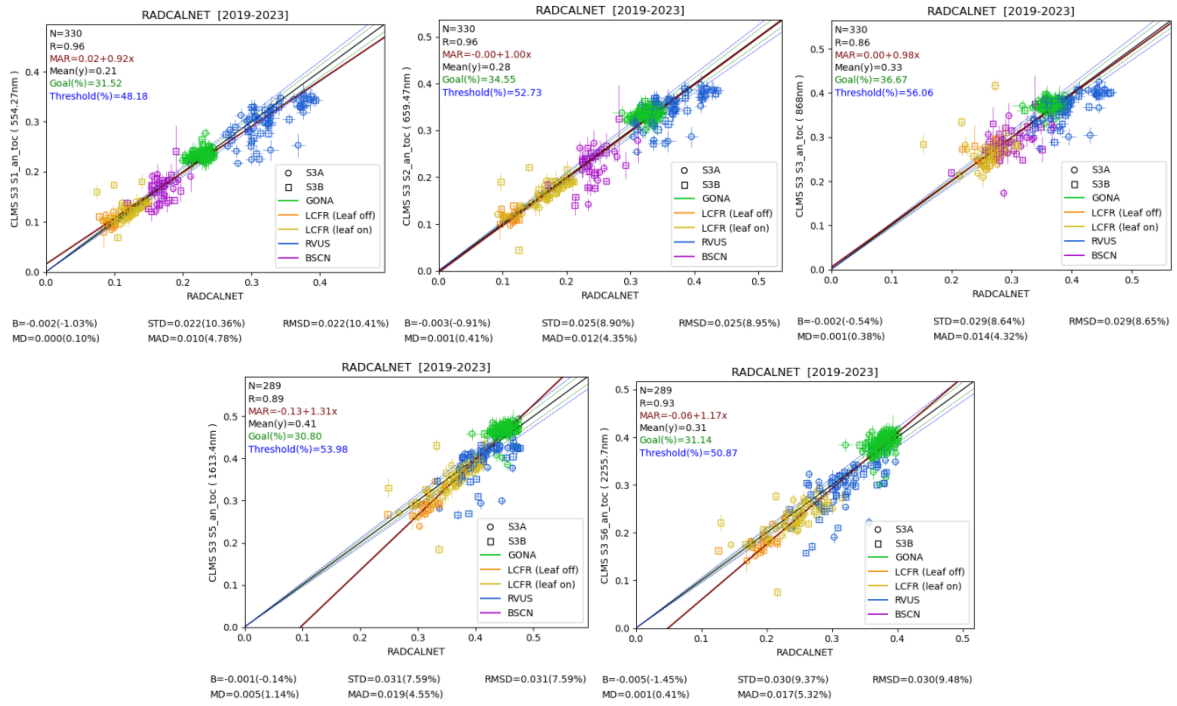


Figure 32: Scatter-plots of CLMS Sentinel-3 TOC V2.3 (VZA<10°) versus RadCalNet measurements (2019-2023) for SLSTR channels. Vertical and horizontal bars represent CLMS TOC V2.3 and RadCalNet associated uncertainties. BSCN does not provide measurements for spectral ranges of S5 and S6. Green and blue lines delimit goal and threshold requirements. Error bars represent the associated uncertainties of CLMS TOC V2.3 and RadCalNet measurements.

Table 11: Performance statistics of CLMS Sentinel-3 TOC V2.3 (VZA<10°) versus RadCalNet measurements (2019-2023) for SLSTR channels. BSCN does not provide measurements for spectral ranges of S5 and S6.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	330	0.96	-0.002 (-1.0%)	<0.001 (0.1%)	0.022 (10.4%)	0.010 (4.8%)	0.022 (10.4%)	0.02 /0.92	31.5 /48.2
S2	330	0.96	-0.003 (-0.9%)	0.001 (0.4%)	0.025 (8.9%)	0.012 (4.4%)	0.025 (9.0%)	0.00 /1.00	34.6 /52.7
S3	330	0.86	-0.002 (-0.5%)	0.001 (0.4%)	0.029 (8.7%)	0.014 (4.3%)	0.029 (8.7%)	0.00 /0.98	36.7 /56.1
S5	289	0.89	-0.001 (-0.1%)	0.005 (1.1%)	0.031 (7.6%)	0.019 (4.6%)	0.031 (7.6%)	-0.13 /1.31	30.8 /54.0
S6	289	0.93	-0.005 (-1.5%)	0.001 (0.4%)	0.030 (9.4%)	0.017 (5.3%)	0.030 (9.5%)	-0.06 /1.17	31.2 /50.9

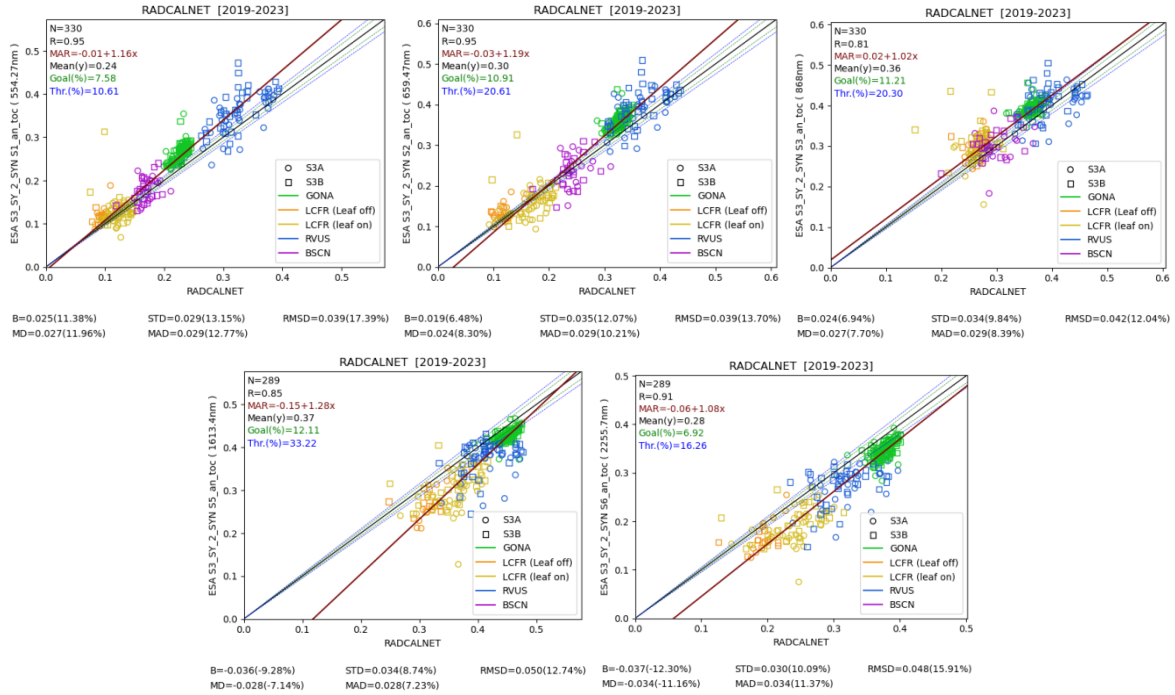


Figure 33: Scatter-plots of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (2019 - 2023) for SLSTR channels. BSCN does not provide measurements for spectral ranges of S5 and S6. Green and blue lines delimit goal and threshold requirements.

Table 12: Performance statistics of ESA Sentinel-3 L2 SYN (VZA<10°) versus RadCalNet measurements (2019-2023) for SLSTR channels. BSCN does not provide measurements for spectral ranges of S5 and S6.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	330	0.95	0.025 (11.4%)	0.027 (12.0%)	0.029 (13.2%)	0.029 (12.8%)	0.039 (17.4%)	-0.01 /1.16	7.6 /10.6
S2	330	0.95	0.019 (6.5%)	0.024 (8.3%)	0.035 (12.1%)	0.029 (10.2%)	0.039 (13.7%)	-0.03 /1.19	10.9 /20.6
S3	330	0.81	0.024 (6.9%)	0.027 (7.7%)	0.034 (9.8%)	0.029 (8.4%)	0.042 (12.0%)	0.02 /1.02	11.2 /20.3
S5	289	0.85	-0.036 (-9.3%)	-0.028 (-7.1%)	0.034 (8.7%)	0.028 (7.2%)	0.050 (12.7%)	-0.15 /1.28	12.1 /33.2
S6	289	0.91	-0.037 (-12.3%)	-0.034 (-11.2%)	0.030 (10.1%)	0.034 (11.4%)	0.048 (15.9%)	-0.06 /1.08	6.9 /16.3

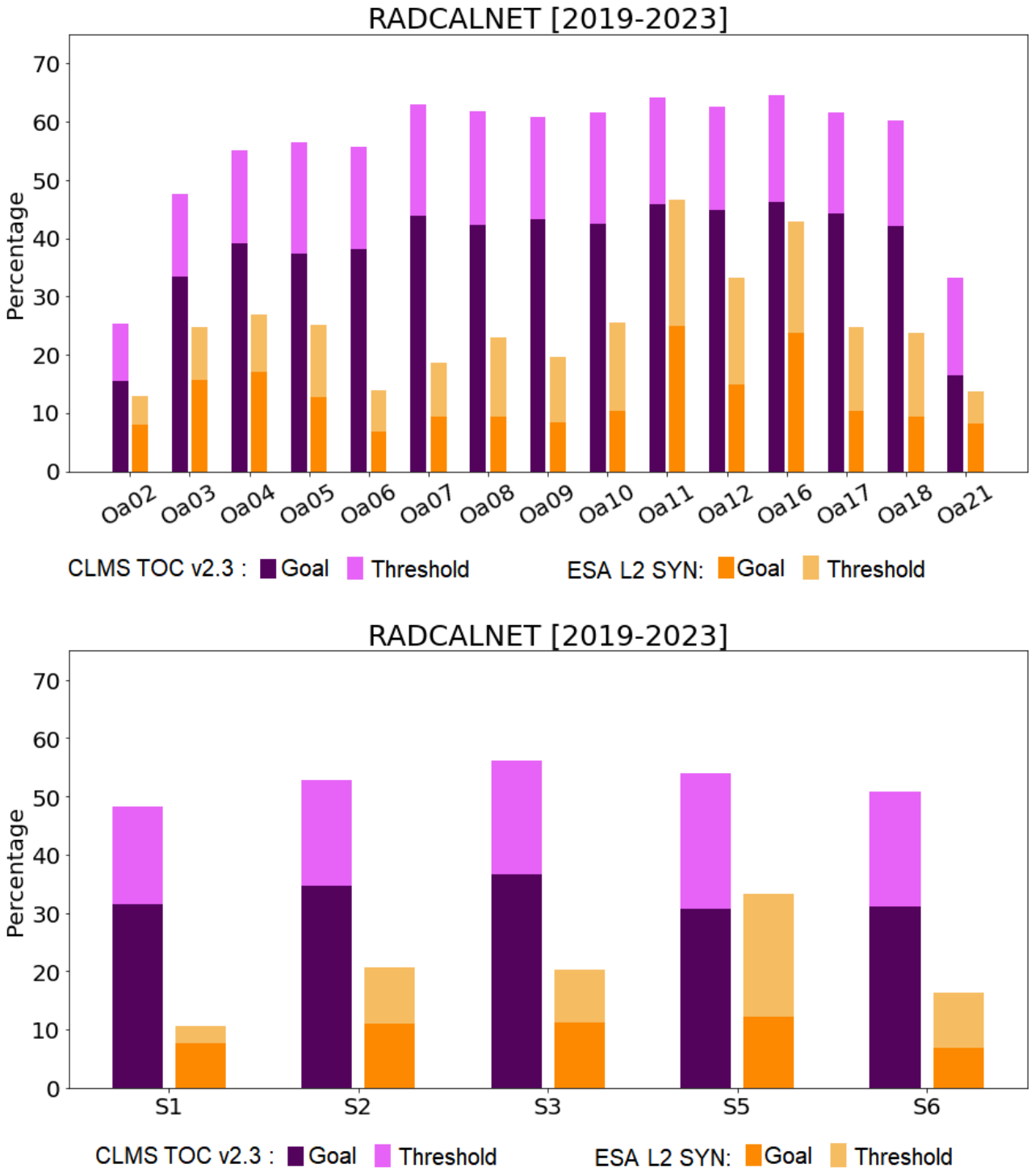


Figure 34: Percentage of cases of CLMS TOC 2.3 (purple) and ESA L2 SYN (orange) reflectances (VZA<10°) versus RadCalNet measurements (2019-2023) within goal (dark colour) and threshold (light colour) requirements. Top and bottom includes OLCI and SLSTR channels.

Main conclusions for Sentinel-3 OLCI channels are:

- CLMS TOC V2.3 (Figure 30, Table 9) shows:
  - Remarkably good accuracy with almost no bias or very low bias (i.e., B and MD typically within ±2%) for most channels [Oa03 – Oa18] and larger for the lower (B of 7.9% for Oa02) and higher (5.4% for Oa21) spectral channels.
  - Overall uncertainty (RMSD) typically better than 0.026 for all channels (except Oa21) which is around 7-9% in relative terms for most of them (Oa07 – Oa18).
- CLMS TOC V2.3 shows improved results than previous TOC V1 [CGLOPS-1_QAR_S3-TOCr300m-V1], where biases typically around ±3% and RMSD around 9-13% were found. The improved results of TOC V2.3 can be partly attributed to the application of OLCI-A calibration factors.
- CLMS TOC V2.3 shows realistic uncertainties, which are similar in magnitude than the uncertainties provided by RadCalNet measurements.
- ESA L2 SYN products show significant worse results (Figure 31, Table 10) than CLMS, with systematic positive biases typically between 3% and 10% (larger for Oa02 with 11.9%) and RMSD up to 0.05 and typically around 11% and 15% in relative terms (except Oa02 and Oa03 with 25.8% and 18.9%, respectively).
  - The systematic positive bias of ESA L2 SYN for all OLCI channels is in line (but higher in magnitude) to that reported for ESA level 1 radiometric annual performance report for 2023 year [OMPC.ACR.APR.004], where OLCI radiances showed an excess of brightness (up to 5%).
- Typically, around 33-46% of CLMS V2.3 pixels are within goal requirement (i.e., 3%) for most channels and typically 55-65% considering threshold level (5%). Significantly better results are found compared with ESA L2 SYN (Figure 34).
- The accuracy of TOC V2.3 (B and MD within ±2%) is also better to that found for other existing AC processors, such as iCOR[8] AC on Sentinel-3/OLCI, where larger positive biases of iCOR surface reflectances within ±8% were found compared with RadCalNet measurements over GONA and RVUS (Wolters et al., 2021).

Main conclusions for Sentinel-3 SLSTR channels are:

- CLMS TOC V2.3 (Figure 32, Table 11) shows:
  - Remarkably good accuracy, with B and MD within ±1.5% for all channels.
  - Overall uncertainty (RMSD) typically better than 0.03 (between 7% and 10% in relative terms).

[8] https://blog.vito.be/remotesensing/icor-for-sentinel-3 (accessed on 2025.02.05)

- CLMS TOC V2.3 shows improved results than previous V1 [CGLOPS-1_QAR_S3-TOCr300m-V1], where the bias was around 1-2% for S1-S3 and up to -15% for S5 and S6. V1 showed uncertainty (RMSD) around 10% for S1-S3 and up to 18% for S5 and S6.
- CLMS TOC V2.3 shows realistic uncertainties for SLSTR, which are similar in magnitude to that of the RadCalNet measurements.
- ESA L2 SYN products (Figure 33, Table 12) show worse performance than CLMS TOC V2.3, with larger positive biases for VNIR channels (11.4%, 6.5% and 6.9% for S1, S2 and S3) and larger negative bias for SWIR (-9.3% and -12.3% for S5 and S6). The uncertainty (RMDS) of ESA L2 SYN ranges from 12% (S3) to 17.4% (S1).
- CLMS TOC V2.3 shows around 30-37% of pixels within goal requirement and 48-56% considering the threshold level. ESA L2 SYN shows lower percentage of cases within goal (up to 12%) and threshold requirements (up to 33%) (see Figure 34).
- Note than CLMS V2.3 incorporates the recommended ESA calibration adjustment factors (Section 3.2.1), which significantly improves the accuracy for SWIR domain (i.e., S5 and S6 channels) compared to TOC V1 and ESA L2 SYN for the period under evaluation (2019-2023). Note that ESA L2 SYN incorporated the calibration adjustment factors from July 2023 onwards [S3.PN-SYN-L2.10].

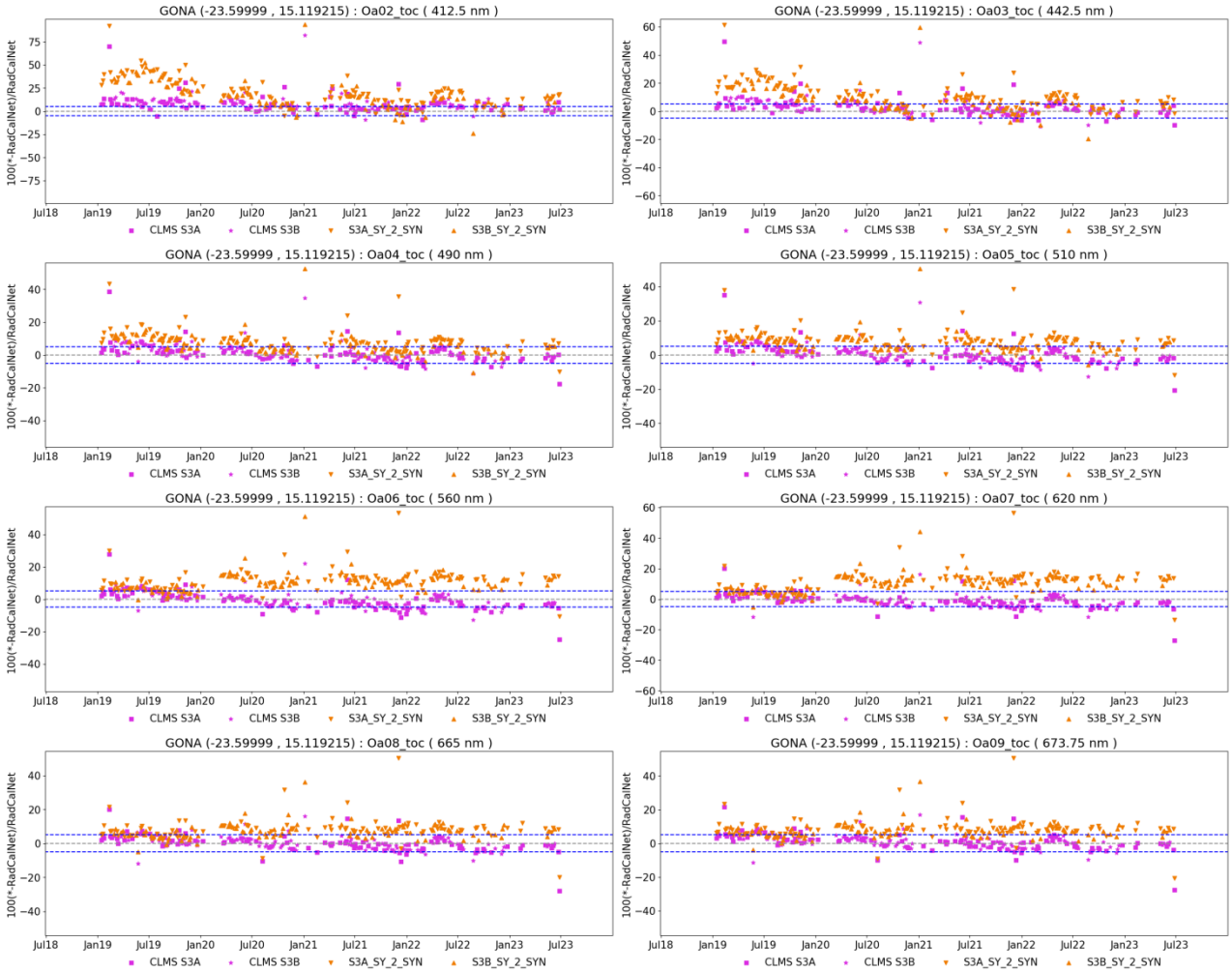
In order to evaluate the evolution of the bias over time, temporal series of relative bias respect to RadCalNet are displayed in Figure 35 for OLCI and Figure 36 for SLSTR. For that purpose, GONA pseudo-invariant site is selected as for the rest of sites no systematic trend in terms of sign of the bias was found.

Main conclusions for OLCI are:

- CLMS TOC V2.3 shows a slightly different trend of the bias depending on the period: for most channels, TOC V2.3 shows slight positive bias for January 2019 to June 2020 and bias close to zero or slightly negative for July 2020 onwards. Note that most of the observations are within the ±5% threshold level. The exception is Oa21 where systematic positive bias is found.
- The relative bias of CLMS product is systematically lower than the relative bias of ESA L2 SYN products, which is mainly positive (ESA > RadCalNet). For Oa02, Oa03, Oa4 and Oa21 the bias of ESA L2 SYN is larger in 2019 compared with the rest of the period. The opposite trend is observed for channels from Oa06 to Oa10, with larger bias from January 2020 onwards compared with 2019. This could be explained by changes in configuration parameters of ESA SYN Level 1 from January 2020 onwards [S3_SYN_DataHub].

Main conclusions for SLSTR are:

- The relative bias of CLMS TOC V2.3 shows the same behaviour than for OLCI reflectances with, however, enhanced seasonality that can be related to slight seasonal variations in the surface reflectance values due to changes in sun zenith angle. The relative bias values are within +/-5% threshold level for VNIR bands while they are systematically positive for SWIR bands.
- The relative bias of ESA L2 SYN displays the same seasonality, but with larger values than CLMS TOC reflectances. For SWIR S5 and S6 channels, ESA L2 SYN and CLMS TOC V2.3 products show opposite sign of relative bias.



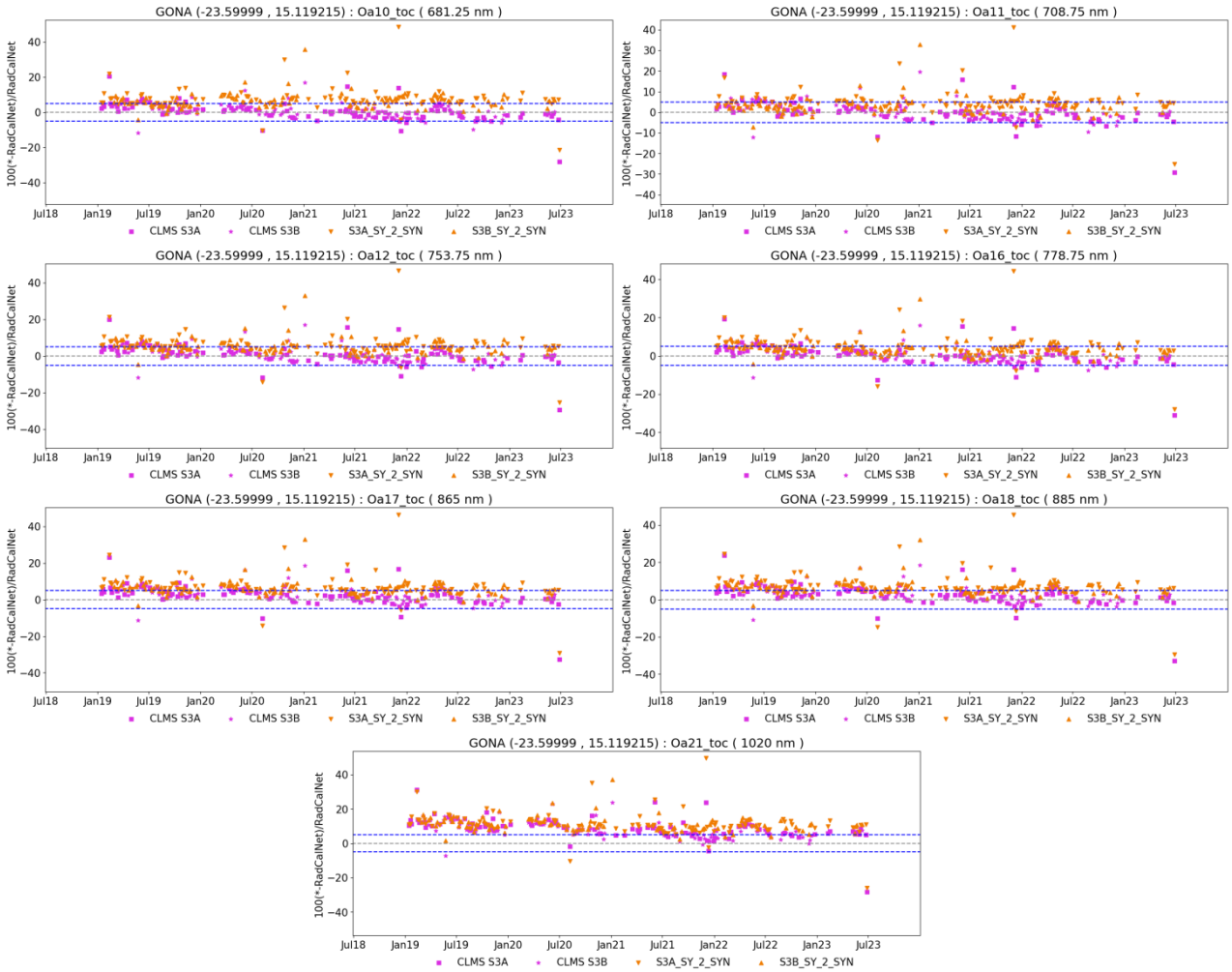


Figure 35: Temporal profiles over (2019-2023 period) of relative bias of surface reflectance products (OLCI channels, VZA<10º) versus RadCalNet measurements calculated as 100*(product - RadCalNet) / RadCalNet. Relative bias of CLMS TOC V2.3 is represented with purple squares and dots for Sentinel-3A and –3B, and relative bias ESA L2 SYN with orange down and up-pointing triangles for Sentinel-3A and -3B. Horizontal blue dashed lines correspond to ±5% threshold level.

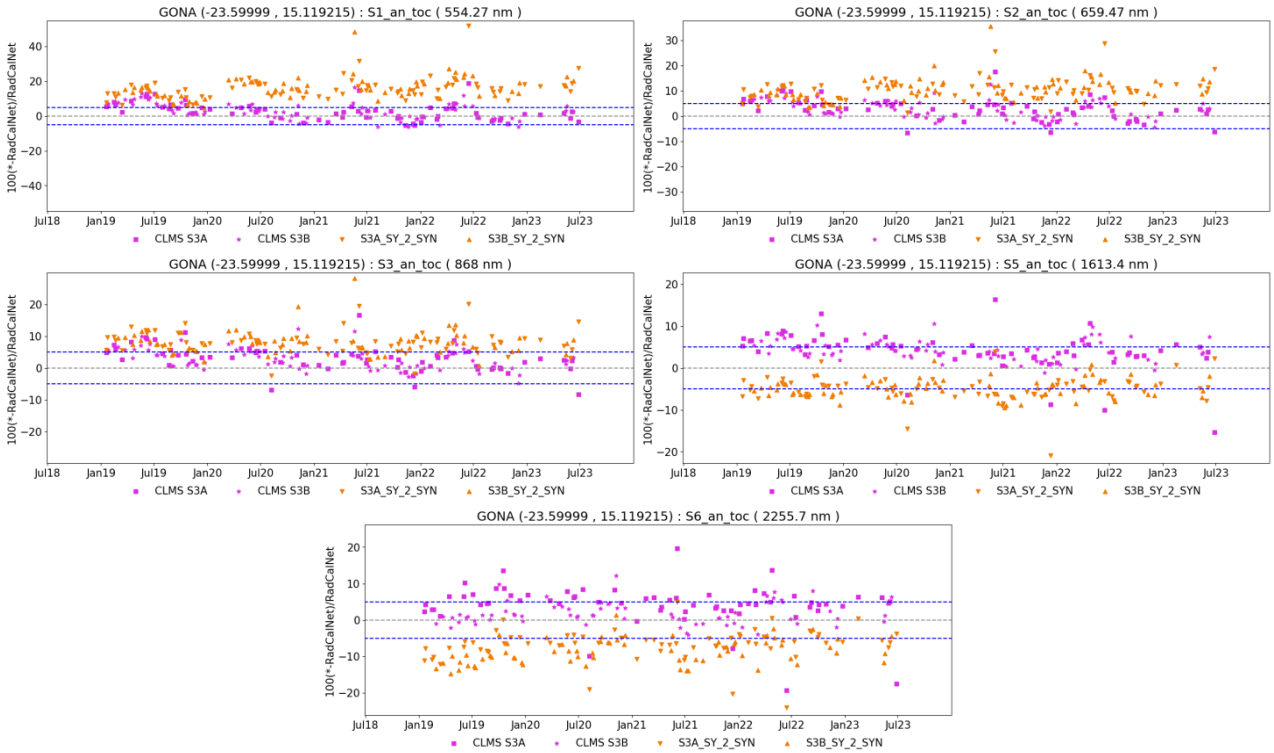
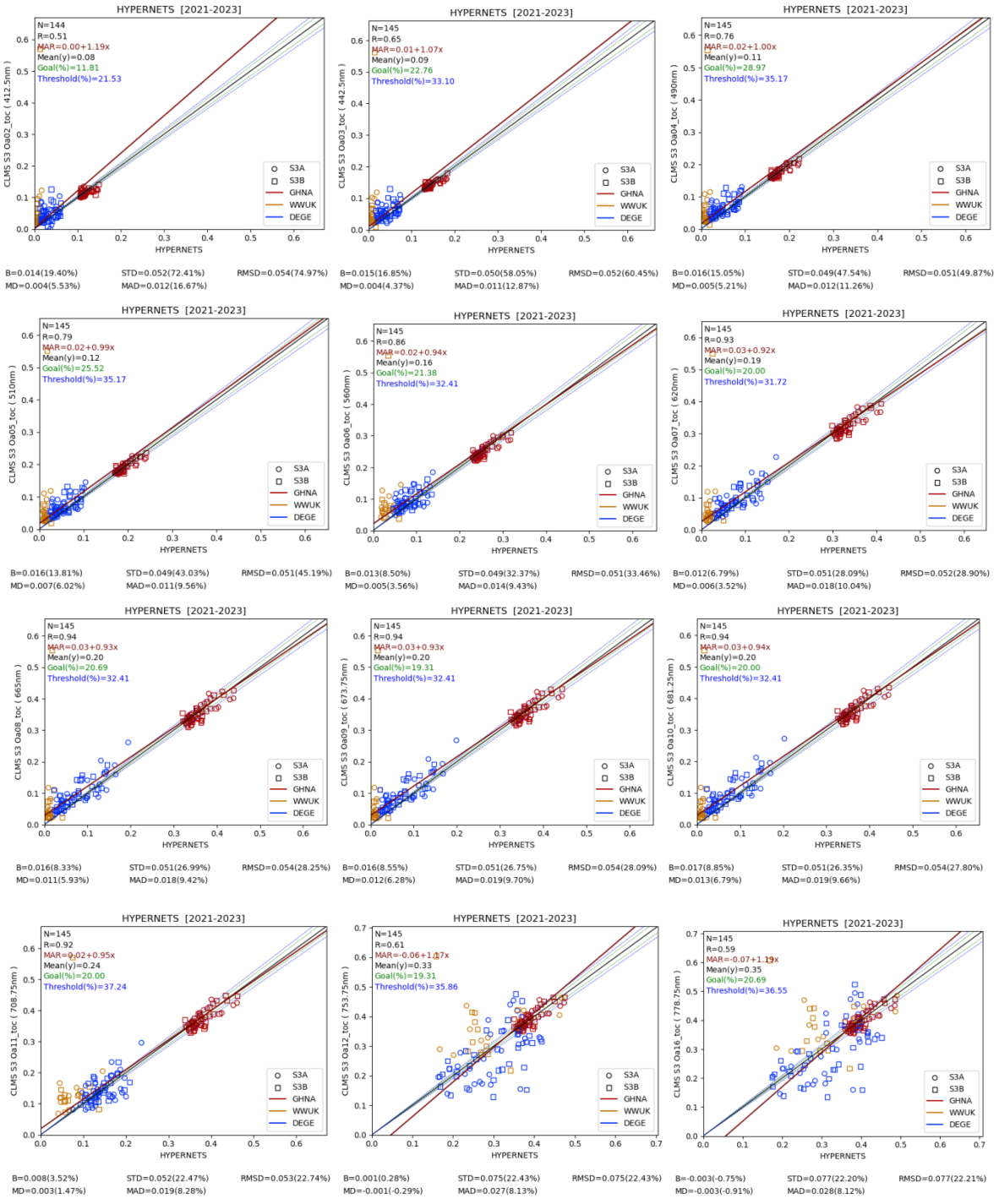


Figure 36: As in Figure 35 but for SLSTR channels.

4.4.2 Comparison with HYPERNETS measurements

Scatter-plots and validation metrics of Sentinel-3 CLMS TOC V2.3 versus HYPERNETS measurements are performed for OLCI (Figure 37, Table 13) and SLSTR (Figure 38, Table 14) channels for the period of 2021-2023 (limited to HYPERNETS availability, see Table 6). Sentinel-3 TOC reflectance retrievals and HYPERNETS observations are compared under similar viewing geometries, only for observations with differences in viewing angles lower than 5°. Note that only Sentinel-3 pixels showing good quality according to Table 3 were considered. Additionally, in case of SLSTR, '**pixel_classif_flags**' corresponding to IdePix are also used, as several outliers were identified due to under detection of clouds.


HYPERNETS [2021-2023]
N=144
R=0.51
MAR=0.00+1.19x
Mean(y)=0.08
Goal(%)=11.81
Threshold(%)=21.53
CLMS S3 Oa02_toc (412.5nm)
B=0.014(19.40%) STD=0.052(72.41%) RMSD=0.054(74.97%)
MD=0.004(5.53%) MAD=0.012(16.67%)
HYPERNETS [2021-2023]
N=145
R=0.65
MAR=0.01+1.07x
Mean(y)=0.09
Goal(%)=22.76
Threshold(%)=33.10
CLMS S3 Oa03_toc (442.5nm)
B=0.015(16.85%) STD=0.050(58.05%) RMSD=0.052(60.45%)
MD=0.004(4.37%) MAD=0.011(12.87%)
HYPERNETS [2021-2023]
N=145
R=0.76
MAR=0.02+1.00x
Mean(y)=0.11
Goal(%)=28.97
Threshold(%)=35.17
CLMS S3 Oa04_toc (490nm)
B=0.016(15.05%) STD=0.049(47.54%) RMSD=0.051(49.87%)
MD=0.005(5.21%) MAD=0.012(11.26%)
HYPERNETS [2021-2023]
N=145
R=0.79
MAR=0.02+0.99x
Mean(y)=0.12
Goal(%)=25.52
Threshold(%)=35.17
CLMS S3 Oa05_toc (510nm)
B=0.016(13.81%) STD=0.049(43.03%) RMSD=0.051(45.19%)
MD=0.007(6.02%) MAD=0.011(9.56%)
HYPERNETS [2021-2023]
N=145
R=0.86
MAR=0.02+0.94x
Mean(y)=0.16
Goal(%)=21.38
Threshold(%)=32.41
CLMS S3 Oa06_toc (560nm)
B=0.013(8.50%) STD=0.049(32.37%) RMSD=0.051(33.46%)
MD=0.005(3.56%) MAD=0.014(9.43%)
HYPERNETS [2021-2023]
N=145
R=0.93
MAR=0.03+0.92x
Mean(y)=0.19
Goal(%)=20.00
Threshold(%)=31.72
CLMS S3 Oa07_toc (620nm)
B=0.012(6.79%) STD=0.051(28.09%) RMSD=0.052(28.90%)
MD=0.006(3.52%) MAD=0.018(10.04%)
HYPERNETS [2021-2023]
N=145
R=0.94
MAR=0.03+0.93x
Mean(y)=0.20
Goal(%)=20.69
Threshold(%)=32.41
CLMS S3 Oa08_toc (665nm)
B=0.016(8.33%) STD=0.051(26.99%) RMSD=0.054(28.25%)
MD=0.011(5.93%) MAD=0.018(9.42%)
HYPERNETS [2021-2023]
N=145
R=0.94
MAR=0.03+0.93x
Mean(y)=0.20
Goal(%)=19.31
Threshold(%)=32.41
CLMS S3 Oa09_toc (673.75nm)
B=0.016(8.55%) STD=0.051(26.75%) RMSD=0.054(28.09%)
MD=0.012(6.28%) MAD=0.019(9.70%)
HYPERNETS [2021-2023]
N=145
R=0.94
MAR=0.03+0.94x
Mean(y)=0.20
Goal(%)=20.00
Threshold(%)=32.41
CLMS S3 Oa10_toc (681.25nm)
B=0.017(8.85%) STD=0.051(26.35%) RMSD=0.054(27.80%)
MD=0.013(6.79%) MAD=0.019(9.66%)
HYPERNETS [2021-2023]
N=145
R=0.92
MAR=0.02+0.95x
Mean(y)=0.24
Goal(%)=20.00
Threshold(%)=37.24
CLMS S3 Oa11_toc (708.75nm)
B=0.008(3.52%) STD=0.052(22.47%) RMSD=0.053(22.74%)
MD=0.003(1.47%) MAD=0.019(8.28%)
HYPERNETS [2021-2023]
N=145
R=0.61
MAR=-0.06+1.17x
Mean(y)=0.33
Goal(%)=19.31
Threshold(%)=35.86
CLMS S3 Oa12_toc (753.75nm)
B=0.001(0.28%) STD=0.075(22.43%) RMSD=0.075(22.43%)
MD=-0.001(-0.29%) MAD=0.027(8.13%)
HYPERNETS [2021-2023]
N=145
R=0.59
MAR=-0.07+1.19x
Mean(y)=0.35
Goal(%)=20.69
Threshold(%)=36.55
CLMS S3 Oa16_toc (778.75nm)
B=-0.003(-0.75%) STD=0.077(22.20%) RMSD=0.077(22.21%)
MD=-0.003(-0.91%) MAD=0.028(8.12%)
S3A
S3B
GHNA
WWUK
DEGE
HYPERNETS

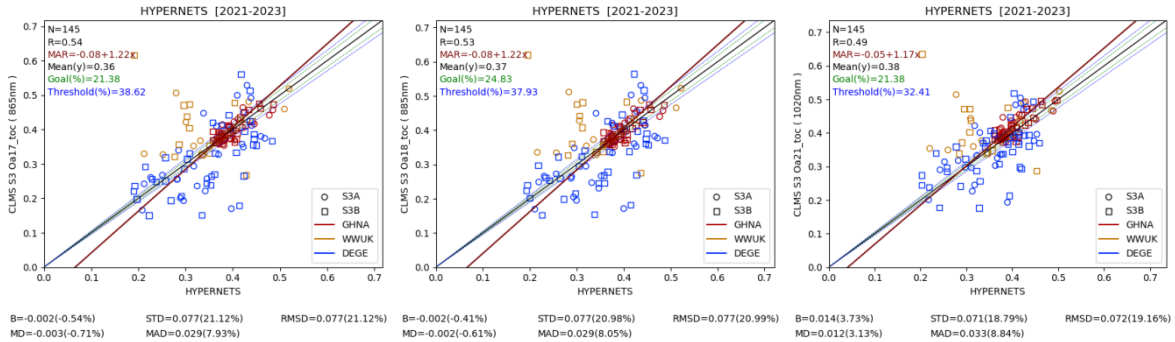


Figure 37: Scatter-plots of CLMS Sentinel-3 TOC V2.3 versus HYPERNETS measurements (ΔVZA < 5º; 2021-2023) for OLCI channels. Green and blue lines delimit goal and threshold requirements.

Table 13: Performance statistics of CLMS Sentinel-3 TOC V2.3 versus HYPERNETS measurements (ΔVZA < 5º; 2021-2023) for OLCI channels.

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Oa02	144	0.51	0.014 (19.4%)	0.004 (5.5%)	0.052 (72.4%)	0.012 (16.7%)	0.054 (75.0%)	0.00 /1.19	11.8 /21.5
Oa03	145	0.65	0.015 (16.9%)	0.004 (4.4%)	0.050 (58.1%)	0.011 (12.8%)	0.052 (60.5%)	0.01 /1.07	22.8 /33.1
Oa04	145	0.76	0.016 (15.1%)	0.005 (5.2%)	0.049 (47.5%)	0.012 (11.3%)	0.051 (49.9%)	0.02 /1.00	29.0 /35.2
Oa05	145	0.79	0.016 (13.8%)	0.007 (6.0%)	0.049 (43.0%)	0.011 (9.6%)	0.051 (45.2%)	0.02 /0.99	25.5 /35.2
Oa06	145	0.86	0.013 (8.5%)	0.005 (3.6%)	0.049 (32.4%)	0.014 (9.4%)	0.051 (33.5%)	0.02 /0.94	21.4 /32.4
Oa07	145	0.93	0.012 (6.8%)	0.006 (3.5%)	0.051 (28.1%)	0.018 (10.0%)	0.052 (28.9%)	0.03 /0.92	20.0 /31.7
Oa08	145	0.94	0.016 (8.3%)	0.011 (5.9%)	0.051 (27.0%)	0.018 (9.4%)	0.054 (28.3%)	0.03 /0.93	20.7 /32.4
Oa09	145	0.94	0.016 (8.6%)	0.012 (6.3%)	0.051 (26.8%)	0.019 (9.7%)	0.054 (28.1%)	0.03 /0.93	19.3 /32.4
Oa10	145	0.94	0.017 (8.9%)	0.013 (6.8%)	0.051 (26.4%)	0.019 (9.7%)	0.054 (27.8%)	0.03 /0.94	20.0 /32.4
Oa11	145	0.92	0.008 (3.5%)	0.003 (1.5%)	0.052 (22.5%)	0.019 (8.3%)	0.053 (22.7%)	0.02 /0.95	20.0 /37.2
Oa12	145	0.61	0.001 (0.3%)	-0.001 (-0.3%)	0.075 (22.4%)	0.027 (8.1%)	0.075 (22.4%)	-0.06 /1.17	19.3 /35.9
Oa16	145	0.59	-0.003 (-0.8%)	-0.003 (-0.9%)	0.077 (22.2%)	0.028 (8.1%)	0.077 (22.2%)	-0.07 /1.19	20.7 /36.6
Oa17	145	0.54	-0.002 (-0.5%)	-0.003 (-0.7%)	0.077 (21.1%)	0.029 (7.9%)	0.077 (21.1%)	-0.08 /1.22	21.4 /38.6
Oa18	145	0.53	-0.002 (-0.4%)	-0.002 (-0.6%)	0.077 (21.0%)	0.029 (8.1%)	0.077 (21.0%)	-0.08 /1.22	24.8 /37.9
Oa21	145	0.49	0.014 (3.7%)	0.012 (3.1%)	0.071 (18.8%)	0.033 (8.8%)	0.072 (19.2%)	-0.05 /1.17	21.4 /32.4

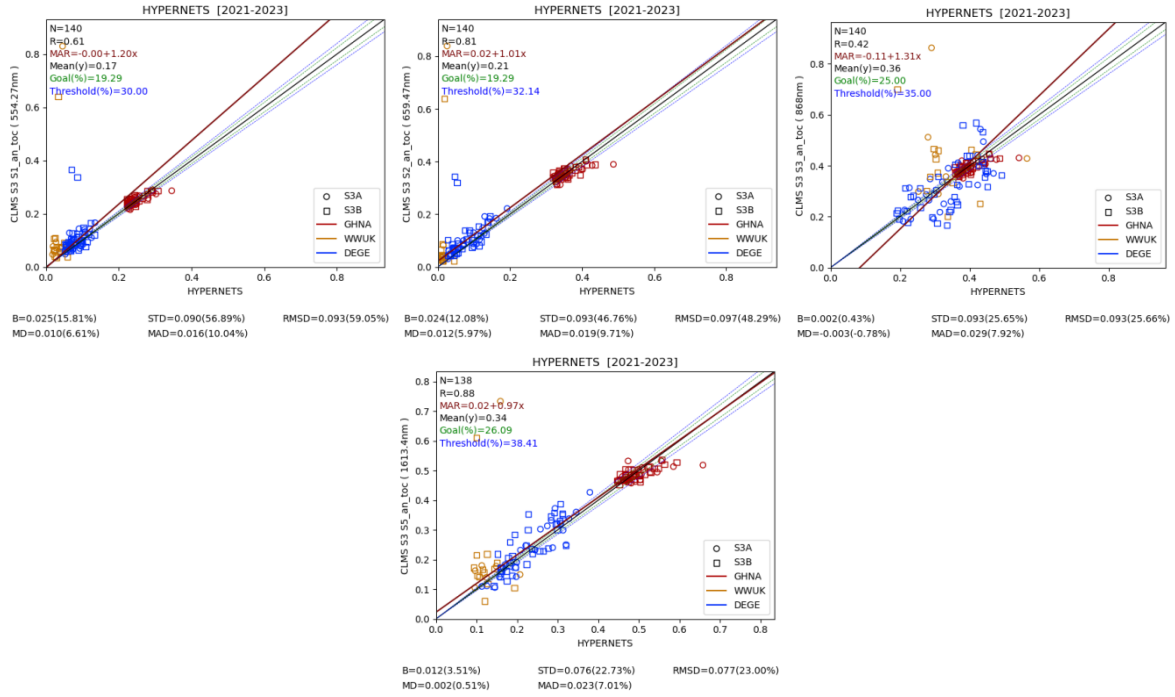


Figure 38: Scatter-plots of CLMS Sentinel-3 TOC V2.3 versus HYPERNETS measurements (ΔVZA < 5°;2021-2023) for SLSTR channels. Green and blue lines delimit goal and threshold requirements.

Table 14: Performance statistics of CLMS Sentinel-3 TOC V2.3 versus HYPERNETS measurements (ΔVZA < 5°;2021-2023) for SLSTR channels.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
S1	140	0.61	0.025 (15.8%)	0.010 (6.6%)	0.009 (56.9%)	0.016 (10.0%)	0.093 (59.1%)	0.00 /1.2	19.3 /30.0
S2	140	0.81	0.024 (12.1%)	0.012 (6.0%)	0.093 (46.8%)	0.019 (9.7%)	0.097 (48.3%)	0.02 /1.01	19.3 /32.1
S3	140	0.42	0.002 (0.4%)	-0.003 (-0.8%)	0.093 (25.7%)	0.029 (7.9%)	0.093 (25.7%)	-0.11 /1.31	25.0 /35.0
S5	138	0.88	0.012 (3.5%)	0.002 (0.5%)	0.076 (22.7%)	0.023 (7.0%)	0.077 (23.0%)	0.02 /0.97	26.1 /38.4

Main conclusions for Sentinel-3 OLCI TOC V2.3 (Figure 37, Table 13) are:

- Positive median deviation (MD) is found for channels from Oa02 to Oa11 and for Oa21, typically lower than 5% (up to 6.8% for Oa10). Note that MD is more suitable than B in those cases due to the low number of samples and the influence of several outliers. The positive deviation is mainly due to the overestimation over WWUK forest site, where TOC V2.3 shows

larger values than expected, which is not observed over other forest AOIs (see section 4.5.3). In case of channels from Oa12 to Oa18, slight negative deviation (MD within -1%) is found.

- The overall uncertainty (RMSD) is around 0.05 for channels in the visible domain (Oa02-Oa11) and around 0.075 for NIR channels (Oa12-Oa21, up to 22% in relative terms).
- Around 20-30% of pixels are typically within goal requirement (i.e., 3%) for most channels and typically 30-38% considering threshold level (5%).

Main conclusions for Sentinel-3 SLSTR TOC V2.3 (Figure 38, Table 14) are:

- Positive median deviation for visible domain, around 6% for S1 and S2, almost no deviation for NIR (MD = -0.8% for S3) and SWIR (MD = 0.5% for S5), showing remarkably good accuracy.
- Worse overall uncertainty is found compared with OLCI channels, with RMSD close to 0.1 for VNIR channels.
- 19-26% of pixels are within goal requirement and 30-38% considering threshold level.

4.4.3 Comparison with FRM4Veg measurements

Figure 39 and Figure 40 show the spectral profiles of CLMS Sentinel-3 TOC V2.3 and FRMs for alfalfa and bare soil samples of the ESA FRM4Veg campaign performed in Barrax on 2nd August 2018. Note that in case of Sentinel-3, the view zenith angle on 2nd August was around 31.3º, and is not comparable with the nadir view of FRM4Veg. Consequently, we used the closest date under the most similar viewing configuration: the 5th August where near nadir viewing zenith angles of OLCI and SLSTR were 3.7º and 7.1º, respectively. For this date, Sentinel-3A and -3B flew in tandem, with almost identical viewing geometries. Therefore, acquisitions from both platforms are included (except S5 and S6 channels in Sentinel-3B SLSTR as these data are not available). However, the quantitative analysis of differences is also not possible as even small changes in the atmospheric conditions hamper the comparability of ground and satellite observations. Nevertheless, the spectral profiles which provide insights of the quality of the OLCI and SLSTR reflectances are shown.

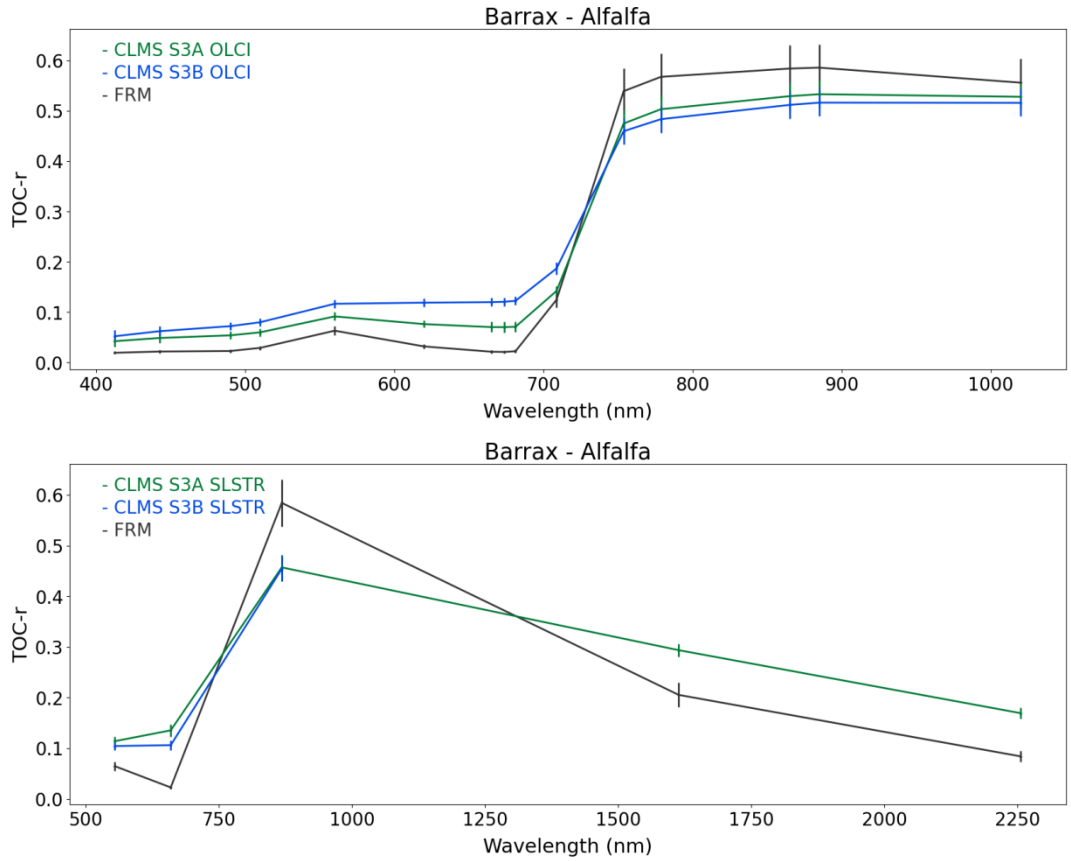


Figure 39: Spectral profiles of CLMS Sentinel-3 TOC V2.3 (green and blue lines for Sentinel-3A and Sentinel-3B, respectively) and FRM (grey) TOC reflectance values, measured over Alfalfa samples in Barrax for OLCI (top) and SLSTR (bottom) channels. Acquisition date of Sentinel-3 was 5nd August, and FRM campaign date was on 2nd August 2018. Vertical bars correspond to uncertainties associated to Sentinel-3 TOC reflectances and FRMs.

For Alfalfa (Figure 39):

- In case of OLCI:
  - CLMS TOC V2.3 reproduces similarly the spectral signature of this dense alfalfa cover, with a sharp transition between red and NIR regions. However, systematic differences of different sign are found in the visible (positive bias, OLCI>FRM) and in the NIR (negative bias, OLCI<FRM), and the signal in the red and red-edge channels is significantly higher. This seems to indicate a potential adjacency effect in the signal, as the border of the alfalfa fields are bare soils (see Figure 4).
  - Sentinel-3A reflectance values are closer to FRM than Sentinel-3B reflectances, whatever the spectral range.

- For SLSTR, CLMS TOC V2.3 shows lower variation between red and NIR channels, which is probably due to a higher impact of the adjacency effect in SLSTR due to its larger effective footprint. CLMS TOC V2.3 shows higher reflectance values than FRM for SWIR channels.

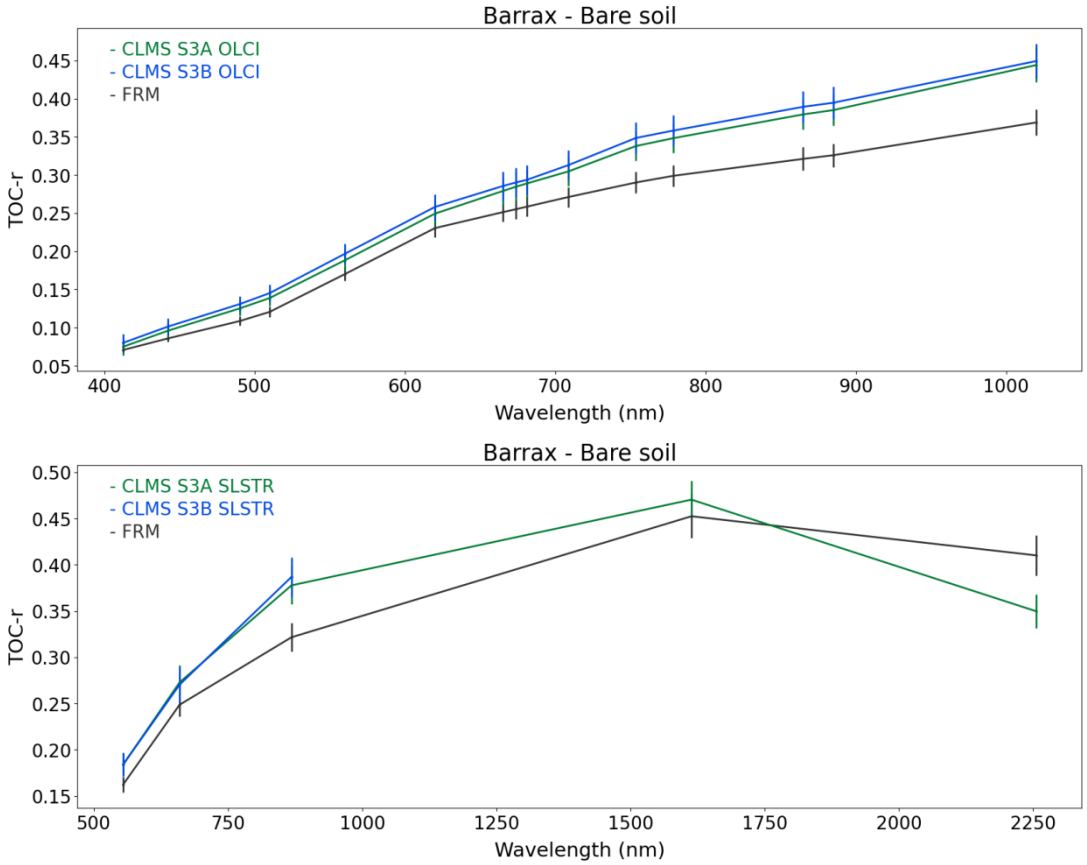


Figure 40: As in Figure 39 for bare soil samples.

For bare soil (Figure 40):

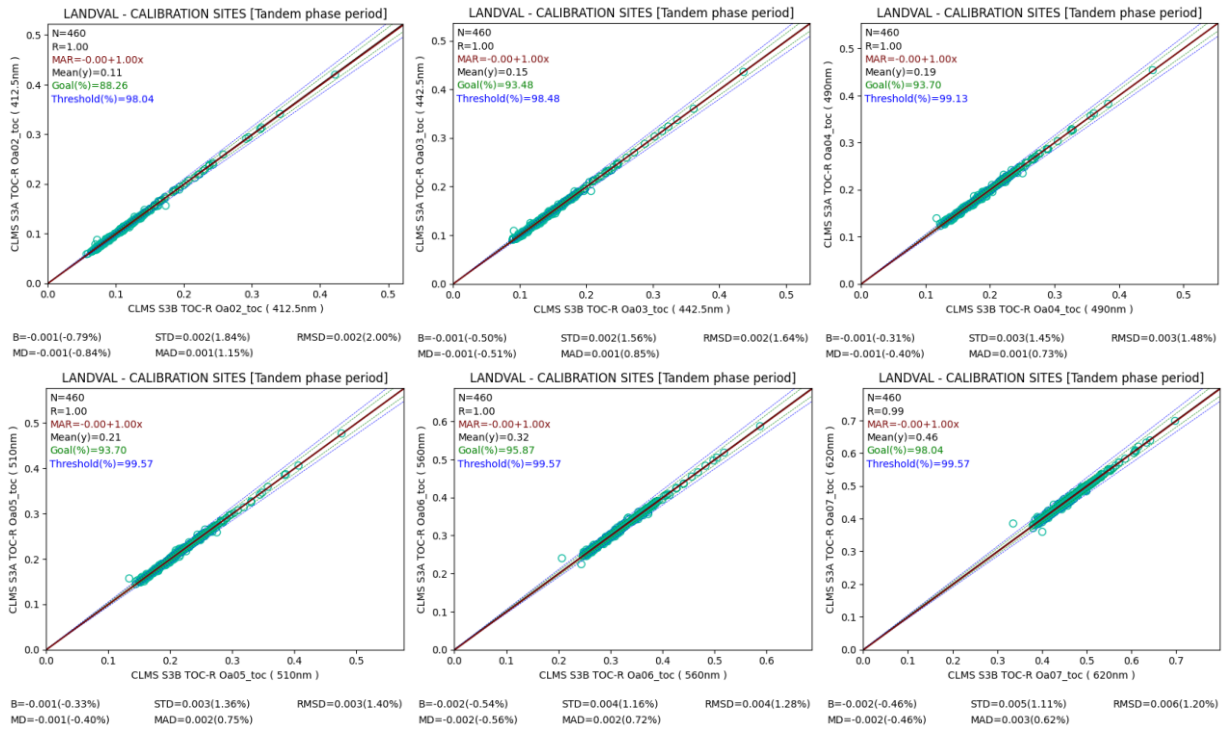
- In case of OLCI:
  - CLMS TOC V2.3 show good agreement and reproduces quite well the spectral FRM of bare area, showing systematically slightly higher values of reflectances. From 700nm onwards, the FRM vs TOC V2.3 differences are higher than their associated uncertainties.
  - Like over Alfalfa, Sentinel-3ATOC reflectances are closer to FRM than Sentinel-3B reflectances.
- For SLSTR, CLMS TOC V2.3 reproduces quite well the spectral signature of the FRM data over this bare area, showing higher values for S1, S2, S3 and S5 and lower values for S6.

It should be noted that CLMS TOC V2.3 associated uncertainties are similar in magnitude than FRM uncertainties (see vertical bars in Figure 39 and Figure 40) which indicates they are realistic.

4.5 PRODUCT INTERCOMPARISON

4.5.1 Error evaluation between Sentinel-3A and Sentinel-3B over calibration sites

The consistency between CLMS TOC reflectance V2.3 from Sentinel-3A versus Sentinel-3B is evaluated in this section. The analysis is performed during the tandem phase (i.e., from 7 June to 16 October 2018) where both satellites observe the same place within 30 seconds (i.e., with the same acquisition geometry) so being perfect to inter-calibrate very accurately the two satellites. Scatter-plots and validation metrics are presented in Figure 41, Table 15 for OLCI and Figure 42, Table 16 for SLSTR. Note that quality flags (Table 3) were used to exclude pixels with suboptimal quality.



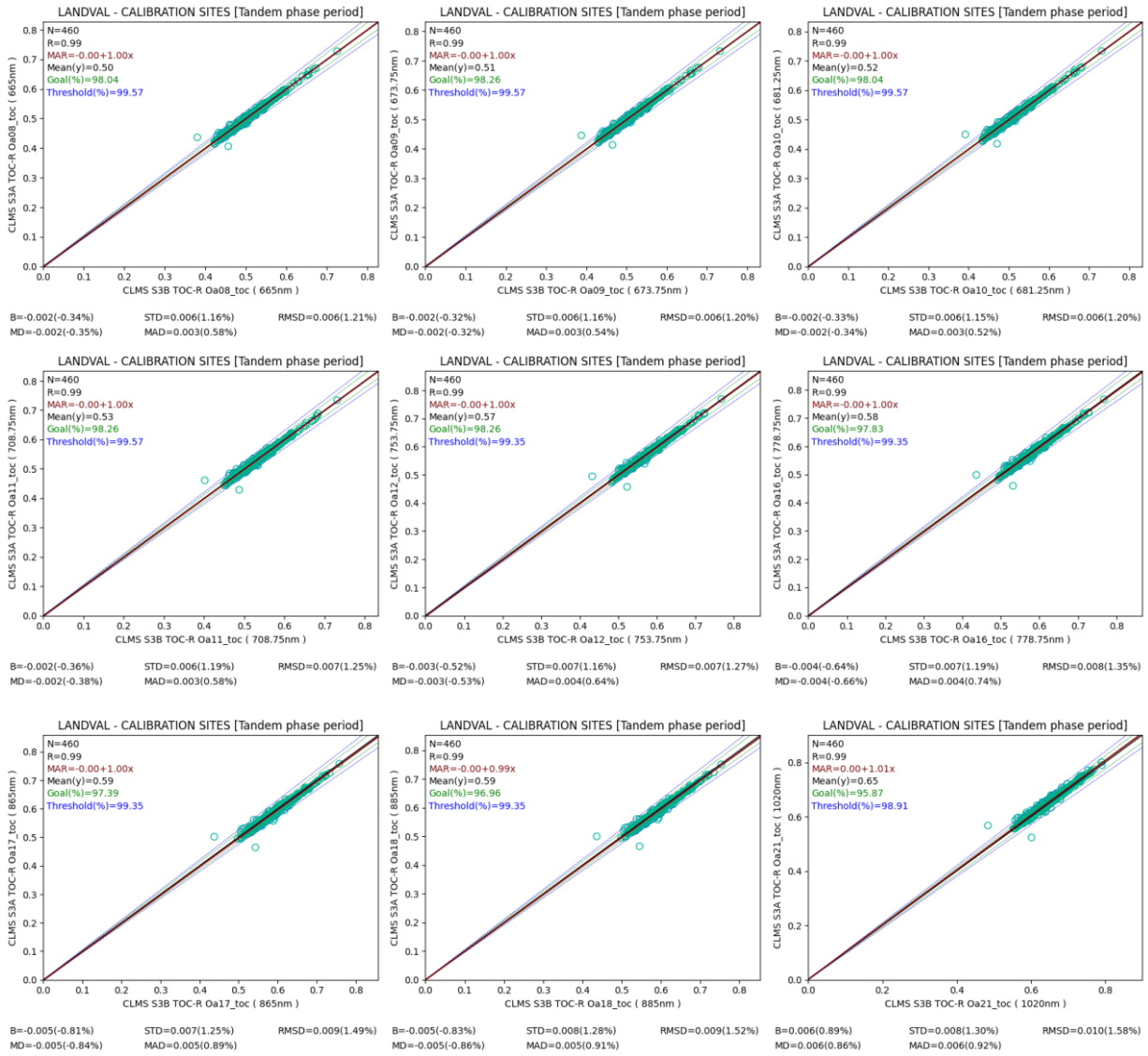


Figure 41: Scatter-plots of CLMS Sentinel-3A versus Sentinel-3B TOC V2.3 for OLCI bands over calibration sites for the tandem phase period (same acquisition geometry). Green and blue lines delimit goal and threshold requirements.

Table 15: Validation statistics of CLMS Sentinel-3A versus Sentinel3-B TOC V2.3 for OLCI bands over calibration sites for the tandem phase period (same acquisition geometry).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	460	1.00	-0.001 (-0.8%)	-0.001 (-0.8%)	0.002 (1.8%)	0.001 (1.2%)	0.002 (2.0%)	0.00 /1.00	88.3 /98.0
Oa03	460	1.00	-0.001 (-0.5%)	-0.001 (-0.5%)	0.002 (1.6%)	0.001 (0.9%)	0.002 (1.6%)	0.00 /1.00	93.5 /98.5
Oa04	460	1.00	-0.001 (-0.3%)	-0.001 (-0.4%)	0.003 (1.5%)	0.001 (0.7%)	0.003 (1.5%)	0.00 /1.00	93.7 /99.1
Oa05	460	1.00	-0.001 (-0.3%)	-0.001 (-0.4%)	0.003 (1.4%)	0.002 (0.8%)	0.003 (1.4%)	0.00 /1.00	93.7 /99.6
Oa06	460	1.00	-0.002 (-0.5%)	-0.002 (-0.6%)	0.004 (1.2%)	0.002 (0.7%)	0.004 (1.3%)	0.00 /1.00	95.9 /99.6
Oa07	460	1.00	-0.002 (-0.5%)	-0.002 (-0.5%)	0.005 (1.1%)	0.003 (0.6%)	0.006 (1.2%)	0.00 /1.00	98.0 /99.6
Oa08	460	0.99	-0.002 (-0.3%)	-0.002 (-0.4%)	0.006 (1.2%)	0.003 (0.6%)	0.006 (1.2%)	0.00 /1.00	98.0 /99.6
Oa09	460	0.99	-0.002 (-0.3%)	-0.002 (-0.3%)	0.006 (1.2%)	0.003 (0.5%)	0.006 (1.2%)	0.00 /1.00	98.3 /99.6
Oa10	460	0.99	-0.002 (-0.3%)	-0.002 (-0.3%)	0.006 (1.2%)	0.003 (0.5%)	0.006 (1.2%)	0.00 /1.00	98.0 /99.6
Oa11	460	0.99	-0.002 (-0.4%)	-0.002 (-0.4%)	0.006 (1.2%)	0.003 (0.6%)	0.007 (1.3%)	0.00 /1.00	98.3 /99.6
Oa12	460	0.99	-0.003 (-0.5%)	-0.003 (-0.5%)	0.007 (1.2%)	0.004 (0.6%)	0.007 (1.3%)	0.00 /1.00	98.3 /99.4
Oa16	460	0.99	-0.004 (-0.6%)	-0.004 (-0.7%)	0.007 (1.2%)	0.004 (0.7%)	0.008 (1.4%)	0.00 /1.00	97.8 /99.4
Oa17	460	0.99	-0.005 (-0.8%)	-0.005 (-0.8%)	0.007 (1.3%)	0.005 (0.9%)	0.009 (1.5%)	0.00 /1.00	97.4 /99.4
Oa18	460	0.99	-0.005 (-0.8%)	-0.005 (-0.9%)	0.008 (1.3%)	0.005 (0.9%)	0.009 (1.5%)	0.00 /0.99	97.0 /99.4
Oa21	460	0.99	0.006 (0.9%)	0.006 (0.9%)	0.008 (1.3%)	0.006 (0.9%)	0.010 (1.6%)	0.00 /1.01	95.9 /98.9

Main conclusions for OLCI channels (Figure 41, Table 15) are:

- Optimal agreement is found between Sentinel-3A and Sentinel-3B surface reflectances: with correlations higher than 0.99, linear MAR relationships on the 1:1 line and mean bias and MD better than - 0.5% for channels from Oa02 to Oa12 and within ±0.9% for channels Oa01 and from Oa16 to Oa21. These results of TOC V2.3 retrievals are better than the radiometric differences obtained during the tandem phase (Clerc et al., 2020; Lamquin et al., 2020), where TOA radiometric biases between 2% (in the blue) and 1% (in the NIR) were found. Then these

results demonstrate that the bright bias previously observed in OLCI-A TOC V1.1 is properly corrected in TOC V2.3.

- The precision (STD) and overall uncertainty (RMSD) is typically better than 1.5%, showing slightly large values (up to 2%) for the lower spectral channels (Oa02, Oa03).
- Typically, more than 93% (98%) of samples are within the goal (threshold) requirement.

Regarding the comparison for SLSTR (Figure 42, Table 16):

- Good consistency is found between Sentinel-3A and Sentinel-3B TOC V2.3 SLSTR reflectance values with bias between -0.3% and 1.5%. Sentinel-3A SLSTR tends to provide slightly higher values than Sentinel-3B for NIR and SWIR channels (S3, S5 and S6).
- In terms of precision, STD of around 1.5% is found, with slight worse results in case of S1 (2.4%). Similar results are found in terms of uncertainty, with RMSD values around 1.5%-2.5%.
- Typically, more than 92% of SLSTR cases are within goal uncertainty requirement and more than 97% within the threshold.

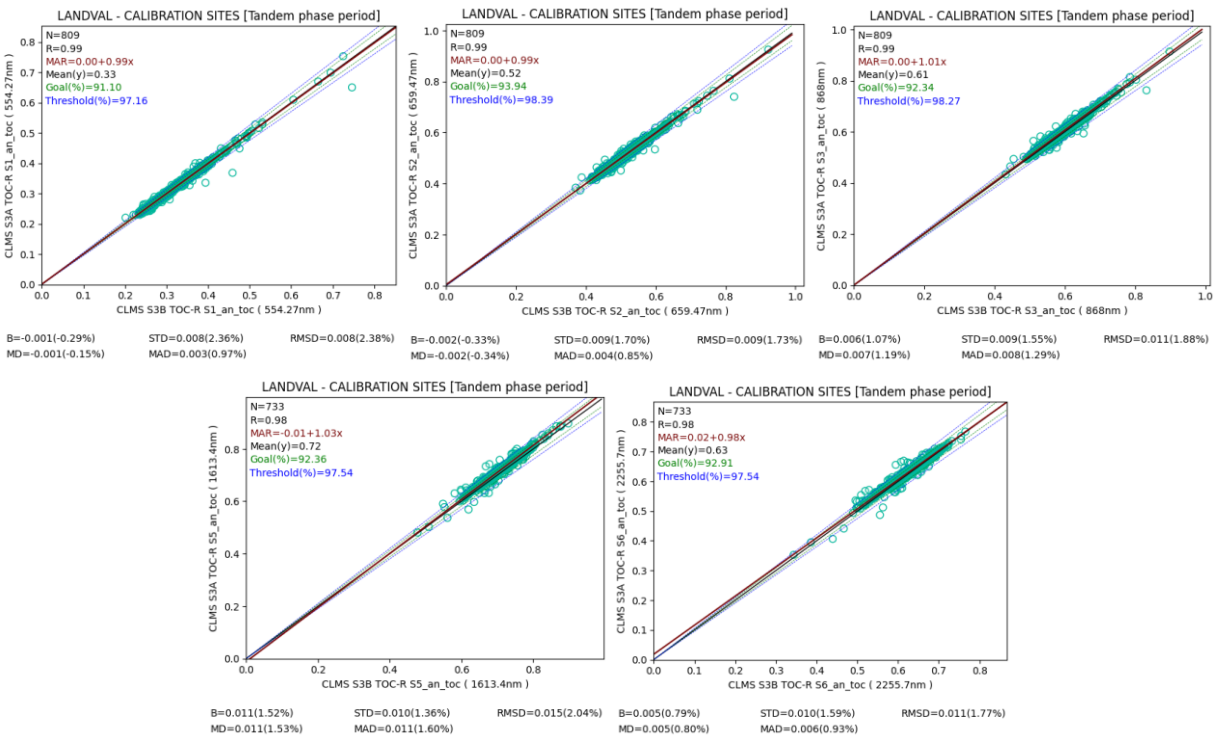


Figure 42: Scatter-plots of CLMS Sentinel-3A versus Sentinel-3B TOC reflectance V2.3 for SLSTR bands over calibration sites for the tandem phase period. Green and blue lines delimit goal and threshold requirements.

Table 16: Validation statistics of CLMS Sentinel-3A versus Sentinel-3B TOC reflectance V2.3 for SLSTR bands over calibration sites for the tandem phase period.

SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	809	0.99	-0.001 (-0.3%)	-0.001 (-0.2%)	0.008 (2.4%)	0.003 (1.0%)	0.008 (2.4%)	0.00 /0.99	91.1 /97.2
S2	809	0.99	-0.002 (-0.3%)	-0.002 (-0.3%)	0.009 (1.7%)	0.004 (0.9%)	0.009 (1.7%)	0.00 /0.99	93.9 /98.3
S3	809	0.99	0.006 (1.1%)	0.007 (1.2%)	0.009 (1.6%)	0.008 (1.3%)	0.011 (1.9%)	0.00 /1.01	92.3 /98.3
S5	733	0.98	0.011 (1.5%)	0.011 (1.5%)	0.010 (1.4%)	0.011 (1.6%)	0.015 (2.0%)	-0.01 /1.03	92.4 /97.5
S6	733	0.98	0.005 (0.8%)	0.005 (0.8%)	0.010 (1.6%)	0.006 (0.9%)	0.011 (1.8%)	0.02 /0.98	92.9 /97.5

4.5.2 Error evaluation between OLCI and SLSTR equivalent channels over calibration sites

The consistency between CLMS TOC V2.3 from OLCI and SLSTR equivalent channels is evaluated over desert calibration sites for the period from July 2018 to December 2022. Scatter-plots and validation metrics are presented in Figure 51 and Table 17. Note that quality flags (Table 3) are used to exclude pixels with suboptimal quality.

Main conclusions for the comparison of CLMS TOC V2.3 OLCI vs. SLSTR are:

- OLCI Oa06 shows higher values than equivalent SLSTR S1 channel, with mean bias and MD around 3.5% for both Sentinel-3A and –3B platforms. Remarkably good accuracy is found in the comparison with OLCI Oa08 versus SLSTR S2 (B and MD within ±0.3%) and OLCI Oa17 versus SLSTR S3 (B and MD within ±0.8%).
- Good precision (STD) around 2-3% is found, showing low scattering.
- In terms of uncertainty, RMSD values of around 4.5% are found in the comparison of OLCI Oa06 versus SLSTR S1, and around 2-2.5% for the comparisons of OLCI Oa08 versus SLSTR S2 and OLCI Oa17 versus SLSTR S3.
- Around 85-90% of samples are within goal requirement (~95-96% considering threshold) for OLCI Oa08 versus SLSTR S2 and OLCI Oa17 versus SLSTR S3. Low percentage of compliance is found in the comparison of OLCI Oa06 versus SLSTR S1 (32-35% goal and 71-72% threshold).

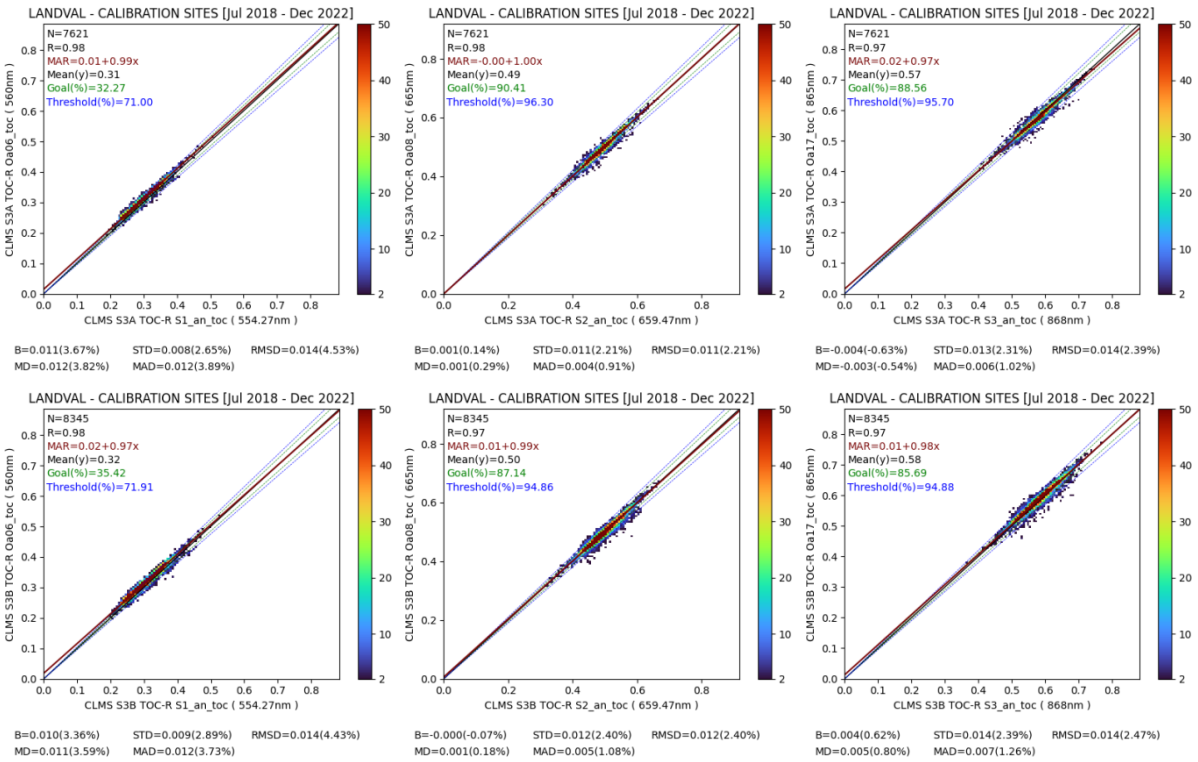


Figure 43: Scatter-plots of CLMS Sentinel-3A (top) and Sentinel-3B (bottom) TOC reflectance V2.3 OLCI versus SLSTR equivalent channels over calibration sites for the period from July 2018 to December 2022. Green and blue lines delimit goal and threshold requirements.

Table 17: Validation statistics of CLMS Sentinel-3 TOC reflectance V2.3 OLCI versus SLSTR equivalent channels over calibration sites for the period from July 2018 to December 2022.

	OLCI Band	SLSTR Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	%Goal /Threshold
Sentinel-3A	Oa06	S1	7621	0.98	0.011 (3.7%)	0.012 (3.8%)	0.008 (2.7%)	0.012 (3.9%)	0.014 (4.5%)	0.01 /0.99	32.3 /71.0
	Oa08	S2	7621	0.98	0.001 (0.1%)	0.001 (0.3%)	0.011 (2.2%)	0.004 (0.9%)	0.011 (2.2%)	0.00 /1.00	90.4 /96.3
	Oa017	S3	7621	0.97	-0.004 (-0.6%)	-0.003 (-0.5%)	0.013 (2.3%)	0.006 (1.0%)	0.014 (2.4%)	0.02 /0.97	88.6 /95.7
Sentinel-3B	Oa06	S1	8345	0.98	0.010 (3.4%)	0.011 (3.6%)	0.009 (2.9%)	0.012 (3.7%)	0.014 (4.4%)	0.02 /0.97	35.4 /71.9
	Oa08	S2	8345	0.97	>-0.001 (-0.1%)	0.001 (0.2%)	0.012 (2.4%)	0.005 (1.1%)	0.012 (2.4%)	0.01 /0.99	87.1 /94.9
	Oa017	S3	8345	0.97	0.004 (0.6%)	0.005 (0.8%)	0.014 (2.4%)	0.007 (1.3%)	0.014 (2.5%)	0.01 /0.98	85.7 /94.9

4.5.3 Consistency between CLMS TOC V2.3 and ESA L2 SYN over AOIs

The consistency between CLMS TOC reflectance V2.3 and ESA L2 SYN reflectance products is evaluated over different AOIs (25 km x 25 km area) representative of the main biome types (see section 3.4) covering different dates and platforms (i.e., Sentinel-3A or -3B). In order to present an overall overview of the bias between CLMS TOC reflectance V2.3 and ESA L2 SYN, the relative mean bias for the different AOI and channels is displayed in Figure 44 for OLCI and Figure 45 for SLSTR.

The error evaluation for each AOI is presented in Annex VI. The spatial consistency between both products (CLMS TOC reflectance V2.3 and ESA L2 SYN) is also evaluated: maps of values and difference maps are displayed in Annex VII.

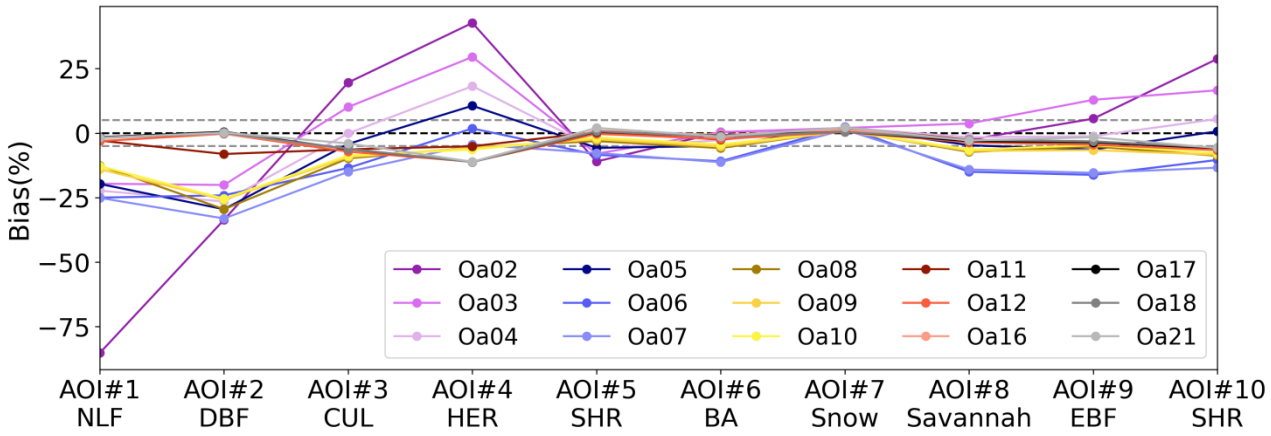


Figure 44: Relative mean bias (%) between CLMS TOC reflectance V2.3 and ESA L2 SYN for the different AOIs and OLCI channels. Dashed lines correspond to 0% and ±5% thresholds.

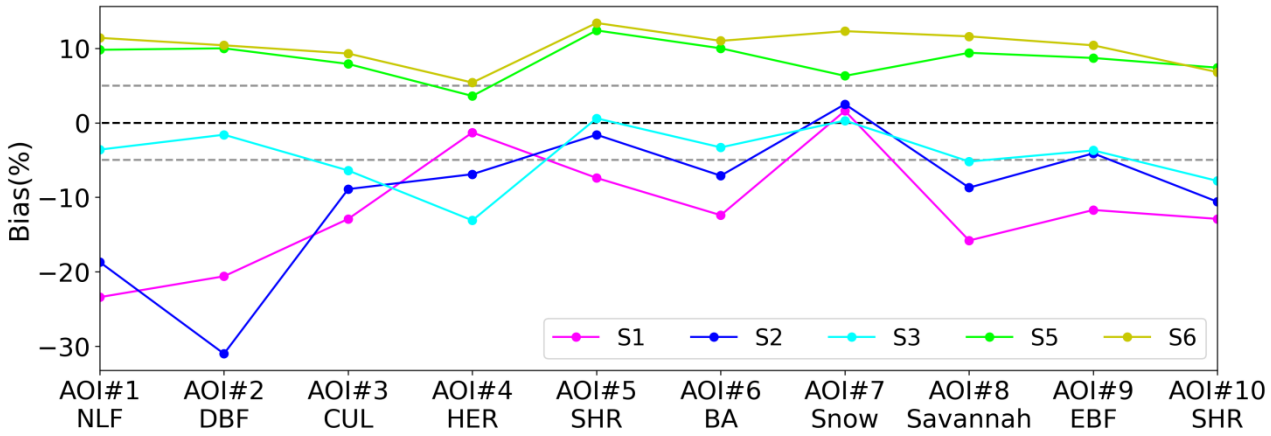


Figure 45: Relative mean bias (%) between CLMS TOC reflectance V2.3 and ESA L2 SYN for the different AOIs and SLSTR channels. Dashed lines correspond to 0% and ±5% thresholds.

Main conclusions of the analysis of the relative mean bias CLMS TOC V2.3 and ESA LS SYN for the different AOIs are:

For OLCI channels (Figure 44):

- TOC V2.3 typically tends to provide lower values (around -3% to -15%) than ESA L2 SYN in most situations, showing larger relative negative bias for NLF and DBF mainly for the lower spectral channels. Per spectral channel, negative bias is found for visible channels and slight negative or close to zero for NIR. The main exceptions of negative bias are:
  - The snow area (AOI#7), where low positive bias is found (up to 2.5%) for all channels.
  - The lower spectral channels, where positive bias is found in some cases: Oa02 and Oa03 for AOI#3 (CUL) and AOI#9 (EBF); Oa02 to Oa06 for AOI#4 (HER); Oa02 to Oa05 for AOI#6 (desert calibration site) and AOI#10 (SHR); Oa02, Oa03 and Oa04 for AOI#8 (Savannah).
- The overall uncertainty (RMSD) is typically similar to bias in magnitude (i.e., error is systematic with no random contribution, see Annex VI).
- Remarkably optimal agreement is found for NIR channels (from Oa011 to Oa21), with R=1.0 (or close to 1), STD<2% (i.e., no or very low scattering) and most or all pixels within goal requirement level (see Annex VI).
- The spatial consistency (see maps in Annex VII) shows same spatial patterns, similar values and low differences (typically within ±0.05).

For SLSTR channels (Figure 45):

- TOC V2.3 shows systematically lower values than ESA L2 SYN for visible (S1, S2) and NIR (S3) channels for all AOIs, except for the snow case (AOI#7) with large reflectance values in the visible where low positive bias (<3%) is found. The negative bias is typically lower for S3 (up -13.1% for AOI#4 - HER) compared with S1 and S2 (up to -31% AOI#2 -DBF).
- In case for SWIR channels (S5, S6), TOC V2.3 shows systematically higher values than ESA L2 SYN with bias between 3.6% (AOI#4, HER) and 13.4% (AOI#5, SHR), mainly explained by the incorporation of recommended ESA calibration adjustment factors (Table 4) in CLMS TOC V2.3.
- Worse overall agreement is generally found between CLMS and ESA TOC reflectances compared to that found for OLCI (Annex VI), with large scattering which could be probably explained by the projection approach used to generate SLSTR products (500 m of nominal spatial resolution) into 300 m of spatial resolution OLCI grid.
- The overall uncertainty (RMSD) ranges from 2.5% (S3 channel for AOI#5 – SHR) to 34% (S2 for AOI#2 - DBF) (see Annex VI).
- The spatial consistency (Annex VII) shows same spatial patterns but some spatial noise in the difference maps (i.e., different sign of bias is found over randomly distributed pixels in some

cases) and differences typically between -0.05 and +0.05 for VNIR and up to +0.1 for SWIR channels.

4.5.4 Consistency between Sentinel-3 TOC V2.3 OLCI and SLSTR over AOIs

This section presents the consistency between CLMS TOC reflectance V2.3 OLCI and SLSTR equivalent channels over the 10 selected AOIs (25 km x 25 km area) representative of different biome types (see section 3.4). From Figure 46 to Figure 55, scatter-plots and associated validation metrics (APU) are presented for each AOI covering different dates and platforms (i.e., Sentinel-3A or -3B). Validation metrics are summarized in Table 18 (Oa06 vs. S1), Table 19 (Oa08 vs. S2) and Table 20 (Oa17 vs. S3).

The comparison of OLCI Oa06 versus SLSTR S1 channel shows:

- An accuracy ranging from -4.4% (AOI#8, Savannah) to 5% (AOI#10, SHR), showing slightly large bias for the EBF case (-8.8%, AOI#9).
- The correlation is generally higher than 0.85 for most cases (80% of AOIs).
- The overall uncertainty is lower than 0.02 for snow-free cases and 0.03 for snow, which is translated in values typically from 3% to 11% in relative terms (up to 17% for NLF).

The comparison of OLCI Oa08 versus SLSTR S2 channel shows:

- The mean bias is ranging from -6.6% (AOI#2, DBF) to 3.5% (AOI#4, HER), showing large bias, again, for the EBF case (-12.5%, AOI#9).
- The correlation is generally higher than 0.75 for most cases, except for snow (0.70).
- The overall uncertainty (RMSD) is lower than (or around) 0.03, translated in relative RMSD values from 1.4% for desert up to 22.7% for NLF in relative terms.

Finally, the comparison of OLCI Oa17 versus SLSTR S3 channel shows:

- Oa17 tends to provide typically higher values than S3, with positive bias up to 6% (AOI#1, NLF). The exceptions are AOI#8 (savannah) and #9 (EBF) with negative biases of -1.1% and -3.2%.
- The correlation is ranging from 0.7 (AOI#7, snow) to 0.96 (AOI#6, BA).
- The overall uncertainty is lower than 0.03, which is translated in relative values up to 8%, except for AOI#1 (NLF, 14%).

Additionally, similar scatter-plots for same AOIs and dates are generated for TOC reflectance V1 and provided in Annex VIII. These results demonstrate the improved co-registration between OLCI and SLSTR of TOC V2.3, as significant better correlations and improved APU metrics are found for all AOIs compared to that found for TOC reflectance V1. The application of calibration adjustment

factors on OLCI-A is another factor that can contribute, at a lesser extent than the co-registration, to a better OLCI/SLSTR consistency

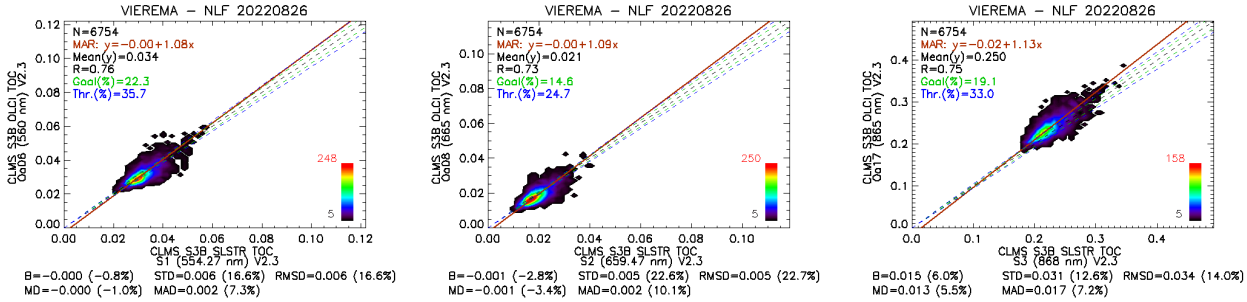


Figure 46: Scatter-plots of Sentinel-3 TOC V2.3 (2022.08.26) OLCI versus SLSTR equivalent bands over the AOI#1 (Vieremä, Finland – NLF).

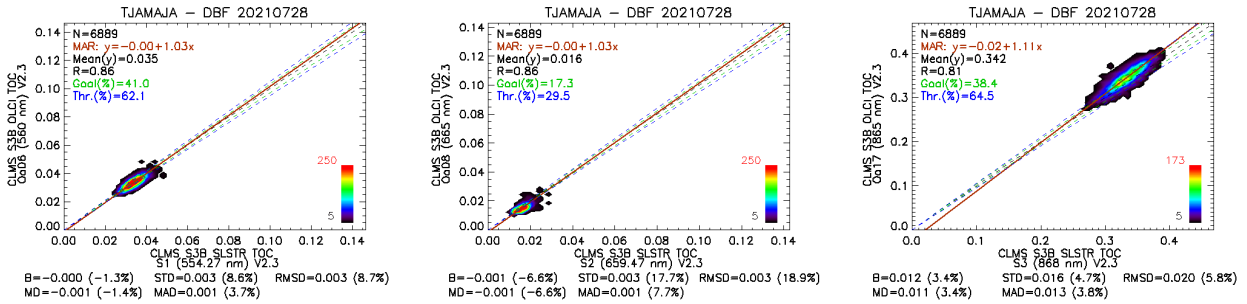


Figure 47: Scatter-plots of Sentinel-3 TOC V2.3 (2018.10.12) OLCI versus SLSTR equivalent bands over the AOI#2 (Tjamaja, Russia – DBF).

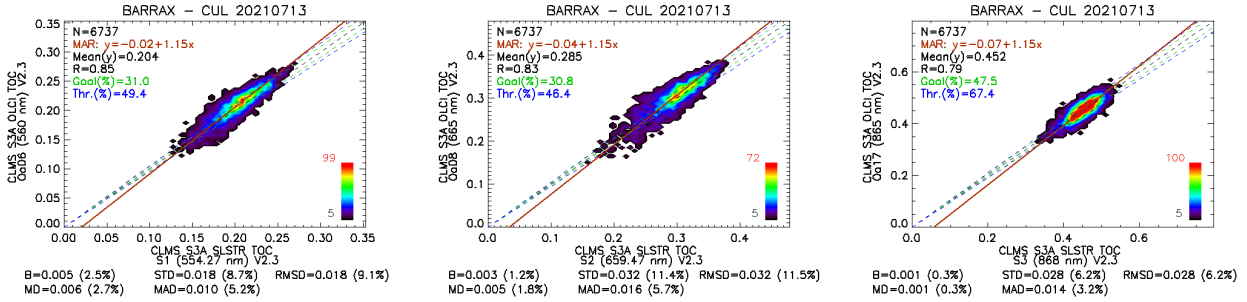


Figure 48: Scatter-plots of Sentinel-3 TOC V2.3 (2021.07.13) OLCI versus SLSTR equivalent bands over the AOI#3 (Barrax, Spain – Cultivated).

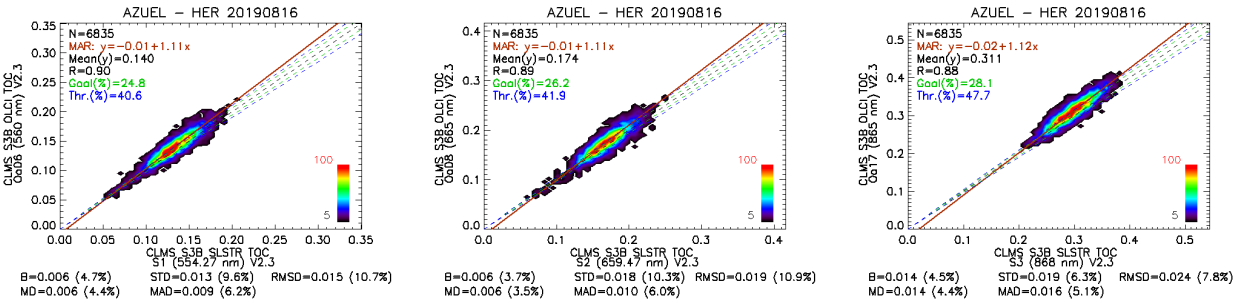


Figure 49: Scatter-plots of Sentinel-3 TOC V2.3 (2019.08.16) OLCI versus SLSTR equivalent bands over the AOI#4 (Azuel, Spain – Herbaceous).

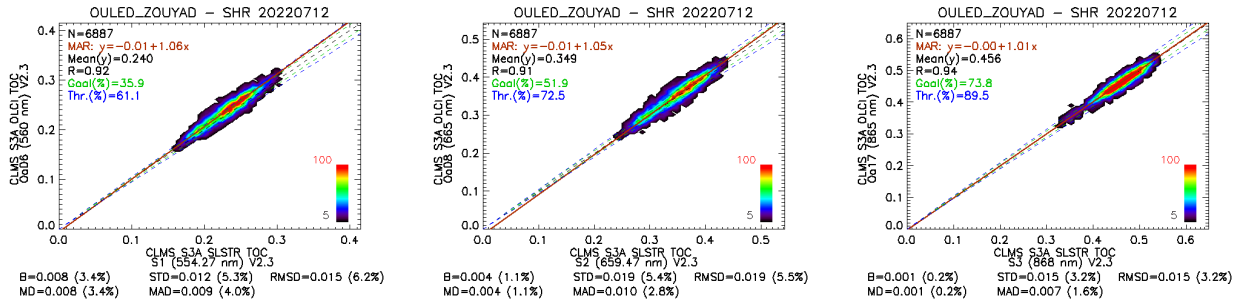


Figure 50: Scatter-plots of Sentinel-3 TOC V2.3 (2022.07.12) OLCI versus SLSTR equivalent bands over the AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

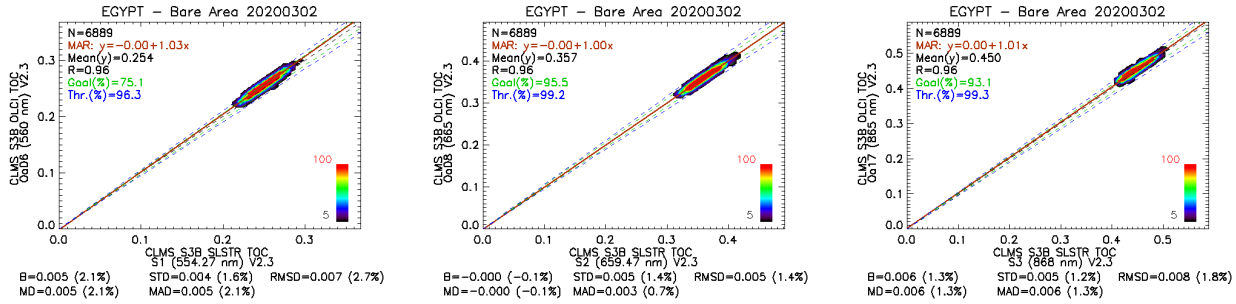


Figure 51: Scatter-plots of Sentinel-3 TOC V2.3 (2020.03.03) OLCI versus SLSTR equivalent bands over the AOI#6 (Calibration Site, Egypt – Bare Areas).

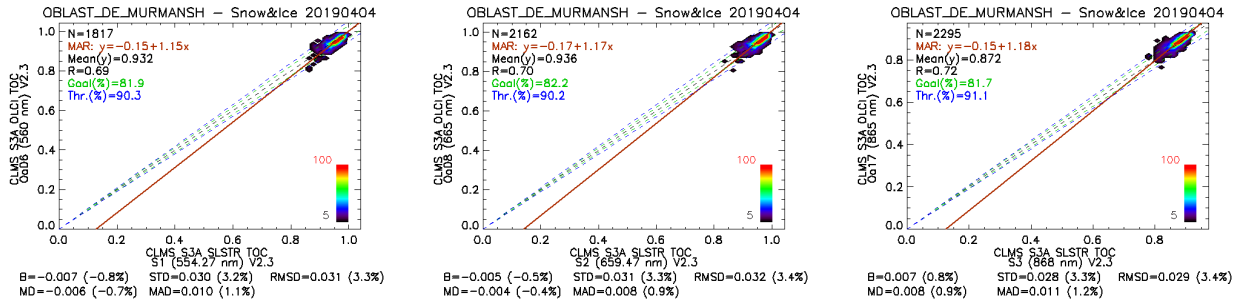


Figure 52: Scatter-plots of Sentinel-3 TOC V2.3 (2019.04.04) OLCI versus SLSTR equivalent bands over the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

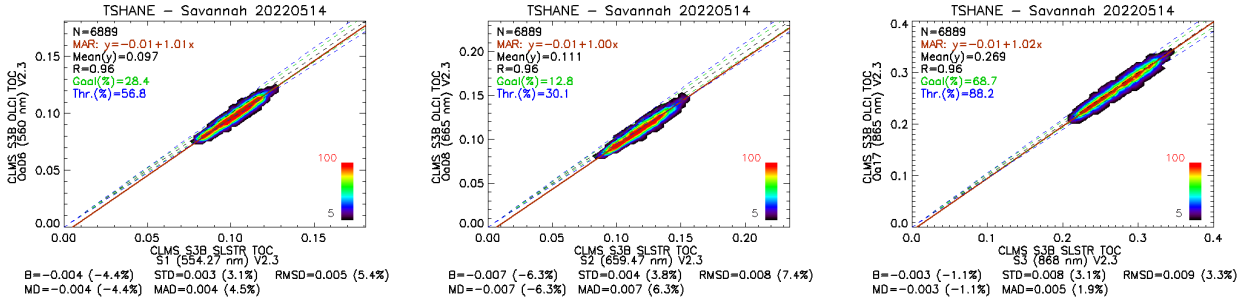


Figure 53: Scatter-plots of Sentinel-3 TOC V2.3 (2022.05.14) OLCI versus SLSTR equivalent bands over the AOI#8 (Tshane, Botswana – Savannah).

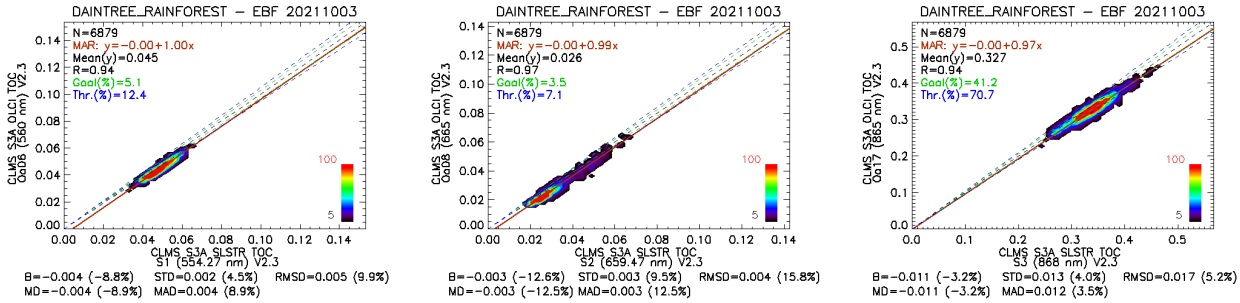


Figure 54: Scatter-plots of Sentinel-3 TOC V2.3 (2021.10.03) OLCI versus SLSTR equivalent bands over the AOI#9 (Daintree Rainforest, Australia - EBF).

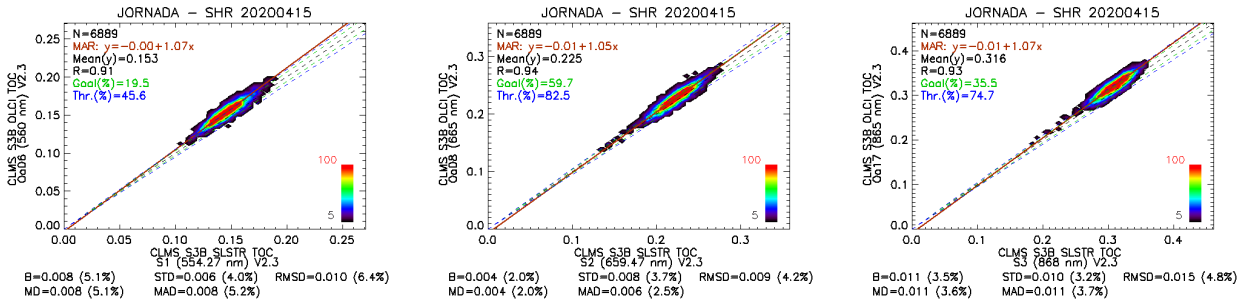


Figure 55: Scatter-plots of Sentinel-3 TOC V2.3 (2020.04.15) OLCI versus SLSTR equivalent bands over the AOI#10 (Jornada, New Mexico - Shrublands).

Table 18: Validation statistics of the comparison of Sentinel-3 TOC V2.3 OLCI Oa06 versus SLSTR S1 bands for different AOIs.

Sentinel-3 TOC V2.3 OLCI Oa06 vs. SLSTR S1									
OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Thres.
AOI#1 (NLF)	6754	0.76	<0.001 (0.8%)	>-0.001 (1.0%)	0.006 (16.6%)	0.002 (7.3%)	0.006 (16.6%)	0.00 /1.08	22.3 /35.7
AOI#2 (DBF)	6889	0.86	>-0.001 (-1.3%)	-0.001 (-1.4%)	0.003 (8.6%)	0.001 (3.7%)	0.003 (8.7%)	0.00 /1.03	41.0 /62.1
AOI#3 (CUL)	6737	0.85	0.005 (2.5%)	0.006 (2.7%)	0.018 (8.7%)	0.010 (5.2%)	0.018 (9.1%)	-0.02 /1.15	31.0 /49.4
AOI#4 (HER)	6835	0.90	0.006 (4.7%)	0.006 (4.4%)	0.013 (9.6%)	0.009 (6.2%)	0.015 (10.7%)	-0.01 /1.11	24.8 /40.6
AOI#5 (SHR)	6887	0.92	0.008 (3.4%)	0.008 (3.4%)	0.012 (5.3%)	0.009 (4.0%)	0.015 (6.2%)	-0.01 /1.06	35.9 /61.1
AOI#6 (BA)	6889	0.96	0.005 (2.1%)	0.005 (2.1%)	0.004 (1.6%)	0.005 (2.1%)	0.007 (2.7%)	0.00 /1.03	75.1 /96.3
AOI#7 (Snow)	1817	0.69	-0.007 (-0.8%)	-0.006 (-0.7%)	0.030 (3.2%)	0.010 (1.1%)	0.031 (3.3%)	0.15 /1.15	81.9 /90.3
AOI#8 (Savannah)	6889	0.96	-0.004 (-4.4%)	-0.004 (-4.4%)	0.003 (3.1%)	0.004 (4.5%)	0.005 (5.4%)	-0.01 /1.01	28.4 /56.8
AOI#9 (EBF)	6879	0.94	-0.004 (-8.8%)	-0.004 (-8.9%)	0.002 (4.5%)	0.004 (8.9%)	0.005 (9.9%)	0.00 /1.00	5.1 /12.4
AOI#10 (SHR)	6889	0.91	0.008 (5.1%)	0.008 (5.1%)	0.006 (4.0%)	0.008 (5.2%)	0.010 (6.4%)	0.00 /1.07	19.5 /45.6

Table 19: Validation statistics of the comparison of Sentinel-3 TOC V2.3 OLCI Oa08 versus SLSTR S2 bands for different AOIs.

Sentinel-3 TOC V2.3 OLCI Oa08 vs. SLSTR S2									
OLCI Band	**N**	**R**	**Bias (%)**	**MD (%)**	**STD (%)**	**MAD (%)**	**RMSD (%)**	**Offset /Slope**	**% Goal /Thres.**
AOI#1 (NLF)	6754	0.73	-0.001 (-2.8%)	-0.001 (-3.4%)	0.005 (22.6%)	0.002 (10.1%)	0.005 (22.7%)	0.00 /1.09	14.6 /24.7
AOI#2 (DBF)	6889	0.86	-0.001 (-6.6%)	-0.001 (-6.6%)	0.003 (17.7%)	0.001 (7.7%)	0.003 (18.9%)	0.00 /1.03	17.3 /29.5
AOI#3 (CUL)	6737	0.83	0.003 (1.2%)	0.005 (1.8%)	0.032 (11.4%)	0.016 (5.7%)	0.032 (11.5%)	-0.04 /1.15	30.8 /46.4
AOI#4 (HER)	6835	0.89	0.006 (3.7%)	0.006 (3.5%)	0.018 (10.3%)	0.010 (6.0%)	0.019 (10.9%)	-0.01 /1.11	26.2 /41.9
AOI#5 (SHR)	6887	0.91	0.004 (1.1%)	0.004 (1.1%)	0.019 (5.4%)	0.010 (2.8%)	0.019 (5.5%)	-0.01 /1.05	51.9 /72.5
AOI#6 (BA)	6889	0.96	<-0.001 (-0.1%)	<-0.001 (-0.1%)	0.005 (1.4%)	0.003 (0.7%)	0.005 (1.4%)	0.00 /1.00	95.5 /99.2
AOI#7 (Snow)	2162	0.70	-0.005 (-0.5%)	-0.004 (-0.4%)	0.031 (3.3%)	0.008 (0.9%)	0.032 (3.4%)	0.17 /1.17	82.2 /90.2
AOI#8 (Savannah)	6889	0.96	-0.007 (-6.3%)	-0.007 (-6.3%)	0.004 (3.8%)	0.007 (6.3%)	0.008 (7.4%)	-0.01 /1.00	12.8 /30.1
AOI#9 (EBF)	6879	0.97	-0.003 (-12.6%)	-0.003 (-12.5%)	0.003 (9.5%)	0.003 (12.5%)	0.004 (15.8%)	0.00 /0.99	3.5 /7.1
AOI#10 (SHR)	6889	0.94	0.004 (2.0%)	0.004 (2.0%)	0.008 (3.7%)	0.006 (2.5%)	0.009 (4.2%)	-0.01 /1.05	59.7 /82.5

Table 20: Validation statistics of the comparison of Sentinel-3 TOC V2.3 OLCI Oa17 versus SLSTR S3 bands for different AOIs.

Sentinel-3 TOC V2.3 OLCI Oa17 vs. SLSTR S3									
OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Thres.
AOI#1 (NLF)	6754	0.75	0.015 (6.0%)	0.013 (5.5%)	0.031 (12.6%)	0.017 (7.2%)	0.034 (14.0%)	-0.02 /1.13	19.1 /33.0
AOI#2 (DBF)	6889	0.81	0.012 (3.4%)	0.011 (3.4%)	0.016 (4.7%)	0.013 (3.8%)	0.020 (5.8%)	-0.02 /1.11	38.4 /64.5
AOI#3 (CUL)	6737	0.79	0.001 (0.3%)	0.001 (0.3%)	0.028 (6.2%)	0.014 (3.2%)	0.028 (6.2%)	-0.07 /1.15	47.5 /67.4
AOI#4 (HER)	6835	0.88	0.014 (4.5%)	0.014 (4.4%)	0.019 (6.3%)	0.016 (5.1%)	0.024 (7.8%)	-0.02 /1.12	28.1 /47.7
AOI#5 (SHR)	6887	0.94	0.001 (0.2%)	0.001 (0.2%)	0.015 (3.2%)	0.007 (1.6%)	0.015 (3.2%)	0.00 /1.01	73.8 /89.5
AOI#6 (BA)	6889	0.96	0.006 (1.3%)	0.006 (1.3%)	0.005 (1.2%)	0.006 (1.3%)	0.008 (1.8%)	0.00 /1.01	93.1 /99.3
AOI#7 (Snow)	2295	0.72	0.007 (0.8%)	0.008 (0.9%)	0.028 (3.3%)	0.011 (1.2%)	0.029 (3.4%)	0.15 /1.18	81.7 /91.1
AOI#8 (Savannah)	6889	0.96	-0.003 (-1.1%)	-0.003 (-1.1%)	0.008 (3.1%)	0.005 (1.9%)	0.009 (3.3%)	-0.01 /1.02	68.7 /88.2
AOI#9 (EBF)	6879	0.94	-0.011 (-3.2%)	-0.011 (-3.2%)	0.013 (4.0%)	0.012 (3.5%)	0.017 (5.2%)	0.00 /0.97	41.2 /70.7
AOI#10 (SHR)	6889	0.93	0.011 (3.5%)	0.011 (3.6%)	0.010 (3.2%)	0.011 (3.7%)	0.015 (4.8%)	-0.01 /1.07	35.5 /74.7

5 CONCLUSIONS

In this study, the quality assessment of CLMS Sentinel-3 surface reflectance V2.3 products derived from OLCI and SLSTR is performed for the period from July 2018 to December 2023. The methodology is based on two main validation approaches: direct validation (i.e., comparison of satellite products with in situ measurements) and indirect validation (or product intercomparison). The following criteria of performance and metrics are assessed: completeness, spatial consistency, temporal consistency and error evaluation (APU). The product intercomparison is performed using Sentinel-3 ESA L2 SYN products, whereas the direct validation is evaluated against ground data from CEOS RadCalNet and complemented with a limited dataset of HYPERNETS and ESA FRM4Veg. Moreover, the consistency of surface reflectance V2.3 derived from different satellites (Sentinel-3A versus Sentinel-3B during the tandem phase) and different sensors (OLCI versus SLSTR similar spectral channels) is evaluated.

5.1 SUMMARY OF RESULTS

5.1.1 Product content

The visual inspection of global maps at reduced resolution and selected tiles and AOIs at full resolution show consistent distribution of values. Illumination and viewing angles values (SZA, VZA, SAA, VAA) are also consistent. Associated reflectance errors for OLCI and SLSTR are typically ranging between 2% and 6%, except for blue spectral domain (large relative errors).

Regarding the recommendations of use of ancillary quality layers for OLCI (**quality_flags**, **pixel_classif_flags** and **AC_process_flag**) and SLSTR (**cloud_an** and **AC_process_flag**):

- **pixel_classif_flags** (output of IdePix snow/cloud/shadow identification module) is found the most effective to remove clouds but with two main limitations: i) conservative approach and, ii) misidentification of clouds and snow. In case of snow targets, the IdePix_SNOW_ICE flag slightly improves the identification of snow pixels compared with previous version due to updated IdePix version (IdePix 7.0.5 in V1 versus IdePix 9.0.6 in V2.3). The updated IdePix version also performs better on thin, semi-transparent clouds and cloud edges identification [CGLOPS-1_QAR_S3-CloudMask].
- Cloud_an (cloud flags from SLSTR Level 1B) systematically removes observations for northern latitudes in winter time, whilst clouds are not properly removed in the equatorial belt. The border of the clouds is also not well detected. Consequently, in addition to Cloud_an, it is recommended the use of same cloud screening identification used for OLCI (i.e., **pixel_classif_flags**) to remove several outliers and cloud contamination.
- Few pixels are removed when **quality_flags** (band saturation indicator from OLCI Level 1B) and **AC_process_flag** (output of atmospheric correction module) are used, except for the bit

3 of **AC_process_flag** (SZA > 65º) which discard large areas in northern and southern latitudes during wintertime.

5.1.2 Product completeness of snow-free best quality pixels

Excluding pixels with SZA larger than 65° creates long data gaps over northern latitudes and southern latitudes during wintertime. On the other hand, SLSTR surface reflectances provide higher number of valid good quality observations than OLCI surface reflectances, due to less restrictive ancillary quality flags.

5.1.3 Temporal consistency

CLMS Sentinel-3 surface reflectances V2.3 are consistent with ESA L2 SYN and RadCalNet measurements reproducing properly the temporal variations. CLMS OLCI and SLSTR reflectances display similar temporal trends for similar spectral channels.

5.1.4 Direct validation

The direct validation of surface reflectances V2.3 retrievals close to nadir (VZA < 10º) compared with CEOS RadCalNet multi-temporal data from 4 sites (desert and sparse vegetation) shows:

- Good accuracy for OLCI reflectances with mean bias and MD typically within ±2% for most channels (slightly large bias for Oa02 and Oa21). Overall uncertainty (RMSD) around 7-9% is found, with typically around 33-46% of pixels within goal requirement (i.e., 3%) and 55-65% for threshold level (5%).
    - reflectance V2.3 shows improved results than V1 [CGLOPS-1_QAR_S3-TOCr300m-V1] (B around ±3% and RMSD around 9-13%).
    - reflectance V2.3 shows better results than ESA L2 SYN (positive B around 3-10%, RMSD around 11-15%).
- Remarkably good accuracy for SLTR reflectances V2.3, with mean bias and MD typically within ±1.5% for all spectral channels. Overall uncertainty (RMSD) around 7-10% is found with around 30-37% of pixels within goal requirement and 48-56% within threshold level.
    - reflectance V2.3 shows improved results than V1 [CGLOPS-1_QAR_S3-TOCr300m-V1] (B up to 2% for VNIR and worse than -10% for SWIR).
    - reflectance V2.3 shows better results than ESA L2 SYN (positive B > 6% for VNIR and negative bias up to -12% for SWIR; RMSD larger than 12%).

- The evolution of bias of surface reflectance V2.3 over time, over GONA pseudo-invariant site, shows a slightly different trend for OLCI depending on the period: for most channels, slight positive bias for January 2019 to June 2020 and bias close to zero or slightly negative for July 2020 onwards. The bias of SLSTR reflectances for VNIR channels shows the same behaviour than OLCI reflectances with, however, enhanced seasonality. On the other hand, systematic positive bias is found for SLSTR SWIR channels.

The direct validation of Sentinel-3 surface reflectance V2.3 retrievals compared with HYPERNETS multi-temporal and multi-angular data from 3 sites (desert, agricultural and DBF) shows:

- Positive differences for most OLCI channels (mainly observed for the forest case), with MD typically lower than 5%. Slight negative differences (MD within -1%) for channels from Oa12 to Oa18. RMSD around 0.05 for visible channels and around 0.075 for NIR. Around 20-30% and around 30-38% of pixels are within goal and threshold requirements, respectively.
- Positive bias (MD around 6%) for visible SLSTR channels, and almost no bias for NIR (S3) and SWIR (S5). Overall RMSD lower than 0.01. Around 19-26% and 30-38% of pixels are within goal and requirements, respectively.

The comparison with ESA FRM4Veg data for bare soil shows qualitatively good agreement for OLCI and SLSTR surface reflectances V3.2, displaying similar spectral signatures. For alfalfa, both OLCI and SLSTR TOC reflectances V3.2 show lower contrast between red and NIR bands, probably due to an adjacency effect (larger in SLSTR due to distinct effective spatial resolution). The spectral signatures of Sentinel-3A and -3B surface reflectance V3.2 are quite similar.

5.1.5 Product intercomparison

The comparison between surface reflectance V2.3 from Sentinel-3A versus Sentinel-3B over desert calibration sites during the tandem phase shows:

- Very good agreement for OLCI (R>0.99, B and MD within ±0.9%, STD and RMSD <1.5%). More than 93% and 98% of cases are within goal and threshold requirements, respectively.
- Good agreement for SLSTR, with B and MD between -0.3% and 1.5%, STD around 1.5% and RMSD around 1.5-2.5%. More than 92% and 97% of cases are within optimal and threshold requirements, respectively.

The consistency between CLMS TOC reflectance V2.3 from OLCI and SLSTR shows:

- Remarkably good accuracy over desert calibration sites for OLCI Oa08 vs. SLSTR S2 (B and MD within ±0.3%) and for OLCI Oa17 vs. SLSTR S3 (B and MD within ±0.8%). OLCI Oa06 shows higher values than SLSTR S1 (MD < 4%).

- The error evaluation over different AOIs shows:
  - OLCI Oa06 vs. SLSTR S1: Accuracy (B, MD) typically within ±5% (except for EBF, -8.8%) and uncertainty (RMSD) typically from 3% to 10%.
  - OLCI Oa08 vs. SLSTR S2: Accuracy from -6.6% to 3.5% (except for EBF, -12.5%) and low uncertainty (RMSD = 1.2%) for desert or sparse vegetation showing large uncertainty for some forests.
  - OLCI Oa17 vs. SLSTR S3: OLCI tends to provide positive bias up to 6% (except for savannah and EBF) and overall uncertainty typically up to 8%.
- CLMS Sentinel-3 surface reflectances V2.3 show remarkably good spatial consistency between OLCI and SLSTR products (R > 0.85 for most cases) over different AOIs. Thanks to an improved OLCI-SLSTR co-registration, better APU metrics are obtained compared to surface reflectances V1.

The comparison of CLMS Sentinel-3 TOC reflectance V2.3 versus ESA L2 SYN over selected AOIs covering the different main biome types shows:

- CLMS surface reflectance V2.3 and ESA L2 SYN display very similar spatial patterns
- CLMS OLCI surface reflectance V2.3 tends to provide negative bias (i.e., CLMS < ESA) for visible channels and slight negative bias or bias close to zero for NIR channels. The exceptions are mainly for blue OLCI channels where large positive bias is found in some AOIs, and for the snow and ice case (high reflectance values) where a good agreement is found (bias <2.5%).
- Negative bias is found for SLSTR visible channels, negative bias or close to zero for NIR and larger positive bias for SWIR channels. The exception is for snow where good agreement is found for VNIR channels.

5.2 CONCLUDING REMARKS AND RECOMMENDATIONS

The main conclusions of CLMS Sentinel-3 TOC reflectance V2.3 quality assessment and validation are:

- Reliable product content at global scale (surface reflectance retrievals, errors and angles).
- Similar values and spatial patterns than equivalent ESA L2 SYN product at local scale over different areas of interest covering different biome types (forests, cultivated areas, herbaceous, shrublands and savannas, desert and snow).
- Good temporal consistency between different OLCI and SLSTR spectral reflectances and with CEOS RadCalNet ground measurements.

- Observations from different satellites (Sentinel-3A versus -3B) and different sensors (OLCI and SLSTR) are consistent. The agreement of surface reflectances V2.3 between OLCI and SLSTR is considerably better than in V1.
- Direct validation versus CEOS RadCalNet measurements shows accuracy within ±2% for most OLCI channels and within ±1.5% for SLSTR. These are improved results compared to the surface reflectance V1. CLMS surface reflectance V2.3 also performs significantly better than ESA L2 SYN. Consequently, the percentage of cases meeting the CLMS goal and threshold uncertainty requirements is significantly higher. The CLMS surface reflectances V2.3 agree well with in-situ data from other initiatives: similar spectral signatures than ESA FRMs and median deviation within -1% to 6% in comparison with HYPERNETS multi-angular measurements.

In summary, CLMS TOC reflectance V2.3 products are reliable for public dissemination and for the production of biophysical products. Several improvements are noticed respect to surface reflectance V1:

- Optimal agreement between Sentinel-3A and -3B: correlations greater than 0.98 and mean biases within ±0.9% for OLCI and within ±1.5% for SLSTR.
- More realistic values for SLSTR S5 and S6 channels thanks to the application of ESA calibration adjustment factors.
- Improved agreement between OLCI and SLSTR reflectances due to a better co-registration.
- Improved accuracy compared with RadCalNet, partly attributed to the application of ESA calibration adjustment factors for OLCI-A and SLSTR
- Slightly improved cloud and snow identification due to updated IdePix processor.

However, some limitations remain:

- The use of Pixel_classif_flags (output of IdePix classification module) is the most effective quality indicator to remove cloud contamination, but provides a very conservative approach. It is recommended to use this ancillary layer for SLSTR reflectances in order to remove potential cloud contamination.
- The identification of snow pixels by IdePix processor is still limited, and misidentifies snow targets as clouds. This is a main limitation for snow applications and for generating other products (i.e., surface albedo).
- Finally, discarding pixels with SZA larger than 65° (where the atmospheric correction is not optimal) can be a strong limitation for applications during wintertime. Using or not this flag is left at each user's judgement.

As the limitations are mainly attributed to the cloud and snow masking, it is recommended to Sentinel-3 mission to improve cloud discrimination and snow identification on both OLCI and SLSTR images.

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ANNEX I: SPECTRAL BANDS OF SENTINEL-3 OLCI AND SLSTR SENSORS

Table 21: Sentinel-3A OLCI spectral bands. The third column, so-called "corrected", indicates if the TOA signal in the band is corrected from atmospheric effect and thus used in the processing.

Band	Central wavelength (nm)	FWHM (nm)	Corrected
Oa01	399.9	14.0	no
Oa02	411.8	9.8	yes
Oa03	442.9	9.9	yes
Oa04	490.4	9.9	yes
Oa05	510.4	9.9	yes
Oa06	560.4	9.9	yes
Oa07	620.4	9.9	yes
Oa08	665.2	9.9	yes
Oa09	674.0	7.4	yes
Oa10	681.6	7.4	yes
Oa11	709.1	9.9	yes
Oa12	754.2	7.4	yes
Oa13	761.7	2.5	no [1]
Oa14	764.8	3.7	no [1]
Oa15	767.9	2.5	no [1]
Oa16	779.2	15.0	yes
Oa17	865.6	19.8	yes
Oa18	884.4	9.9	yes
Oa19	899.4	9.9	no [2]
Oa20	939.3	19.7	no [2]
Oa21	1012.9	26.9	yes

[1] *This band is highly contaminated by oxygen absorption*

[2] *This band is highly contaminated by water vapour*

Similarly, for Sentinel-3B OLCI, bands Oa01, Oa13, Oa14, Oa15, Oa19 and Oa20 are not corrected from atmospheric effects.

Table 22: same as Table 21 but for Sentinel-3A SLSTR.

Band	Central wavelength (nm)	FWHM (nm)	Corrected
S1	554.3	19.2	yes
S2	659.5	19.2	yes
S3	868.0	20.6	yes
S4	1374.8	20.4	No [1]
S5	1613.4	66.4	yes
S6	2255.5	51.0	yes

[1] *This band is used for the detection of cirrus.*

Similarly, for Sentinel-3B SLSTR, band S4 is not corrected from atmospheric effects.

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ANNEX II: ANNOTATION FLAGS OF SENTINEL-3 TOC REFLECTANCE V2.3

The Sentinel-3 TOC reflectance V2.3 products contain 4 quality layers:

- **cloud_an**: cloud flags from SLSTR Level 1B, stored as 16-bit integer values (Table 23)
- **quality_flags**: quality flags from OLCI Level 1B, stored as 32-bit unsigned integer values (Table 24)
- **pixel_classif_flags**: output of OLCI-IdePix, stored as 16-bit integer values (Table 25)
- **AC_process_flag**: output of Atmospheric Correction, as defined in the ATBD of Atmospheric Corrections [CGLOPS1_ATBD_S3-AC-V1.1], stored as one-byte values (Table 26)

Table 23: cloud_an flag encoding[9]

Bit	Name	Value
0	Visible channels cloud test	1
1	1.37 µm threshold test	2
2	1.6 µm small-scale histogram test	4
3	1.6 µm large-scale histogram test	8
4	2.25 µm small-scale histogram test	16
5	2.25 µm large-scale histogram test	32
6	11 µm spatial coherence test	64
7	12 µm gross cloud test	128
8	11 µm/12 µm thin cirrus test	256
9	3.7 µm/12 µm medium/high level test	512
10	11 µm/3.7 µm fog/low stratus test	1024
11	11 µm/12 µm view difference test (uses both views)	2048
12	3.7 µm /11 µm view difference test (uses both views)	4096
13	11 µm/12 µm thermal histogram test	8192
	DN no data value	-1

[9] See https://sentinel.esa.int/documents/247904/1872792/Sentinel-3-SLSTR-Product-Data-Format-Specification-Level-1 page 24

Table 24: quality_flags encoding[10]

Byte/Bit	Name	Value
0/0	saturated_Oa21	1
0/1	saturated_Oa20	2
0/2	saturated_Oa19	4
0/3	saturated_Oa18	8
0/4	saturated_Oa17	16
0/5	saturated_Oa16	32
0/6	saturated_Oa15	64
0/7	saturated_Oa14	128
1/0	saturated_Oa13	256
1/1	saturated_Oa12	512
1/2	saturated_Oa11	1024
1/3	saturated_Oa10	2048
1/4	saturated_Oa09	4096
1/5	saturated_Oa08	8192
1/6	saturated_Oa07	16384
1/7	saturated_Oa06	32768
2/0	saturated_Oa05	65536
2/1	saturated_Oa04	131072
2/2	saturated_Oa03	262144
2/3	saturated_Oa02	524288
2/4	saturated_Oa01	1048576
2/5	dubious	2097152
2/6	sun_glint_risk	4194304
2/7	duplicated	8388608
3/0	cosmetic	16777216
3/1	invalid	33554432
3/2	straylight_risk	67108864
3/3	bright	134217728
3/4	tidal_region	268435456
3/5	fresh_inland_water	536870912
3/6	coastline	1073741824
3/7	land	2147483648

[10] See https://sentinel.esa.int/documents/247904/1872756/Sentinel-3-OLCI-Product-Data-Format-Specification-OLCI-Level-1 page 47-50

Table 25: pixel_classif_flags encoding

Bit	Name	Value
0	IDEPIX_INVALID	1
1	IDEPIX_CLOUD	2
2	IDEPIX_CLOUD_AMBIGUOUS	4
3	IDEPIX_CLOUD_SURE	8
4	IDEPIX_CLOUD_BUFFER	16
5	IDEPIX_CLOUD_SHADOW	32
6	IDEPIX_SNOW_ICE	64
7	IDEPIX_BRIGHT	128
8	IDEPIX_WHITE	256
9	IDEPIX_COASTLINE	512
10	IDEPIX_LAND	1024
11	IDEPIX_MOUNTAIN_SHADOW	2048
	DN no data value	-1

Please note that these flags are **not** mutual exclusive and that multiple combinations are possible. For instance, a pixel can have the value 192: IDEPIX_SNOW_ICE (64) and IDEPIX_BRIGHT (128).

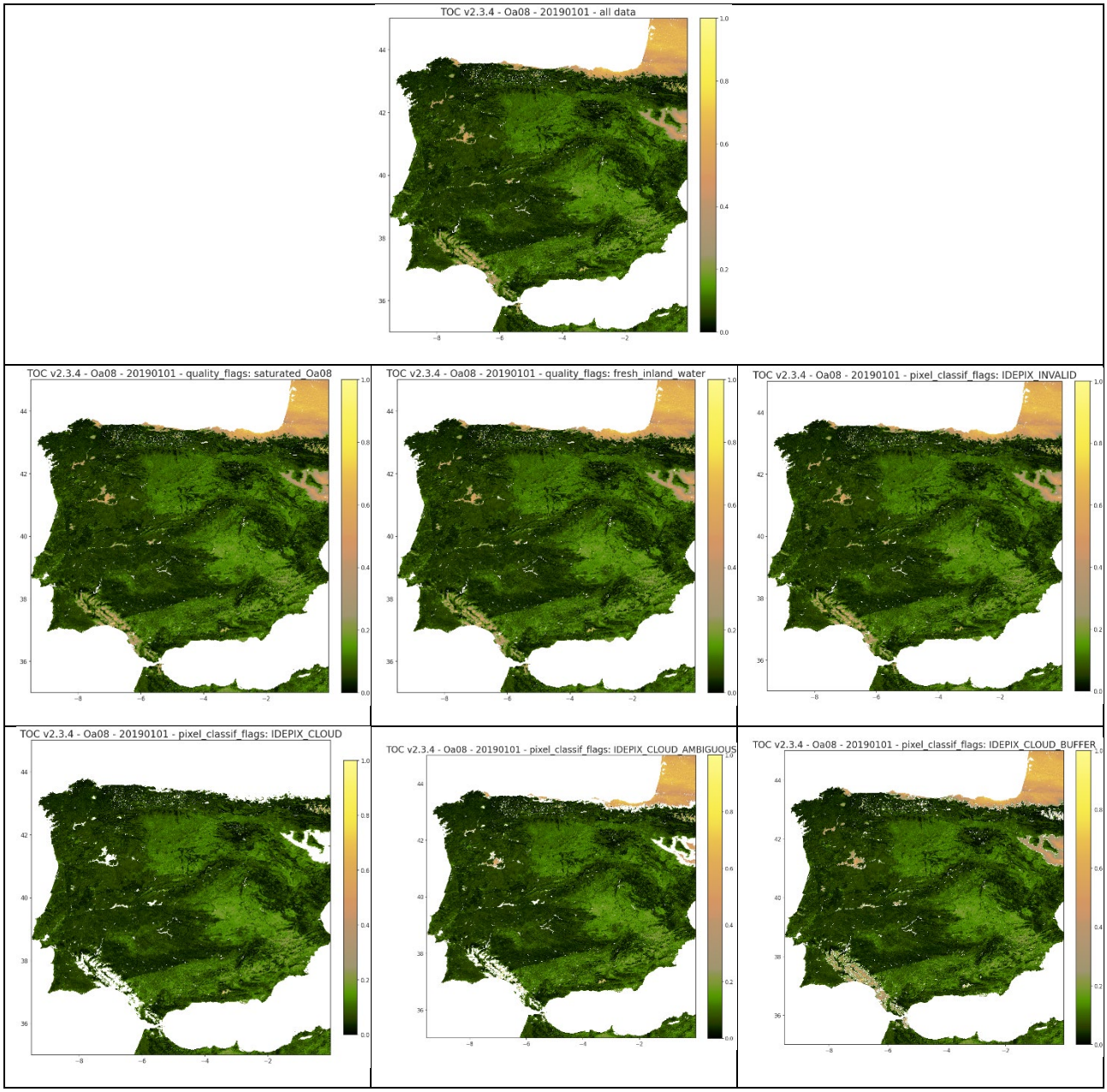
Table 26: AC_process_flag encoding

Bit	Name	Flag value
0	Reserved – not implemented	0 fixed
2,1	Bit 2: 0, Bit 1: 0 Nominal AOT value (AOT ≤0.5): High confidence level	0
	Bit 2: 0, Bit 1: 1 High AOT value (0.5 < AOT ≤1.0): Moderate confidence level	2
	Bit 2:1, Bit 1: 0 Very high AOT value (1.0 < AOT ≤1.5): Low confidence level	4
	Bit 2:1, Bit 1: 1 Critical AOT value (1.5 < AOT): Very low confidence level	6
3	High solar zenith angle (SZA>65°)[11]: Low confidence level	8
4	High viewing zenith angle (VZA>65°)[12]: Low confidence level	16
5	Use of a climatology (This flag value indicates the use, as backup, of the MERRA-2 climatology in case of missing CAMS NRT ancillary data)	32
6	Not used	64
7	Not used	128
	DN no data value	-1

[11] Since only SLSTR nadir images are used, the SZA and VZA of OLCI and SLSTR are very close.
[12] Note that a VZA > 65° is never reached on OLCI and SLSTR nadir images.

ANNEX III: FULL RESOLUTION MAPS OVER SELECTED AREAS WITH EACH FLAG APPLIED SEPARATELY

The following figures show the maps for the OLCI Oa08 and SLSTR S2 channels for of X17Y04 tile, X18Y03 tile, X20Y02 tile and X05Y01 tile. For each figure, the map of all pixels Sentinel-3 TOC reflectance pixels is displayed at the top. The rest of images in the figures show the map when different quality layer recommendations are used bit per bit, as indicated on top of each subfigure.



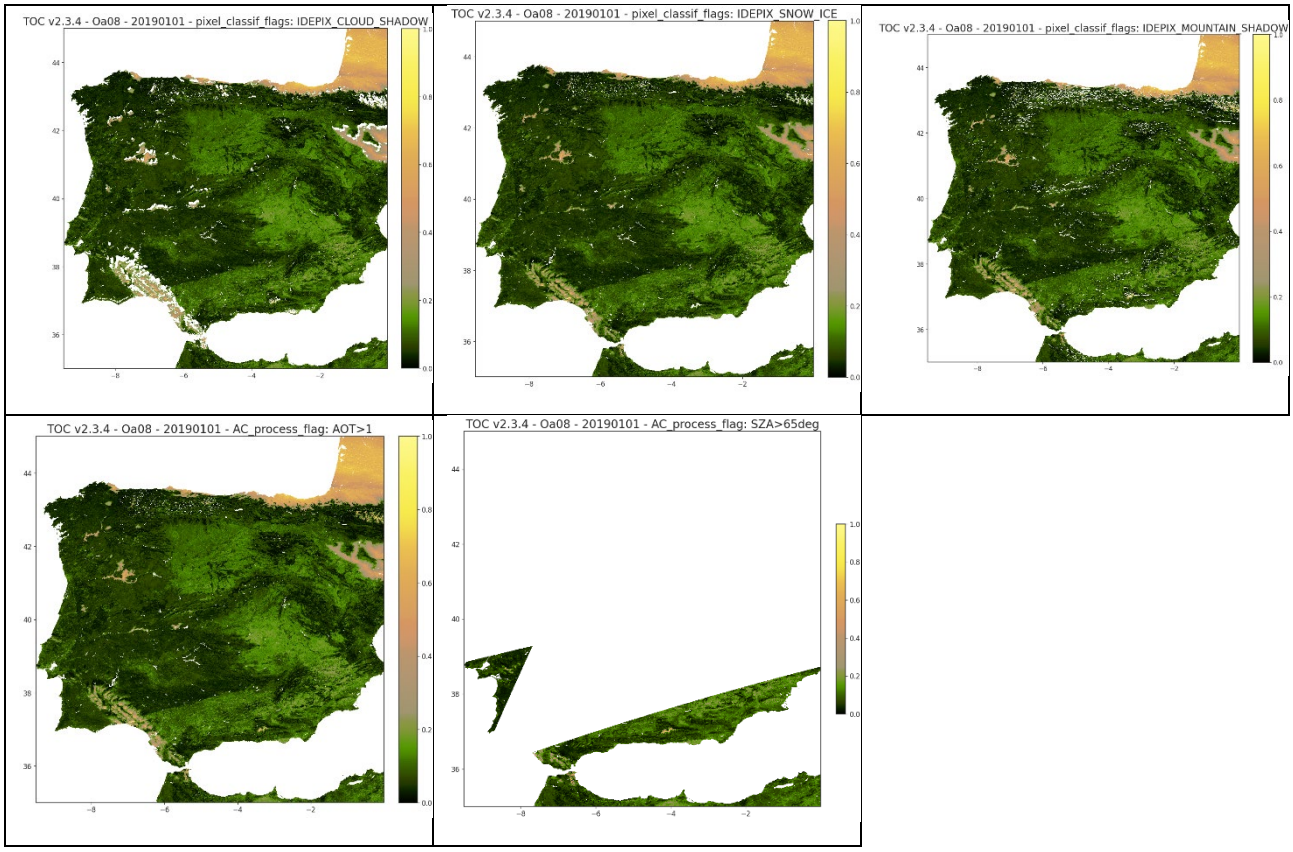
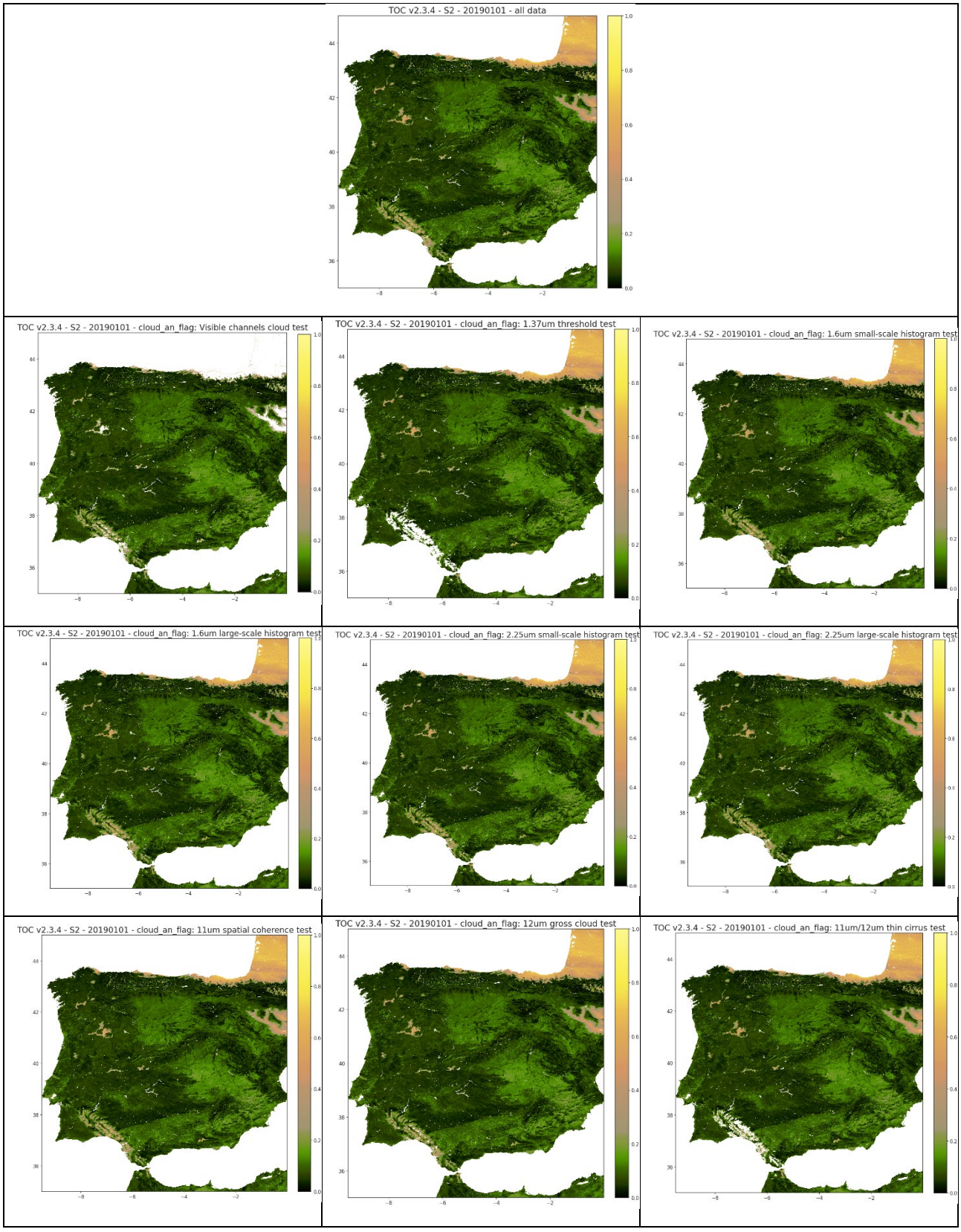


Figure 56: *Top*: Map of Sentinel-3 OLCI (Oa08) TOC reflectance (all pixels) for 01/01/2019 at full resolution for the X17Y04 tile. Other images show the same tile when each bit of the quality layers is applied to exclude pixels.


TOC v2.3.4 - S2 - 20190101 - all data
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: Visible channels cloud test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 1.37um threshold test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 1.6um small-scale histogram test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 1.6um large-scale histogram test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 2.25um small-scale histogram test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 2.25um large-scale histogram test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 11um spatial coherence test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 12um gross cloud test
TOC v2.3.4 - S2 - 20190101 - cloud_an_flag: 11um/12um thin cirrus test

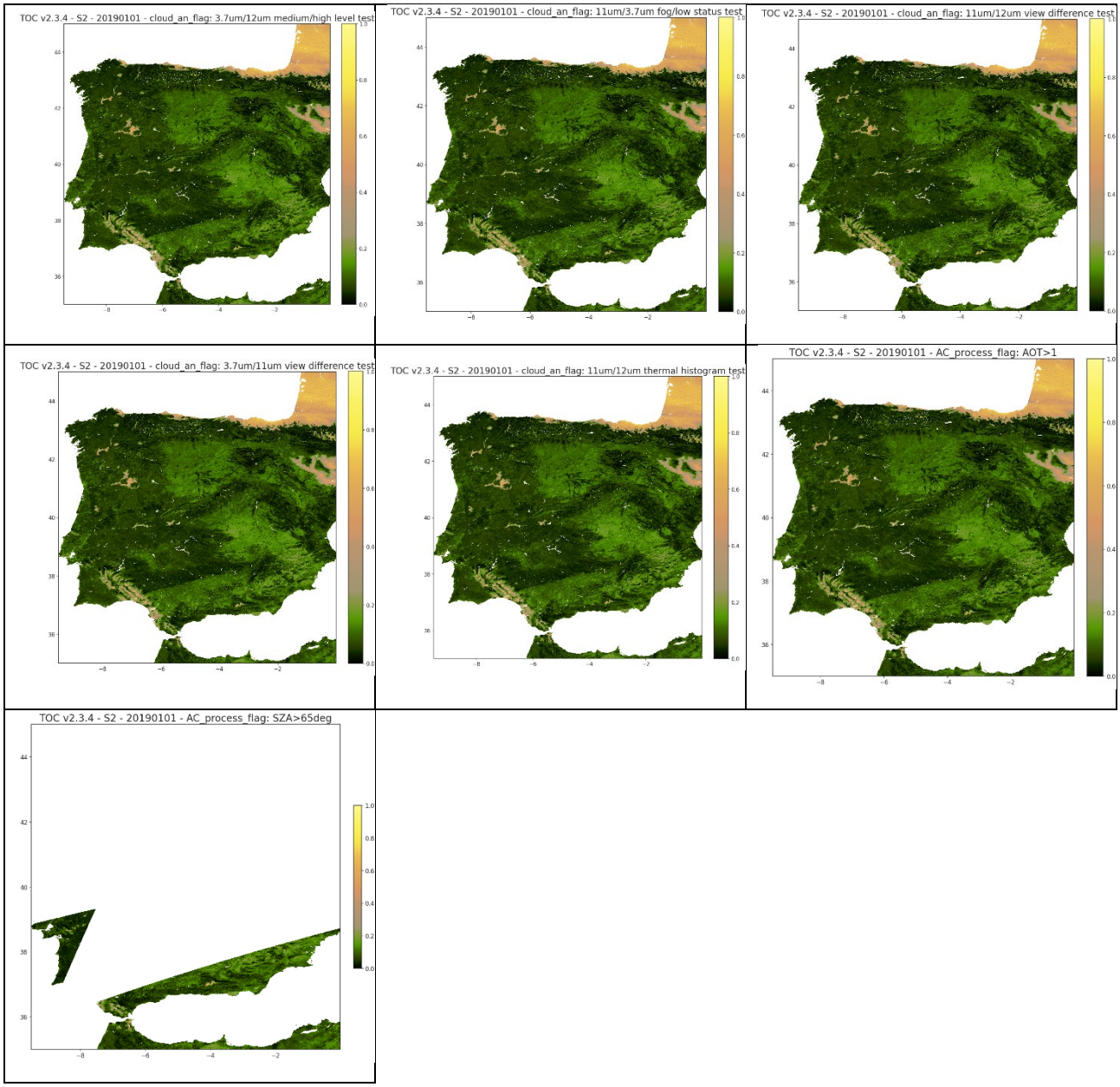
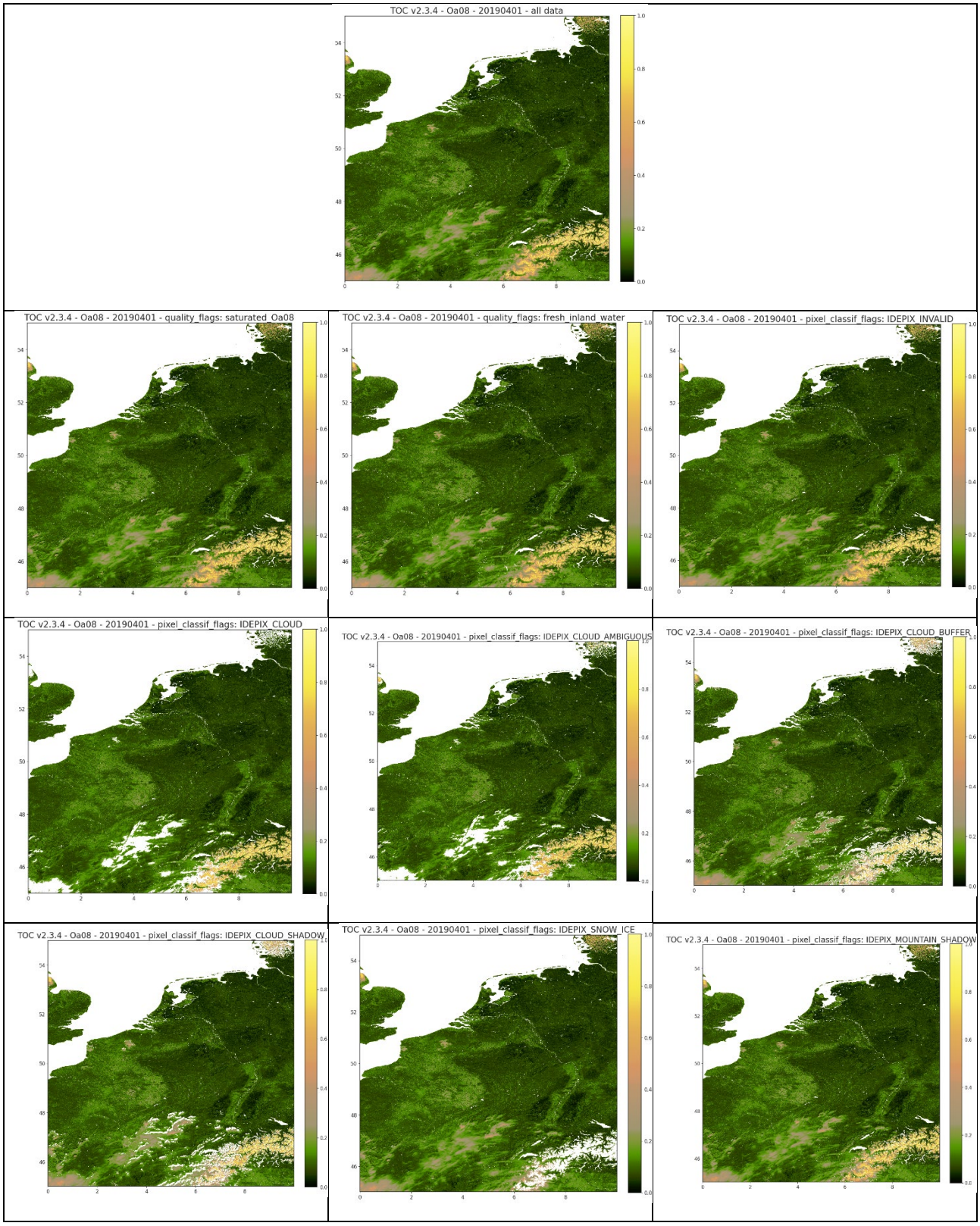


Figure 57: Top: Map of Sentinel-3 SLSTR (S2) TOC reflectance (all pixels) for 01/01/2019 at full resolution for the X17Y04 tile. Other images show the same tile when each bit of the quality layers is applied to exclude pixels.


TOC v2.3.4 - Oa08 - 20190401 - all data
TOC v2.3.4 - Oa08 - 20190401 - quality_flags: saturated_Oa08
TOC v2.3.4 - Oa08 - 20190401 - quality_flags: fresh_inland_water
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_INVALID
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_CLOUD
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_CLOUD_AMBIGUOUS
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_CLOUD_BUFFER
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_CLOUD_SHADOW
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_SNOW_ICE
TOC v2.3.4 - Oa08 - 20190401 - pixel_classif_flags: IDEPIX_MOUNTAIN_SHADOW

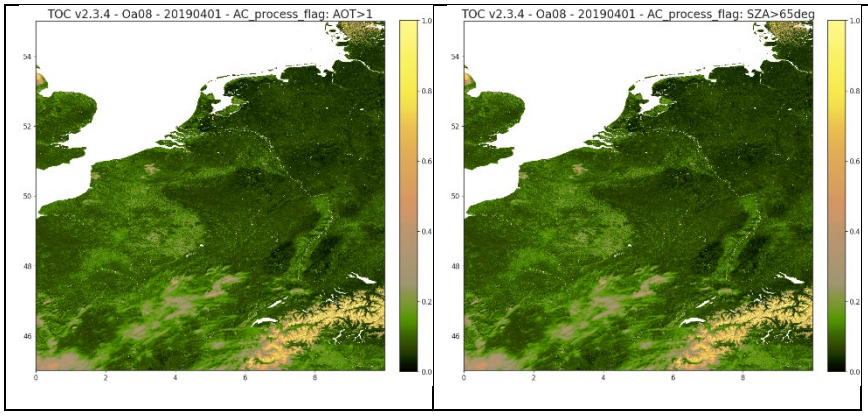
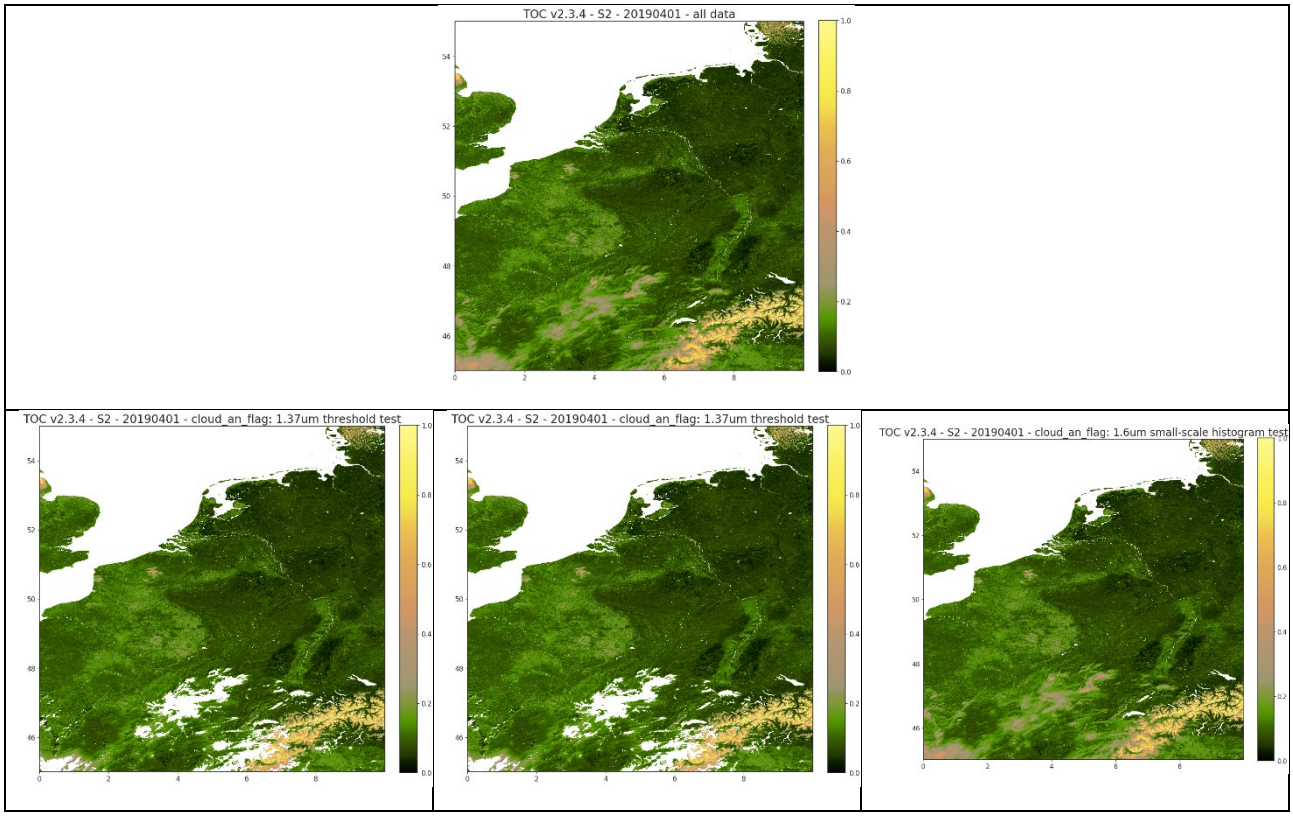
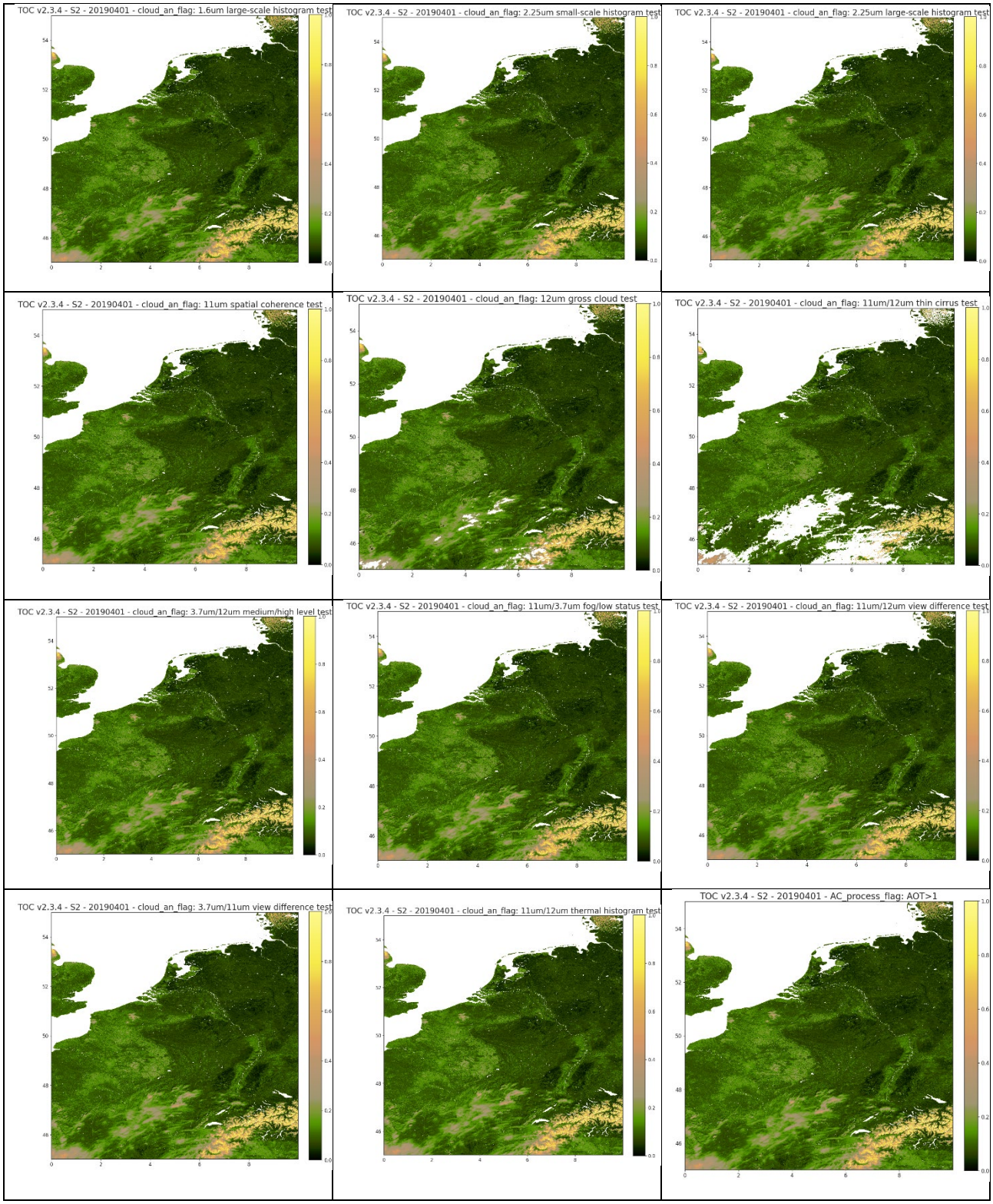


Figure 58: Same as Figure 64 but for the X18Y03 tile for date 01/04/2019.




TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 1.6um large-scale histogram test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 2.25um small-scale histogram test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 2.25um large-scale histogram test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 11um spatial coherence test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 12um gross cloud test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 11um/12um thin cirrus test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 3.7um/12um medium/high level test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 11um/3.7um fog/low status test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 11um/12um view difference test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 3.7um/11um view difference test
TOC v2.3.4 - S2 - 20190401 - cloud_an_flag: 11um/12um thermal histogram test
TOC v2.3.4 - S2 - 20190401 - AC_process_flag: AOT>1

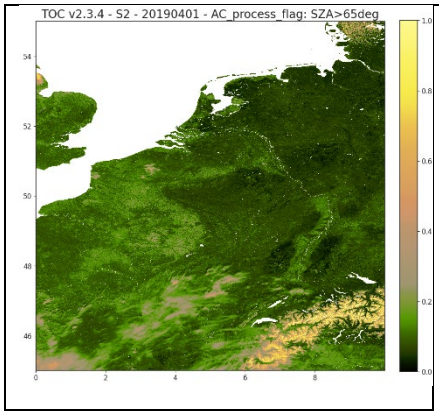
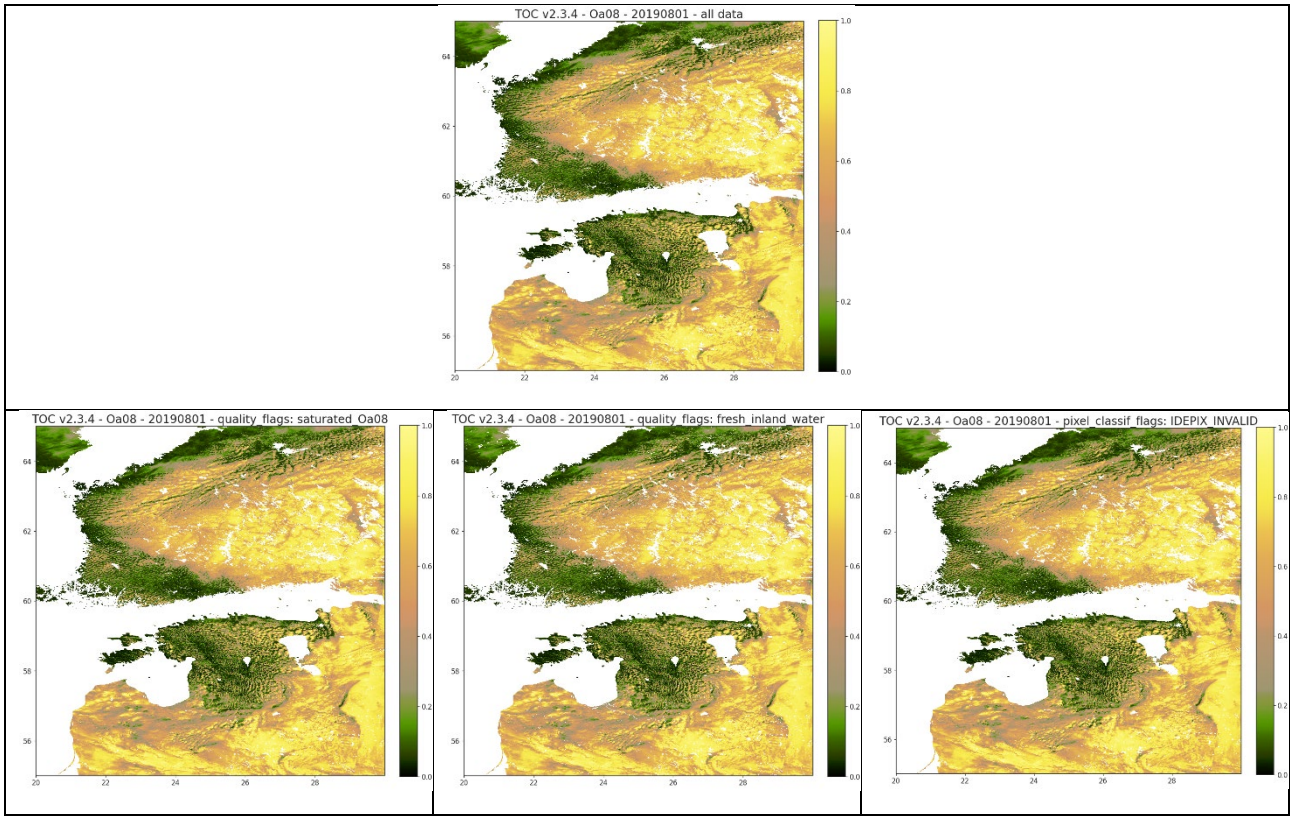


Figure 59: Same as Figure 65 but for the X18Y03 tile for date 01/04/2019.



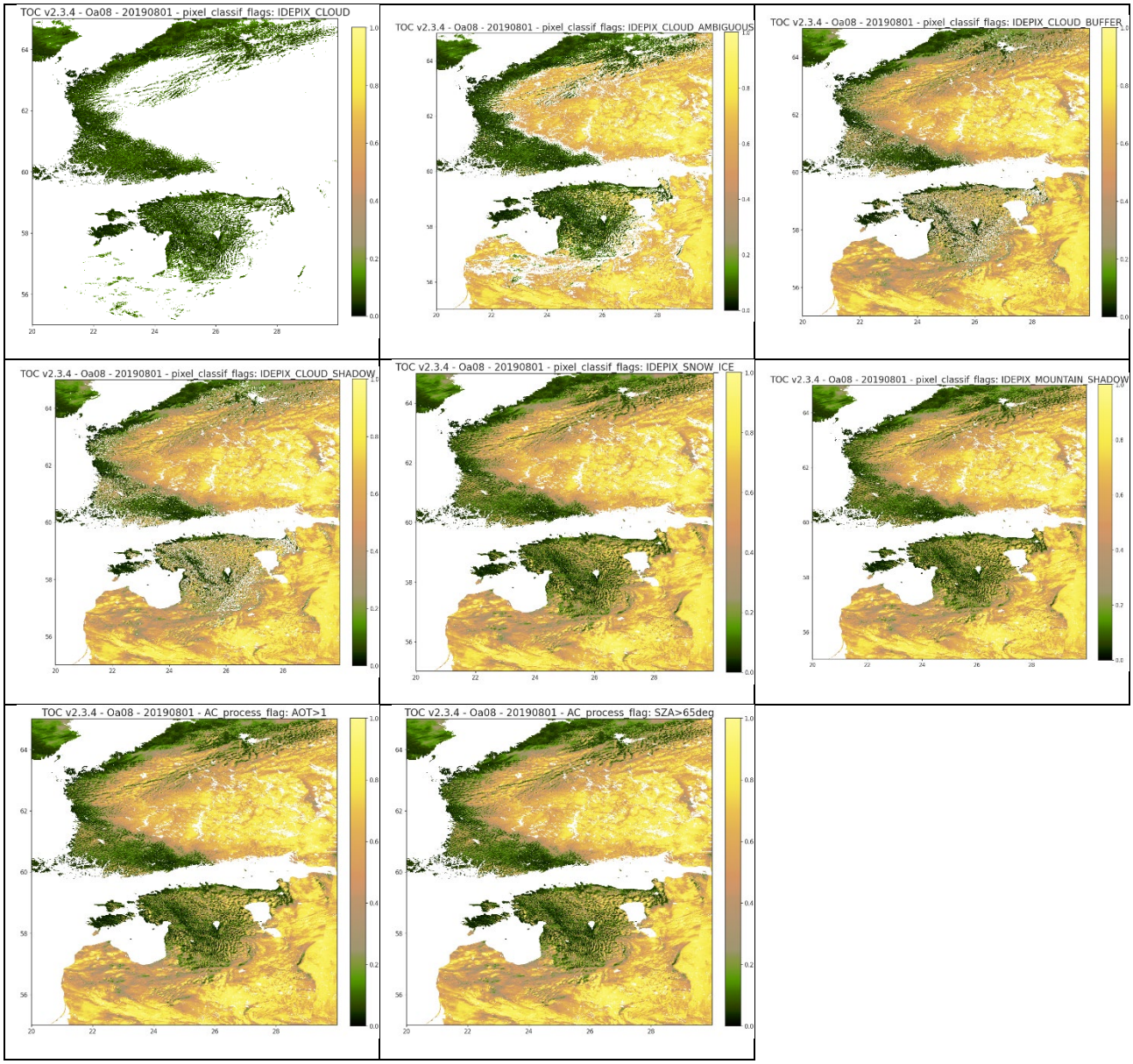
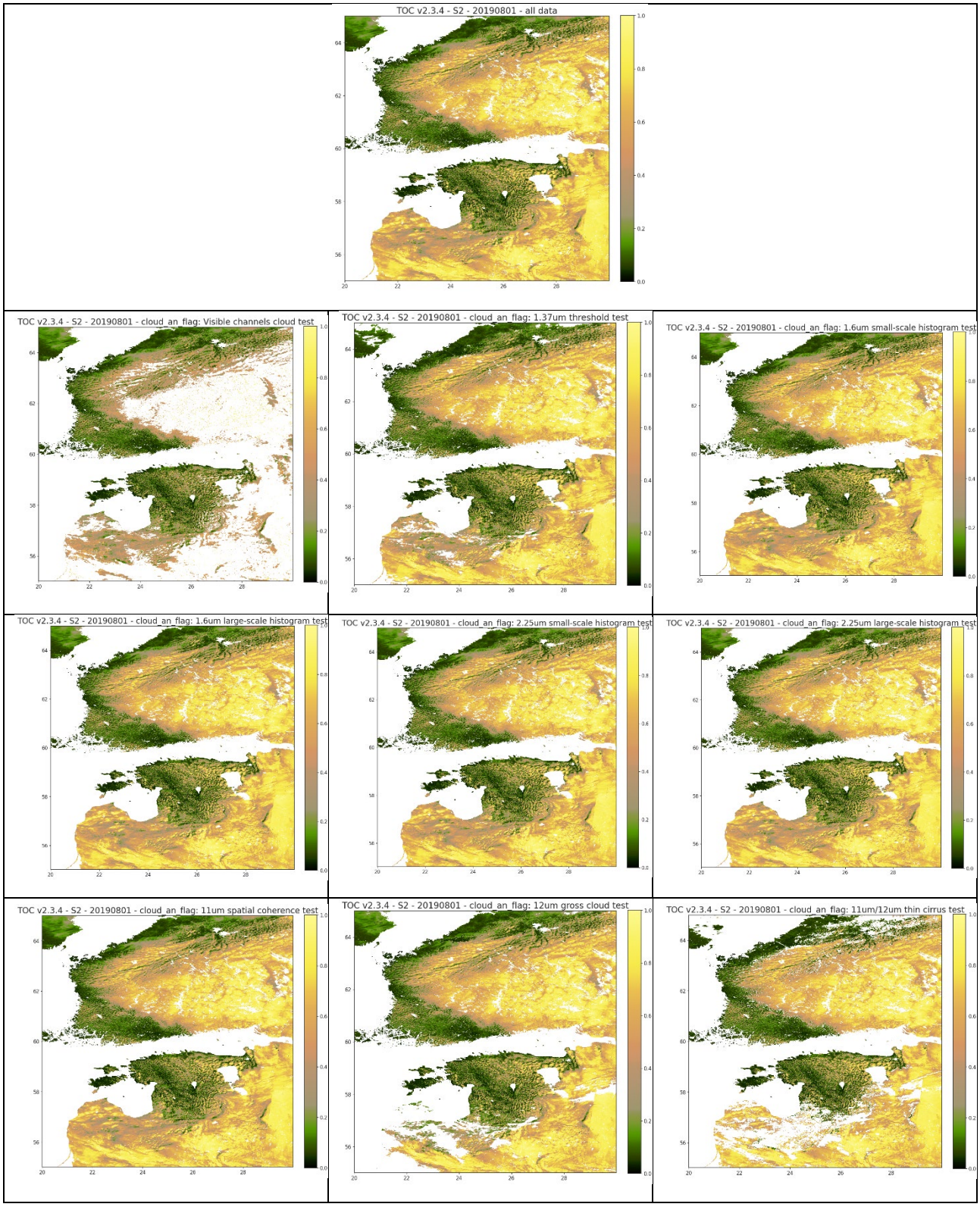


Figure 60: Same as Figure 64 but for the X20Y02 tile for date 01/08/2019.


TOC v2.3.4 - S2 - 20190801 - all data
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: Visible channels cloud test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 1.37um threshold test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 1.6um small-scale histogram test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 1.6um large-scale histogram test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 2.25um small-scale histogram test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 2.25um large-scale histogram test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 11um spatial coherence test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 12um gross cloud test
TOC v2.3.4 - S2 - 20190801 - cloud_an_flag: 11um/12um thin cirrus test

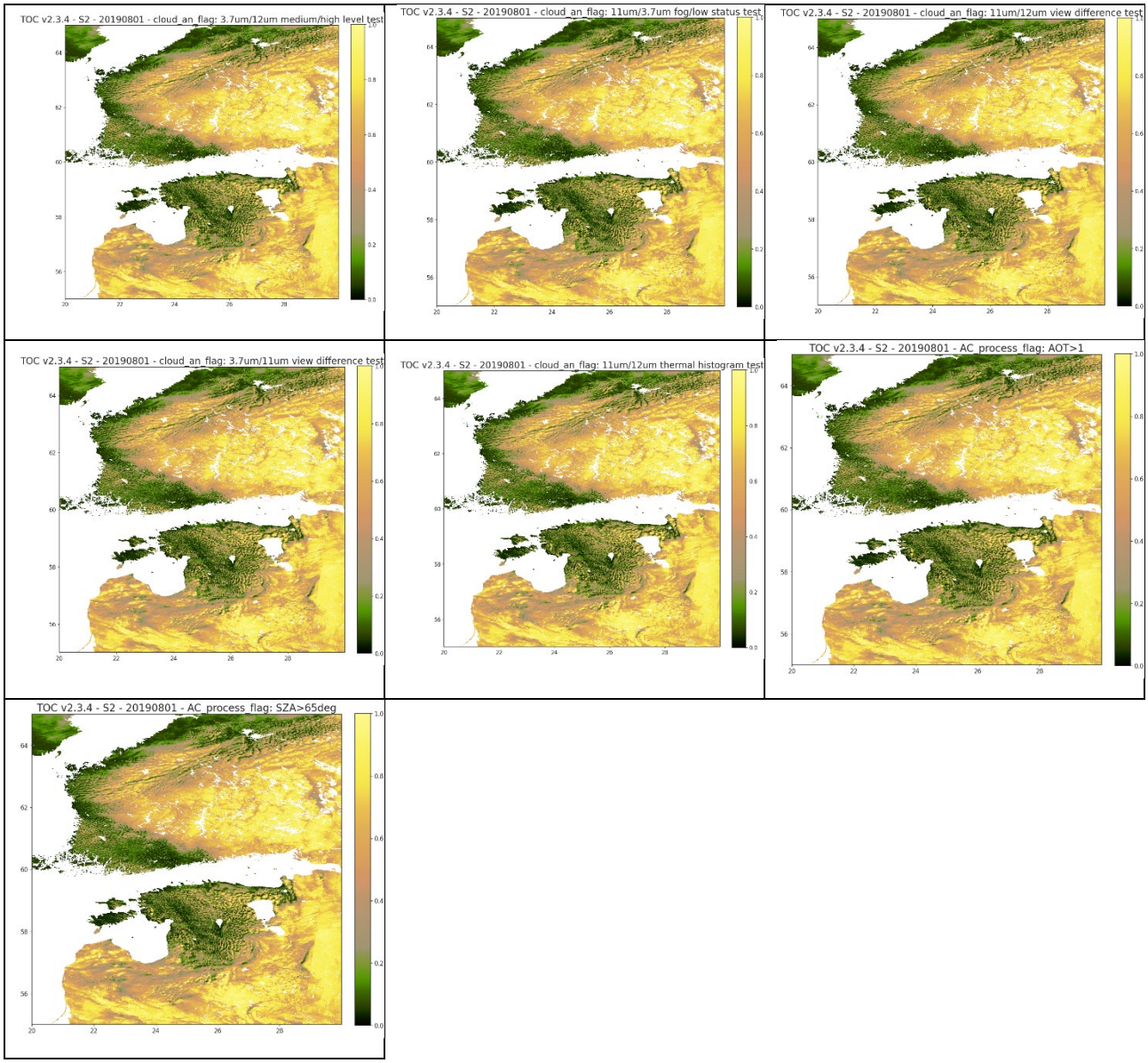
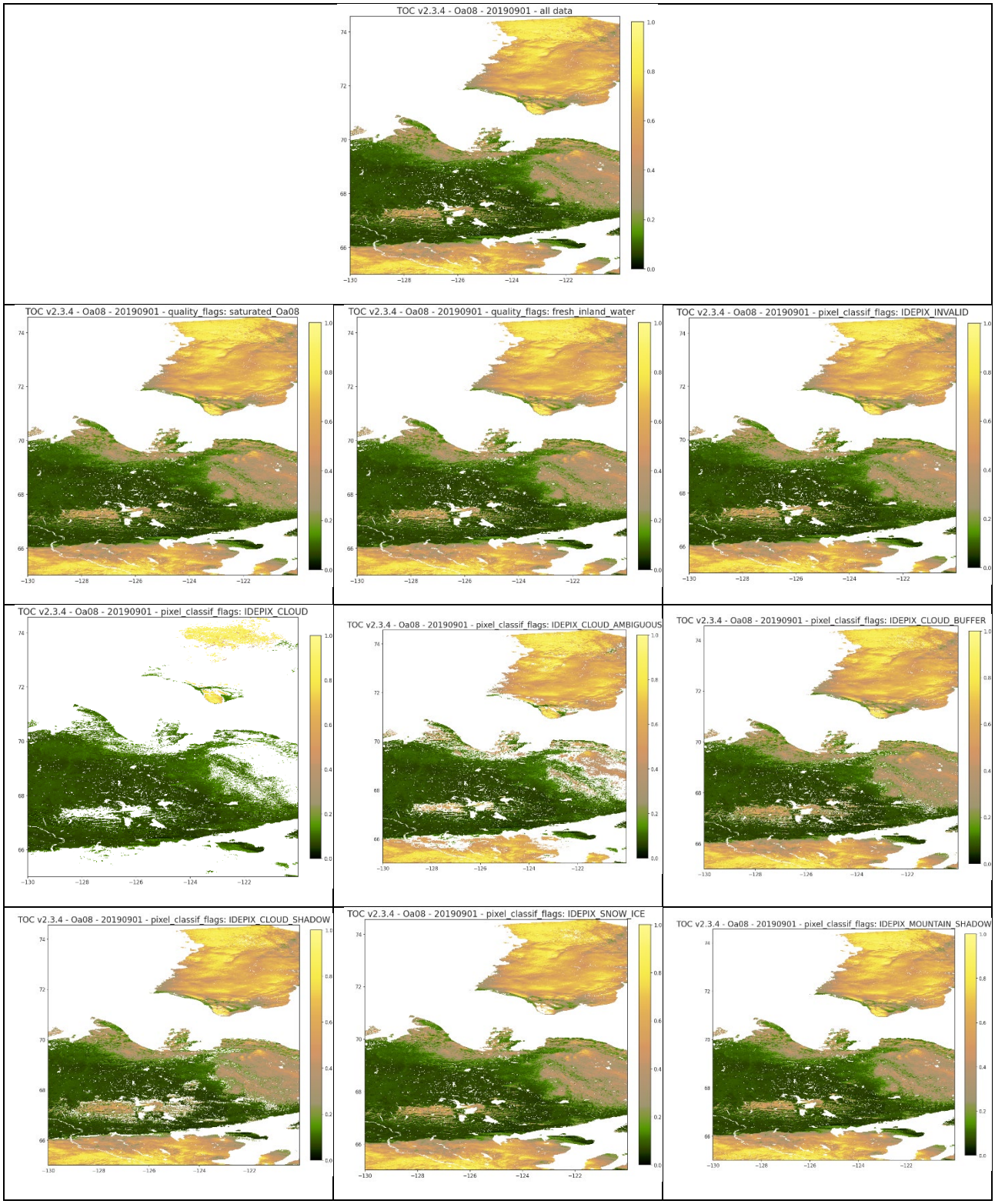


Figure 61: Same as Figure 65 but for the X20Y02 tile for date 01/08/2019.


TOC v2.3.4 - Oa08 - 20190901 - all data
TOC v2.3.4 - Oa08 - 20190901 - quality_flags: saturated_Oa08
TOC v2.3.4 - Oa08 - 20190901 - quality_flags: fresh_inland_water
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_INVALID
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_CLOUD
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_CLOUD_AMBIGUOUS
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_CLOUD_BUFFER
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_CLOUD_SHADOW
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_SNOW_ICE
TOC v2.3.4 - Oa08 - 20190901 - pixel_classif_flags: IDEPIX_MOUNTAIN_SHADOW

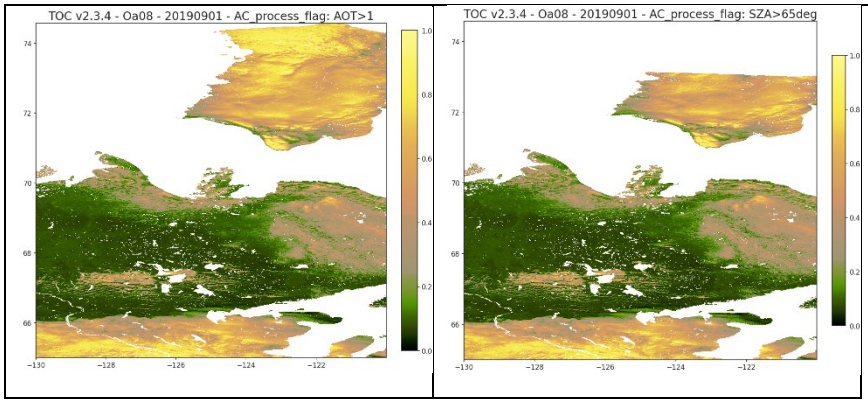
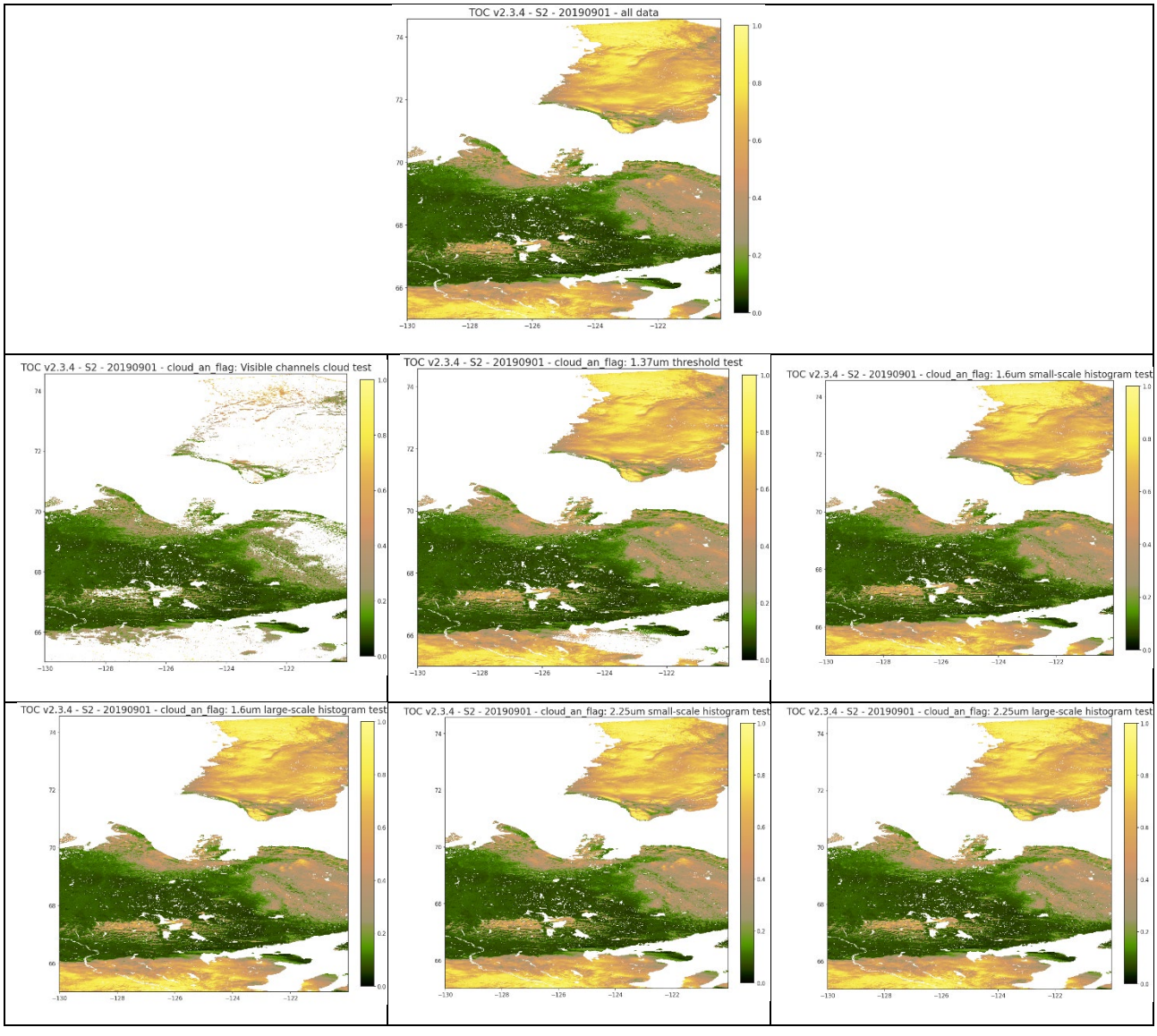


Figure 62: Same as Figure 64 but for the X05Y01 tile for date 01/09/2019.



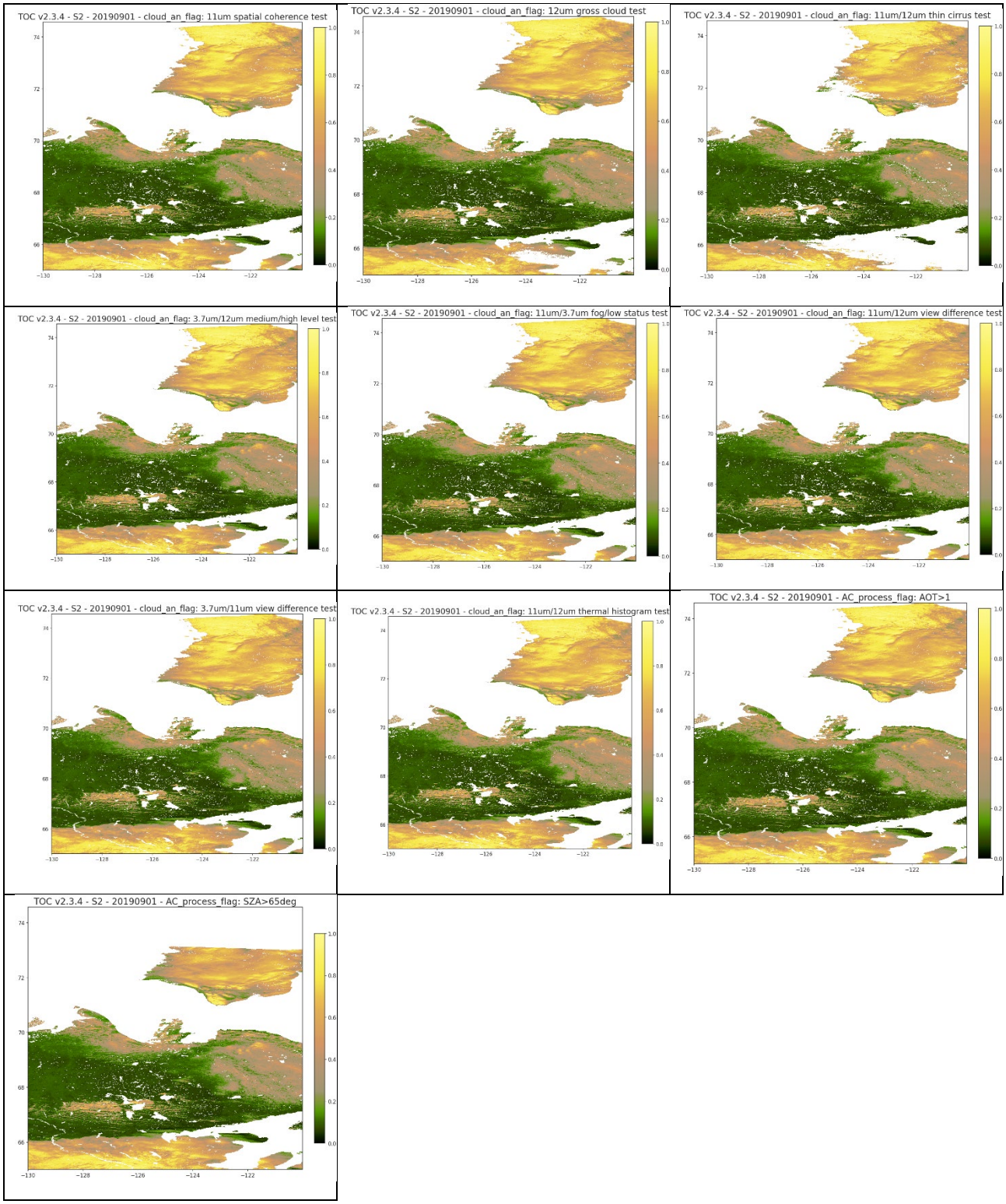


Figure 63: Same as Figure 65 but for the X05Y01 tile for date 01/09/2019.

ANNEX IV: TEMPORAL PROFILES OF SENTINEL-3 OLCI VS. RADCALNET

BSCN

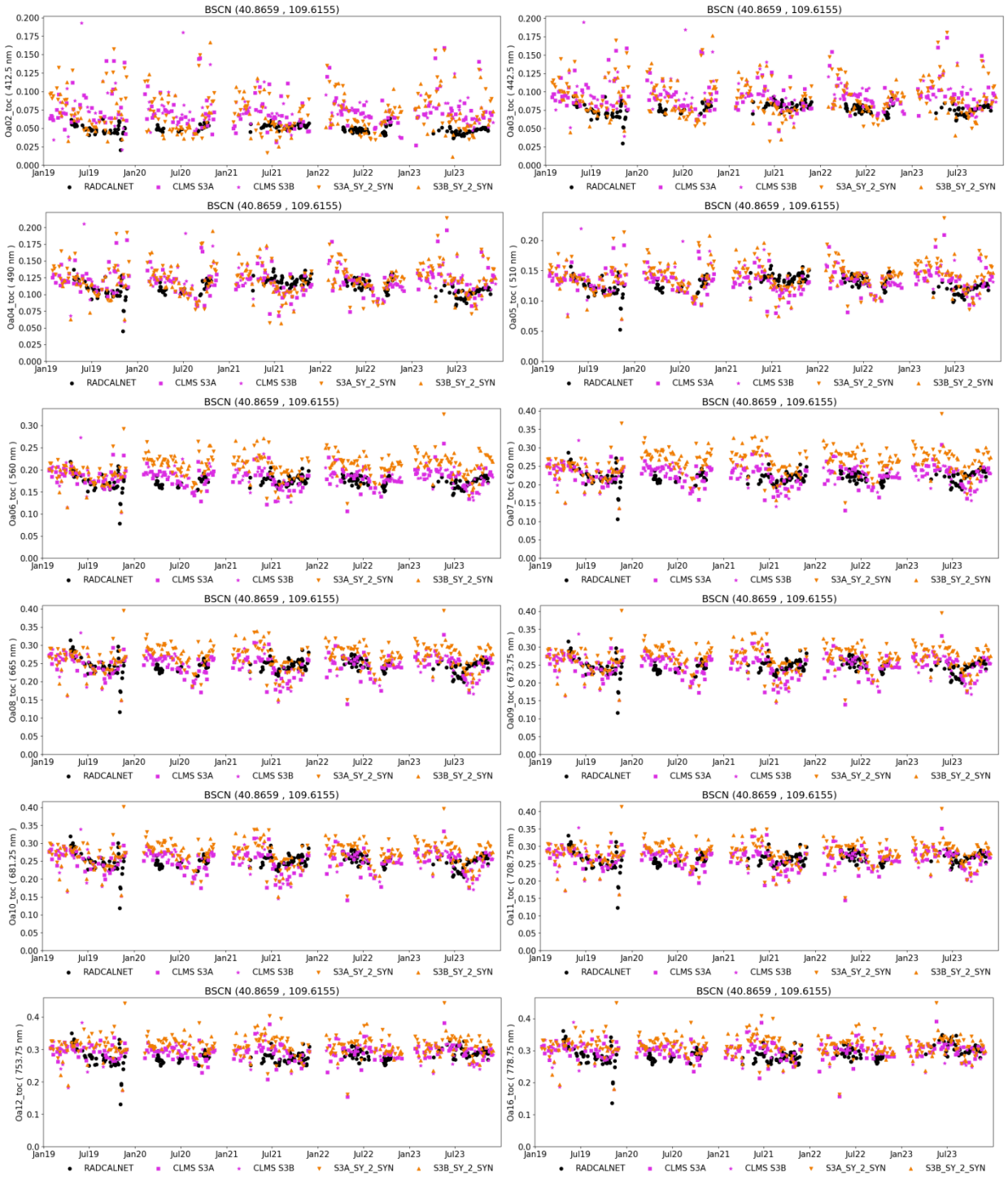
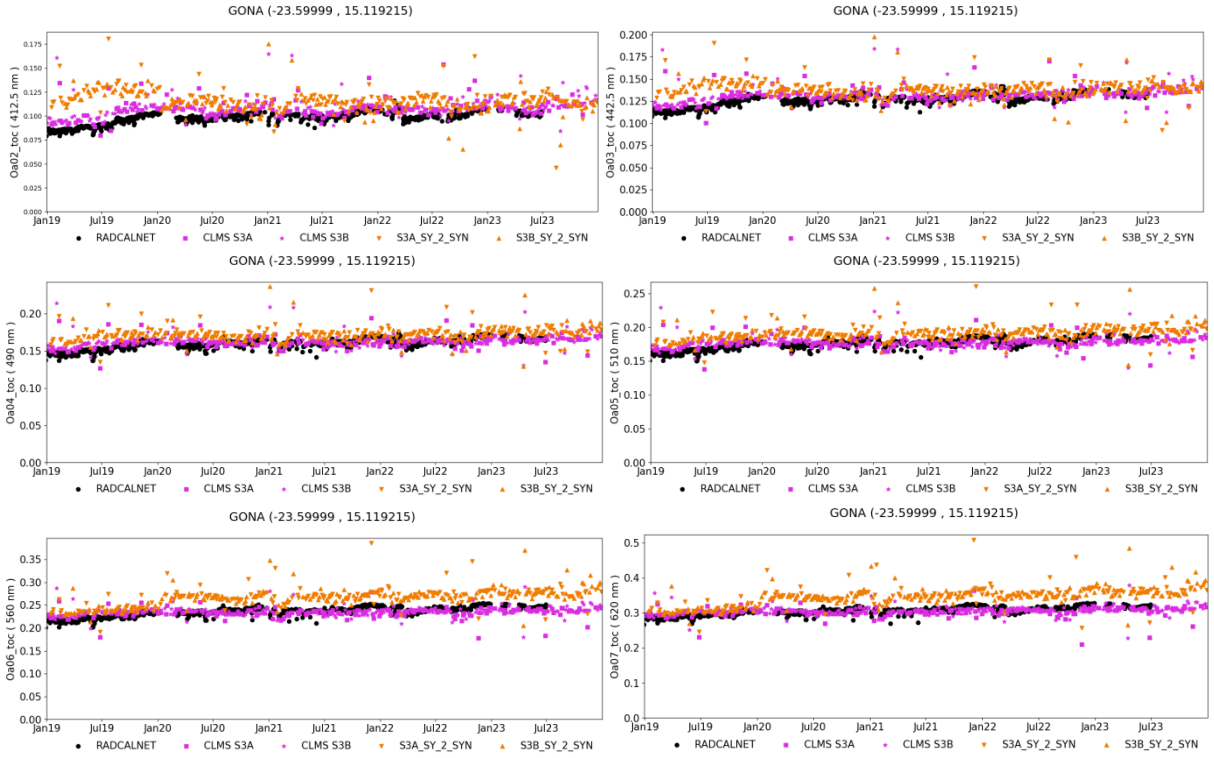


Figure 64: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over BSCN site and Sentinel-3 OLCI products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

GONA



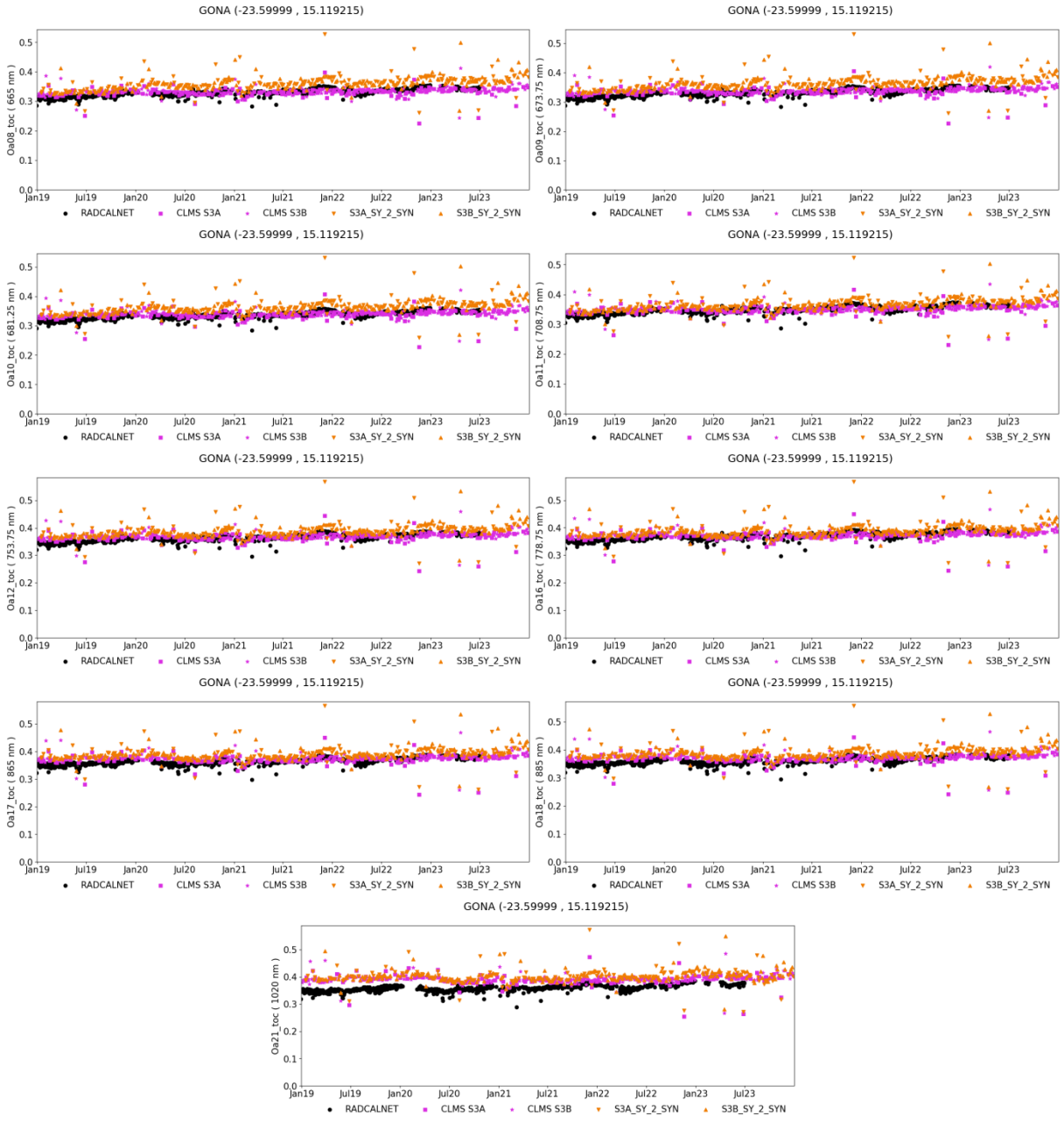
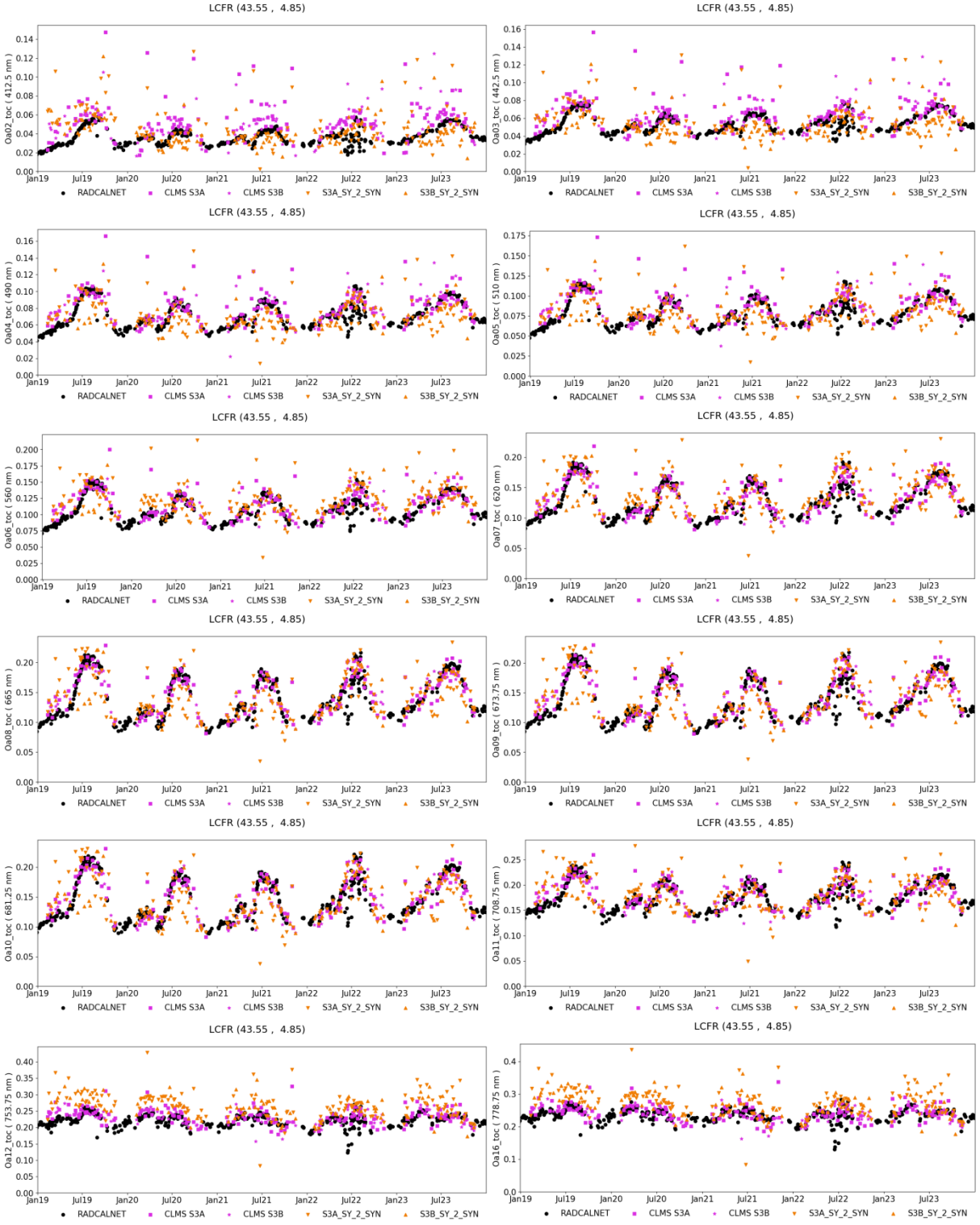


Figure 65: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over GONA site and Sentinel-3 OLCI products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

LCFR


LCFR (43.55 , 4.85)
Oa02_toc (412.5 nm)
Oa03_toc (442.5 nm)
Oa04_toc (490 nm)
Oa05_toc (510 nm)
Oa06_toc (560 nm)
Oa07_toc (620 nm)
Oa08_toc (665 nm)
Oa09_toc (673.75 nm)
Oa10_toc (681.25 nm)
Oa11_toc (708.75 nm)
Oa12_toc (753.75 nm)
Oa16_toc (778.75 nm)
RADCALNET
CLMS S3A
CLMS S3B
S3A_SY_2_SYN
S3B_SY_2_SYN

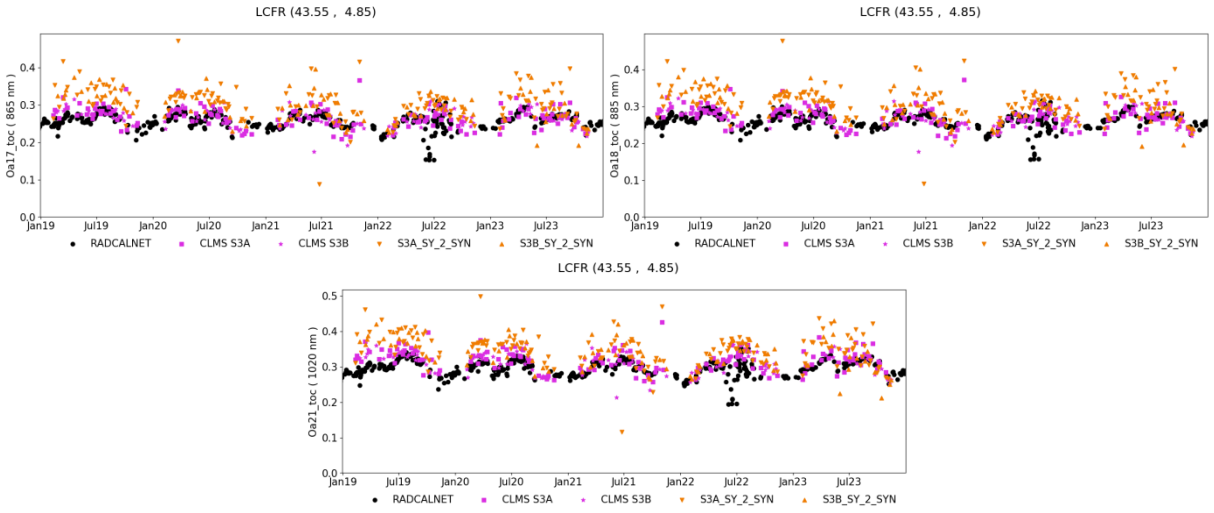
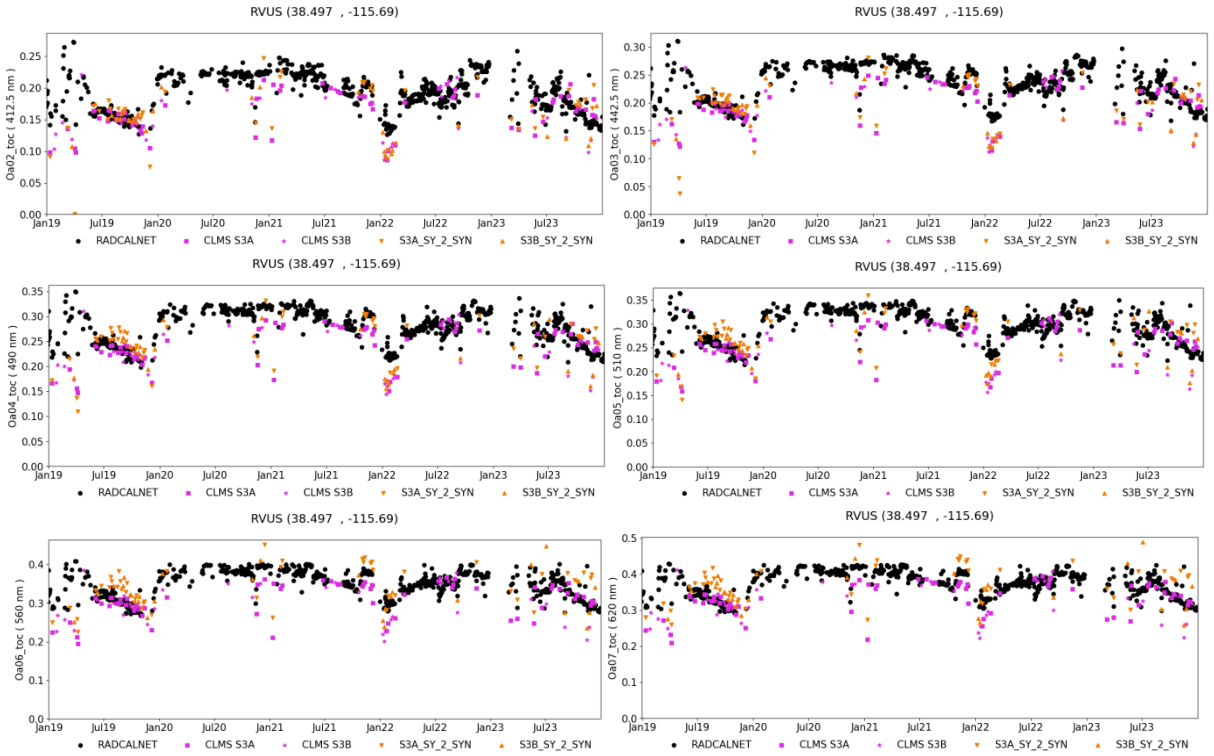


Figure 66: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over LCFR site and Sentinel-3 OLCI products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

RVUS



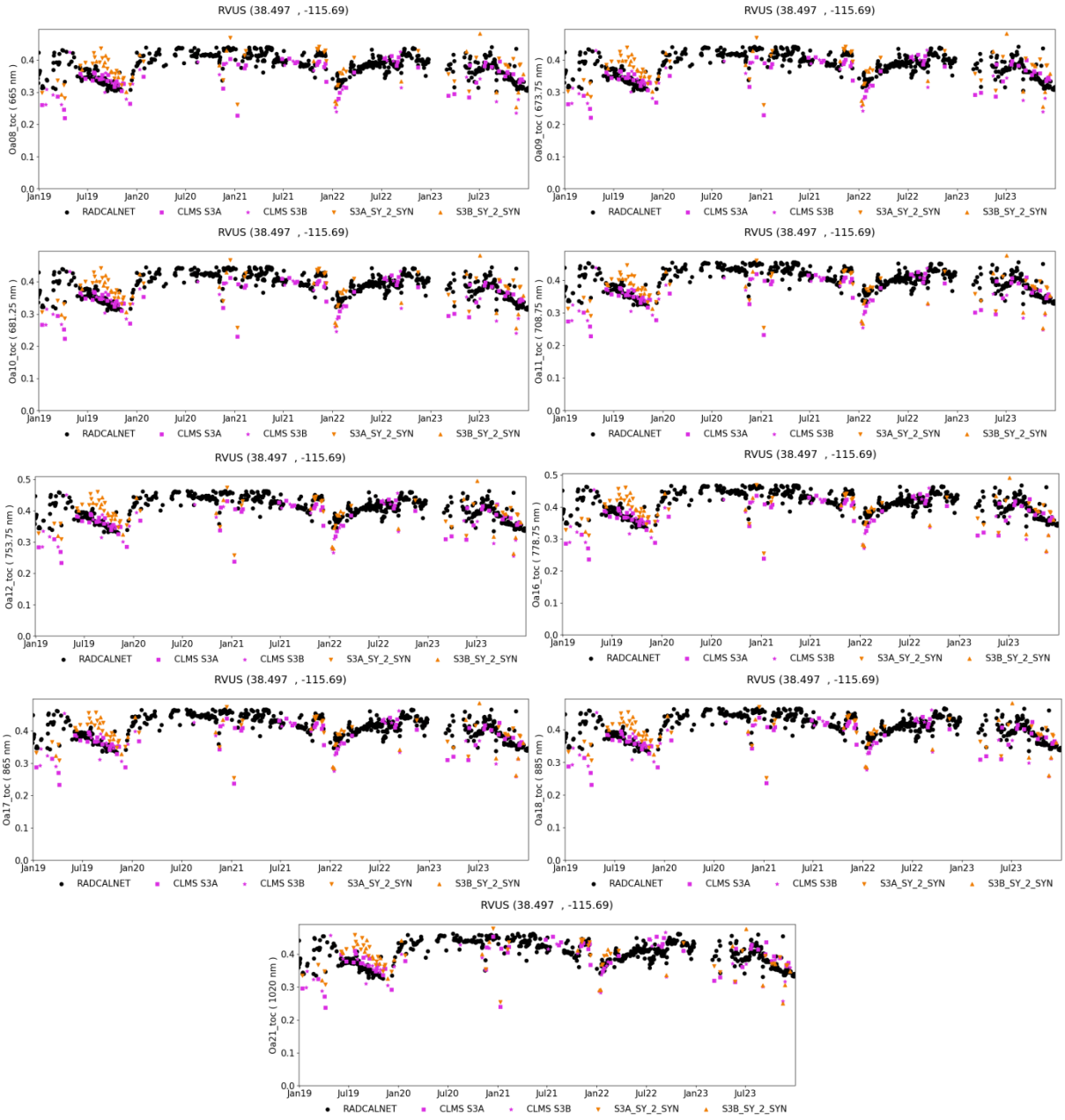


Figure 67: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over RVUS site and Sentinel-3 OLCI products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

ANNEX V: TEMPORAL PROFILES OF SENTINEL-3 SLSTR VS. RADCALNET

BSCN

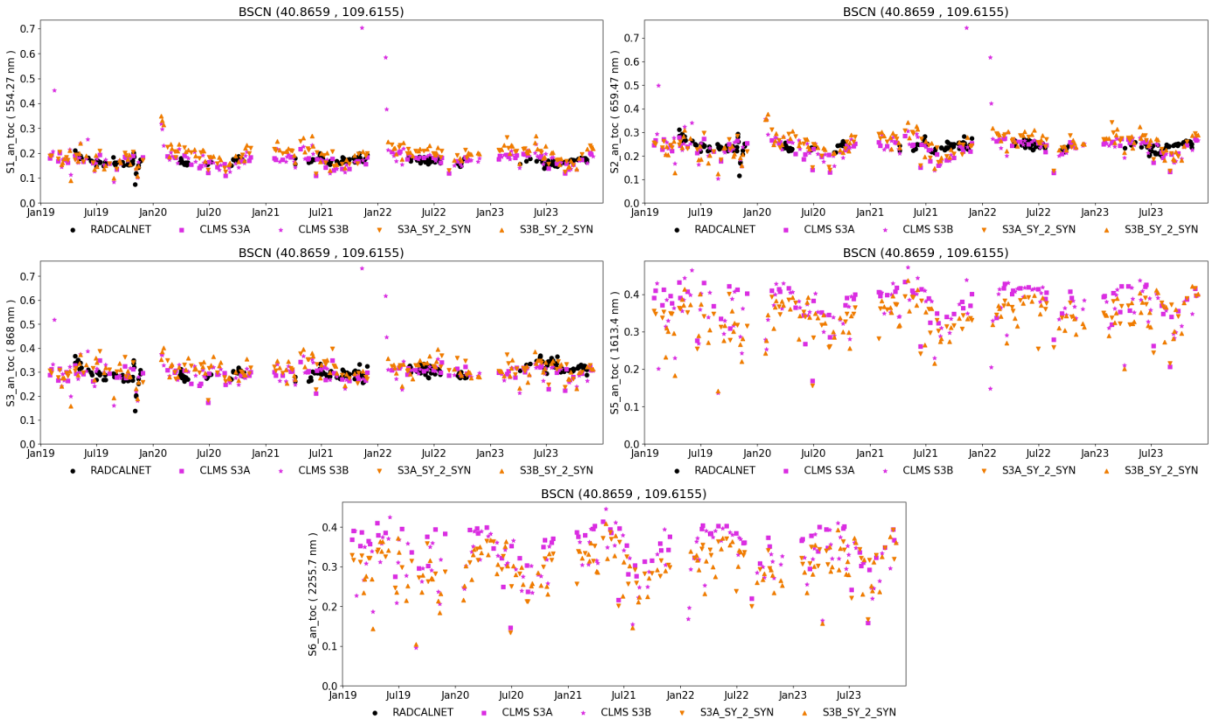


Figure 68: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over BSCN site and Sentinel-3 SLSTR products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

GONA

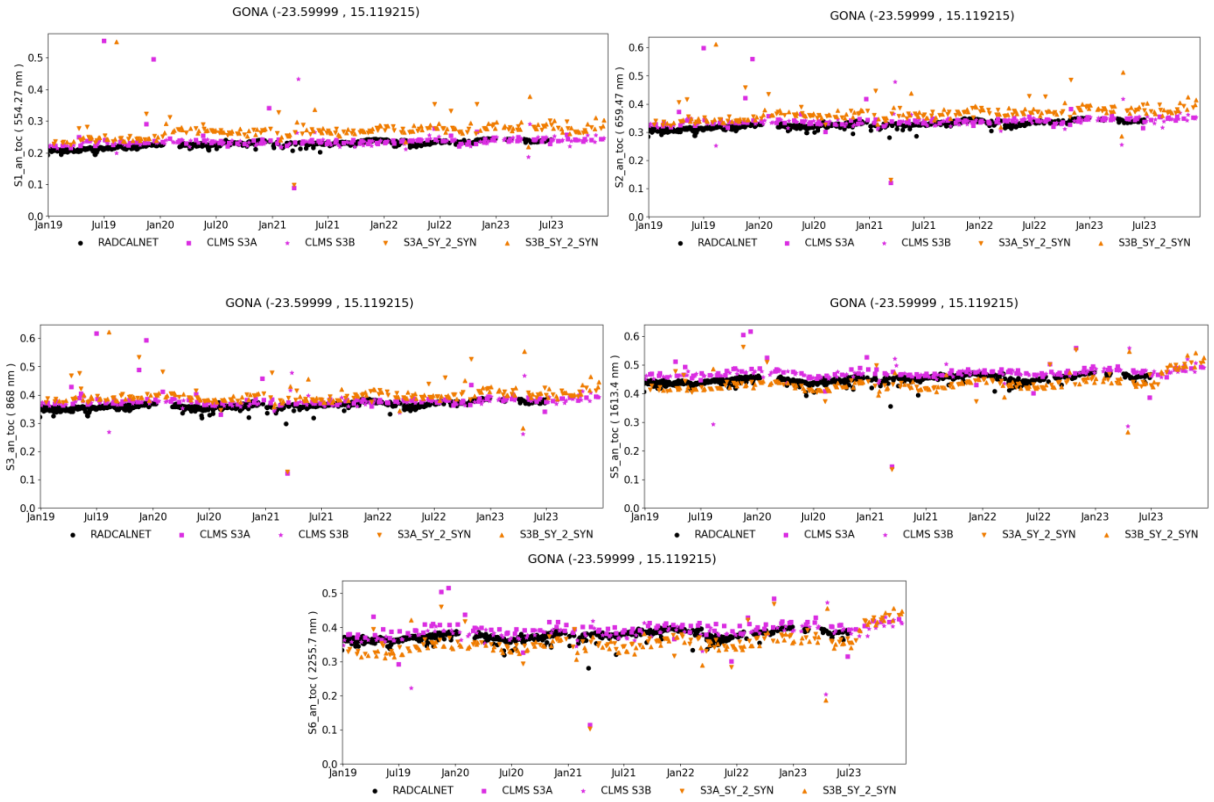


Figure 69: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over GONA site and Sentinel-3 SLSTR products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

LCFR

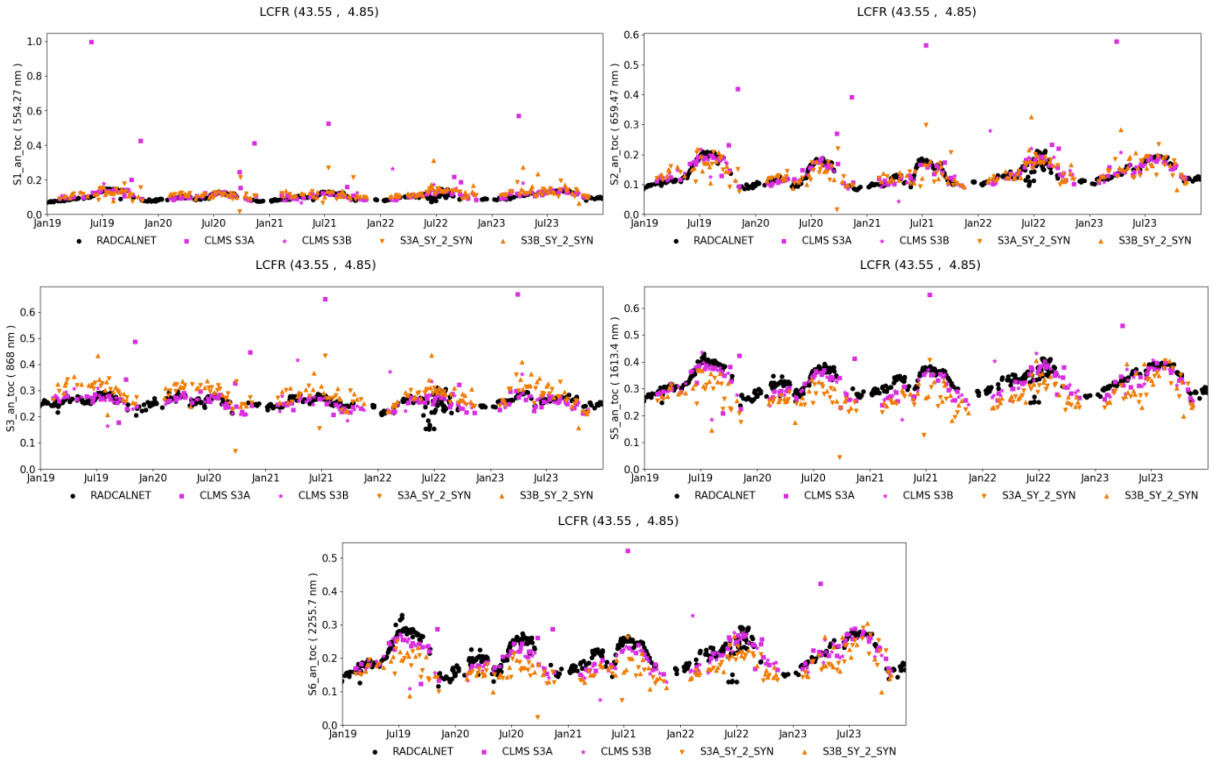


Figure 70: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over LCFR site and Sentinel-3 SLSTR products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

RVUS

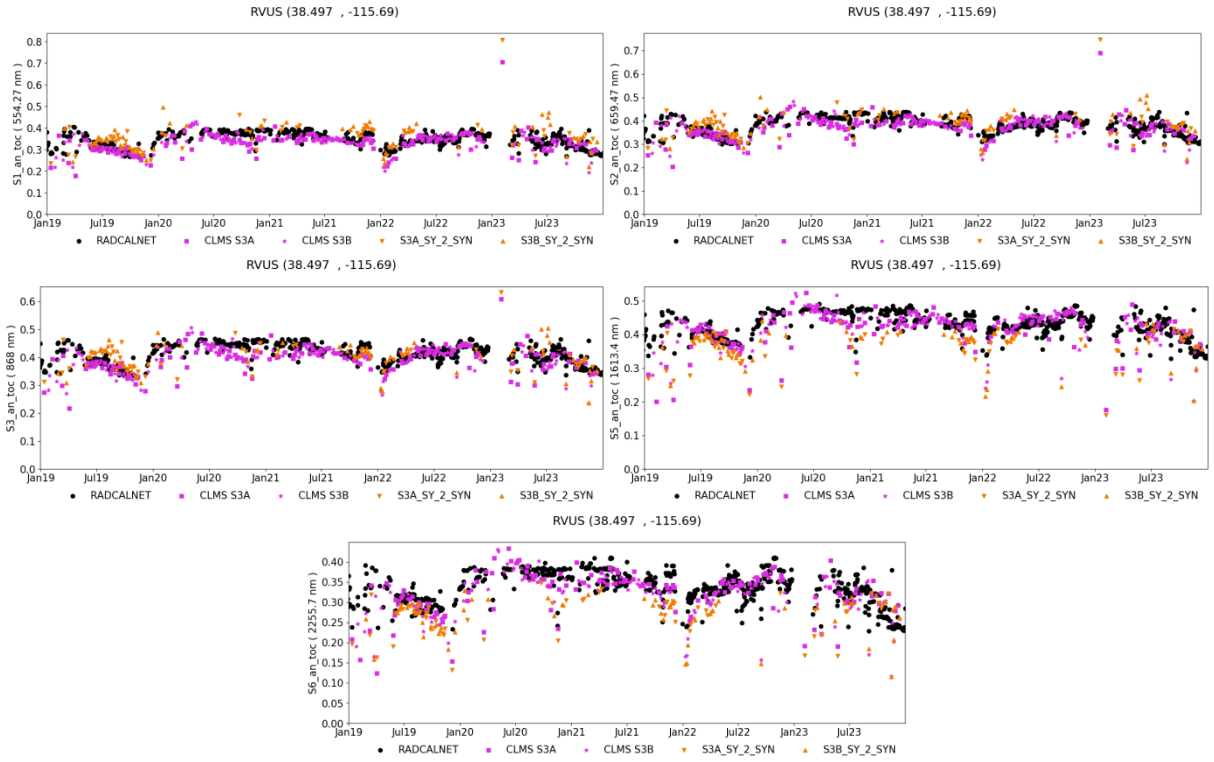


Figure 71: Temporal profiles (2019-2023 period) of RadCalNet measurements (black dots) over RVUS site and Sentinel-3 SLSTR products (VZA<10°): CLMS TOC V2.3 (purple squares and dots for Sentinel-3A and Sentinel-3B) and ESA L2 SYN (orange down and up-pointing triangles for Sentinel-3A and Sentinel-3B).

ANNEX VI: ERROR EVALUATION BETWEEN CLMS TOC V2.3 AND ESA L2 SYN OVER AOIS

For each AOI, the following graphs are presented:

- Spectral profiles of Sentinel-3 OLCI and SLSTR CLMS TOC V2.3 and ESA L2 SYN products for the pixel located at the central coordinate of each AOI.
- Scatter-plots and associated validation metrics (APU) for each OLCI and SLSTR channel.

AOI#1 VIEREMÄ (FINLAND) – NLF

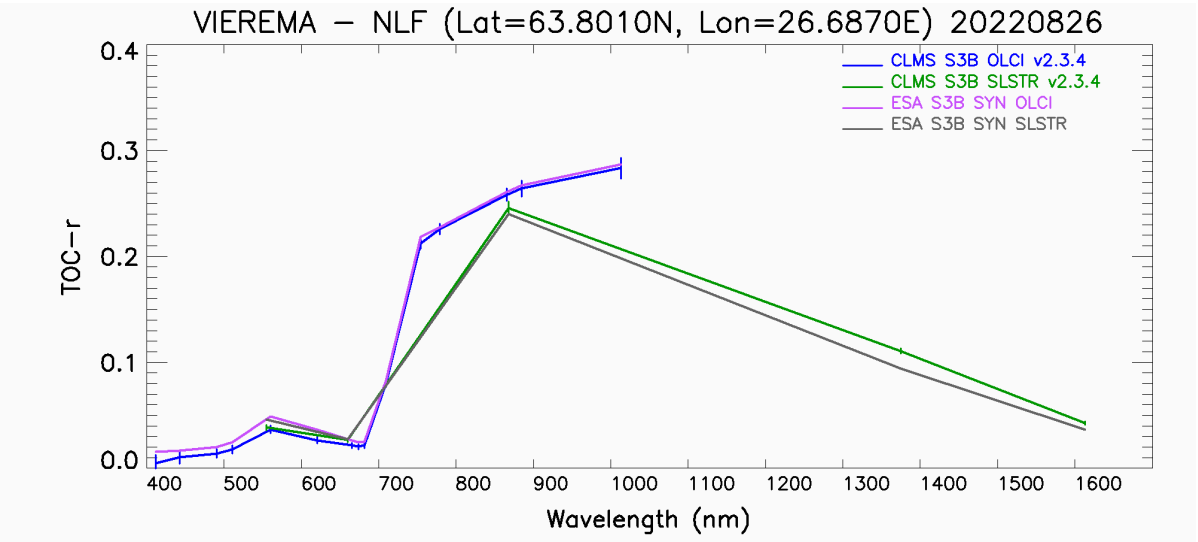


Figure 72: Spectral profile of Sentinel-3B (2022.08.26) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#1 Vieremä (Finland) – NLF.

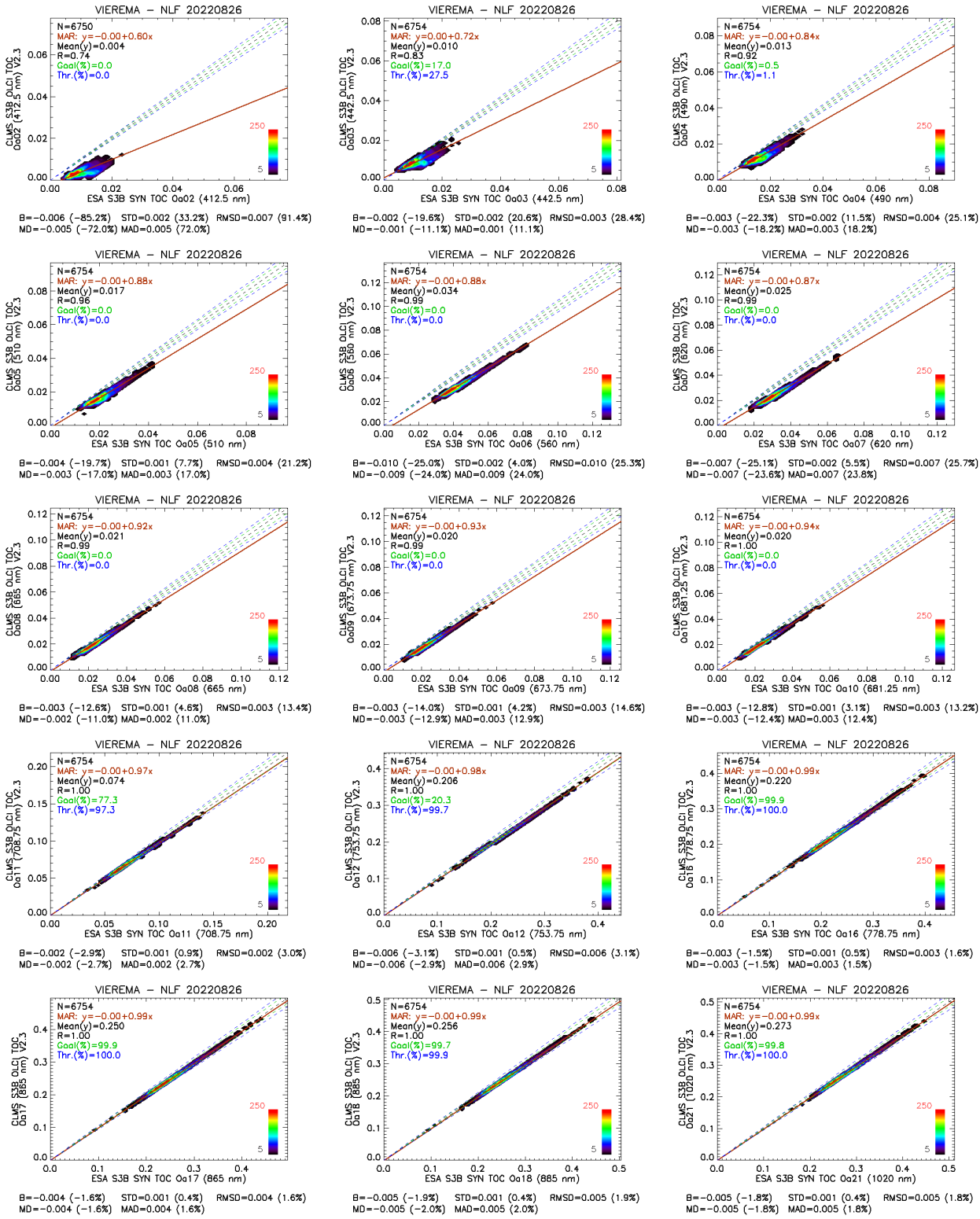


Figure 73: Scatter-plots of Sentinel-3B (2022.08.26) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#1 (Vieremä, Finland – NLF).

Table 27: Validation statistics of the comparison of Sentinel-3B (2022.08.26) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#1 (Vieremä, Finland – NLF).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6750	0.74	-0.006 (-85.2%)	-0.005 (-72.0%)	0.002 (33.2%)	0.005 (72.0%)	0.007 (91.4%)	0.00 /0.6	0.0 /0.0
Oa03	6754	0.93	-0.002 (-19.6%)	-0.001 (-11.1%)	0.002 (20.1%)	0.001 (11.1%)	0.003 (28.4%)	0.00 /0.72	17.0 /27.5
Oa04	6754	0.92	-0.003 (-22.3%)	-0.003 (-18.2%)	0.002 (11.5%)	0.003 (18.2%)	0.004 (25.1%)	0.00 /0.84	0.5 /1.1
Oa05	6754	0.96	-0.004 (-19.7%)	-0.003 (-17.0%)	0.001 (7.7%)	0.003 (17.0%)	0.004 (21.2%)	0.00 /0.88	0.0 /0.0
Oa06	6754	0.99	-0.010 (-25.0%)	-0.009 (24.0%)	0.002 (4.0%)	0.009 (24.0%)	0.010 (25.3%)	0.00 /0.88	0.0 /0.0
Oa07	6754	0.99	-0.007 (-25.1%)	-0.007 (23.6%)	0.002 (5.5%)	0.007 (23.8%)	0.007 (25.7%)	0.00 /0.87	0.0 /0.0
Oa08	6754	0.99	-0.003 (-12.6%)	-0.002 (-11.0%)	0.001 (4.6%)	0.002 (11.0%)	0.003 (13.4%)	0.00 /0.92	0.0 /0.0
Oa09	6754	0.99	-0.003 (-14.0%)	-0.003 (12.9%)	0.001 (4.2%)	0.003 (12.9%)	0.003 (14.6%)	0.00 /0.93	0.0 /0.0
Oa10	6754	1.00	-0.003 (-12.8%)	-0.003 (-12.4%)	0.001 (3.1%)	0.003 (12.4%)	0.003 (13.2%)	0.00 /0.94	0.0 /0.0
Oa11	6754	1.00	-0.002 (-2.9%)	-0.002 (-2.7%)	0.001 (0.9%)	0.002 (2.7%)	0.002 (3.0%)	0.00 /0.97	77.3 /97.3
Oa12	6754	1.00	-0.006 (-3.1%)	-0.006 (-2.9%)	0.001 (0.5%)	0.006 (2.9%)	0.006 (3.1%)	0.00 /0.98	20.3 /99.7
Oa16	6754	1.00	-0.003 (-1.5%)	-0.003 (-1.5%)	0.001 (0.5%)	0.003 (1.5%)	0.003 (1.6%)	0.00 /0.99	99.9 /100
Oa17	6754	1.00	-0.004 (-1.6%)	-0.004 (-1.6%)	0.001 (0.4%)	0.004 (1.6%)	0.004 (1.6%)	0.00 /0.99	99.9 /100
Oa18	6754	1.00	-0.005 (-1.9%)	-0.005 (-2.0%)	0.001 (0.4%)	0.005 (2.0%)	0.005 (1.9%)	0.00 /0.99	99.7 /99.9
Oa21	6754	1.00	-0.005 (-1.8%)	-0.005 (-1.8%)	0.001 (0.4%)	0.005 (1.8%)	0.005 (1.8%)	0.00 /0.99	99.8 /100

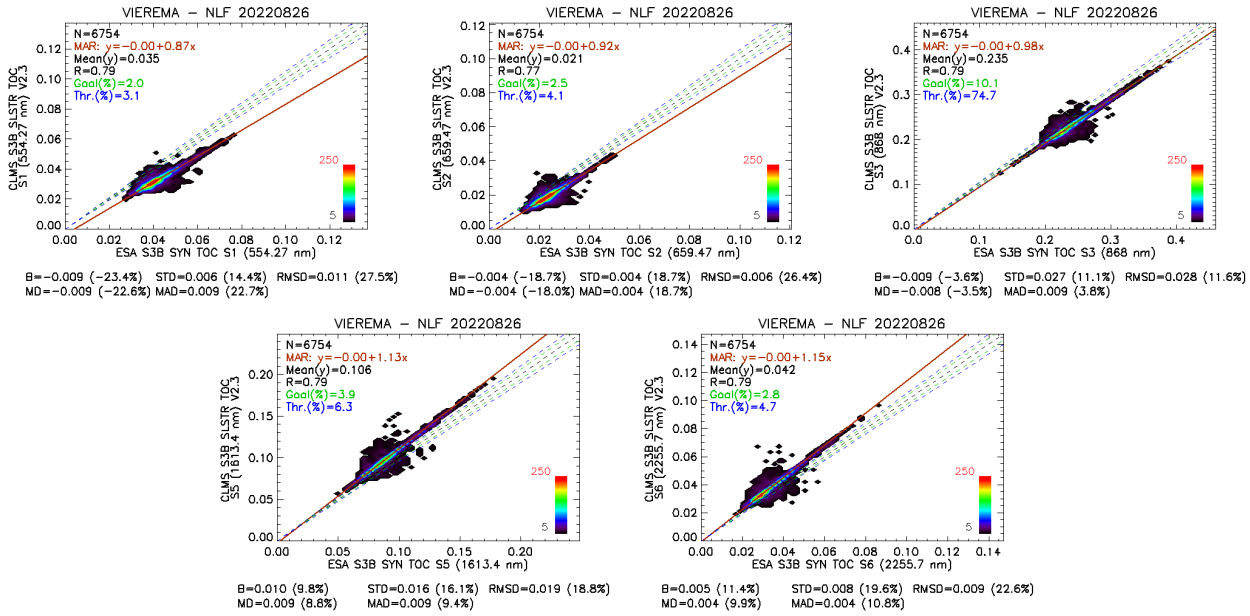


Figure 74: Scatter-plots of Sentinel-3B (2022.08.26) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#1 (Vieremä, Finland – NLF).

Table 28: Validation statistics of the comparison of Sentinel-3B (2022.08.26) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#1 (Vieremä, Finland – NLF).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6754	0.79	-0.009 (-23.4%)	-0.009 (-22.6%)	0.006 (14.4%)	0.009 (22.7%)	0.011 (27.5%)	0.00 /0.87	2.0 /3.1
S2	6754	0.77	-0.004 (-18.7%)	-0.004 (-18.0%)	0.004 (18.7%)	0.004 (18.7%)	0.006 (26.4%)	0.00 /0.92	2.5 /4.1
S3	6754	0.79	-0.009 (-3.6%)	-0.008 (-3.5%)	0.027 (11.1%)	0.009 (3.8%)	0.028 (11.6%)	0.00 /0.98	10.1 /74.7
S5	6754	0.79	0.010 (9.8%)	0.009 (8.8%)	0.016 (16.1%)	0.009 (9.4%)	0.019 (18.8%)	0.00 /1.13	3.9 /6.3
S6	6754	0.79	0.005 (11.4%)	0.004 (9.9%)	0.008 (19.6%)	0.004 (10.8%)	0.009 (22.6%)	0.00 /1.15	2.8 /4.7

AOI#2 TJAMAJA (RUSSIA) – DBF

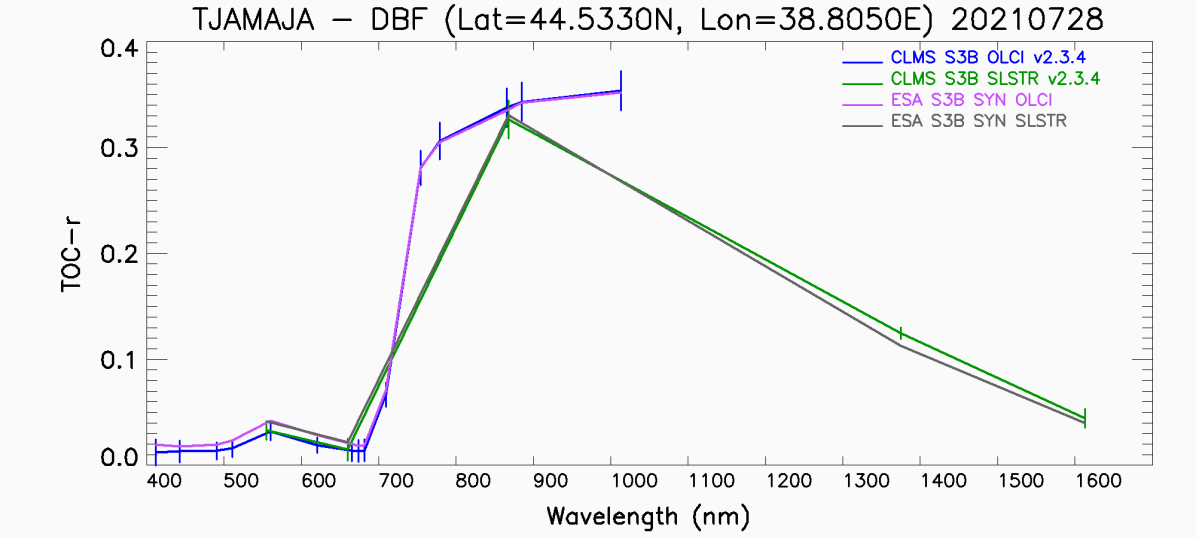


Figure 75: Spectral profile of Sentinel-3B (2021.07.28) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#2 (Tjamaja, Russia – DBF).

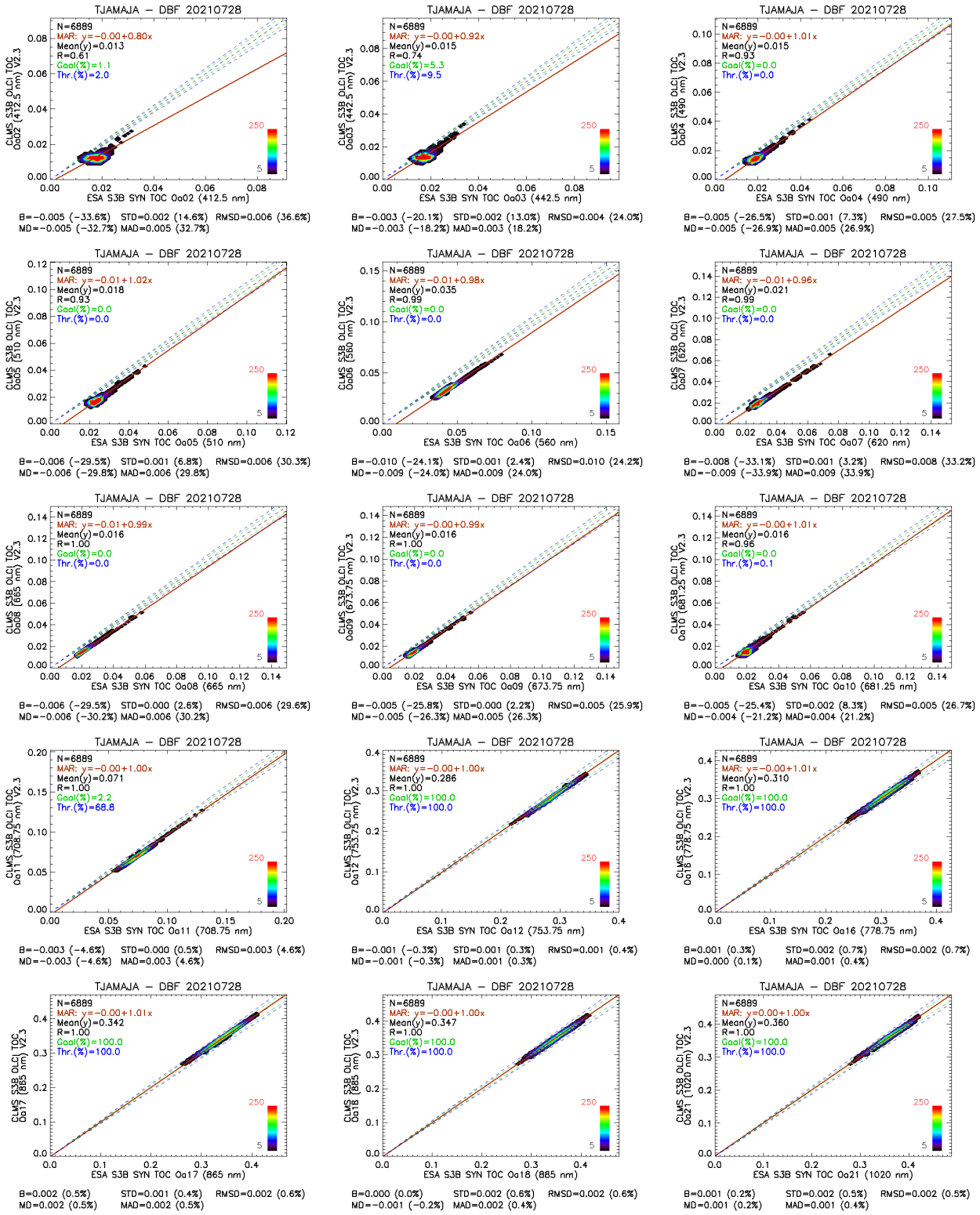


Figure 76: Scatter-plots of Sentinel-3B (2021.07.28) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#2 (Tjamaja, Russia – DBF).

Table 29: Validation statistics of the comparison of Sentinel-3B (2021.07.28) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#2 (Tjamaja, Russia – DBF).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6889	0.61	-0.005 (-33.6%)	-0.005 (-32.7%)	0.002 (14.6%)	0.005 (32.7%)	0.006 (36.6%)	0.00 /0.80	1.1 /2.0
Oa03	6889	0.74	-0.003 (-20.1%)	-0.003 (-18.2%)	0.002 (13.0%)	0.003 (18.2%)	0.004 (24.0%)	0.01 /0.92	5.3 /9.5
Oa04	6889	0.93	-0.005 (-26.5%)	-0.005 (-26.9%)	0.001 (7.3%)	0.005 (26.9%)	0.005 (27.5%)	0.00 /1.01	0.0 /0.0
Oa05	6889	0.93	-0.006 (-29.5%)	-0.006 (-29.8%)	0.001 (6.8%)	0.006 (29.8%)	0.006 (30.3%)	-0.01 /1.02	0.0 /0.0
Oa06	6889	0.99	-0.010 (-24.1%)	-0.009 (-24.0%)	0.001 (2.4%)	0.009 (24.0%)	0.010 (24.2%)	-0.01 /0.98	0.0 /0.0
Oa07	6889	0.99	-0.008 (-33.1%)	-0.009 (-33.9%)	0.001 (3.2%)	0.009 (533.9%)	0.008 (33.2%)	-0.01 /0.96	0.0 /0.0
Oa08	6889	1.00	-0.006 (-29.5%)	-0.006 (-30.2%)	<0.001 (2.6%)	0.006 (30.2%)	0.006 (29.6%)	-0.01 /0.99	0.0 /0.0
Oa09	6889	1.00	-0.005 (-25.8%)	-0.005 (26.3%)	<0.001 (2.2%)	0.005 (26.3%)	0.005 (25.9%)	0.00 /0.99	0.0 /0.0
Oa10	6889	0.96	-0.005 (-25.4%)	-0.004 (-21.2%)	0.002 (8.3%)	0.004 (21.2%)	0.005 (26.7%)	0.00 /1.01	0.0 /0.1
Oa11	6889	1.00	-0.003 (-8.1%)	-0.003 (-4.6%)	<0.001 (0.5%)	0.003 (4.6%)	0.003 (4.6%)	0.00 /1.00	2.2 /68.8
Oa12	6889	1.00	-0.001 (-0.3%)	-0.001 (-0.3%)	0.001 (0.3%)	0.001 (0.3%)	0.001 (0.4%)	0.00 /1.00	100.0 /100.0
Oa16	6889	1.00	-0.001 (-0.3%)	>-0.001 (-0.1%)	0.002 (0.7%)	0.001 (0.4%)	0.002 (0.7%)	0.00 /1.01	100.0 /100.0
Oa17	6889	1.00	0.002 (0.5%)	0.002 (0.5%)	0.001 (0.4%)	0.002 (0.5%)	0.002 (0.6%)	0.00 /1.01	100.0 /100.0
Oa18	6889	1.00	0.000 (0.0%)	-0.001 (-0.2%)	0.002 (0.6%)	0.002 (0.4%)	0.002 (0.6%)	0.00 /1.00	100.0 /100.0
Oa21	6889	1.00	0.001 (0.2%)	0.001 (0.2%)	0.002 (0.5%)	0.001 (0.4%)	0.002 (0.5%)	0.00 /1.00	100.0 /100.0

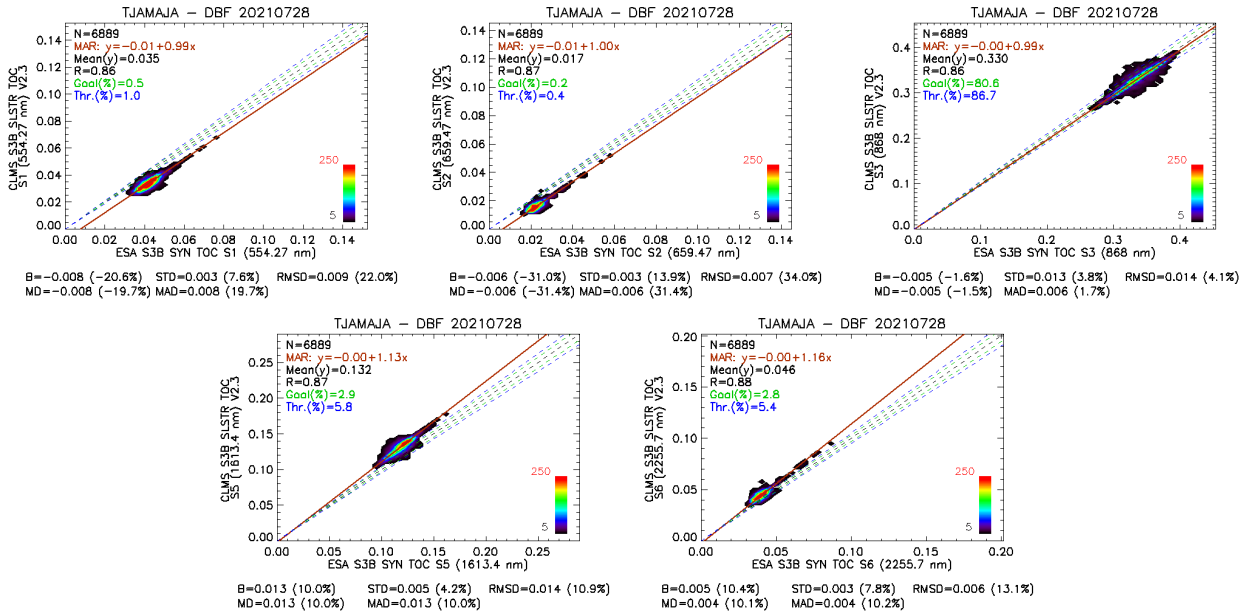


Figure 77: Scatter-plots of Sentinel-3B (2021.07.28) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#2 (Tjamaja, Russia – DBF).

Table 30: Validation statistics of the comparison of Sentinel-3B (2021.07.28) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#2 (Tjamaja, Russia – DBF).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6889	0.86	-0.008 (-20.6%)	-0.008 (-19.7%)	0.003 (7.6%)	0.008 (19.7%)	0.009 (22.0%)	-0.01 /0.99	0.5 /1.0
S2	6889	0.87	-0.006 (-31.0%)	-0.006 (-31.4%)	0.003 (13.9%)	0.006 (31.4%)	0.007 (34.0%)	-0.01 /1.00	0.2 /0.4
S3	6889	0.86	-0.005 (-1.6%)	-0.005 (-1.5%)	0.013 (3.8%)	0.006 (1.7%)	0.014 (4.1%)	0.00 /0.99	80.6 /86.7
S5	6889	0.87	0.013 (10.0%)	0.013 (10.0%)	0.005 (4.2%)	0.013 (10.0%)	0.014 (10.9%)	0.00 /1.13	2.9 /5.8
S6	6889	0.88	0.005 (10.4%)	0.004 (10.1%)	0.003 (7.8%)	0.004 (10.2%)	0.006 (13.1%)	0.00 /1.16	2.8 /5.4

AOI#3 BARRAX (SPAIN) – CULTIVATED

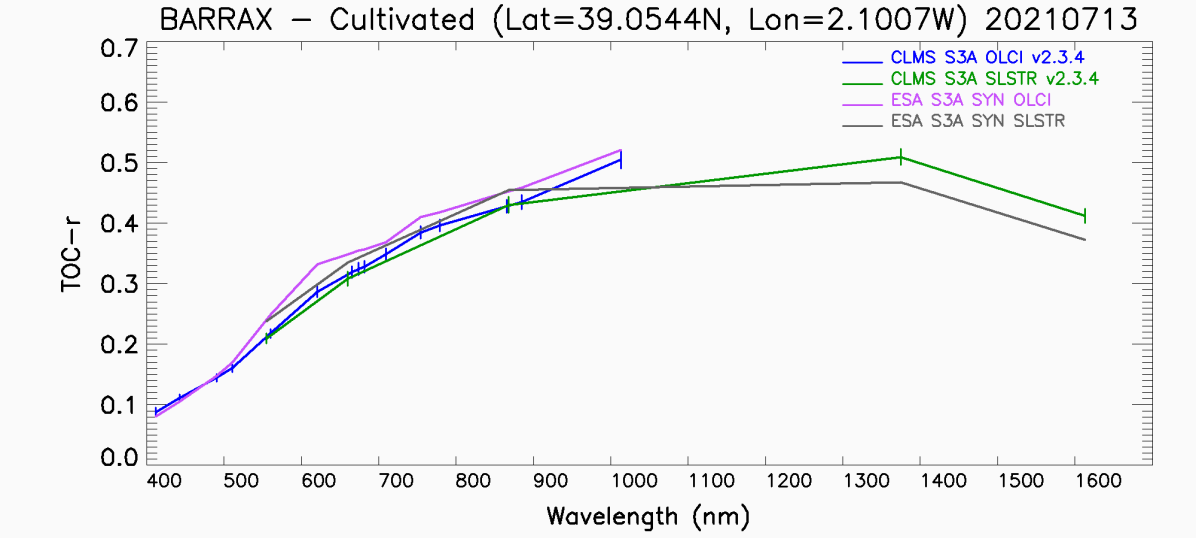


Figure 78: Spectral profile of Sentinel-3A (2021.07.13) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#3 (Barrax, Spain – Cultivated).

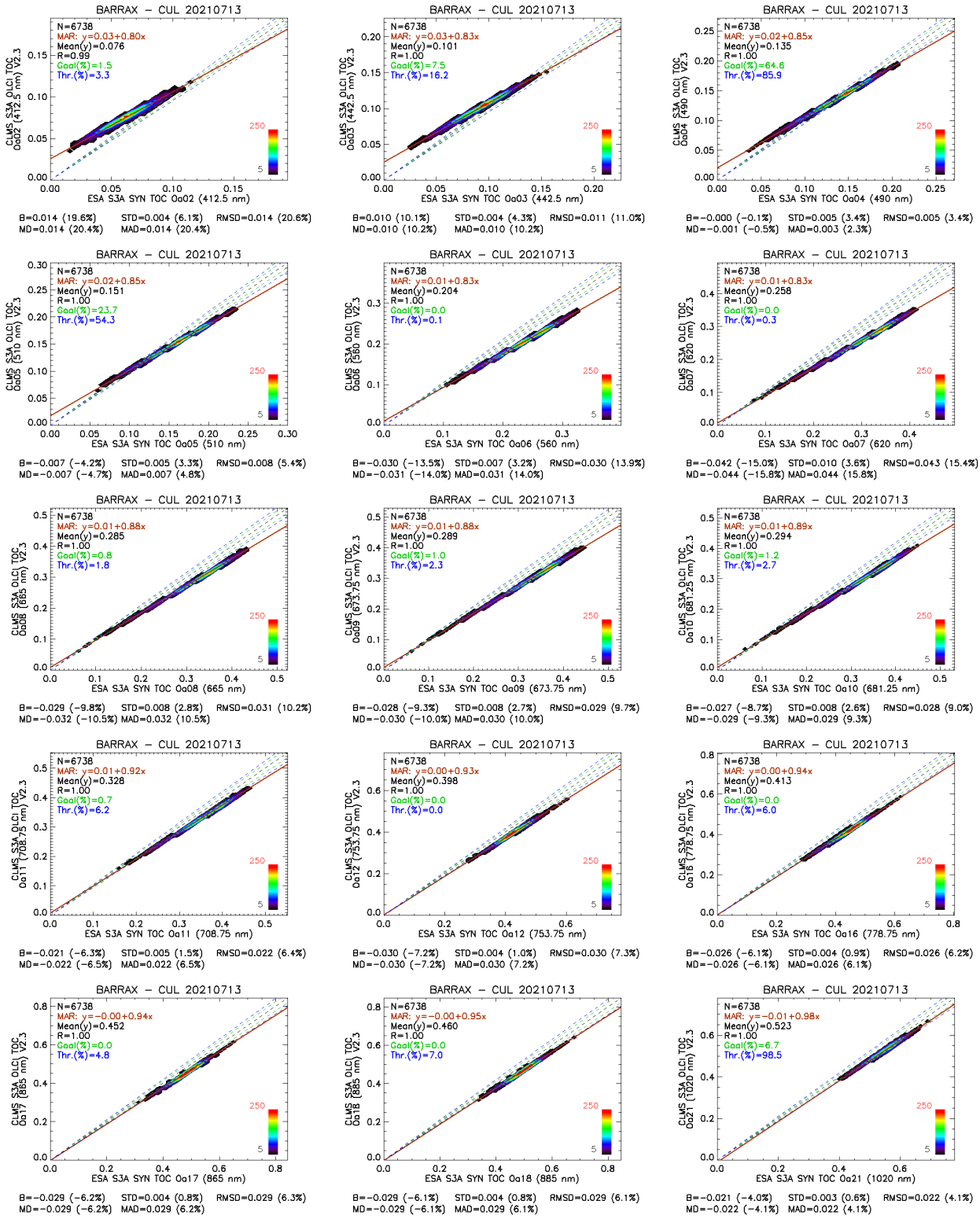


Figure 79: Scatter-plots of Sentinel-3A (2021.07.13) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#3 (Barrax, Spain – Cultivated).

Table 31: Validation statistics of the comparison of Sentinel-3A (2021.07.13) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#3 (Barrax, Spain – Cultivated).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6738	0.99	0.014 (19.6%)	0.014 (20.4%)	0.004 (6.1%)	0.014 (20.4%)	0.014 (20.6%)	0.03 /0.80	1.5 /3.3
Oa03	6738	1.00	0.010 (10.1%)	0.010 (10.2%)	0.004 (4.3%)	0.010 (10.2%)	0.011 (11.0%)	0.03 /0.83	7.5 /16.2
Oa04	6738	1.00	>-0.001 (-0.1%)	-0.001 (-0.5%)	0.005 (3.4%)	0.003 (2.3%)	0.005 (3.4%)	0.02 /0.85	64.6 /85.9
Oa05	6738	1.00	-0.004 (-4.2%)	-0.007 (-4.7%)	0.005 (3.3%)	0.007 (4.8%)	0.008 (5.4%)	0.02 /0.85	23.7 /54.3
Oa06	6738	1.00	-0.030 (-13.5%)	-0.031 (-14.0%)	0.007 (3.2%)	0.031 (14.0%)	0.030 (13.9%)	0.01 /0.83	0.0 /0.1
Oa07	6738	1.00	-0.042 (-15.0%)	-0.044 (-15.8%)	0.010 (3.6%)	0.044 (15.8%)	0.043 (15.4%)	0.01 /0.83	0.0 /0.3
Oa08	6738	1.00	-0.029 (-9.8%)	-0.032 (-10.5%)	0.008 (2.8%)	0.032 (10.5%)	0.031 (10.2%)	0.01 /0.88	0.8 /1.8
Oa09	6738	1.00	-0.028 (-9.3%)	-0.030 (-10.0%)	0.008 (2.7%)	0.030 (10.0%)	0.029 (9.7%)	0.01 /0.88	1.0 /2.3
Oa10	6738	1.00	-0.027 (-8.7%)	-0.029 (-9.3%)	0.008 (2.6%)	0.029 (9.3%)	0.028 (9.0%)	0.01 /0.89	1.2 /2.7
Oa11	6738	1.00	-0.021 (-6.3%)	-0.022 (-6.5%)	0.005 (1.5%)	0.022 (6.5%)	0.022 (6.4%)	0.01 /0.92	0.7 /6.2
Oa12	6738	1.00	-0.030 (-7.2%)	-0.030 (-7.2%)	0.004 (1.0%)	0.030 (7.2%)	0.030 (7.3%)	0.00 /0.93	0.0 /0.0
Oa16	6738	1.00	-0.026 (-6.1%)	-0.026 (-6.1%)	0.004 (0.9%)	0.026 (6.1%)	0.026 (6.2%)	0.00 /0.94	0.0 /6.0
Oa17	6738	1.00	-0.029 (-6.2%)	-0.029 (-6.2%)	0.004 (0.8%)	0.029 (6.2%)	0.029 (6.3%)	0.00 /0.94	0.0 /4.8
Oa18	6738	1.00	-0.029 (-6.1%)	-0.029 (-6.1%)	0.004 (0.8%)	0.029 (6.1%)	0.029 (6.1%)	0.00 /0.95	0.0 /7.0
Oa21	6738	1.00	-0.021 (-4.0%)	-0.022 (-4.1%)	0.003 (0.6%)	0.022 (4.1%)	0.022 (4.1%)	-0.01 /0.98	6.7 /98.5

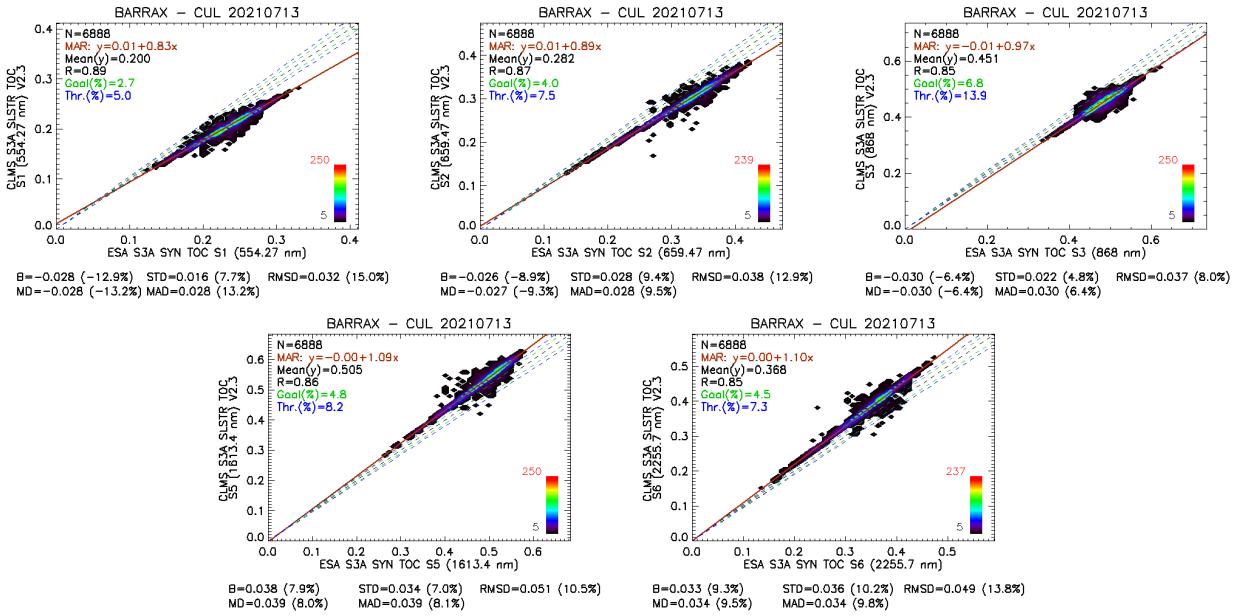


Figure 80: Scatter-plots of Sentinel-3A (2021.07.13) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#3 (Barrax, Spain – Cultivated).

Table 32: Validation statistics of the comparison of Sentinel-3A (2021.07.13) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#3 (Barrax, Spain – Cultivated).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6888	0.89	-0.028 (-12.9%)	-0.028 (-13.2%)	0.016 (7.7%)	0.028 (13.2%)	0.032 (15.0%)	0.01 /0.83	2.7 /5.0
S2	6888	0.87	-0.026 (-8.9%)	-0.027 (-9.3%)	0.028 (9.4%)	0.028 (9.5%)	0.038 (12.9%)	0.01 /0.89	4.0 /7.5
S3	6888	0.85	-0.030 (-6.4%)	-0.030 (-6.4%)	0.022 (4.8%)	0.030 (6.4%)	0.037 (8.0%)	-0.01 /0.97	6.8 /13.9
S5	6888	0.86	0.038 (7.9%)	0.039 (8.0%)	0.034 (7.0%)	0.039 (8.1%)	0.051 (10.5%)	0.00 /1.09	4.8 /8.2
S6	6888	0.85	0.033 (9.3%)	0.034 (9.5%)	0.036 (10.2%)	0.034 (9.8%)	0.049 (13.8%)	0.00 /1.10	4.5 /7.3

AOI#4 AZUEL (SPAIN) – HERBACEOUS

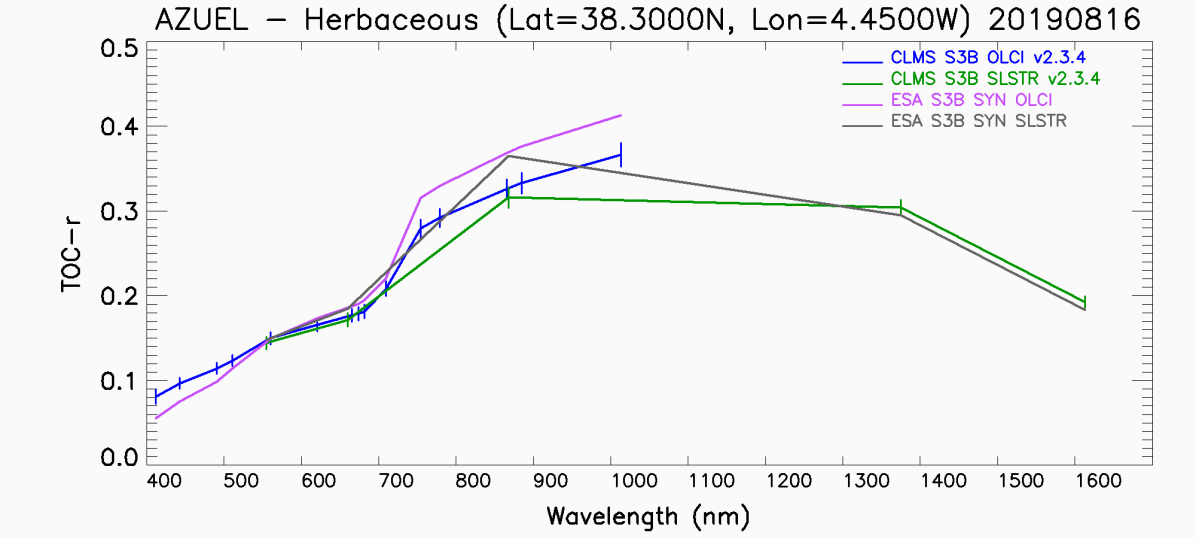


Figure 81: Spectral profile of Sentinel-3B (2019.08.16) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#4 (Azuel, Spain – Herbaceous).

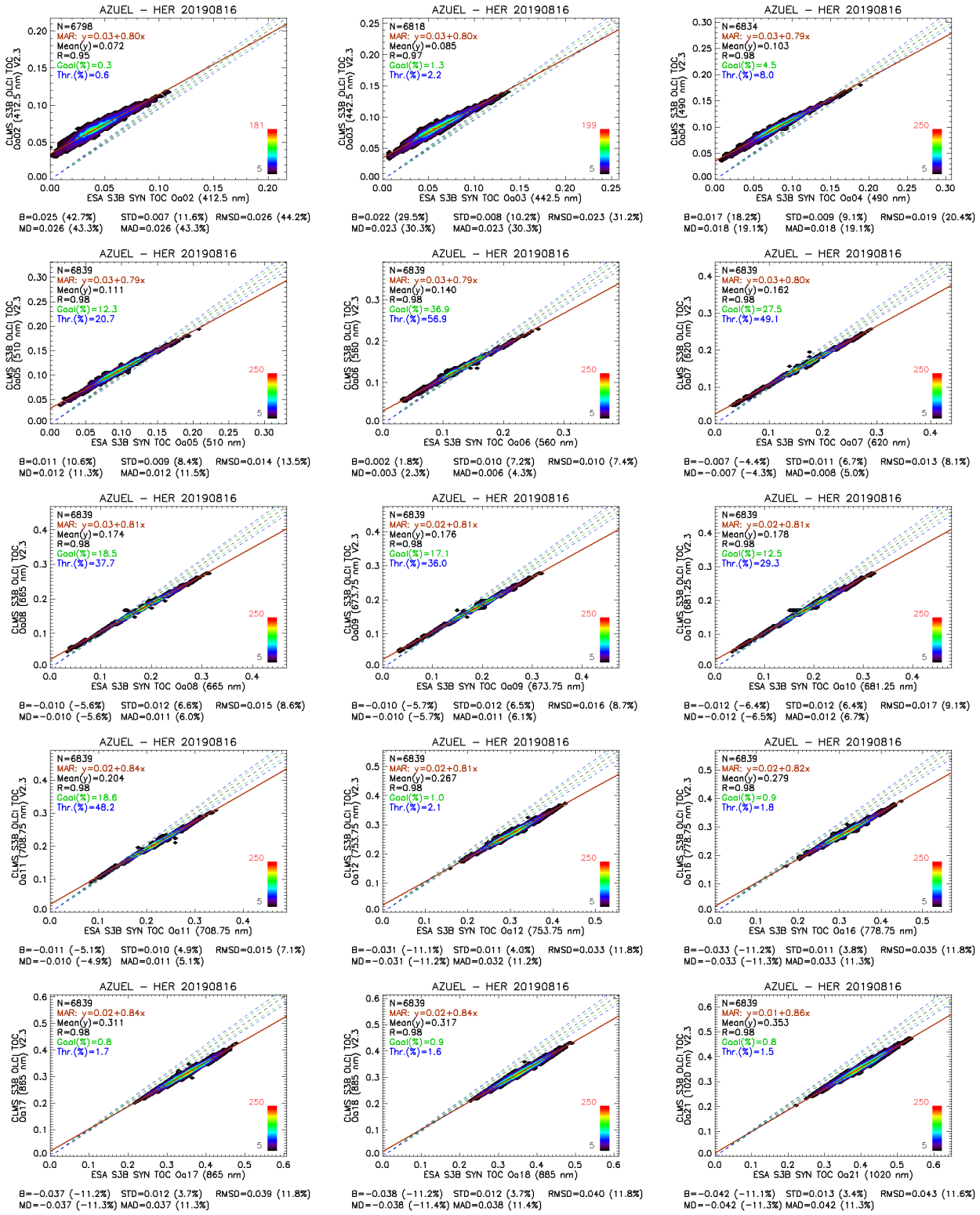


Figure 82: Scatter-plots of Sentinel-3B (2019.08.16) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#4 (Azuel, Spain – Herbaceous).

Table 33: Validation statistics of the comparison of Sentinel-3B (2019.08.16) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#4 (Azuel, Spain – Herbaceous).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6798	0.95	0.025 (42.7%)	0.026 (43.3%)	0.007 (11.6%)	0.026 (43.3%)	0.026 (44.2%)	0.03 /0.80	0.3 /0.6
Oa03	6818	0.97	0.022 (29.5%)	0.023 (30.3%)	0.008 (10.2%)	0.023 (30.3%)	0.023 (31.2%)	0.03 /0.80	1.3 /2.2
Oa04	6834	0.98	0.017 (18.2%)	0.018 (19.1%)	0.009 (9.1%)	0.018 (19.1%)	0.019 (20.4%)	0.03 /0.79	4.5 /8.0
Oa05	6839	0.98	0.011 (10.6%)	0.012 (11.3%)	0.009 (8.4%)	0.012 (11.5%)	0.014 (13.5%)	0.03 /0.79	12.3 /20.7
Oa06	6839	0.98	0.002 (1.8%)	0.003 (2.3%)	0.010 (7.2%)	0.006 (4.3%)	0.010 (7.4%)	0.03 /0.79	36.9 /56.9
Oa07	6839	0.98	-0.007 (-4.4%)	-0.007 (-4.3%)	0.011 (6.7%)	0.008 (5.0%)	0.013 (8.1%)	0.03 /0.80	27.5 /49.1
Oa08	6739	0.98	-0.010 (-5.6%)	-0.010 (-5.6%)	0.012 (6.6%)	0.011 (6.0%)	0.015 (8.6%)	0.03 /0.81	18.5 /37.7
Oa09	6739	0.98	-0.010 (-5.7%)	-0.010 (-5.7%)	0.012 (6.5%)	0.011 (6.1%)	0.016 (8.7%)	0.02 /0.81	17.1 /36.0
Oa10	6739	0.98	-0.012 (-6.4%)	-0.012 (-6.5%)	0.012 (6.4%)	0.012 (6.7%)	0.017 (9.1%)	0.02 /0.81	12.5 /29.3
Oa11	6739	0.98	-0.011 (-5.1%)	-0.010 (-4.9%)	0.010 (4.9%)	0.011 (5.1%)	0.015 (7.1%)	0.02 /0.84	18.6 /48.2
Oa12	6739	0.98	-0.031 (-11.1%)	-0.031 (-11.2%)	0.011 (4.0%)	0.032 (11.2%)	0.033 (11.8%)	0.02 /0.81	1.0 /2.1
Oa16	6739	0.98	-0.033 (-11.2%)	-0.033 (-11.3%)	0.011 (3.8%)	0.033 (11.3%)	0.035 (11.8%)	0.02 /0.82	0.9 /1.8
Oa17	6739	0.98	-0.037 (-11.2%)	-0.037 (-11.3%)	0.012 (3.7%)	0.037 (11.3%)	0.039 (11.8%)	0.02 /0.84	0.8 /1.7
Oa18	6739	0.98	-0.038 (-11.2%)	-0.038 (-11.4%)	0.012 (3.7%)	0.038 (11.4%)	0.040 (11.8%)	0.02 /0.84	0.9 /1.6
Oa21	6739	0.98	-0.042 (-11.1%)	-0.042 (-11.3%)	0.013 (3.4%)	0.042 (11.3%)	0.043 (11.6%)	0.01 /0.86	0.8 /1.5

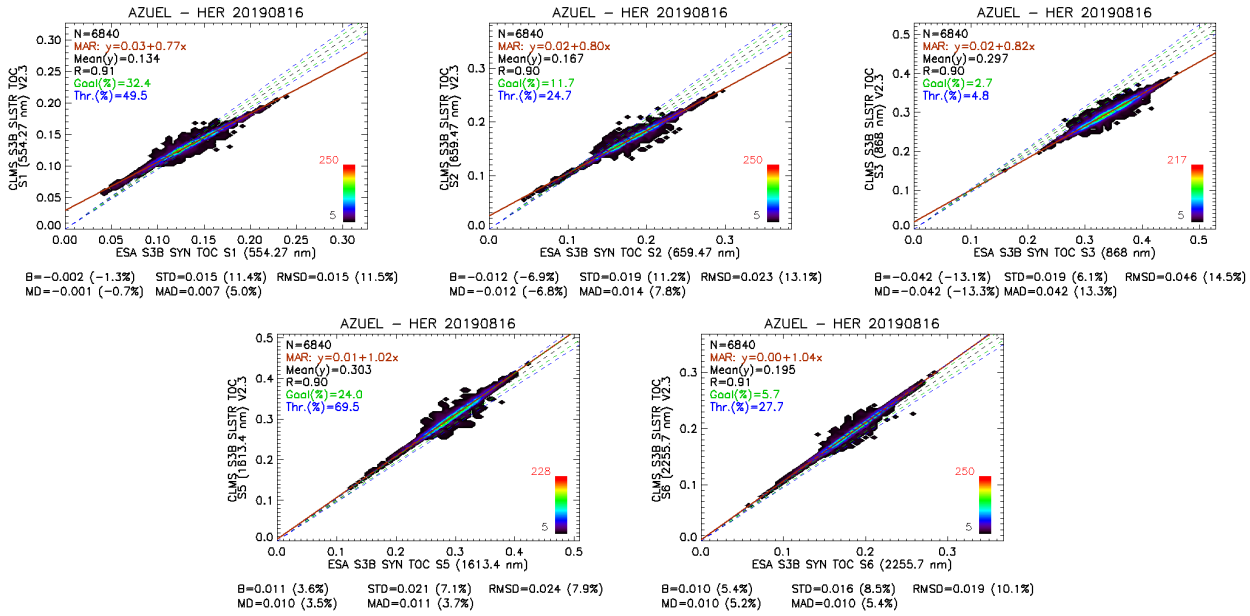


Figure 83: Scatter-plots of Sentinel-3B (2019.08.16) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#4 (Azuel, Spain – Herbaceous).

Table 34: Validation statistics of the comparison of Sentinel-3B (2019.08.16) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#4 (Azuel, Spain – Herbaceous).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6840	0.91	-0.002 (-1.3%)	-0.001 (-0.7%)	0.015 (11.4%)	0.007 (5.0%)	0.015 (11.5%)	0.03 /0.77	32.4 /49.5
S2	6840	0.90	-0.012 (-6.9%)	-0.012 (-6.8%)	0.019 (11.2%)	0.014 (7.8%)	0.023 (13.1%)	0.02 /0.80	11.7 /24.7
S3	6840	0.90	-0.042 (-13.1%)	-0.042 (-13.3%)	0.019 (6.1%)	0.042 (13.3%)	0.046 (14.5%)	0.02 /0.82	2.7 /4.8
S5	6840	0.90	0.011 (3.6%)	0.010 (3.5%)	0.021 (7.1%)	0.011 (3.7%)	0.024 (7.9%)	0.01 /1.02	24.0 /69.5
S6	6840	0.85	0.010 (5.4%)	0.010 (5.2%)	0.016 (8.5%)	0.010 (5.4%)	0.019 (10.1%)	0.00 /1.04	5.7 /27.7

AOI#5 OULED ZOUYAD (TUNISIA) – SHRUBLANDS

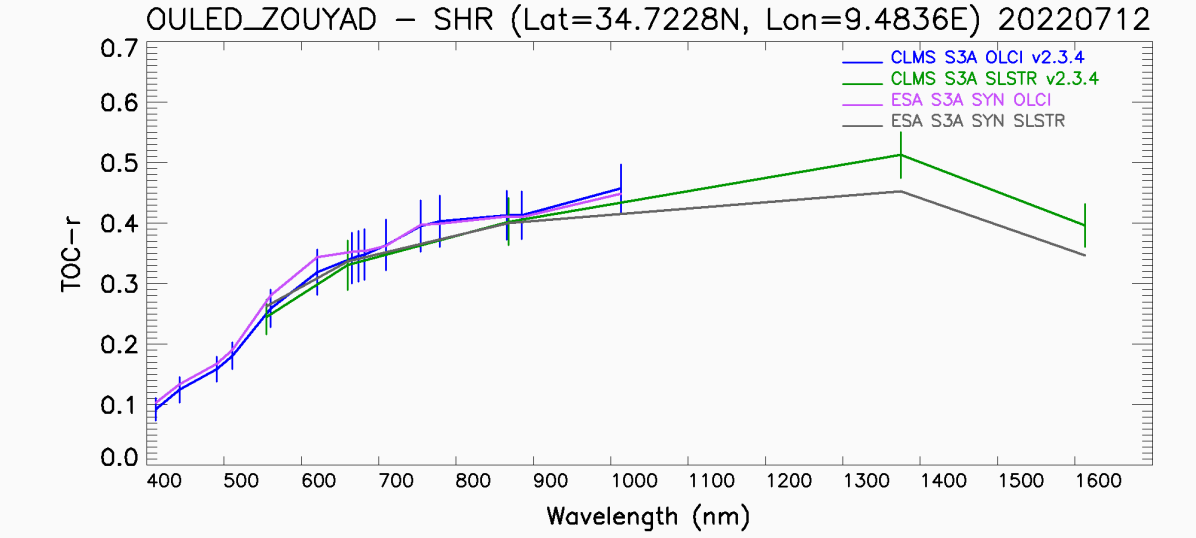


Figure 84: Spectral profile of Sentinel-3A (2022.07.12) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

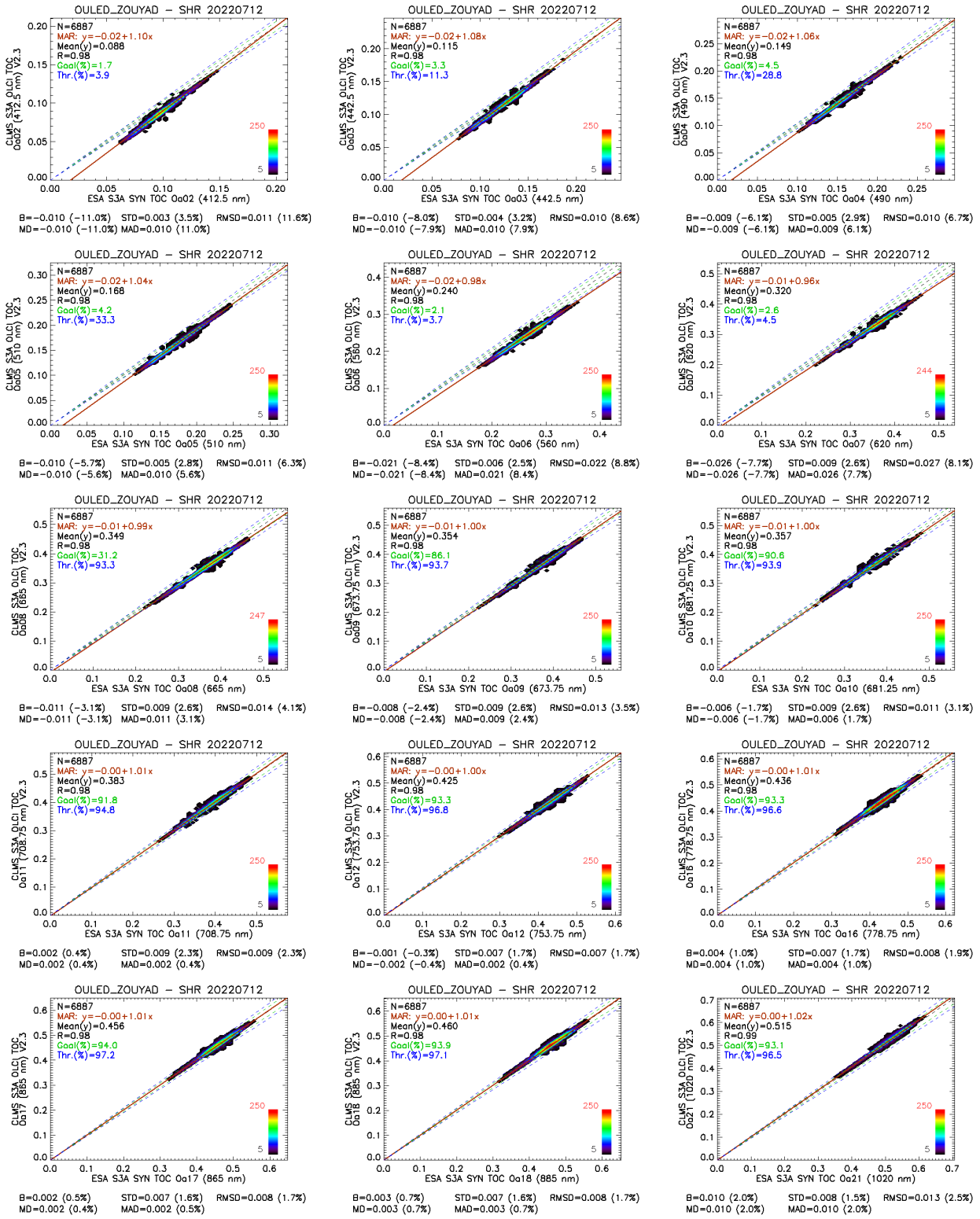


Figure 85: Scatter-plots of Sentinel-3A (2022.07.12) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

Table 35: Validation statistics of the comparison of Sentinel-3A (2022.07.12) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6887	0.98	-0.010 (-11.0%)	-0.010 (-11.0%)	0.003 (3.5%)	0.010 (11.0%)	0.011 (11.6%)	-0.02 /1.10	1.7 /3.9
Oa03	6887	0.98	-0.010 (-8.0%)	-0.010 (-7.9%)	0.004 (3.2%)	0.010 (7.9%)	0.010 (8.6%)	-0.02 /1.08	3.3 /11.3
Oa04	6887	0.98	-0.009 (-6.1%)	-0.009 (-6.1%)	0.005 (2.9%)	0.009 (6.1%)	0.010 (6.7%)	-0.02 /1.06	4.5 /28.8
Oa05	6887	0.98	-0.010 (-5.7%)	0.010 (-5.6%)	0.005 (2.8%)	0.010 (5.6%)	0.011 (6.3%)	-0.02 /1.04	4.2 /33.3
Oa06	6887	0.98	-0.021 (-8.4%)	-0.021 (-8.4%)	0.006 (2.5%)	0.021 (8.4%)	0.022 (8.8%)	-0.02 /0.98	2.1 /3.7
Oa07	6887	0.98	-0.026 (-7.7%)	-0.026 (-7.7%)	0.009 (2.6%)	0.026 (7.7%)	0.027 (8.1%)	-0.01 /0.96	2.6 /4.5
Oa08	6887	0.98	-0.011 (-3.1%)	-0.011 (-3.1%)	0.009 (2.6%)	0.011 (3.1%)	0.014 (4.1%)	-0.01 /0.99	31.2 /93.3
Oa09	6887	0.98	-0.008 (-2.4%)	-0.008 (-2.4%)	0.009 (2.6%)	0.009 (2.4%)	0.013 (3.5%)	-0.01 /1.00	86.1 /93.7
Oa10	6887	0.98	-0.006 (-1.7%)	-0.006 (-1.7%)	0.009 (2.6%)	0.006 (1.7%)	0.011 (3.1%)	-0.01 /1.00	90.6 /93.9
Oa11	6887	0.98	0.002 (0.4%)	0.002 (0.4%)	0.009 (2.3%)	0.002 (0.4%)	0.009 (2.3%)	0.00 /1.01	91.8 /94.8
Oa12	6887	0.98	-0.001 (-0.3%)	-0.002 (-0.4%)	0.007 (1.7%)	0.002 (0.4%)	0.007 (1.7%)	0.00 /1.00	93.3 /96.8
Oa16	6887	0.98	0.004 (1.0%)	0.004 (1.0%)	0.007 (1.7%)	0.004 (1.0%)	0.008 (1.9%)	0.00 /1.01	93.3 /96.6
Oa17	6887	0.98	0.002 (0.5%)	0.002 (0.4%)	0.007 (1.6%)	0.002 (0.5%)	0.008 (1.7%)	0.00 /1.01	94.0 /97.2
Oa18	6887	0.98	0.003 (0.7%)	0.003 (0.7%)	0.007 (1.6%)	0.003 (0.7%)	0.008 (1.7%)	0.00 /1.01	93.9 /97.1
Oa21	6887	0.99	0.010 (2.0%)	0.010 (2.0%)	0.008 (1.5%)	0.010 (2.0%)	0.013 (2.5%)	0.00 /1.02	93.1 /96.5

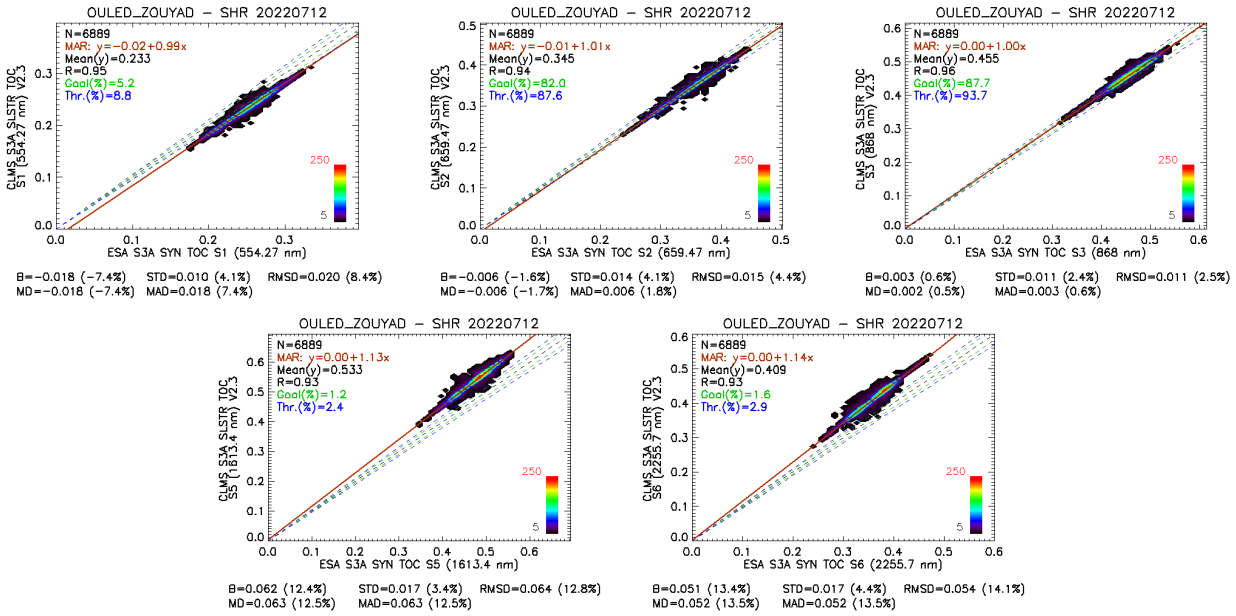


Figure 86: Scatter-plots of Sentinel-3A (2022.07.12) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

Table 36: Validation statistics of the comparison of Sentinel-3A (2022.07.12) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6889	0.95	-0.018 (-7.4%)	-0.018 (-7.4%)	0.010 (4.1%)	0.018 (7.4%)	0.020 (8.4%)	-0.02 /0.99	5.2 /8.8
S2	6889	0.94	-0.006 (-1.6%)	-0.006 (-1.7%)	0.014 (4.1%)	0.006 (1.8%)	0.015 (4.4%)	-0.01 /1.01	82.0 /87.6
S3	6889	0.96	0.003 (0.6%)	0.002 (0.5%)	0.011 (2.4%)	0.003 (0.6%)	0.011 (2.5%)	0.00 /1.00	87.7 /93.7
S5	6889	0.93	0.062 (12.4%)	0.063 (12.5%)	0.017 (3.4%)	0.063 (12.5%)	0.064 (12.8%)	0.00 /1.13	1.2 /2.4
S6	6889	0.93	0.051 (13.4%)	0.052 (13.5%)	0.017 (4.4%)	0.052 (13.5%)	0.054 (14.1%)	0.00 /1.14	1.6 /2.9

AOI#6 CALIBRATION SITE (EGYPT) – BARE AREAS

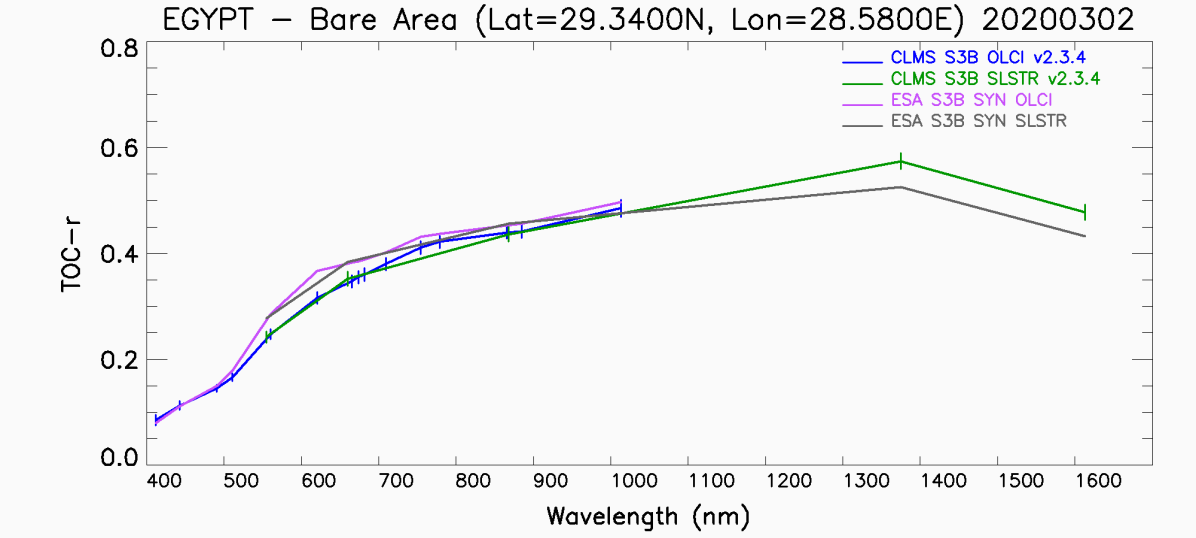


Figure 87: Spectral profile of Sentinel-3B (2023.03.02) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#6 (Calibration Site, Egypt – Bare Areas).

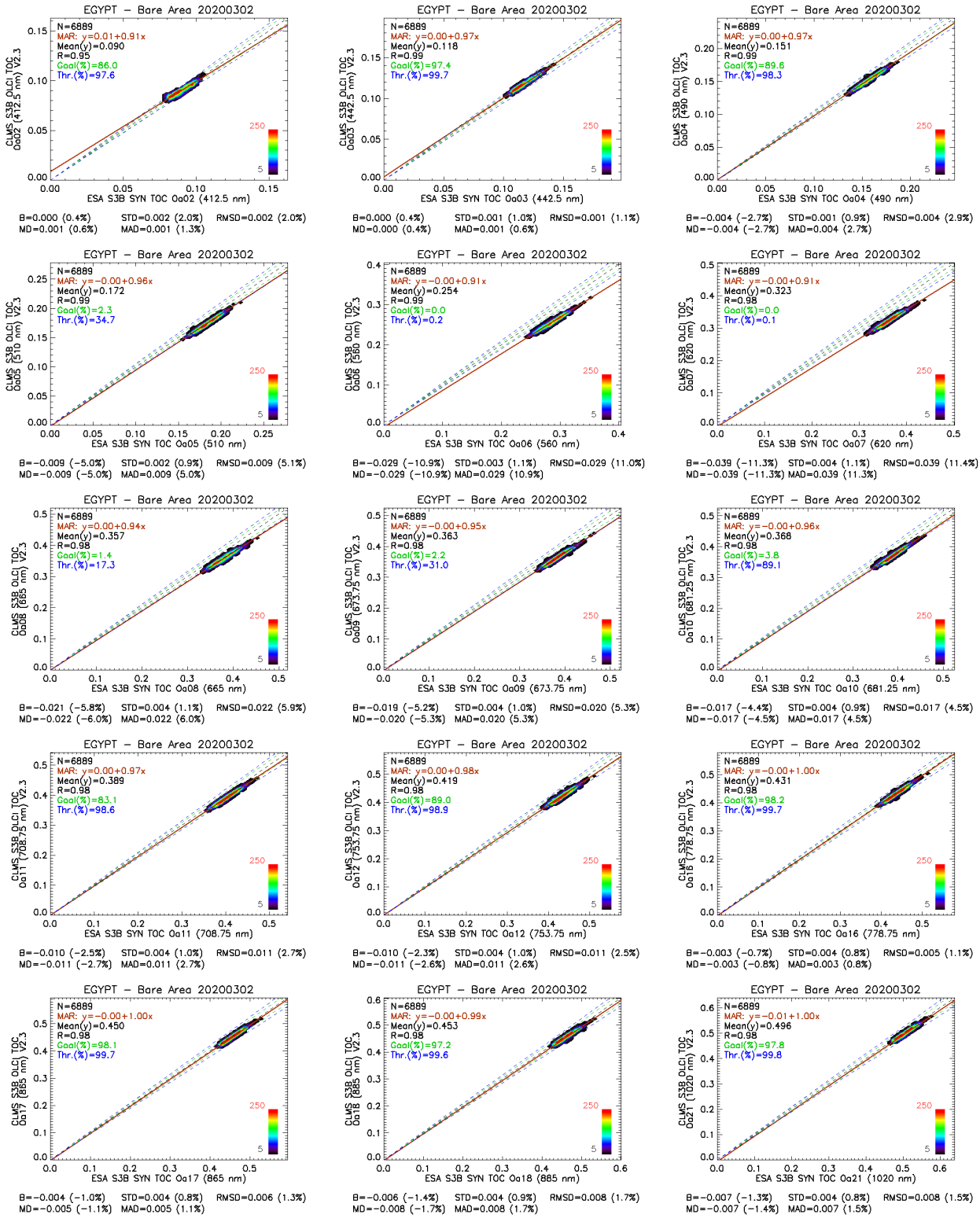


Figure 88: Scatter-plots of Sentinel-3B (2023.03.02) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#6 (Calibration Site, Egypt – Bare Areas).

Table 37: Validation statistics of the comparison of Sentinel-3B (2023.03.02) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#6 (Calibration Site, Egypt – Bare Areas).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6889	0.95	<0.001 (0.4%)	0.001 (0.6%)	0.002 (2.0%)	0.001 (1.3%)	0.002 (2.0%)	0.01 /0.91	86.0 /97.6
Oa03	6889	0.99	<0.001 (0.4%)	<0.001 (0.4%)	0.001 (1.0%)	0.001 (0.6%)	0.001 (1.1%)	0.00 /0.97	97.4 /99.7
Oa04	6889	0.99	-0.004 (-2.7%)	-0.004 (-2.7%)	0.001 (0.9%)	0.004 (2.7%)	0.004 (2.9%)	0.00 /0.97	89.6 /98.3
Oa05	6889	0.99	-0.009 (-5.0%)	-0.009 (-5.0%)	0.002 (0.9%)	0.009 (5.0%)	0.009 (5.1%)	0.00 /0.96	2.3 /34.7
Oa06	6889	0.99	-0.029 (-10.9%)	-0.029 (-10.9%)	0.003 (1.1%)	0.029 (10.9%)	0.029 (11.0%)	0.00 /0.91	0.0 /0.2
Oa07	6889	0.98	-0.039 (-11.3%)	-0.039 (-11.3%)	0.004 (1.1%)	0.039 (11.3%)	0.039 (11.4%)	0.00 /0.91	0.0 /0.1
Oa08	6889	0.98	-0.021 (-5.8%)	-0.022 (-6.0%)	0.004 (1.1%)	0.022 (5.9%)	0.022 (5.9%)	0.00 /0.94	1.4 /17.3
Oa09	6889	0.98	-0.019 (-5.2%)	-0.020 (-5.3%)	0.004 (1.0%)	0.020 (5.3%)	0.020 (5.3%)	0.00 /0.95	2.2 /31.0
Oa10	6889	0.98	-0.017 (-4.4%)	-0.017 (-4.5%)	0.004 (0.9%)	0.017 (4.5%)	0.017 (4.5%)	0.00 /0.96	3.8 /89.1
Oa11	6889	0.98	-0.010 (-2.5%)	-0.011 (-2.7%)	0.004 (1.0%)	0.011 (2.7%)	0.011 (2.7%)	0.00 /0.97	83.1 /98.6
Oa12	6889	0.98	-0.010 (-2.3%)	-0.011 (-2.6%)	0.004 (1.0%)	0.011 (2.6%)	0.011 (2.5%)	0.00 /0.98	89.0 /98.9
Oa16	6889	0.98	-0.003 (-0.7%)	-0.003 (-0.8%)	0.004 (0.8%)	0.003 (0.8%)	0.005 (1.1%)	0.00 /1.00	98.2 /99.7
Oa17	6889	0.98	-0.004 (-1.0%)	-0.005 (-1.1%)	0.004 (0.8%)	0.005 (1.1%)	0.006 (1.3%)	0.00 /1.00	98.1 /99.7
Oa18	6889	0.98	0.006 (-1.4%)	-0.008 (-1.7%)	0.004 (0.9%)	0.008 (1.7%)	0.008 (1.7%)	0.00 /0.99	97.2 /99.6
Oa21	6889	0.98	-0.007 (-1.3%)	-0.007 (-1.4%)	0.004 (0.8%)	0.007 (1.5%)	0.008 (1.5%)	-0.01 /1.00	97.8 /99.8

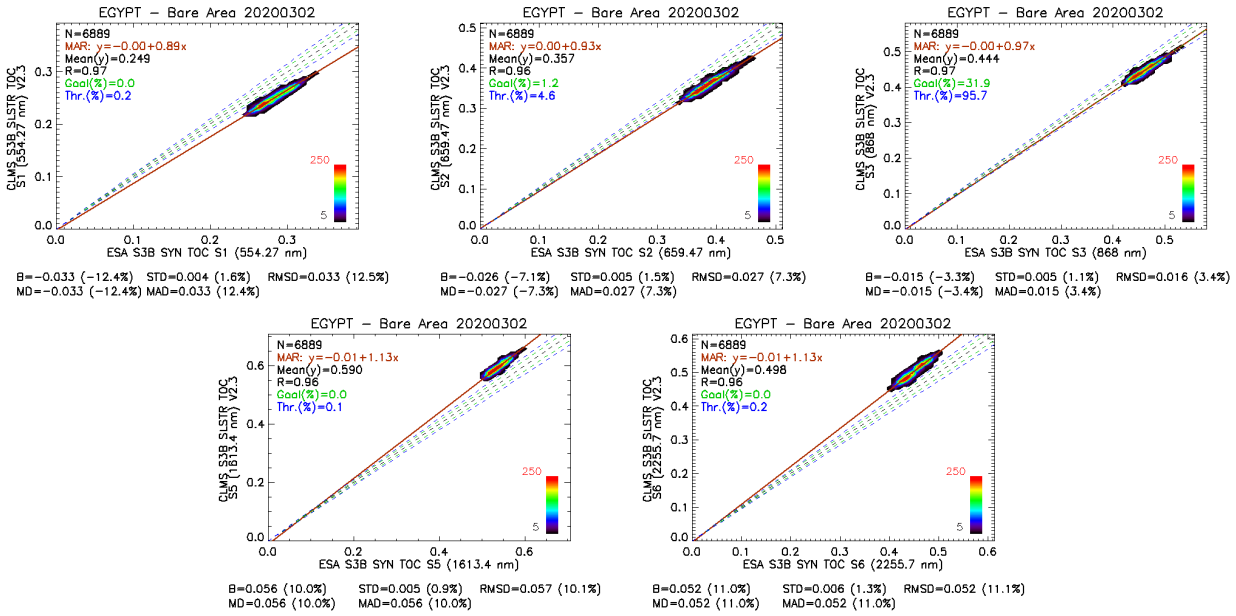


Figure 89: Scatter-plots of Sentinel-3B (2023.03.02) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#6 (Calibration Site, Egypt – Bare Areas).

Table 38: Validation statistics of the comparison of Sentinel-3B (2023.03.02) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#6 (Calibration Site, Egypt – Bare Areas).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6889	0.97	-0.033 (-12.4%)	-0.033 (-12.4%)	0.004 (1.6%)	0.033 (12.4%)	0.033 (12.5%)	0.00 /0.89	0.0 /0.2
S2	6889	0.96	-0.026 (-7.1%)	-0.027 (-7.3%)	0.005 (1.5%)	0.027 (7.3%)	0.027 (7.3%)	0.00 /0.93	1.2 /4.6
S3	6889	0.97	-0.015 (-3.3%)	-0.015 (-3.4%)	0.005 (1.1%)	0.015 (3.4%)	0.015 (3.4%)	0.00 /0.97	31.9 /95.7
S5	6889	0.96	0.056 (10.0%)	0.056 (10.0%)	0.005 (0.9%)	0.056 (10.0%)	0.057 (10.1%)	-0.01 /1.13	0.0 /0.1
S6	6889	0.96	0.052 (11.0%)	0.052 (11.0%)	0.006 (1.3%)	0.052 (11.0%)	0.052 (11.1%)	-0.01 /1.13	0.0 /0.2

AOI#7 OBLAST DE MURMANSH (RUSSIA) – SNOW & ICE

This section shows the results for the AOI#7 (Oblast de Murmansh, Russia) snow and ice area. Note that, for this area some exceptions are considered in the use of annotation flags recommendations:

- In case of OLCI, the IdePix_SNOW_ICE is not used to filter out pixels, as the focus of the analysis over this AOI is to evaluate the agreement over snow. Then, for OLCI all recommendations showed in Table 3 are considered, with the exception of IdePix_SNOW_ICE.
- In case of SLSTR, the use of **cloud_an** annotation flag removes most of the observations around the whole AOI as this flag misidentifies snow as clouds. Then, for this case, same screening approach than OLCI was applied.

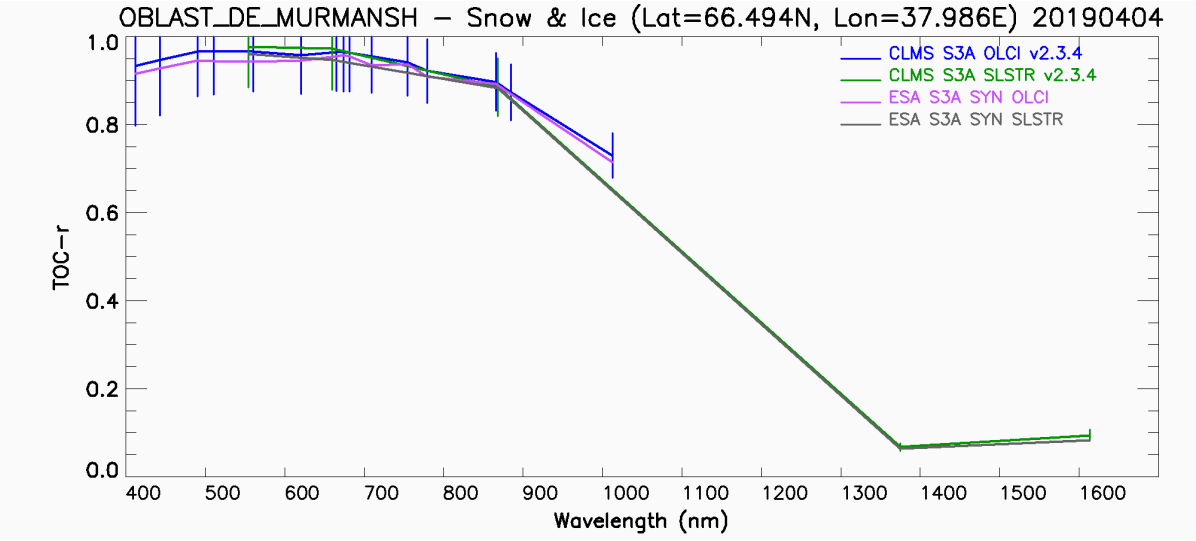


Figure 90: Spectral profile of Sentinel-3A (2019.04.04) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

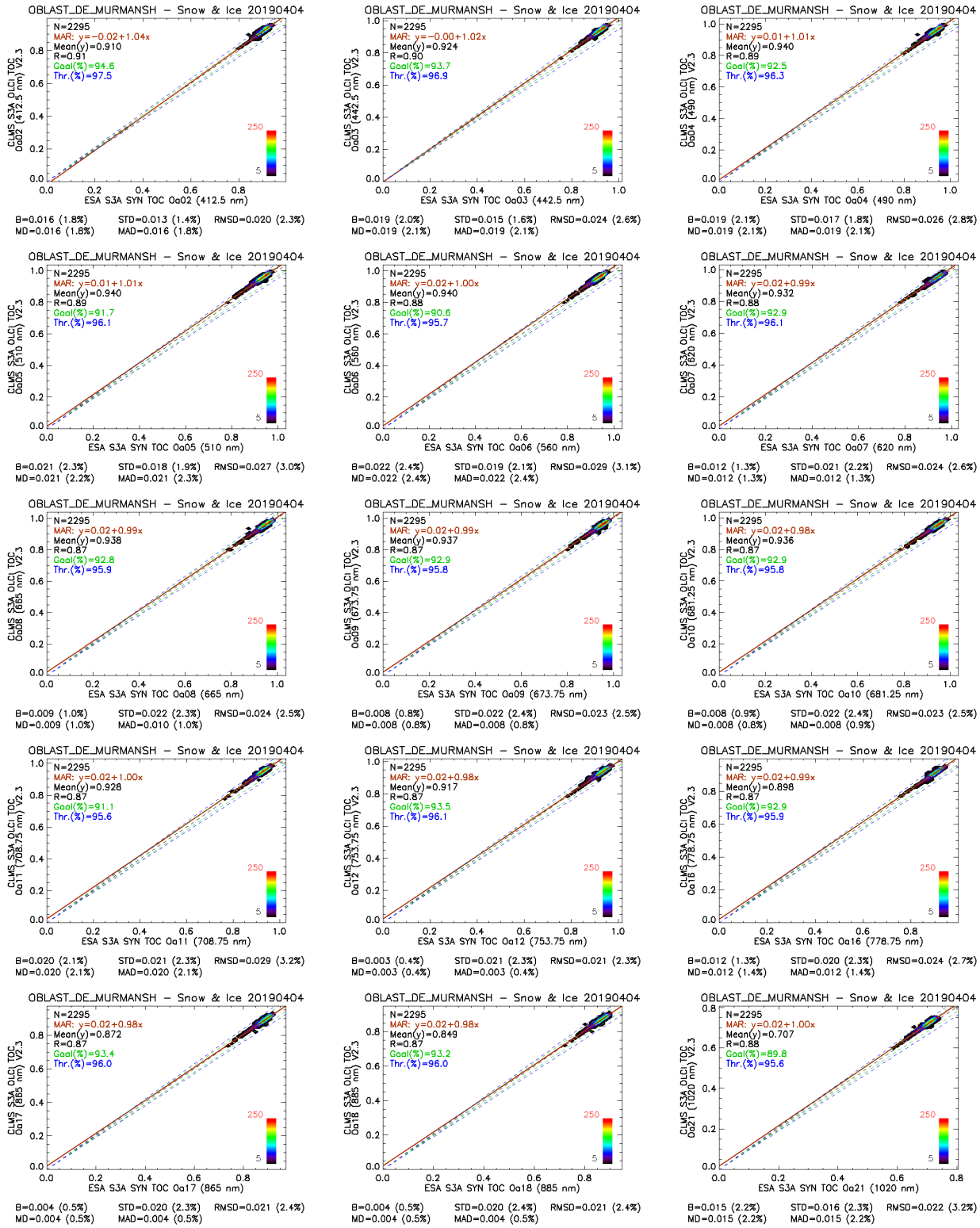


Figure 91: Scatter-plots of Sentinel-3A (2019.04.04) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

Table 39: Validation statistics of the comparison of Sentinel-3A (2019.04.04) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	2295	0.91	0.016 (1.8%)	0.016 (1.8%)	0.013 (1.4%)	0.016 (1.8%)	0.020 (2.3%)	-0.02 /1.04	94.6 /97.5
Oa03	2295	0.90	0.019 (2.0%)	0.019 (2.1%)	0.015 (1.6%)	0.019 (2.1%)	0.024 (2.6%)	0.00 /1.02	93.7 /96.9
Oa04	2295	0.89	0.019 (2.1%)	0.019 (2.1%)	0.017 (1.8%)	0.019 (2.1%)	0.026 (2.8%)	0.01 /1.01	92.5 /96.3
Oa05	2295	0.89	0.021 (2.3%)	0.021 (2.2%)	0.018 (1.9%)	0.021 (2.3%)	0.027 (3.0%)	0.01 /1.01	91.7 /96.1
Oa06	2295	0.88	0.022 (2.4%)	0.022 (2.4%)	0.019 (2.1%)	0.022 (2.4%)	0.029 (3.1%)	0.02 /1.00	90.6 /95.7
Oa07	2295	0.88	0.012 (1.3%)	0.012 (1.3%)	0.021 (2.2%)	0.012 (1.3%)	0.024 (2.6%)	0.02 /0.99	92.9 /96.1
Oa08	2295	0.87	0.009 (1.0%)	0.009 (1.0%)	0.022 (2.3%)	0.010 (1.0%)	0.024 (2.5%)	0.02 /0.99	92.8 /95.9
Oa09	2295	0.87	0.008 (0.8%)	0.008 (0.8%)	0.022 (2.4%)	0.008 (0.8%)	0.023 (2.5%)	0.02 /0.99	92.9 /95.8
Oa10	2295	0.87	0.008 (0.9%)	0.008 (0.8%)	0.022 (2.4%)	0.008 (0.9%)	0.023 (2.5%)	0.02 /0.98	92.9 /95.8
Oa11	2295	0.87	0.020 (2.1%)	0.020 (2.1%)	0.021 (2.3%)	0.020 (2.1%)	0.029 (3.2%)	0.02 /1.00	91.1 /95.6
Oa12	2295	0.87	0.003 (0.4%)	0.003 (0.4%)	0.021 (2.3%)	0.003 (0.4%)	0.021 (2.3%)	0.02 /0.98	93.5 /96.1
Oa16	2295	0.87	0.012 (1.3%)	0.012 (1.4%)	0.020 (2.3%)	0.012 (1.4%)	0.024 (2.7%)	0.02 /0.99	92.9 /95.9
Oa17	2295	0.87	0.004 (0.5%)	0.004 (0.5%)	0.020 (2.3%)	0.004 (0.5%)	0.021 (2.4%)	0.02 /0.98	93.4 /96.0
Oa18	2295	0.87	0.004 (0.5%)	0.004 (0.5%)	0.020 (2.4%)	0.004 (0.5%)	0.021 (2.4%)	0.02 /0.98	93.2 /96.0
Oa21	2295	0.88	0.015 (2.2%)	0.015 (2.2%)	0.016 (2.3%)	0.015 (2.2%)	0.022 (3.2%)	0.02 /1.00	89.8 /95.6

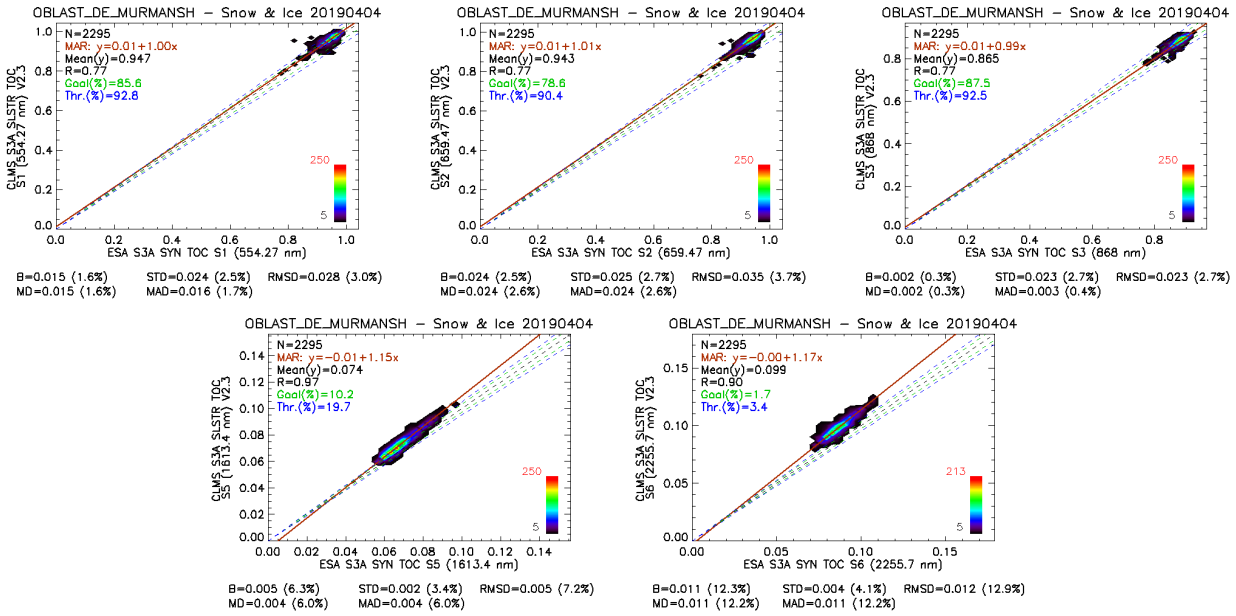


Figure 92: Scatter-plots of Sentinel-3A (2019.04.04) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

Table 40: Validation statistics of the comparison of Sentinel-3A (2019.04.04) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	2295	0.77	0.015 (1.6%)	0.015 (1.6%)	0.024 (2.5%)	0.016 (1.7%)	0.028 (3.0%)	0.01 /1.00	85.6 /92.8
S2	2295	0.77	0.024 (2.5%)	0.024 (2.6%)	0.025 (2.7%)	0.024 (2.6%)	0.035 (3.7%)	0.01 /1.01	78.6 /90.4
S3	2295	0.77	0.002 (0.3%)	0.002 (0.3%)	0.023 (2.7%)	0.003 (0.4%)	0.023 (2.7%)	0.01 /0.99	87.5 /92.5
S5	2295	0.97	0.005 (6.3%)	0.004 (6.0%)	0.002 (3.4%)	0.004 (6.0%)	0.005 (7.2%)	-0.01 /1.15	10.2 /19.7
S6	2295	0.90	0.011 (12.3%)	0.011 (12.2%)	0.004 (4.1%)	0.011 (12.2%)	0.012 (12.9%)	0.00 /1.17	1.7 /3.4

AOI#8 TSHANE (BOTSWANA) – SAVANNAH

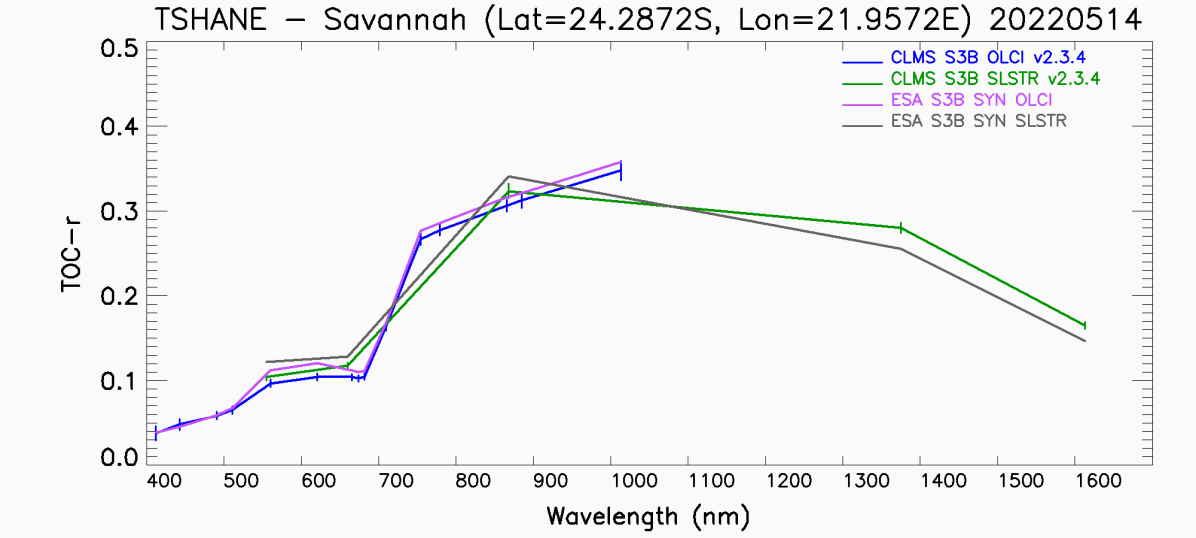


Figure 93: Spectral profile of Sentinel-3B (2022.05.14) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#8 (Tshane, Botswana – Savannah).

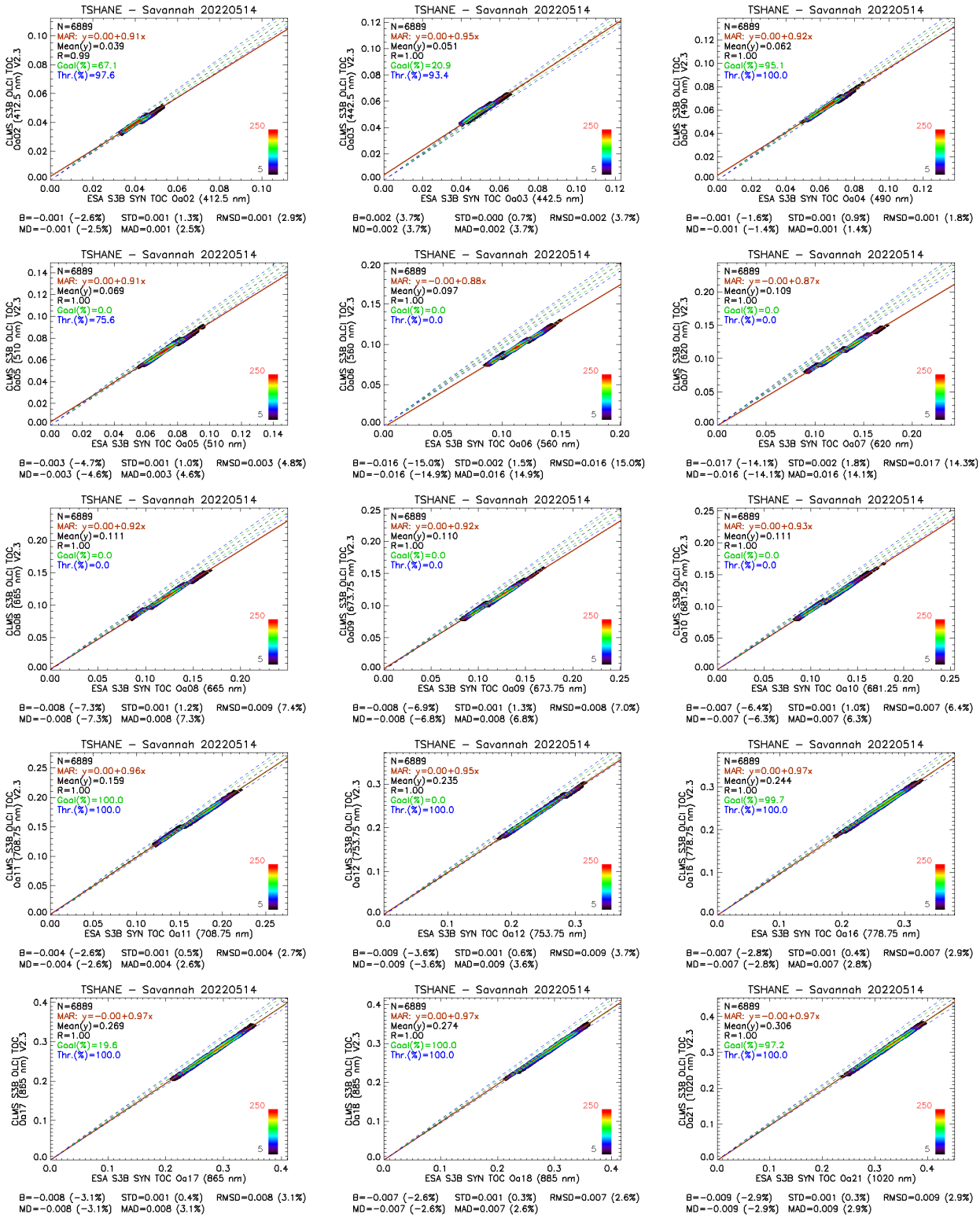


Figure 94: Scatter-plots of Sentinel-3B (2022.05.14) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#8 (Tshane, Botswana – Savannah).

Table 41: Validation statistics of the comparison of Sentinel-3B (2022.05.14) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#8 (Tshane, Botswana – Savannah).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6889	0.99	-0.001 (-2.6%)	-0.001 (-2.5%)	0.001 (1.3%)	0.001 (2.5%)	0.001 (2.9%)	0.00 /0.91	67.1 /97.6
Oa03	6889	1.00	0.002 (3.7%)	0.002 (3.7%)	<0.001 (0.7%)	0.002 (3.7%)	0.002 (3.7%)	0.00 /0.95	20.9 /93.4
Oa04	6889	1.00	-0.001 (-1.6%)	-0.001 (1.4%)	0.001 (0.9%)	0.001 (1.4%)	0.001 (1.8%)	0.00 /0.92	95.1 /100
Oa05	6889	1.00	-0.003 (-4.7%)	-0.003 (-4.6%)	0.001 (1.0%)	0.003 (4.6%)	0.003 (4.8%)	0.00 /0.91	0.0 /75.6
Oa06	6889	1.00	-0.016 (-15.0%)	-0.016 (-14.9%)	0.002 (1.5%)	0.016 (14.9%)	0.016 (15.0%)	0.00 /0.88	0.0 /0.0
Oa07	6889	1.00	-0.017 (-14.1%)	-0.016 (-14.1%)	0.002 (1.8%)	0.016 (14.1%)	0.017 (14.3%)	0.00 /0.87	0.0 /0.0
Oa08	6889	1.00	-0.008 (-7.3%)	-0.008 (-7.3%)	0.001 (1.2%)	0.008 (7.3%)	0.009 (7.4%)	0.00 /0.92	0.0 /0.0
Oa09	6889	1.00	-0.008 (-6.9%)	-0.008 (-6.8%)	0.001 (1.3%)	0.008 (6.8%)	0.008 (7.0%)	0.00 /0.92	0.0 /0.0
Oa10	6889	1.00	-0.007 (-6.4%)	-0.007 (-6.3%)	0.001 (1.0%)	0.007 (6.3%)	0.007 (6.4%)	0.00 /0.92	0.0 /0.0
Oa11	6889	1.00	-0.004 (-2.6%)	-0.004 (-2.6%)	0.001 (0.5%)	0.004 (2.6%)	0.004 (2.7%)	0.00 /0.96	100 /100
Oa12	6889	1.00	-0.009 (-3.6%)	-0.009 (-3.6%)	0.001 (0.6%)	0.009 (3.6%)	0.009 (3.7%)	0.00 /0.95	0.0 /100
Oa16	6889	1.00	-0.007 (-2.8%)	-0.007 (-2.8%)	0.001 (0.4%)	0.007 (2.8%)	0.007 (2.9%)	0.00 /0.97	99.7 /100
Oa17	6889	1.00	-0.008 (-3.1%)	-0.008 (-3.1%)	0.001 (0.4%)	0.008 (3.1%)	0.008 (3.1%)	0.00 /0.97	19.6 /100
Oa18	6889	1.00	-0.007 (-2.6%)	-0.007 (-2.6%)	0.001 (0.3%)	0.007 (2.6%)	0.007 (2.6%)	0.00 /0.97	100 /100
Oa21	6889	1.00	-0.009 (-2.9%)	-0.009 (-2.9%)	0.001 (0.3%)	0.009 (2.9%)	0.009 (2.9%)	0.00 /0.97	97.2 /100

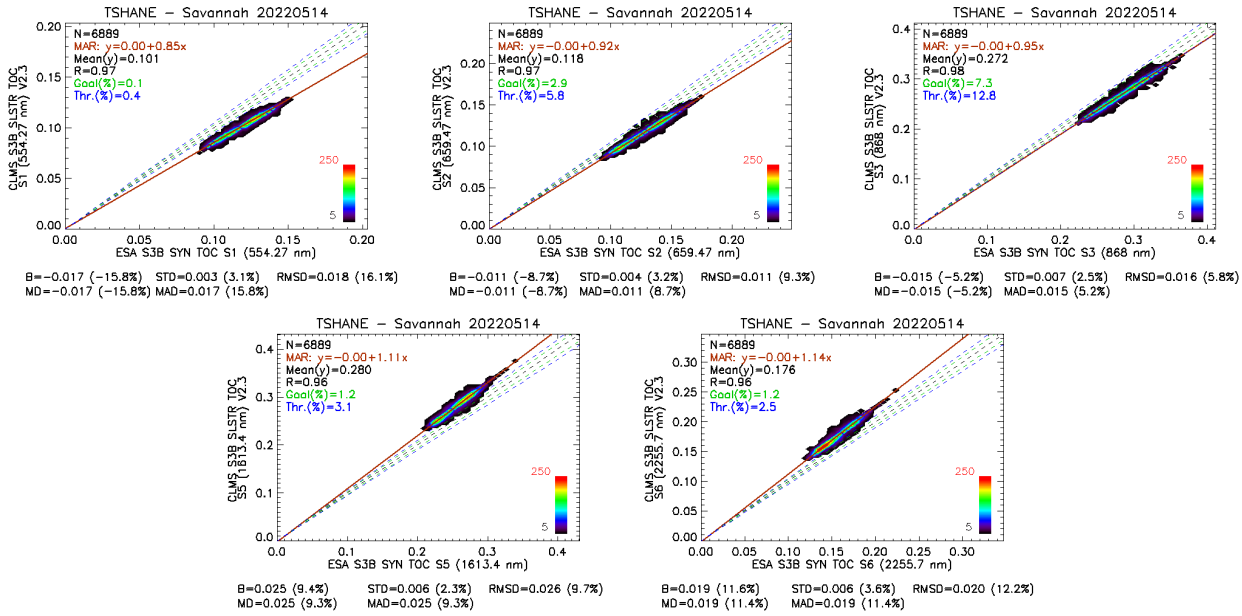


Figure 95: Scatter-plots of Sentinel-3B (2022.05.14) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#8 (Tshane, Botswana – Savannah).

Table 42: Validation statistics of the comparison of Sentinel-3B (2022.05.14) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#8 (Tshane, Botswana – Savannah).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6889	0.97	-0.017 (-15.8%)	-0.017 (-15.8%)	0.003 (3.1%)	0.017 (15.8%)	0.018 (16.1%)	0.00 /0.85	0.1 /0.4
S2	6889	0.97	-0.011 (-8.7%)	-0.011 (-8.7%)	0.004 (3.2%)	0.011 (8.7%)	0.011 (9.3%)	0.00 /0.92	2.9 /5.8
S3	6889	0.98	-0.015 (-5.2%)	-0.015 (-5.2%)	0.007 (2.5%)	0.015 (5.2%)	0.016 (5.8%)	0.00 /0.95	7.3 /12.8
S5	6889	0.96	0.025 (9.4%)	0.025 (9.3%)	0.006 (2.3%)	0.025 (9.3%)	0.026 (9.7%)	0.00 /1.11	1.2 /3.1
S6	6889	0.96	0.019 (11.6%)	0.019 (11.4%)	0.006 (3.6%)	0.019 (11.4%)	0.020 (12.2%)	0.00 /1.14	1.2 /2.5

AOI#9 DAINTREE RAINFOREST (AUSTRALIA) - EBF

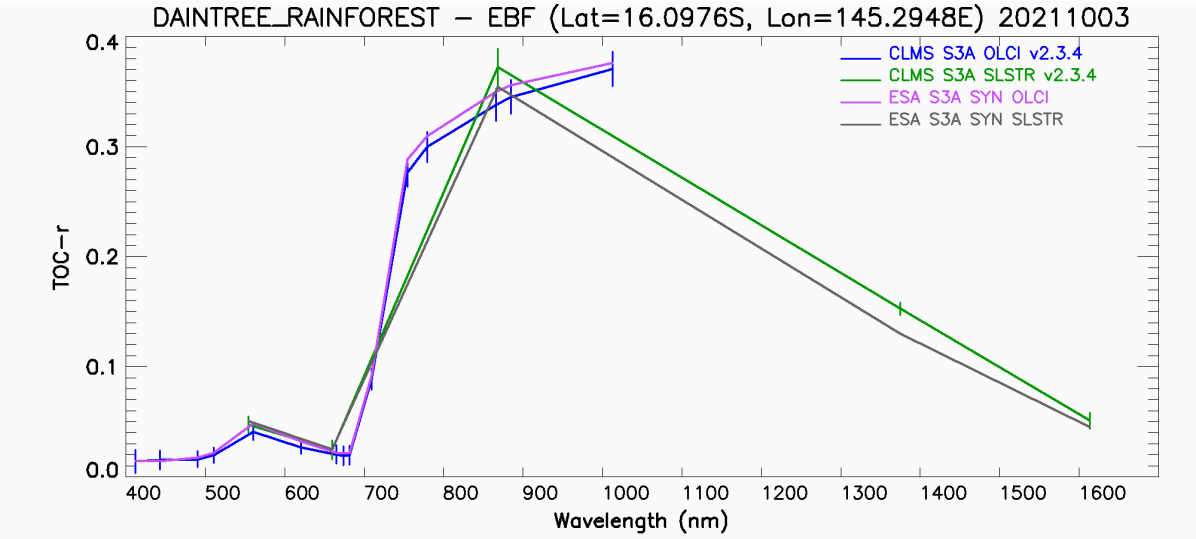


Figure 96: Spectral profile of Sentinel-3A (2021.10.03) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#9 (Daintree Rainforest, Australia - EBF).

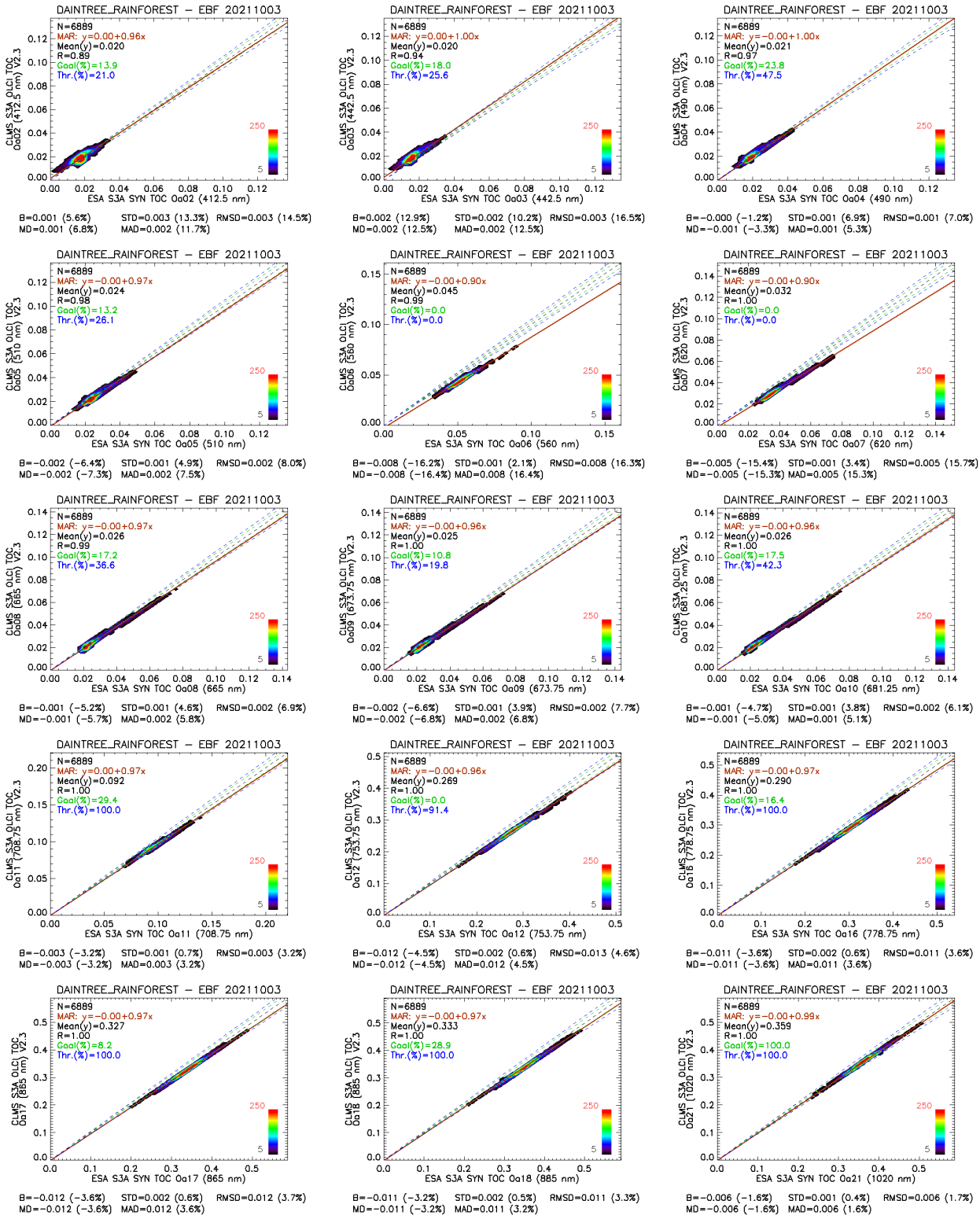


Figure 97: Scatter-plots of Sentinel-3A (2021.10.03) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#9 (Daintree Rainforest, Australia - EBF).

Table 43: Validation statistics of the comparison of Sentinel-3A (2021.10.03) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#9 (Daintree Rainforest, Australia - EBF).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6889	0.89	0.001 (5.6%)	0.001 (6.8%)	0.003 (13.3%)	0.002 (11.7%)	0.003 (14.5%)	0.03 /0.96	13.9 /21.0
Oa03	6889	0.94	0.022 (12.9%)	0.002 (12.5%)	0.002 (10.2%)	0.002 (12.5%)	0.003 (16.5%)	0.00 /1.00	18.0 /25.6
Oa04	6889	0.97	>-0.001 (-1.2%)	-0.001 (-3.3%)	0.001 (6.9%)	0.001 (5.3%)	0.001 (7.0%)	0.00 /1.00	23.8 /47.5
Oa05	6889	0.98	-0.002 (-6.4%)	-0.002 (-7.3%)	0.001 (4.9%)	0.002 (7.5%)	0.002 (8.0%)	0.00 /0.97	13.2 /26.1
Oa06	6889	0.99	-0.008 (-16.2%)	-0.008 (-16.4%)	0.001 (2.1%)	0.008 (16.4%)	0.008 (16.3%)	0.00 /0.90	0.0 /0.0
Oa07	6889	1.00	-0.005 (-15.4%)	-0.005 (-15.3%)	0.001 (3.4%)	0.005 (15.3%)	0.005 (15.7%)	0.00 /0.90	0.0 /0.0
Oa08	6889	0.99	-0.001 (-5.2%)	-0.001 (-5.7%)	0.001 (4.6%)	0.002 (5.8%)	0.002 (6.8%)	0.00 /0.97	17.2 /36.6
Oa09	6889	1.00	-0.002 (-6.6%)	-0.002 (-6.8%)	0.001 (3.9%)	0.002 (6.8%)	0.002 (7.7%)	0.00 /0.96	10.8 /19.8
Oa10	6889	1.00	-0.001 (-4.7%)	-0.001 (-5.0%)	0.001 (3.8%)	0.001 (5.1%)	0.002 (6.1%)	0.00 /0.96	17.5 /42.3
Oa11	6889	1.00	-0.003 (-3.2%)	-0.003 (-3.2%)	0.001 (0.7%)	0.003 (3.2%)	0.003 (3.2%)	0.00 /0.97	29.4 /100
Oa12	6889	1.00	-0.012 (-4.5%)	-0.012 (-4.5%)	0.002 (0.6%)	0.012 (4.5%)	0.013 (4.6%)	0.00 /0.96	0.0 /91.4
Oa16	6889	1.00	-0.011 (-3.6%)	-0.011 (-3.6%)	0.002 (0.6%)	0.011 (3.6%)	0.011 (3.6%)	0.00 /0.97	16.4 /100
Oa17	6889	1.00	-0.012 (-3.6%)	-0.012 (-3.6%)	0.002 (0.6%)	0.012 (3.6%)	0.012 (3.7%)	0.00 /0.97	8.2 /100
Oa18	6889	1.00	-0.011 (-3.2%)	-0.011 (-3.2%)	0.002 (0.5%)	0.011 (3.2%)	0.011 (3.3%)	0.00 /0.97	28.9 /100
Oa21	6889	1.00	-0.006 (-1.6%)	-0.006 (-1.6%)	0.001 (0.4%)	0.006 (1.6%)	0.006 (1.7%)	0.00 /0.99	100 /100

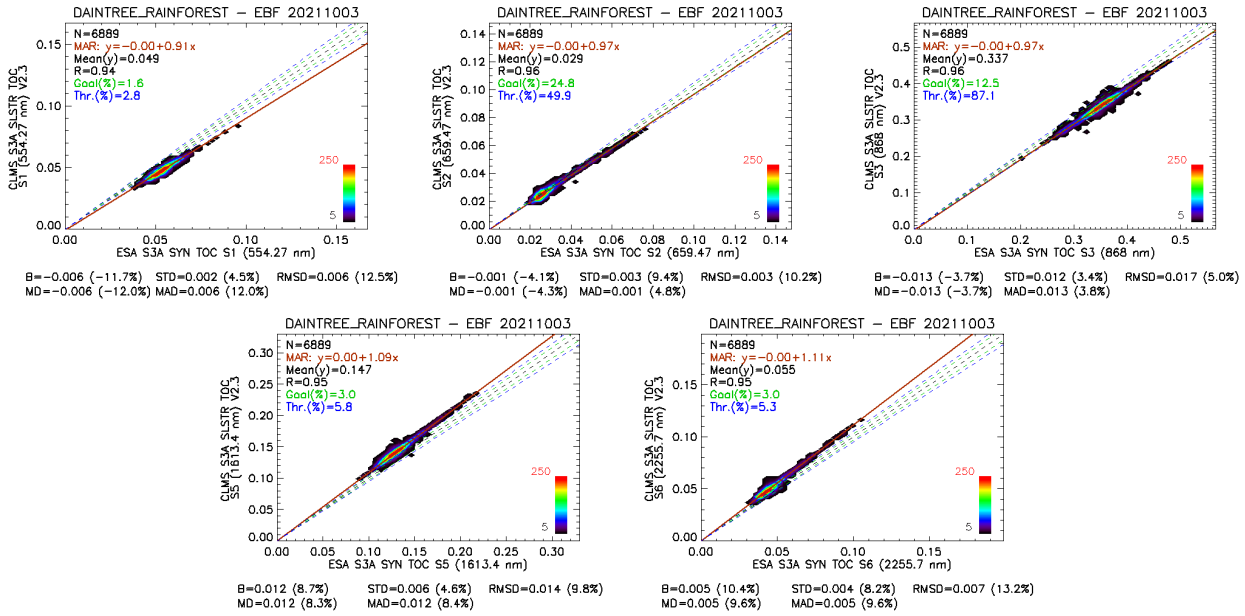


Figure 98: Scatter-plots of Sentinel-3A (2021.10.03) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#9 (Daintree Rainforest, Australia - EBF).

Table 44: Validation statistics of the comparison of Sentinel-3A (2021.10.03) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#9 (Daintree Rainforest, Australia - EBF).

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6889	0.94	-0.006 (-11.7%)	-0.006 (-12.0%)	0.002 (4.5%)	0.006 (12.0%)	0.006 (12.5%)	0.00 /0.91	1.6 /2.8
S2	6889	0.96	-0.001 (-4.1%)	-0.001 (-4.1%)	0.003 (9.4%)	0.001 (4.8%)	0.003 (10.2%)	0.00 /0.97	24.8 /49.9
S3	6889	0.96	-0.013 (-3.7%)	-0.013 (-3.7%)	0.012 (3.4%)	0.013 (3.8%)	0.017 (5.0%)	0.00 /0.97	12.5 /87.1
S5	6889	0.95	0.012 (8.7%)	0.012 (8.3%)	0.006 (4.6%)	0.012 (8.4%)	0.014 (9.8%)	0.00 /1.09	3.0 /5.8
S6	6889	0.95	0.005 (10.4%)	0.005 (9.6%)	0.004 (8.2%)	0.005 (9.6%)	0.007 (13.2%)	0.00 /1.11	3.0 /5.3

AOI#10 JORNADA (NEW MEXICO) - SHRUBLANDS

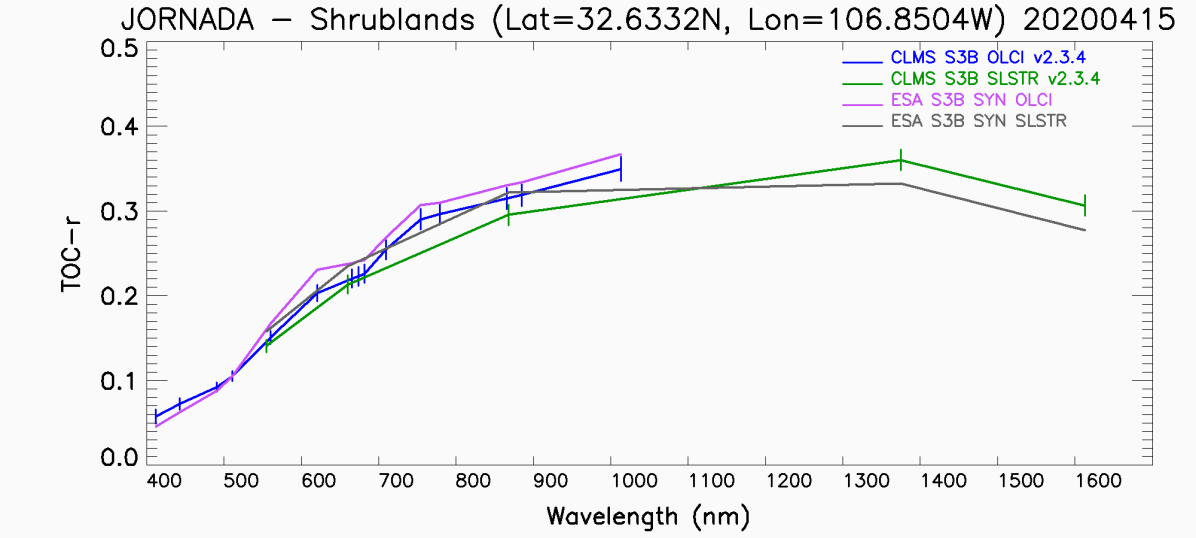


Figure 99: Spectral profile of Sentinel-3B (2020.04.15) OLCI and SLSTR bands for CLMS TOC V2.3 (blue and green) and ESA L2 SYN (purple and grey) products. Computation over the central coordinate of AOI#10 (Jornada, New Mexico - Shrublands).

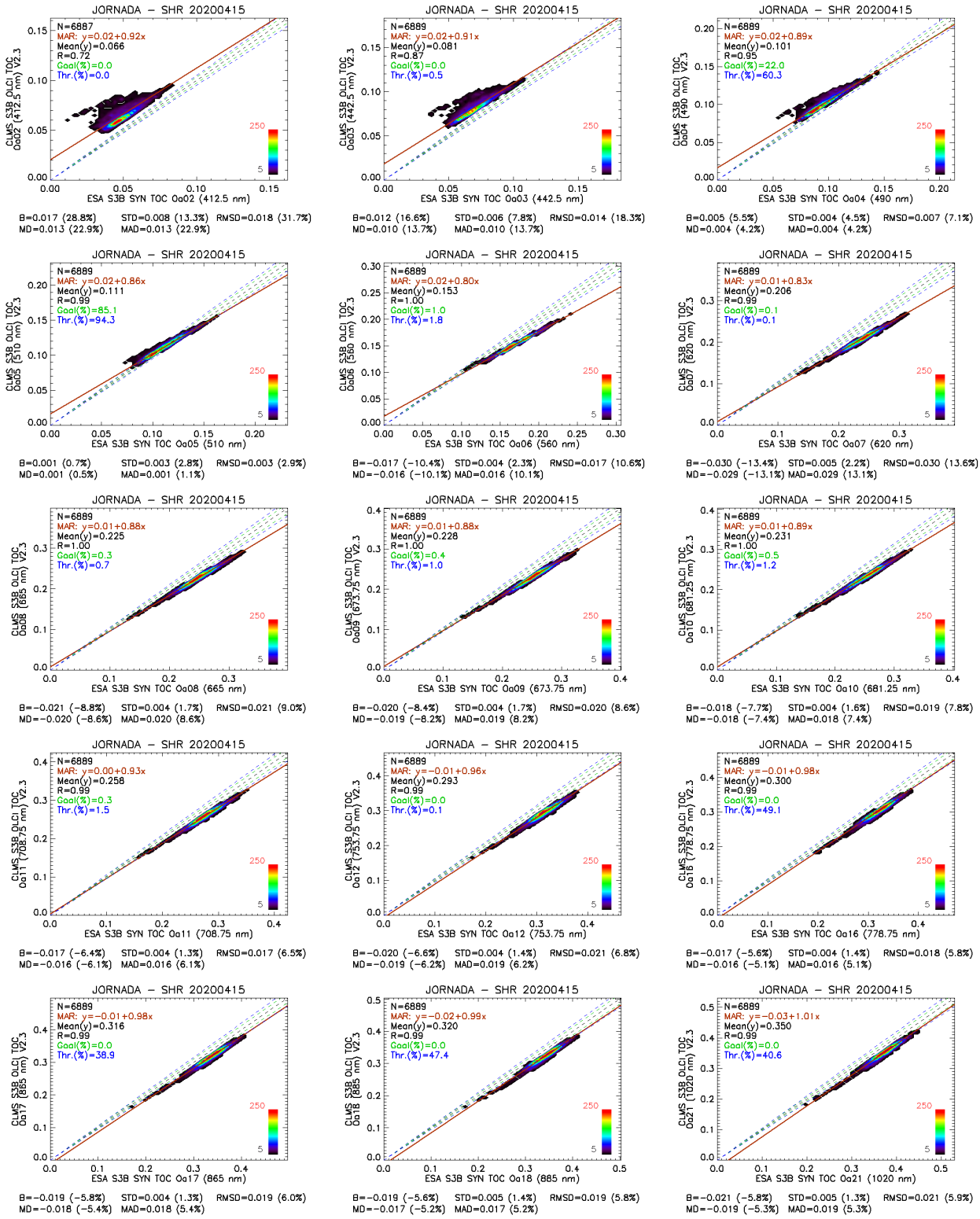


Figure 100: Scatter-plots of Sentinel-3B (2020.04.15) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#10 (Jornada, New Mexico - Shrublands).

Table 45: Validation statistics of the comparison of Sentinel-3B (2020.04.15) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#10 (Jornada, New Mexico - Shrublands

OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
Oa02	6887	0.72	0.017 (28.8%)	0.013 (22.9%)	0.008 (13.3%)	0.013 (22.9%)	0.018 (31.7%)	0.02 /0.92	0.0 /0.0
Oa03	6889	0.87	0.012 (16.6%)	0.010 (13.7%)	0.006 (7.8%)	0.010 (13.7%)	0.014 (18.3%)	0.02 /0.91	0.0 /0.5
Oa04	6889	0.95	0.005 (5.5%)	0.004 (4.2%)	0.004 (4.5%)	0.004 (4.2%)	0.007 (7.1%)	0.02 /0.89	22.0 /60.3
Oa05	6889	0.99	0.001 (0.7%)	0.001 (0.5%)	0.003 (2.8%)	0.001 (1.1%)	0.003 (2.9%)	0.02 /0.86	85.1 /94.3
Oa06	6889	1.00	-0.017 (-10.4%)	-0.016 (-10.1%)	0.004 (2.3%)	0.016 (10.1%)	0.017 (10.6%)	0.02 /0.80	1.0 /1.8
Oa07	6889	0.99	-0.030 (-13.4%)	-0.029 (-13.1%)	0.005 (2.2%)	0.029 (13.1%)	0.030 (13.6%)	0.01 /0.83	0.1 /0.1
Oa08	6889	1.00	-0.021 (-8.8%)	-0.020 (-8.6%)	0.004 (1.7%)	0.020 (8.6%)	0.021 (9.0%)	0.01 /0.88	0.3 /0.7
Oa09	6889	1.00	-0.020 (-8.4%)	-0.019 (-8.2%)	0.004 (1.7%)	0.019 (8.2%)	0.020 (8.6%)	0.01 /0.88	0.4 /1.0
Oa10	6889	1.00	-0.018 (-7.7%)	-0.018 (-7.4%)	0.004 (1.6%)	0.018 (7.4%)	0.019 (7.8%)	0.01 /0.89	0.5 /1.2
Oa11	6889	0.99	-0.017 (-6.4%)	-0.016 (-6.1%)	0.004 (1.3%)	0.016 (6.1%)	0.017 (6.5%)	0.00 /0.93	0.3 /1.5
Oa12	6889	0.99	-0.020 (-6.6%)	-0.019 (-6.2%)	0.004 (1.4%)	0.019 (6.2%)	0.021 (6.8%)	-0.01 /0.96	0.0 /0.1
Oa16	6889	0.99	-0.017 (-5.6%)	-0.016 (-5.1%)	0.004 (1.4%)	0.016 (5.1%)	0.018 (5.8%)	-0.01 /0.98	0.0 /49.1
Oa17	6889	0.99	-0.019 (-5.8%)	-0.018 (-5.4%)	0.004 (1.3%)	0.018 (5.4%)	0.019 (6.0%)	-0.01 /0.98	0.0 /38.9
Oa18	6889	0.99	-0.019 (-5.6%)	-0.017 (-5.2%)	0.005 (1.4%)	0.017 (5.2%)	0.019 (5.8%)	-0.02 /0.99	0.0 /47.4
Oa21	6889	0.99	-0.021 (-5.8%)	-0.019 (-5.3%)	0.005 (1.3%)	0.019 (5.3%)	0.021 (5.9%)	-0.03 /1.01	0.0 /40.6

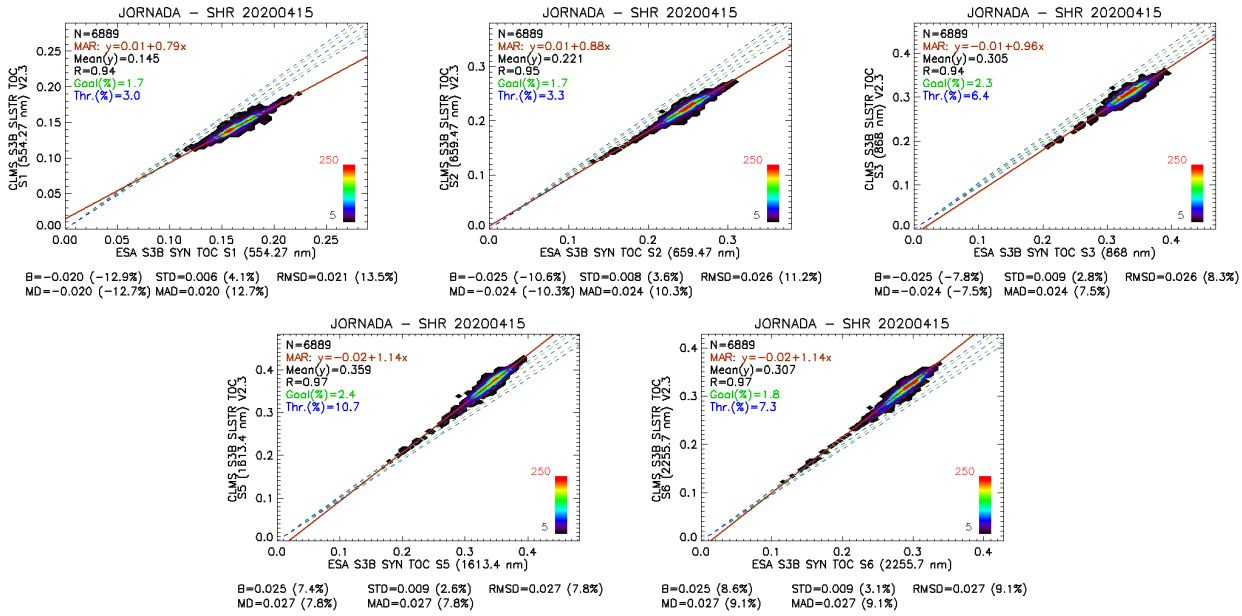


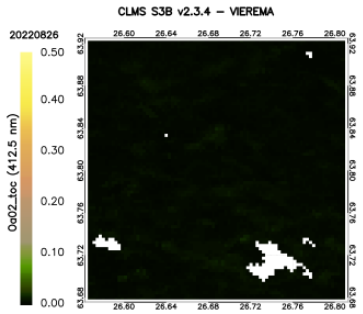
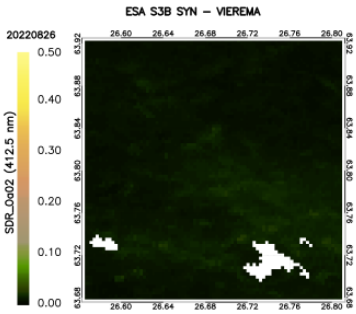
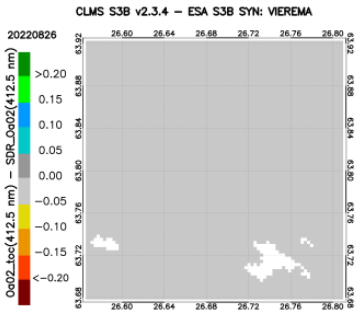
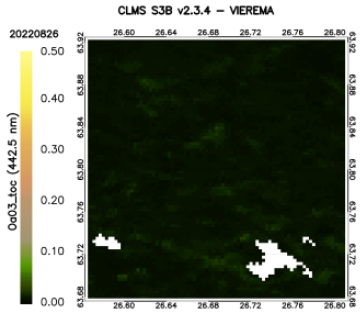
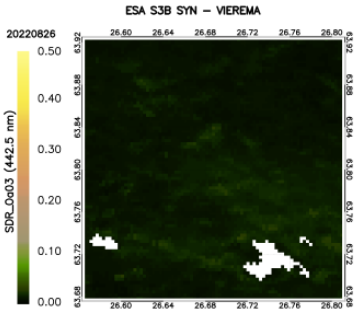
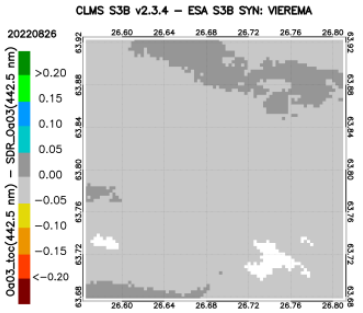
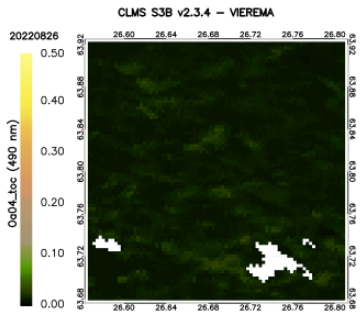
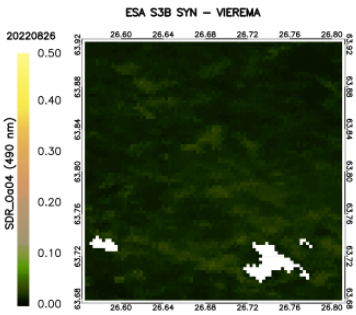
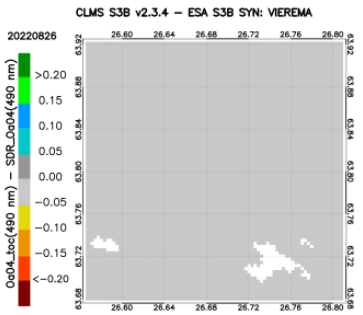
Figure 101: Scatter-plots of Sentinel-3B (2020.04.15) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#10 (Jornada, New Mexico - Shrublands).

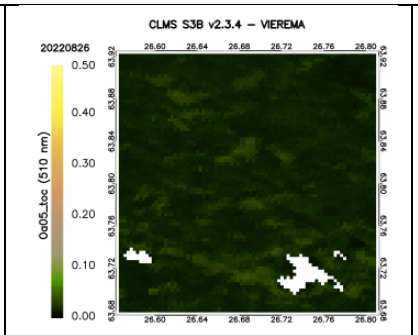
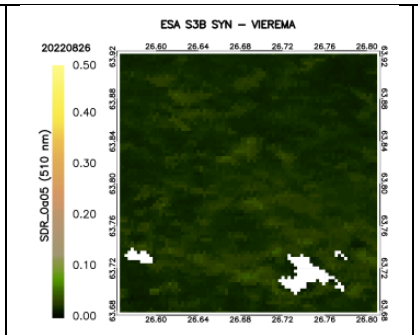
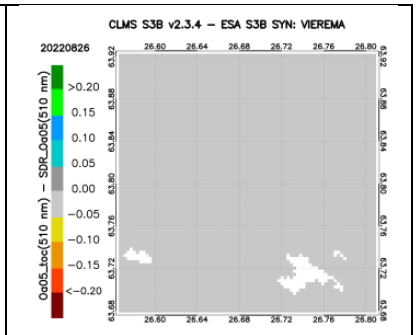
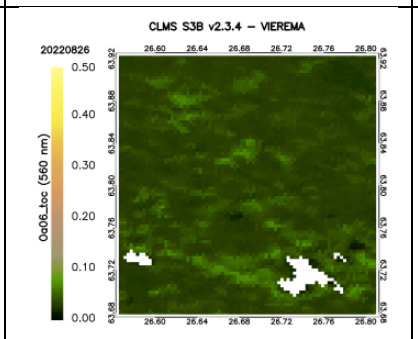
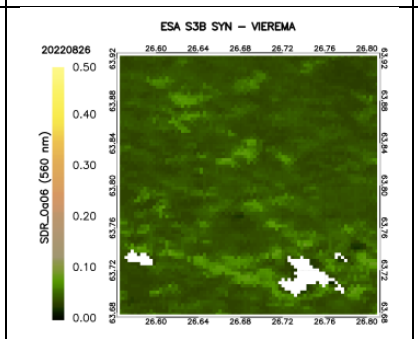
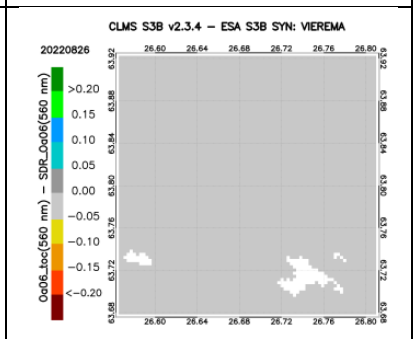
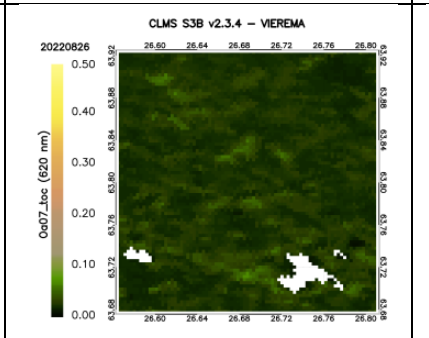
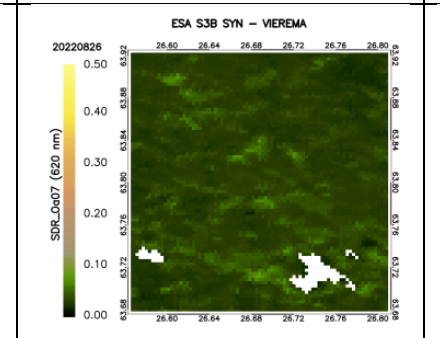
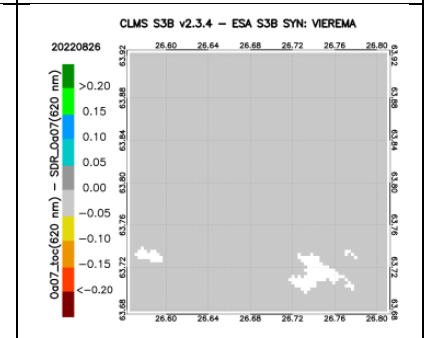
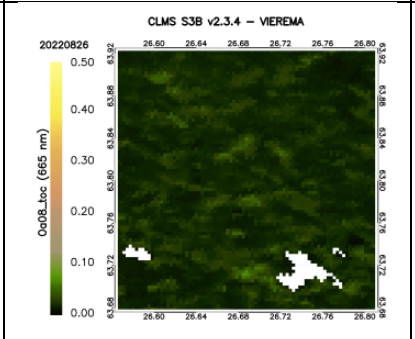
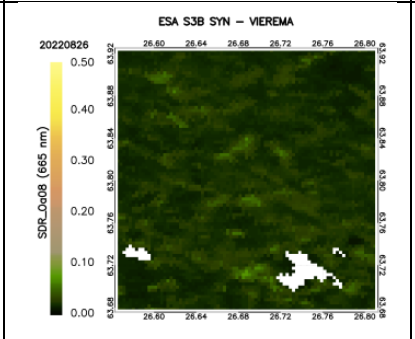
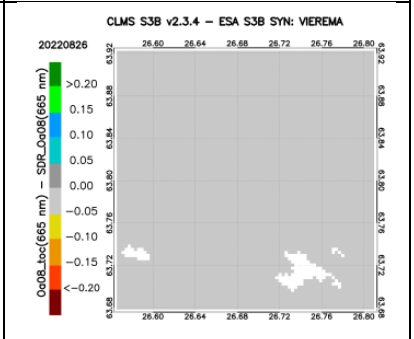
Table 46: Validation statistics of the comparison of Sentinel-3B (2020.04.15) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#10 (Jornada, New Mexico - Shrublands).

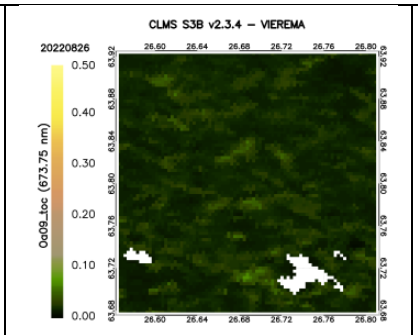
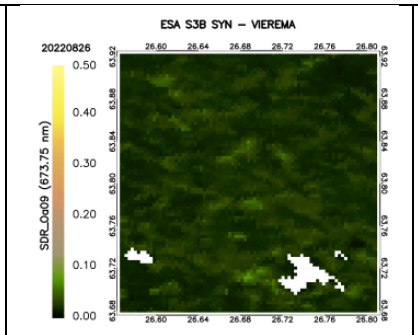
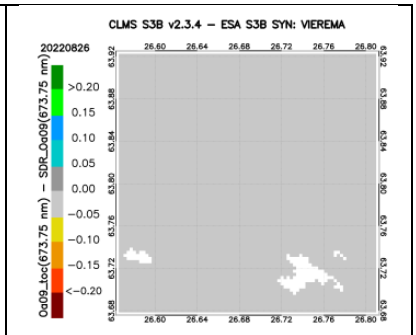
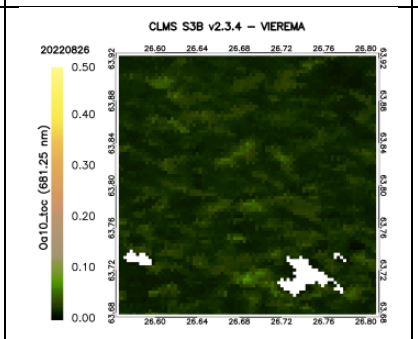
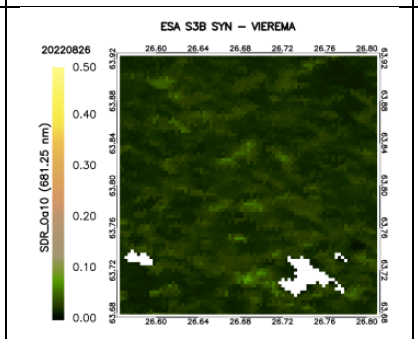
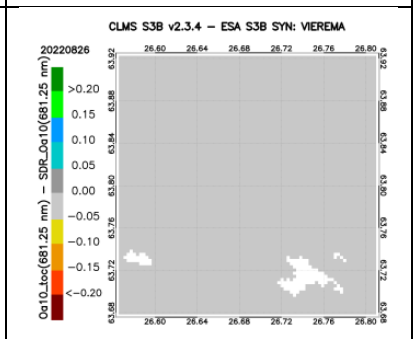
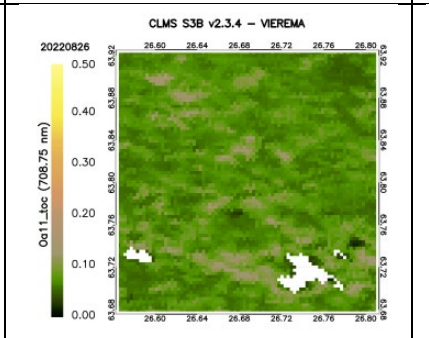
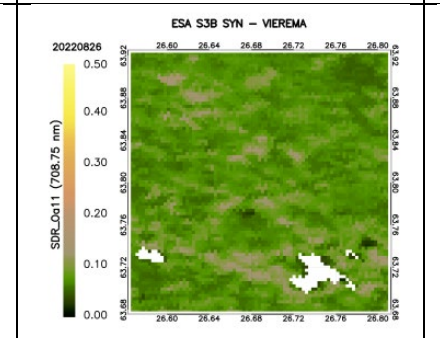
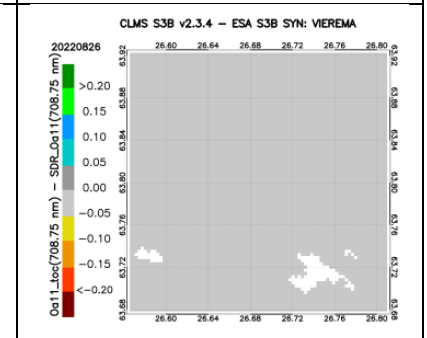
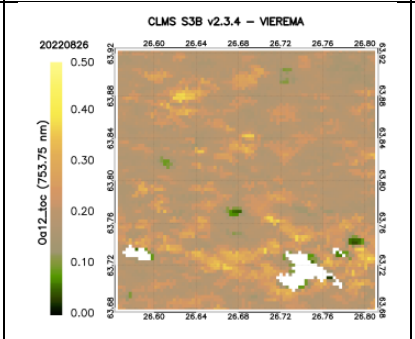
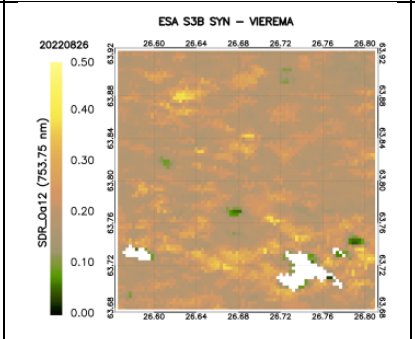
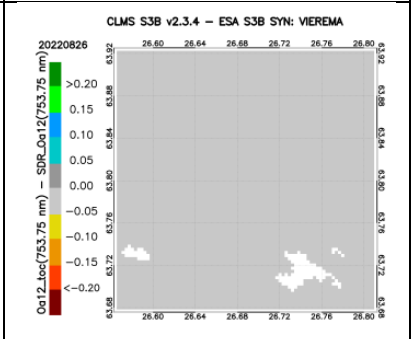
OLCI Band	N	R	Bias (%)	MD (%)	STD (%)	MAD (%)	RMSD (%)	Offset /Slope	% Goal /Threshold
S1	6889	0.94	-0.020 (-12.9%)	-0.020 (-12.7%)	0.006 (4.1%)	0.020 (12.7%)	0.021 (13.5%)	0.01 /0.79	1.7 /3.0
S2	6889	0.95	-0.025 (-10.6%)	-0.024 (-10.3%)	0.008 (3.6%)	0.024 (10.3%)	0.026 (11.2%)	0.01 /0.88	1.7 /3.3
S3	6889	0.94	-0.025 (-7.8%)	-0.024 (-7.5%)	0.009 (2.8%)	0.024 (7.5%)	0.026 (8.3%)	-0.01 /0.96	2.3 /6.4
S5	6889	0.97	0.025 (7.4%)	0.027 (7.8%)	0.009 (2.6%)	0.027 (7.8%)	0.027 (7.8%)	-0.02 /1.14	2.4 /10.7
S6	6889	0.97	0.025 (6.8%)	0.027 (9.1%)	0.009 (3.1%)	0.027 (9.1%)	0.027 (9.1%)	-0.02 /1.14	1.8 /7.3

ANNEX VII: SPATIAL CONSISTENCY BETWEEN CLMS TOC V2.3 AND ESA L2 SYN OVER AOIS

AOI#1 VIEREMÄ (FINLAND) – NLF

	CLMS TOC V2.3 map	ESA L2 SYN map	Difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			

Oa05			
Oa06			
Oa07			
Oa08			

Oa09			
Oa10			
Oa11			
Oa12			

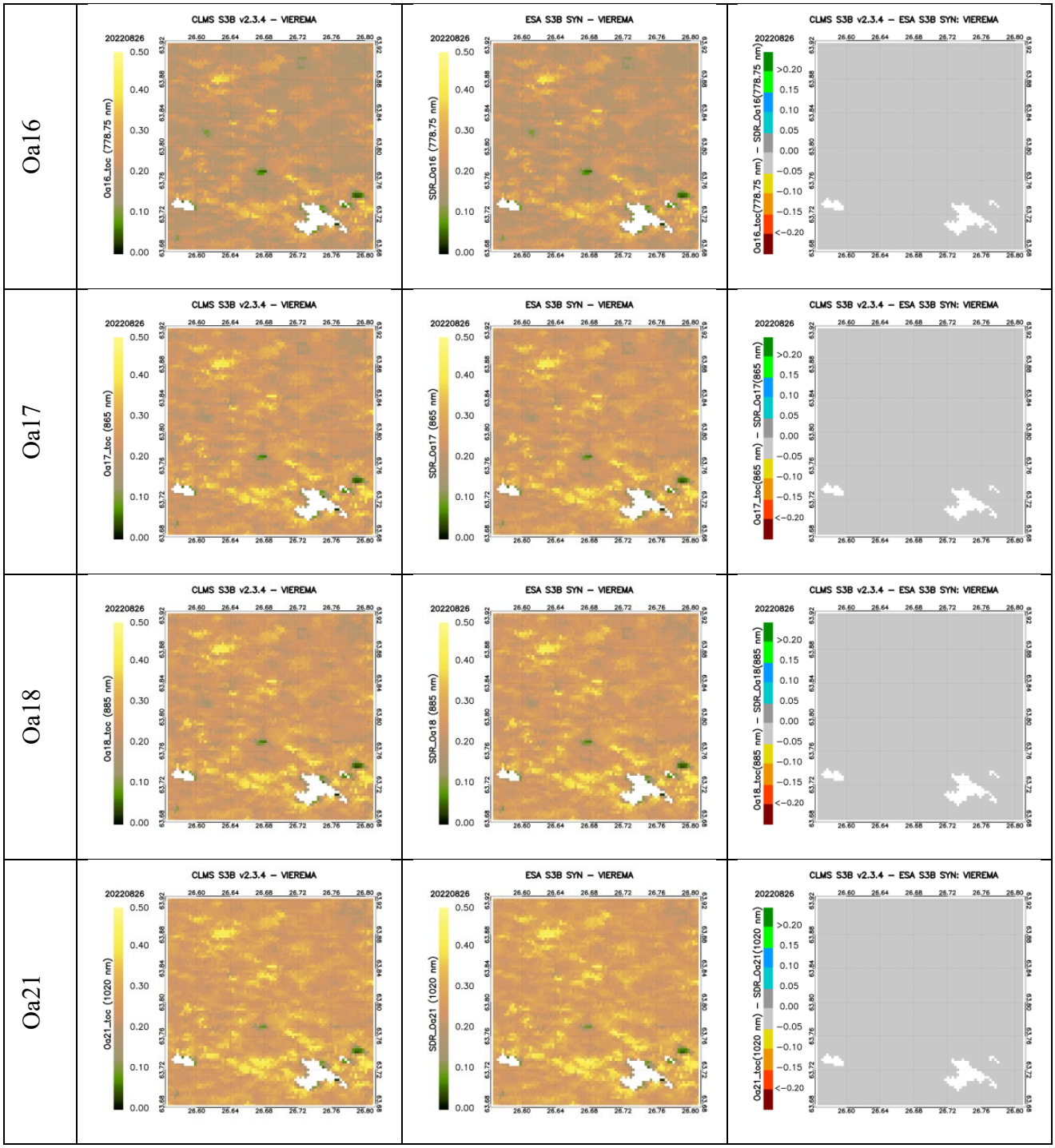
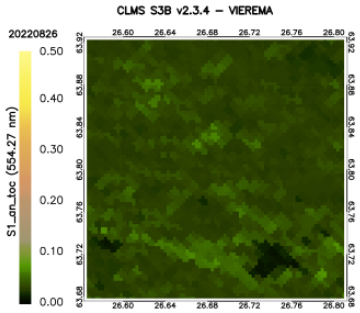
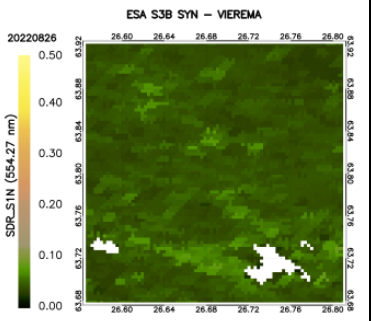
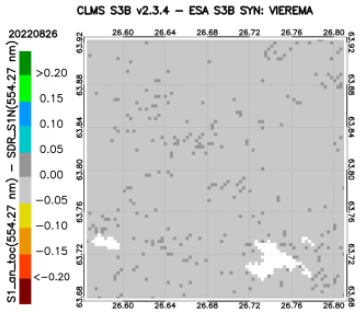
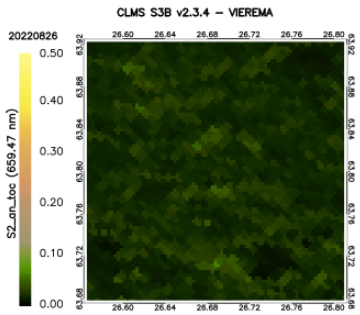
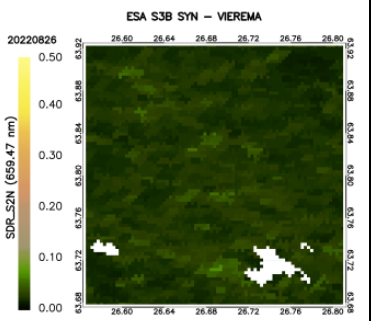
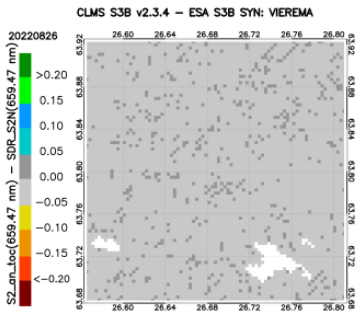
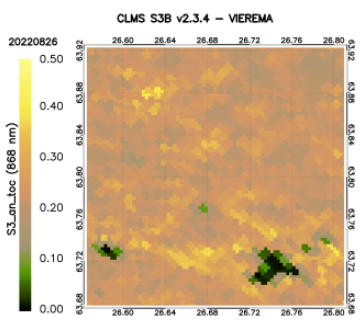
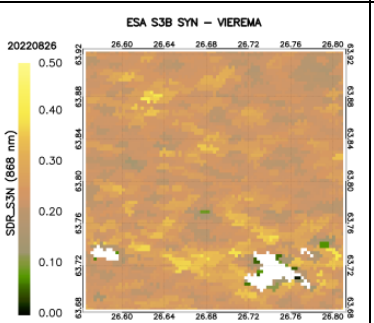
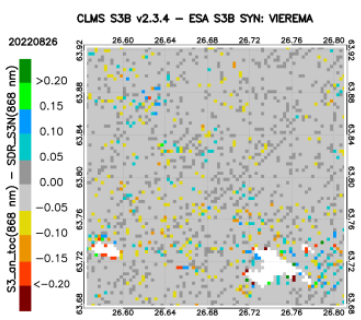
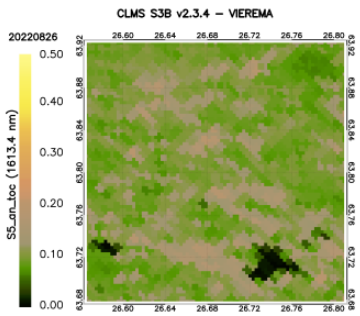
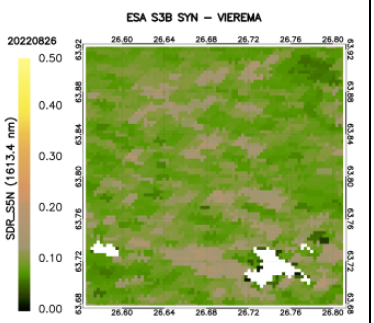
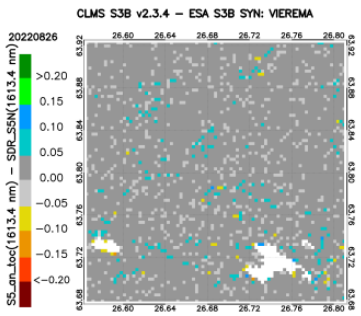


Figure 102: Spatial consistency of Sentinel-3B (2022.08.26) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#1 (Vieremä, Finland – NLF). From left to right: maps of CLMSS and ESA products and difference map (CLMS – ESA).

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
S1			
S2			
S3			
S5			

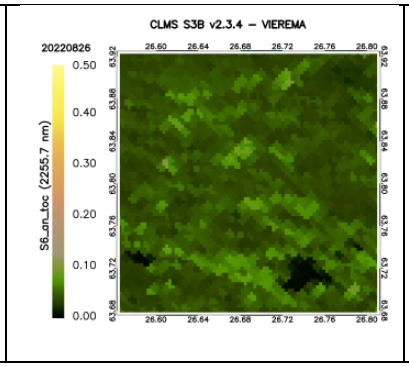
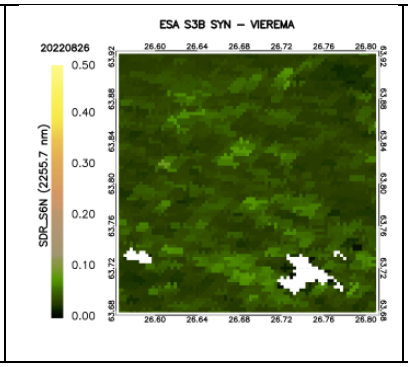
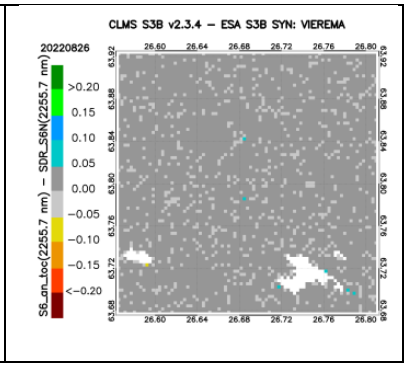
S6			

Figure 103: Spatial consistency of Sentinel-3B (2022.08.26) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#1 (Vieremä, Finland – NLF). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

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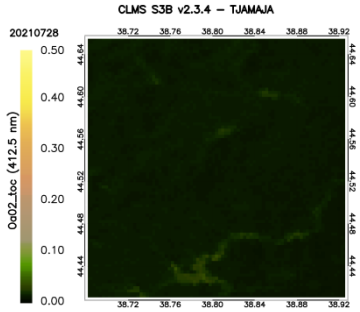
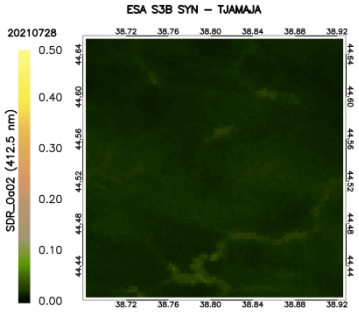
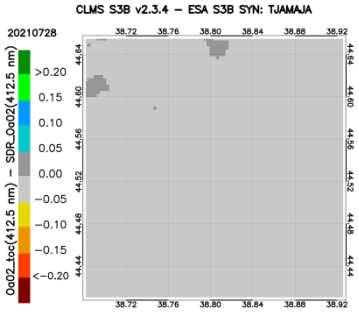
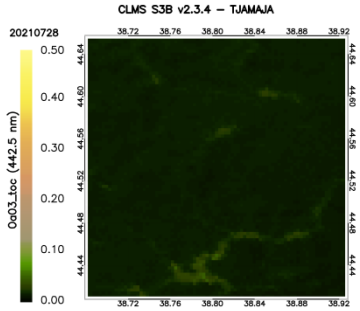
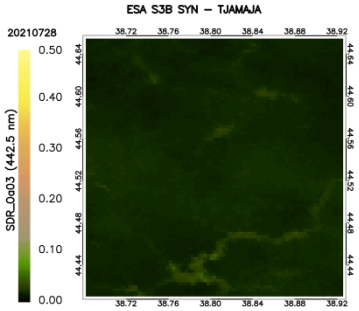
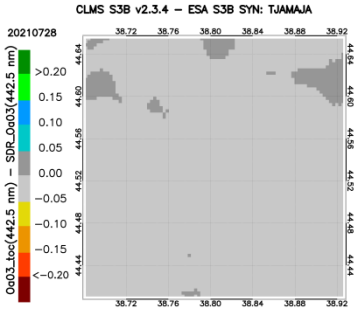
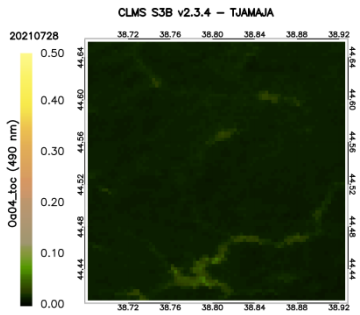
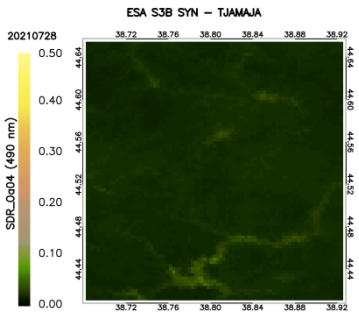
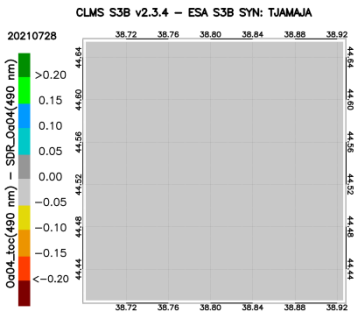
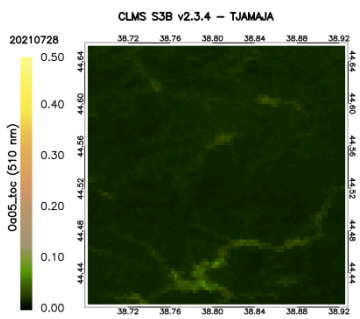
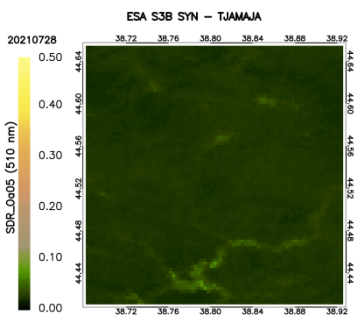
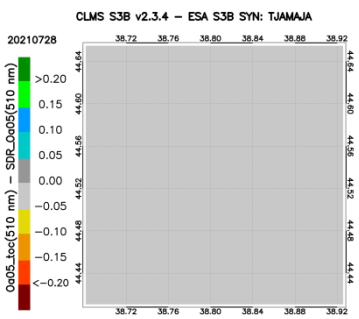
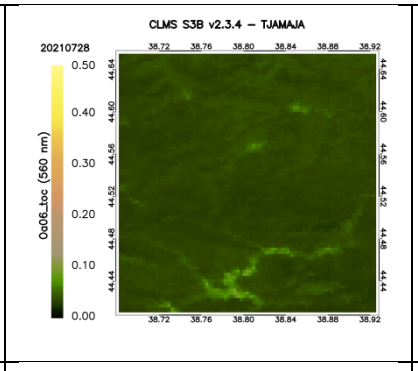
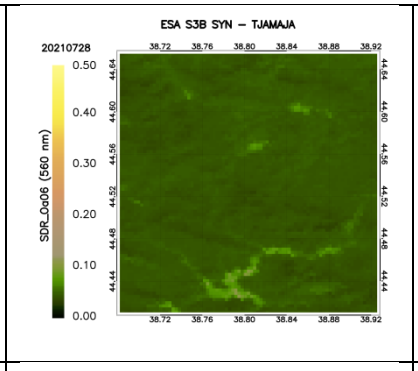
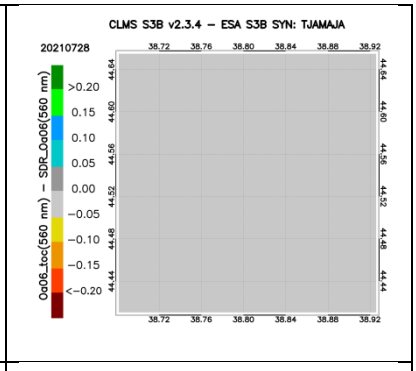
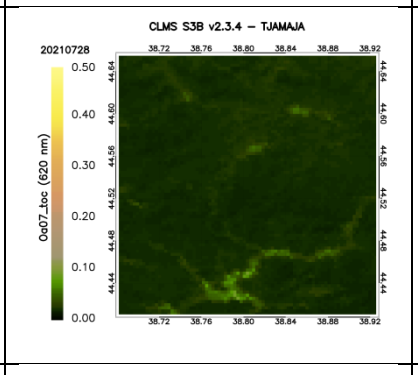
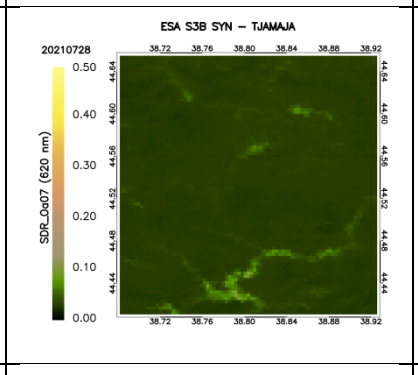
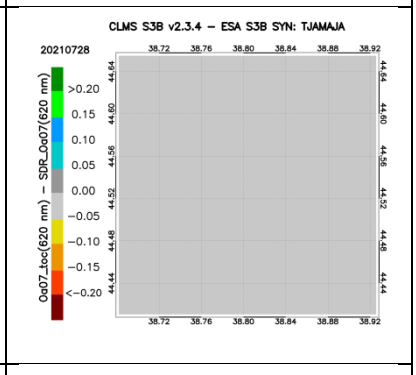
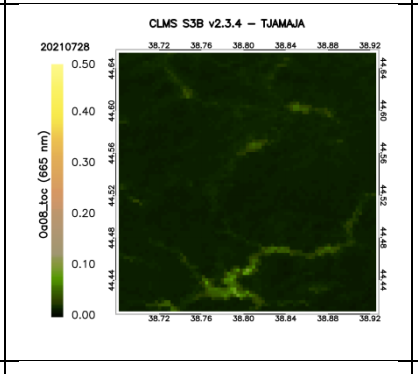
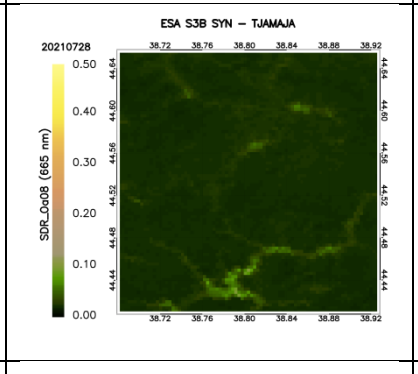
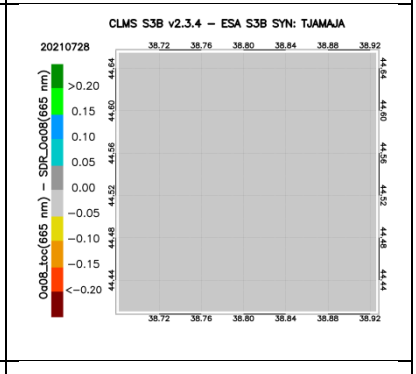
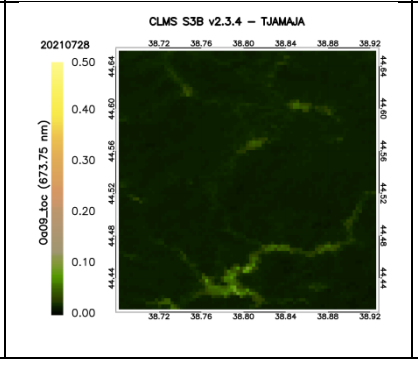
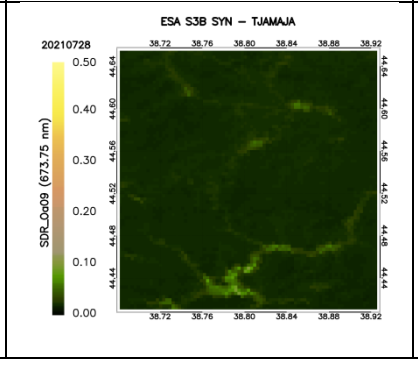
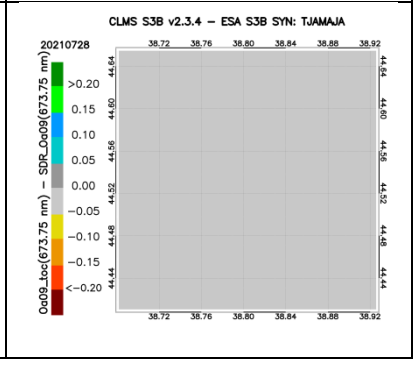
AOI#2 TJAMAJA (RUSSIA) – DBF

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

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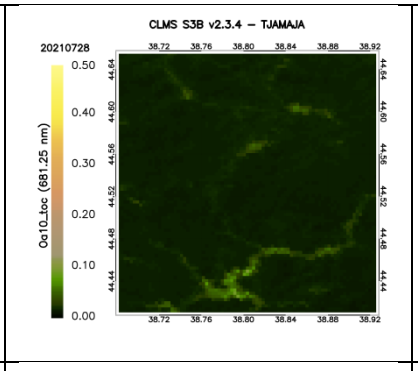
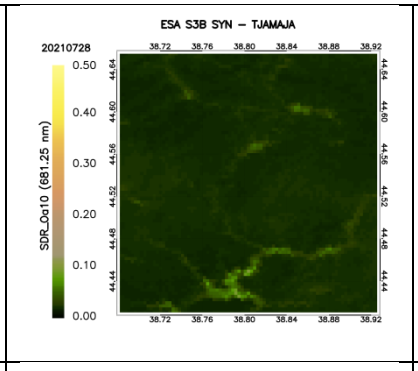
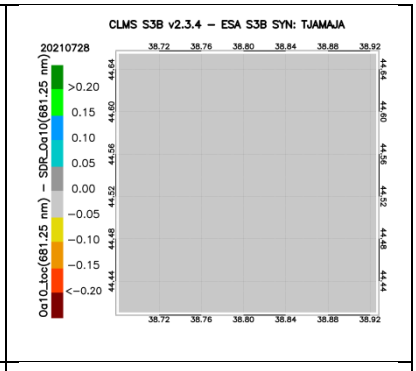
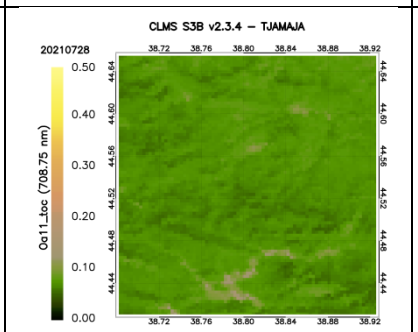
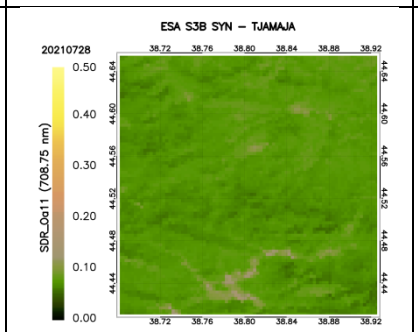
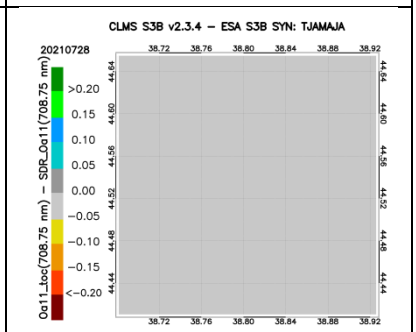
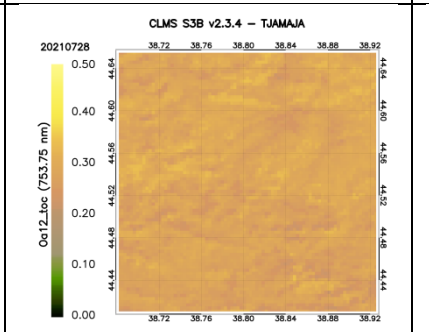
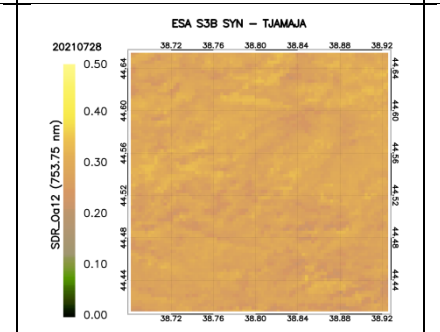
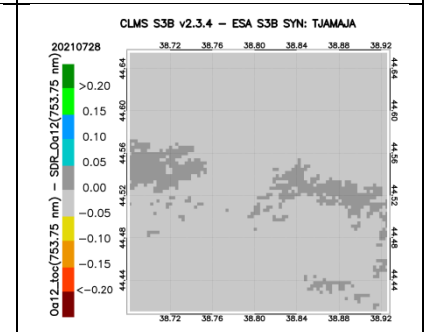
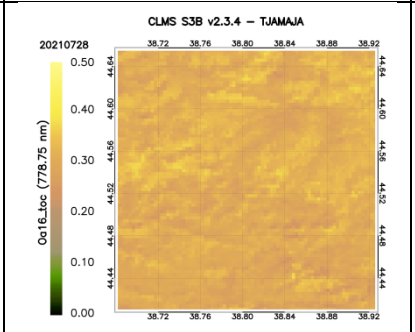
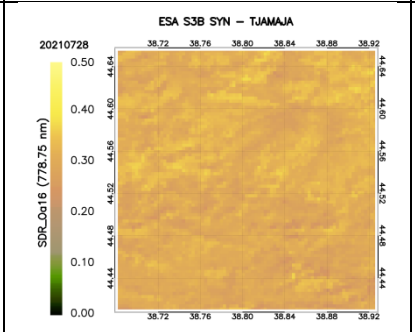
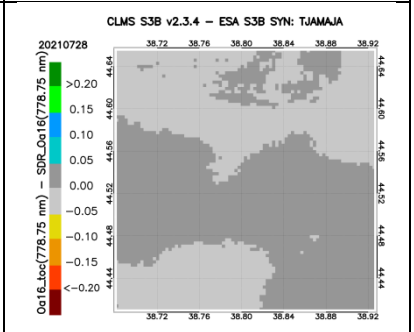
Oa10			
Oa11			
Oa12			
Oa16			

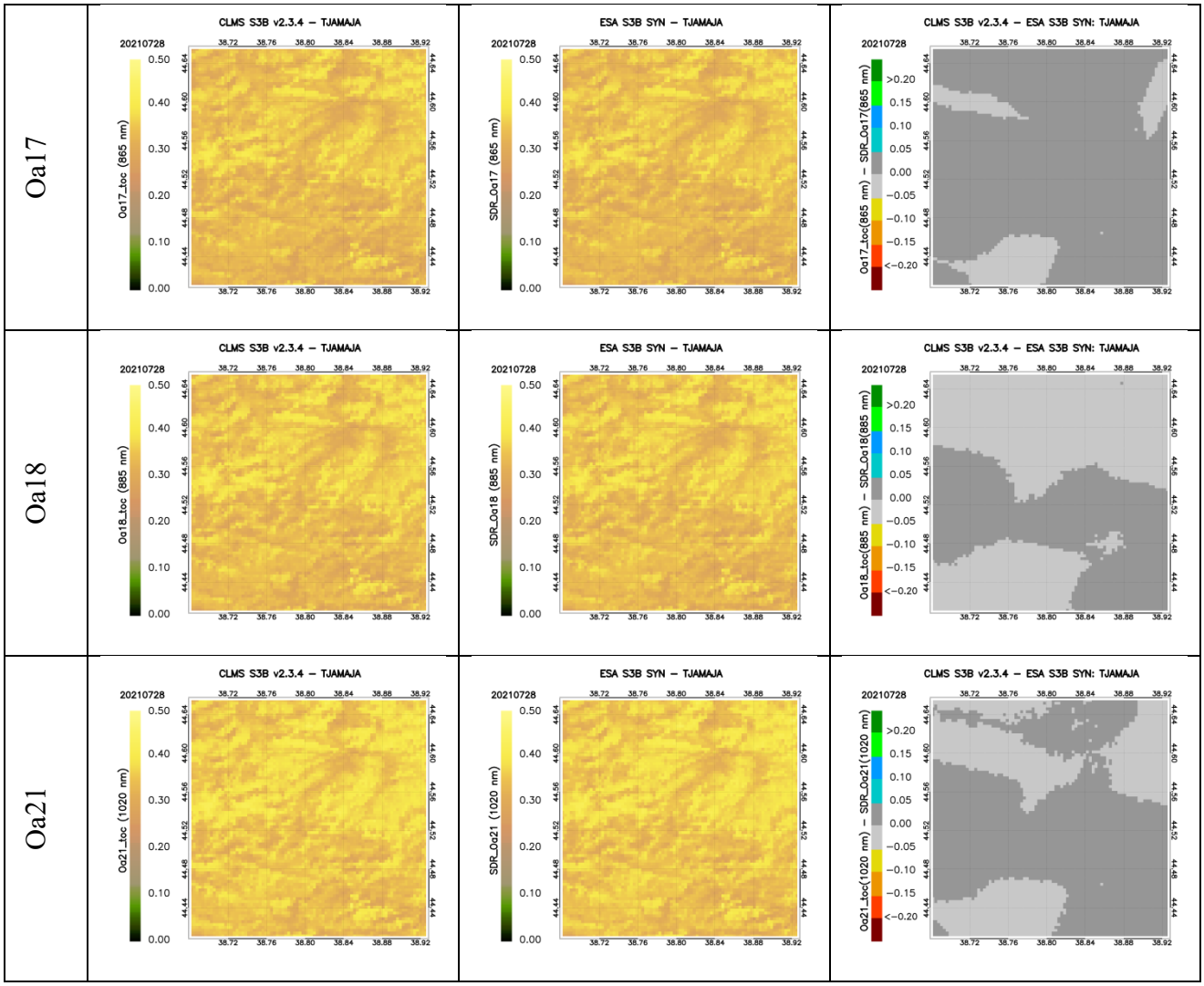
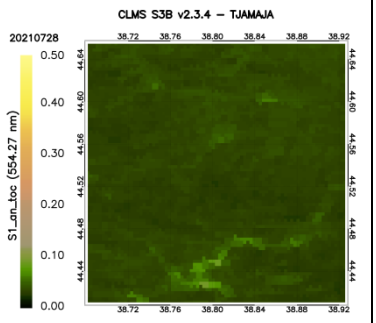
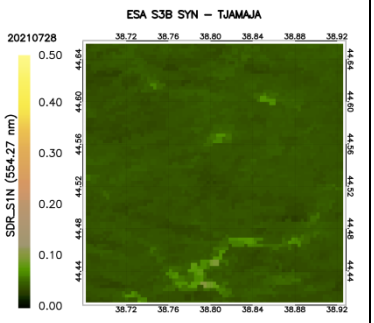
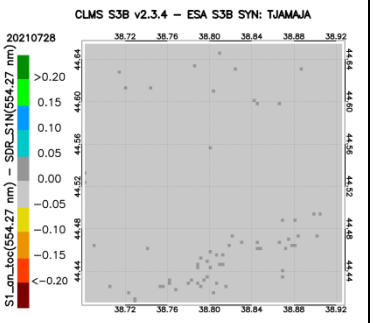
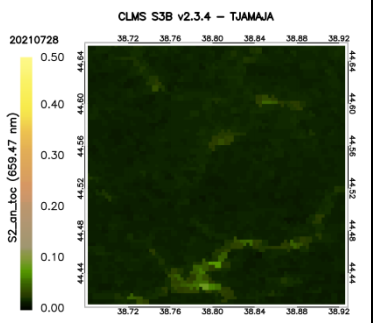
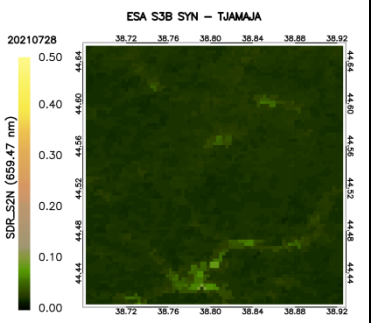
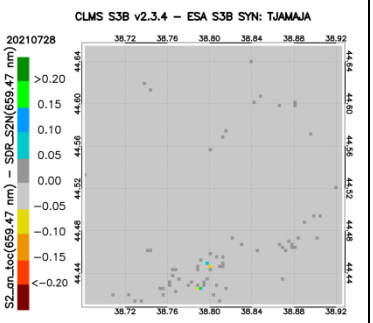
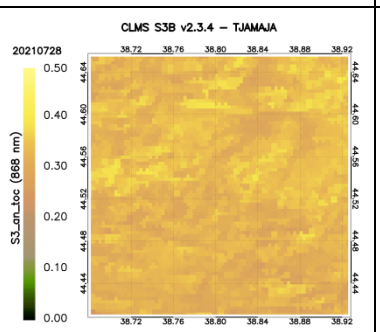
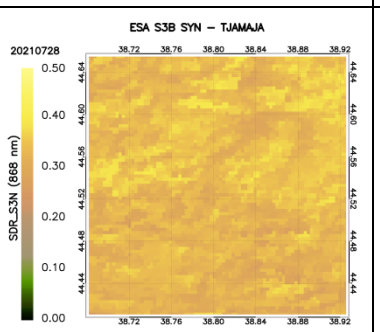
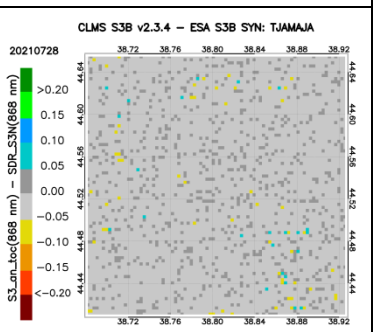
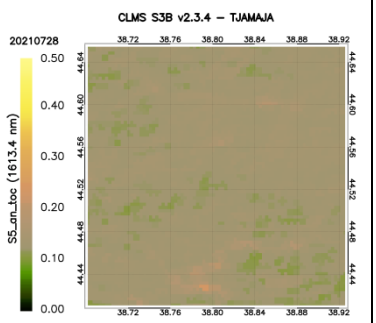
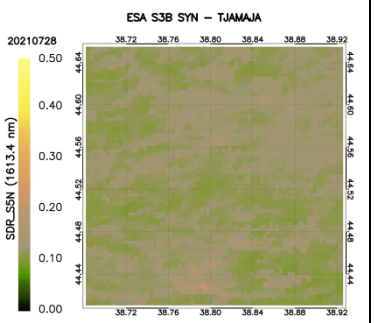
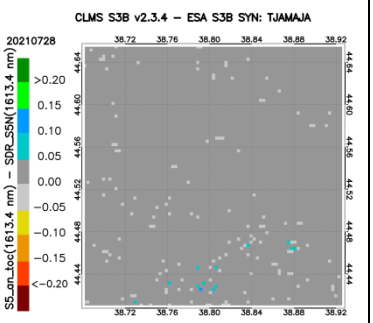


Figure 104: Spatial consistency of Sentinel-3B (2021.07.28) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#2 (Tjamaja, Russia – DBF). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
S1			
S2			
S3			
S5			

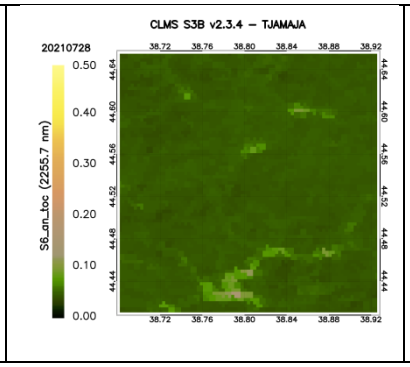
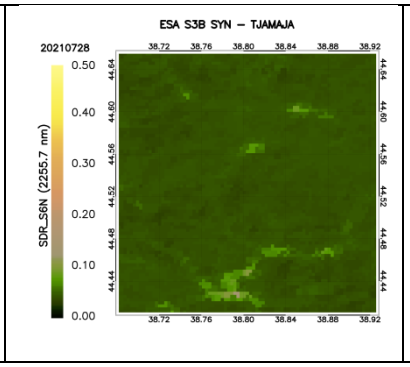
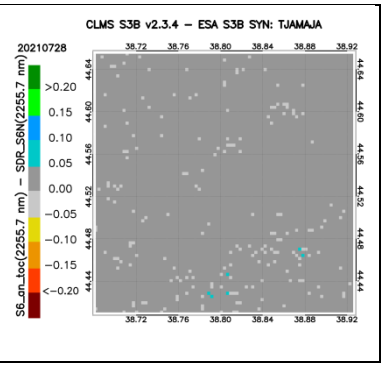
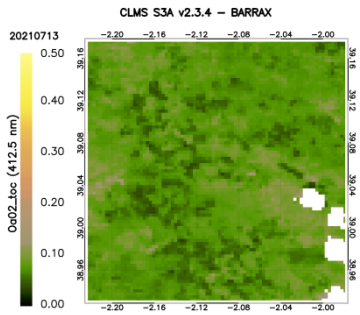
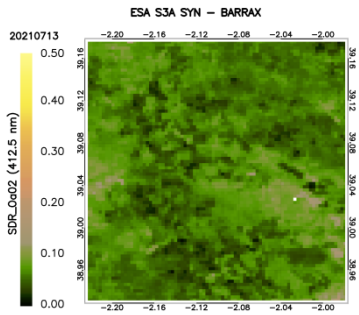
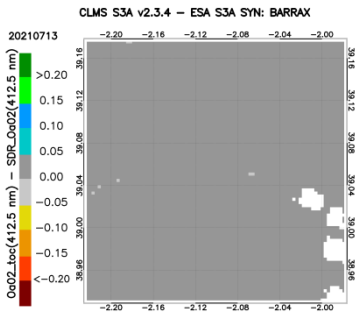
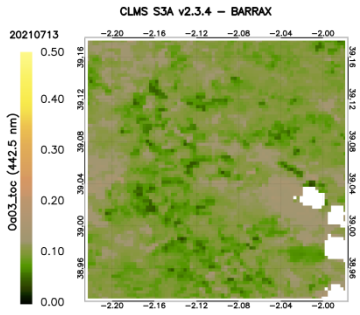
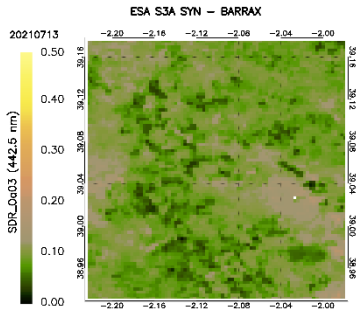
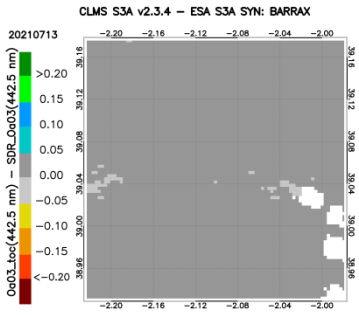
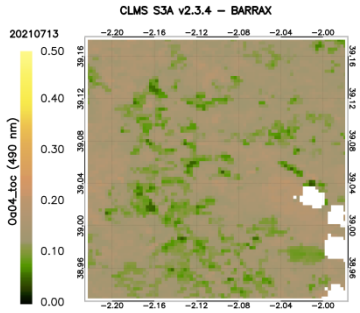
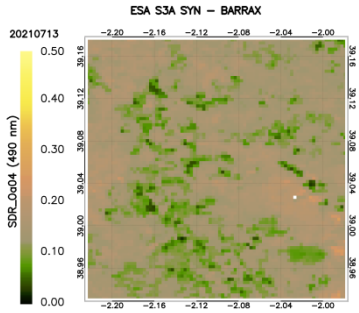
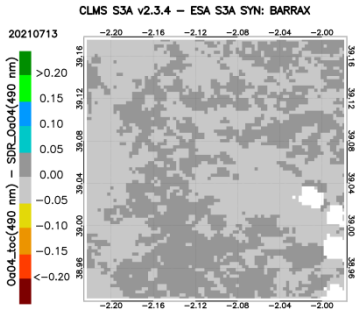
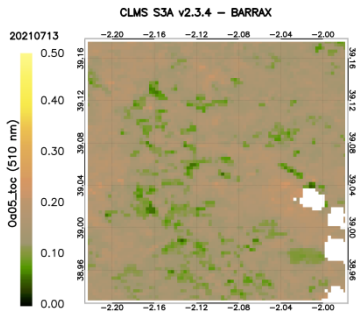
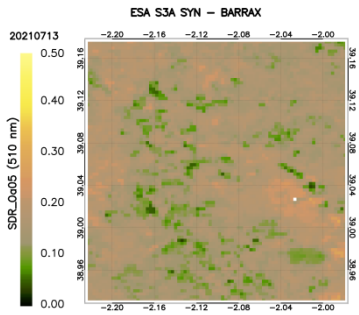
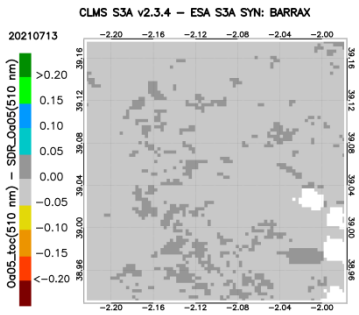
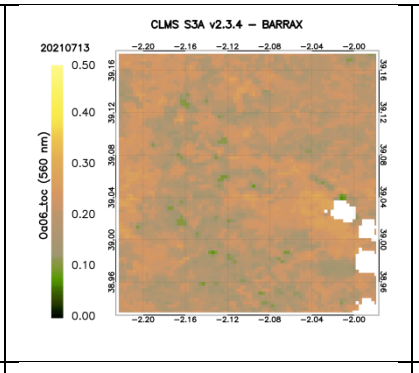
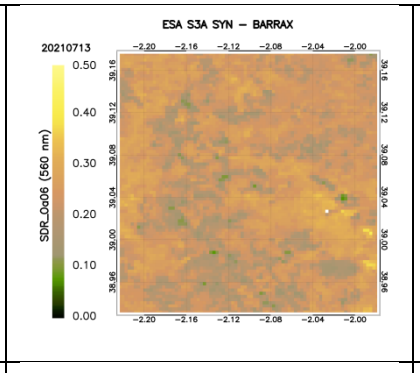
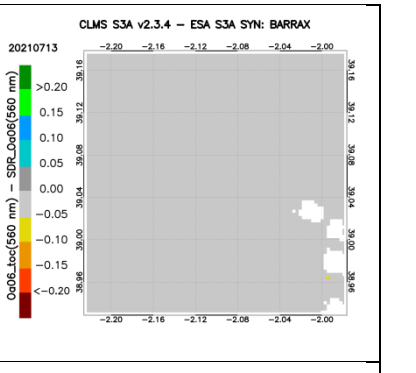
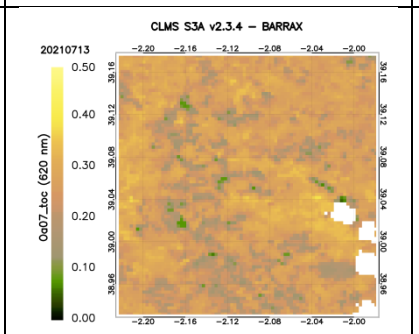
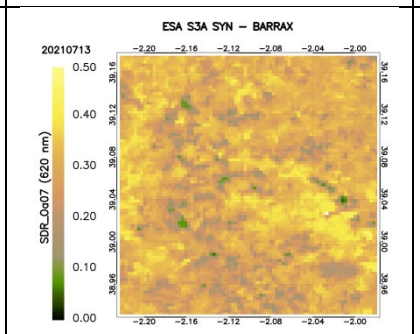
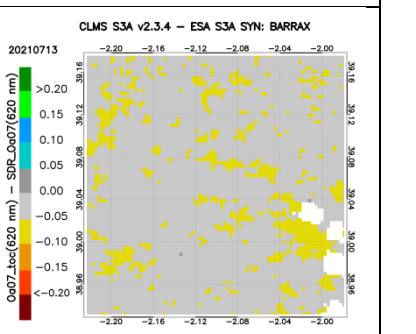
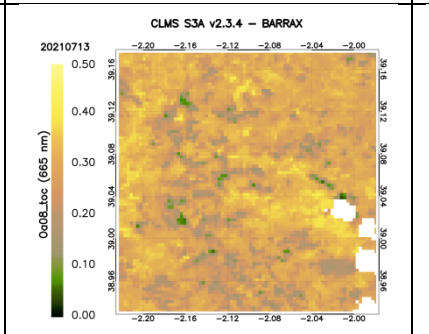
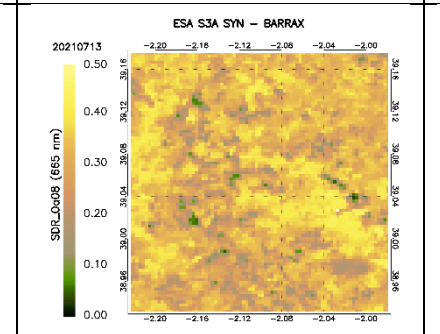
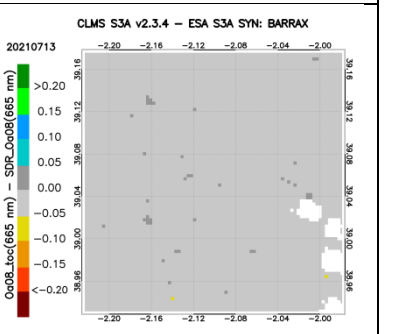
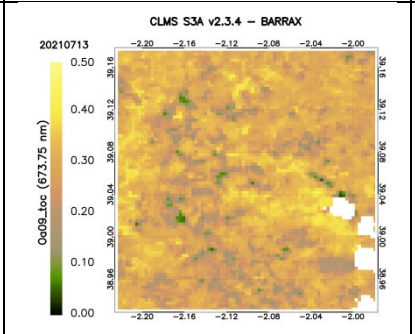
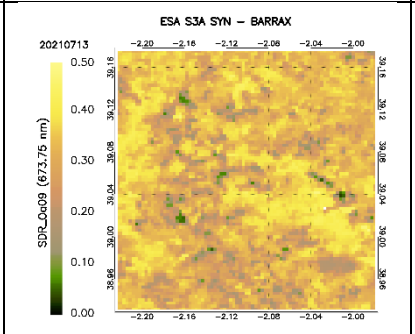
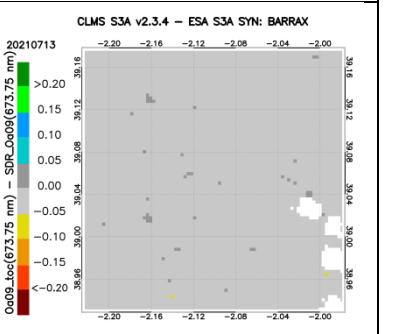
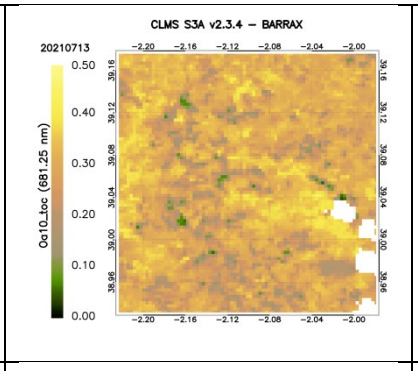
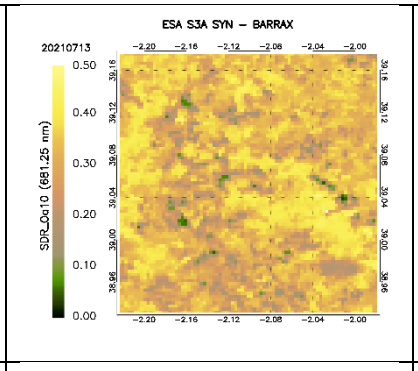
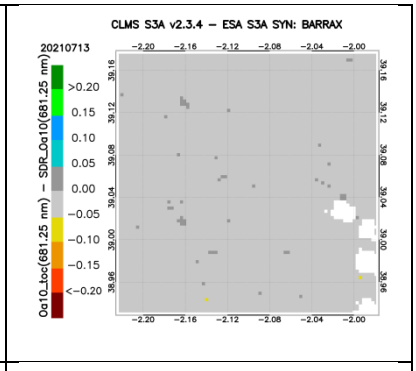
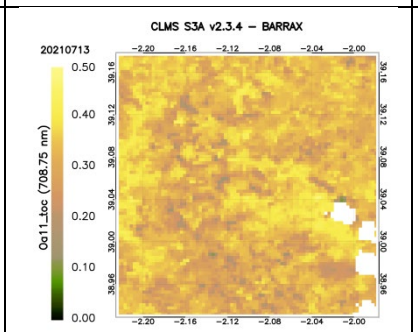
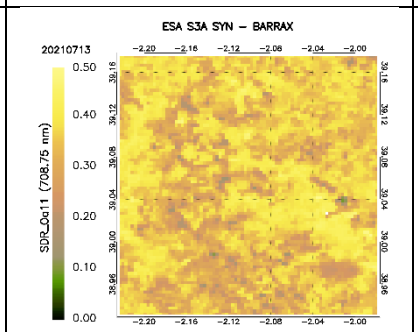
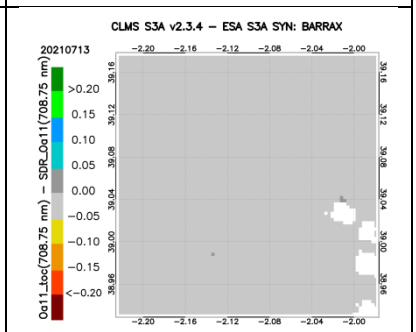
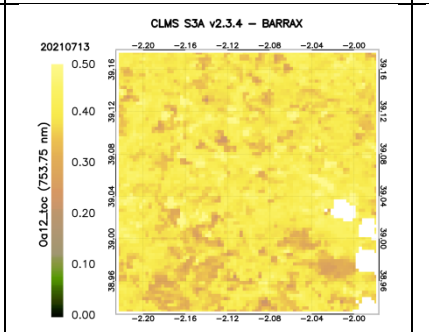
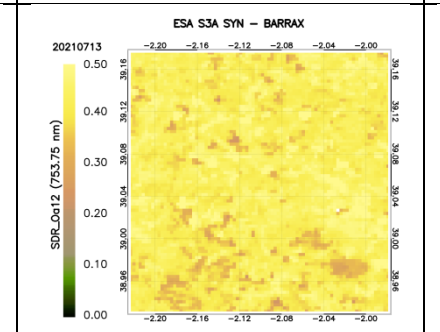
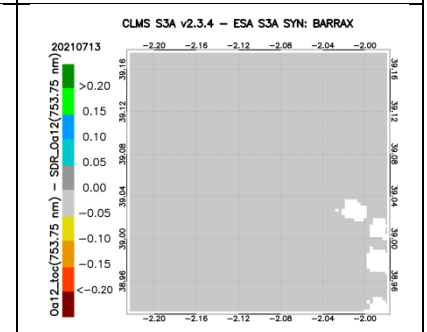
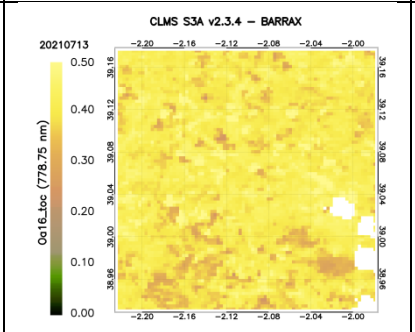
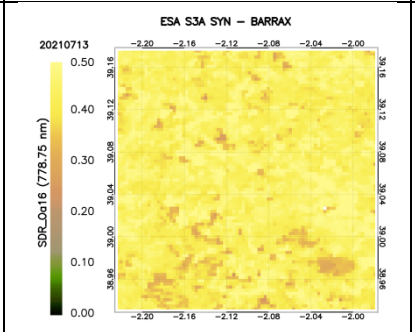
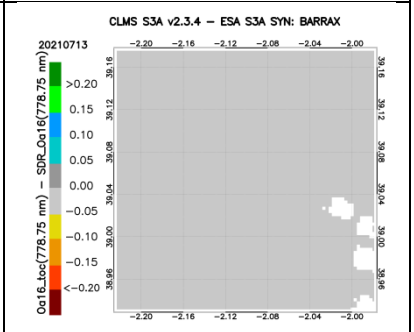
S6			

Figure 105: Spatial consistency of Sentinel-3B (2021.07.28) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#2 (Tjamaja, Russia – DBF). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#3 BARRAX (SPAIN) – CULTIVATED

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

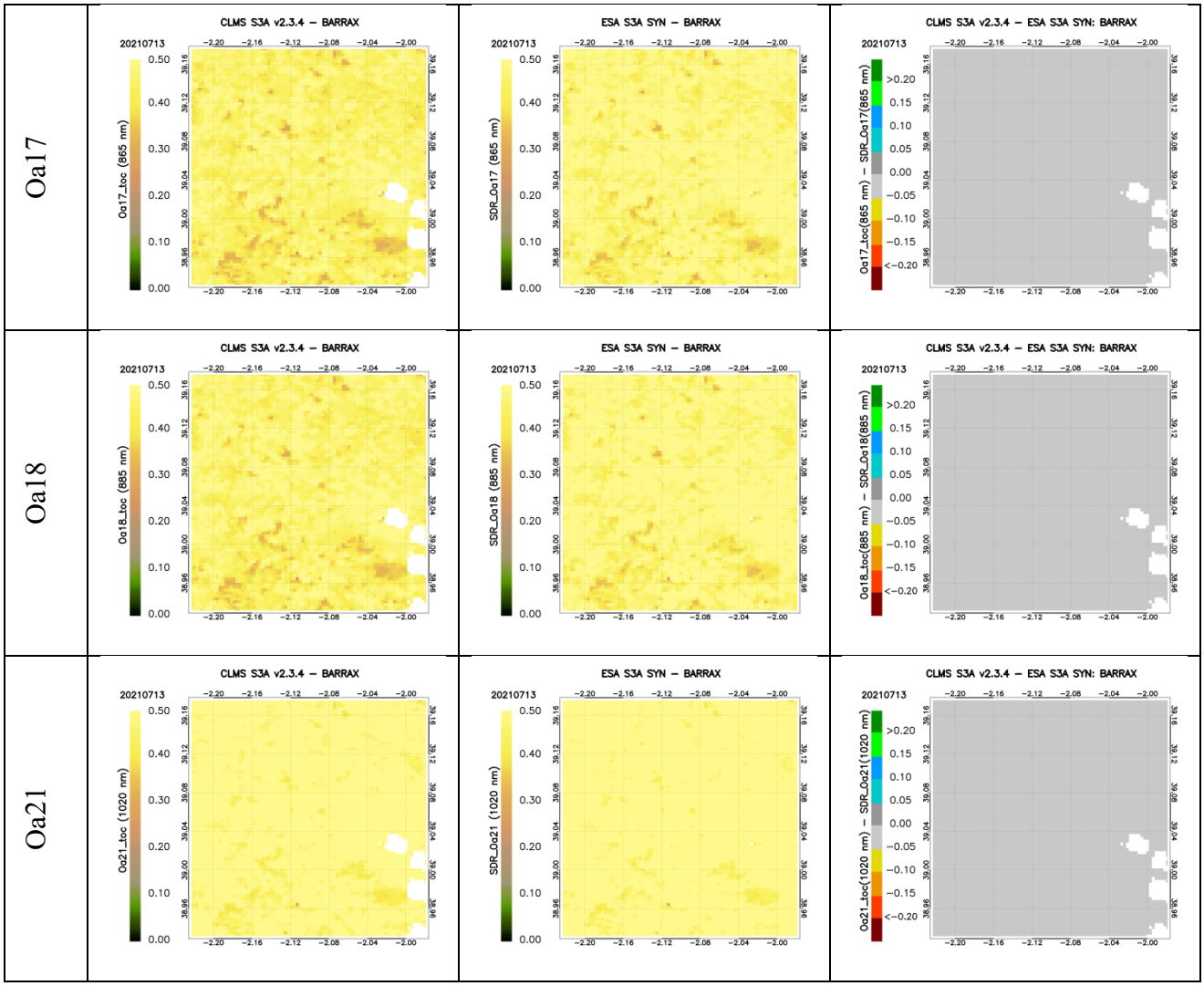
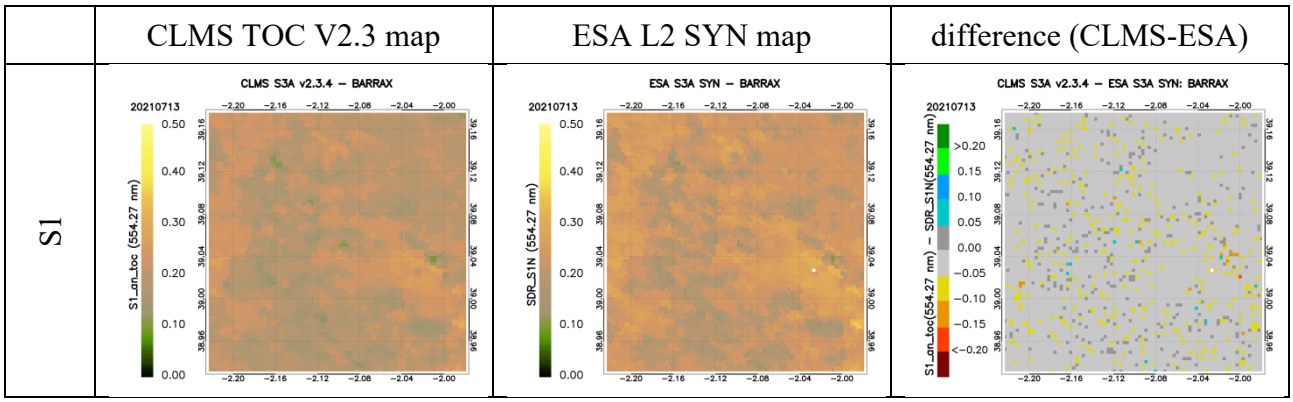


Figure 106: Spatial consistency of Sentinel-3A (2021.07.13) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#3 (Barrax, Spain – Cultivated). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).



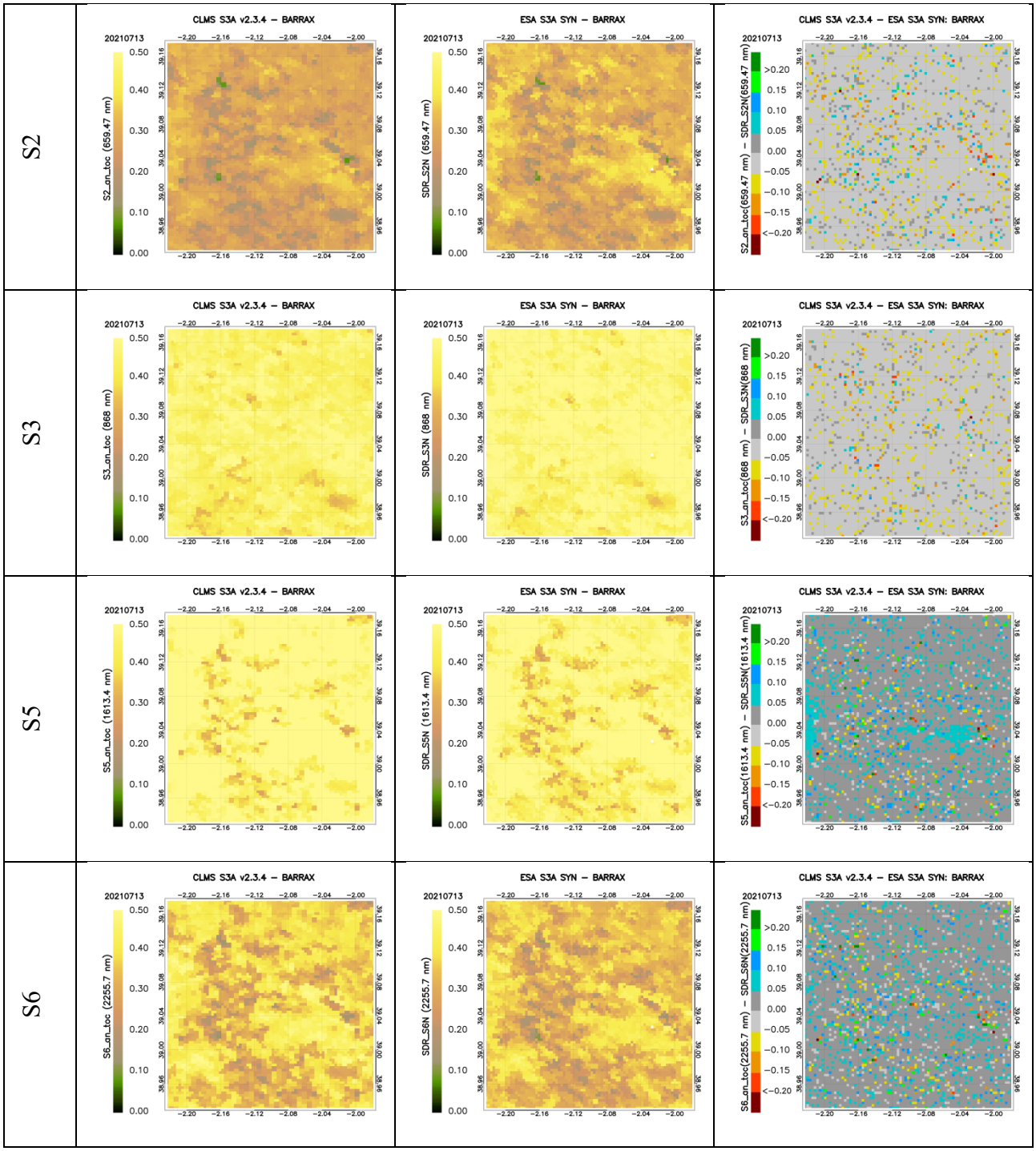
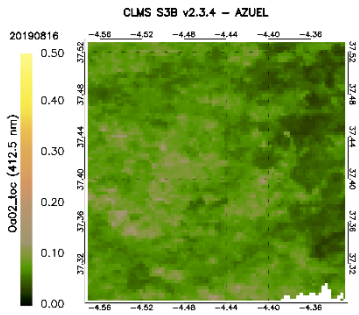
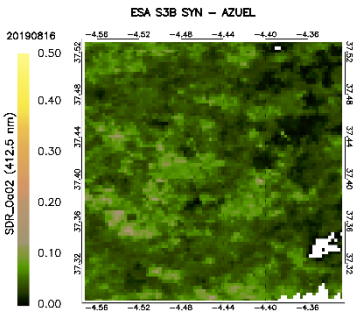
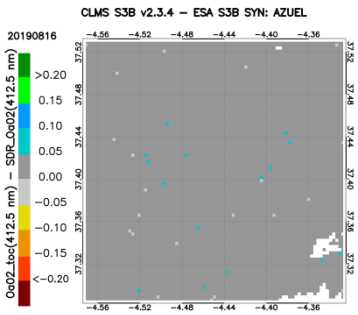
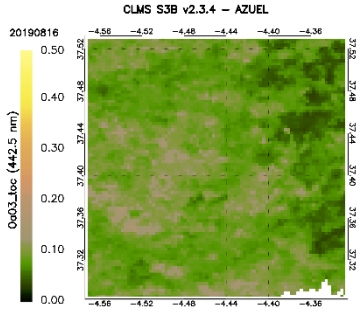
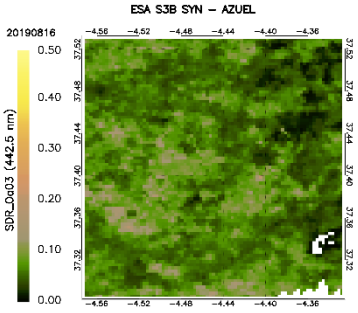
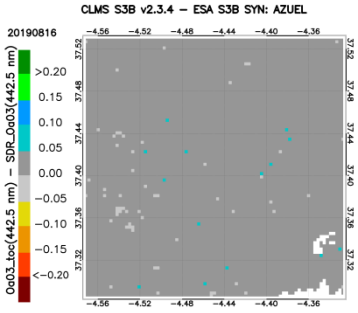
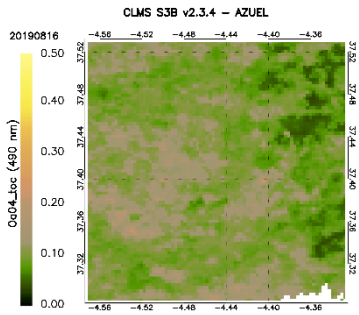
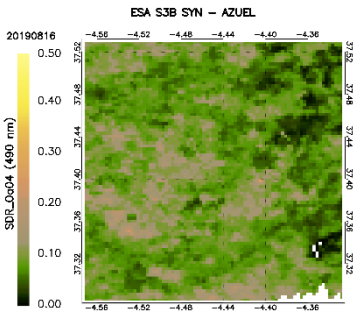
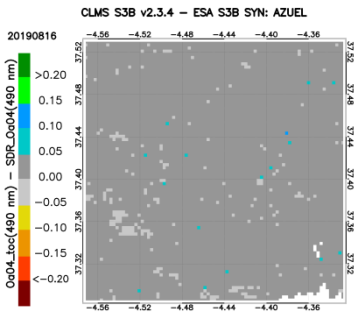
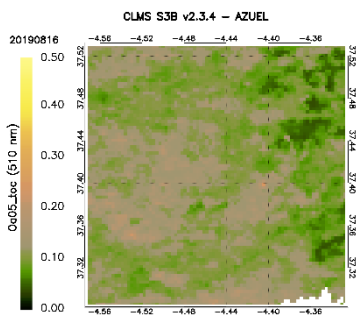
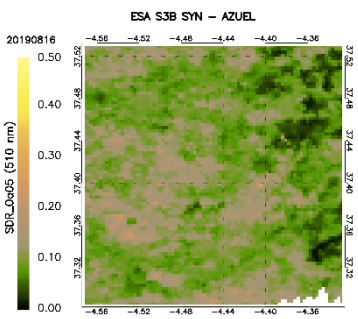
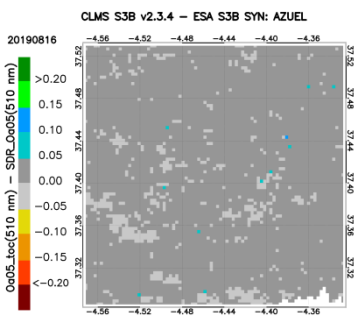
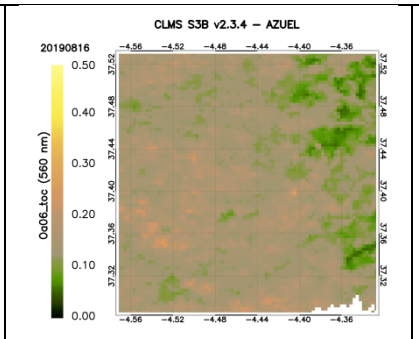
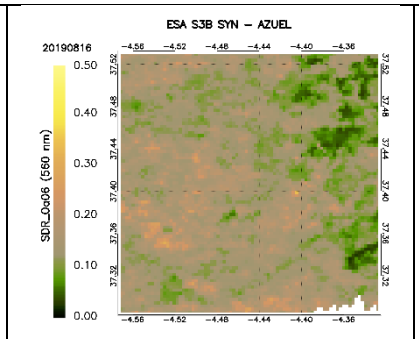
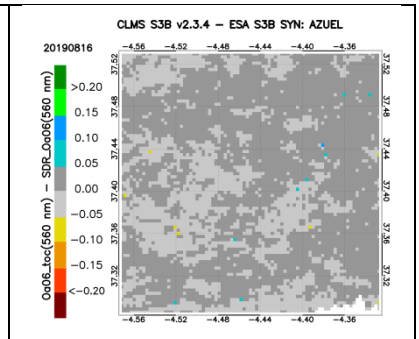
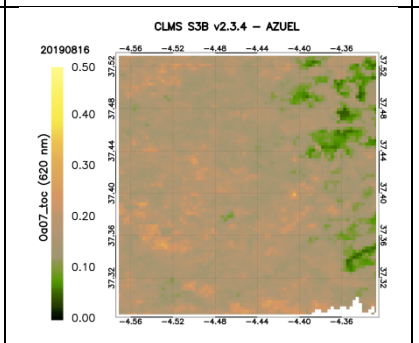
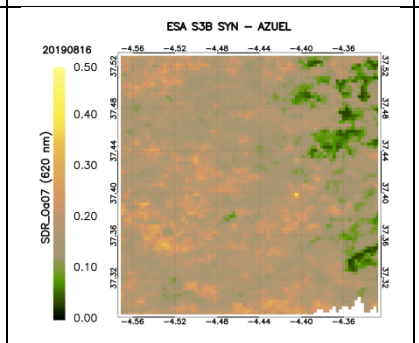
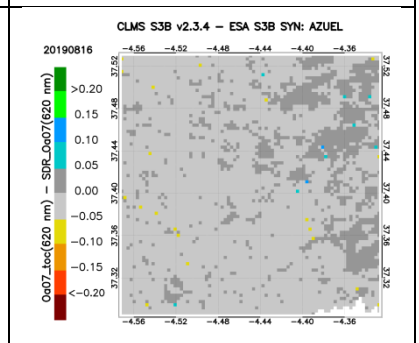
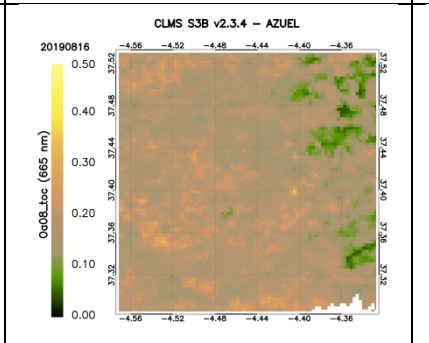
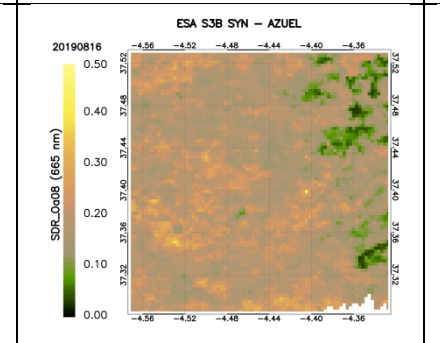
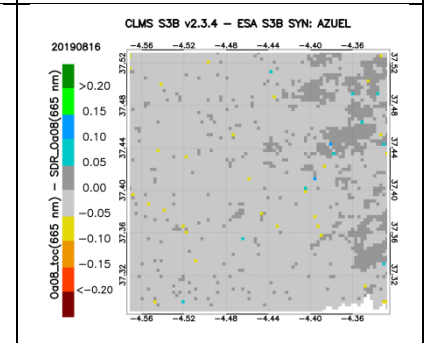
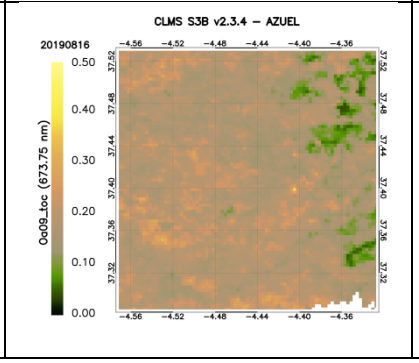
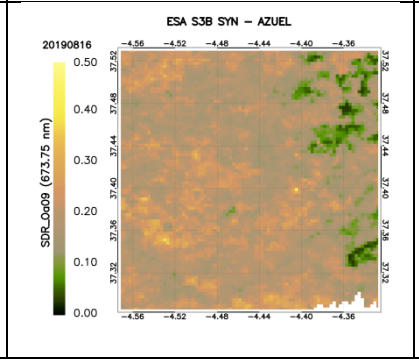
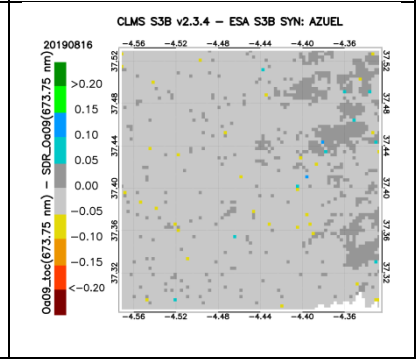
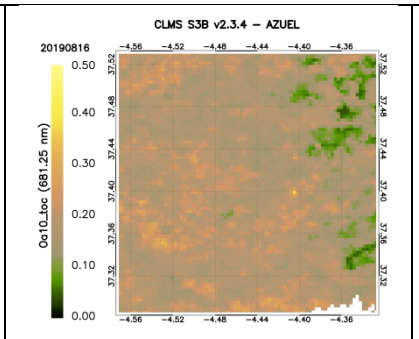
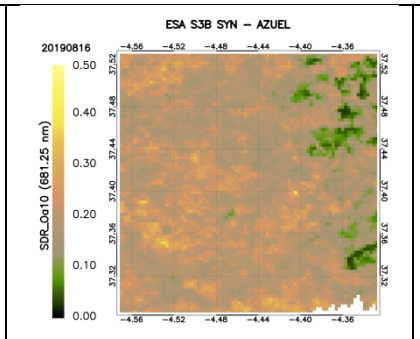
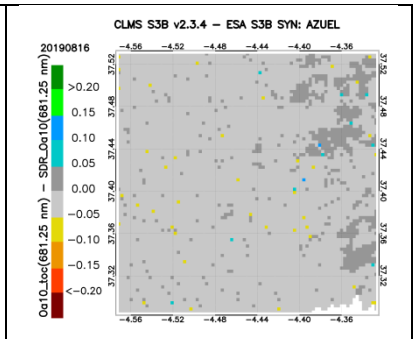
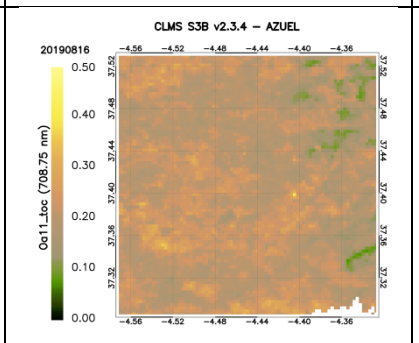
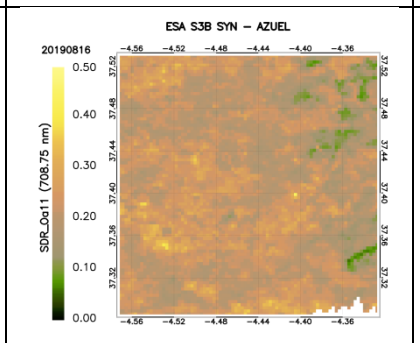
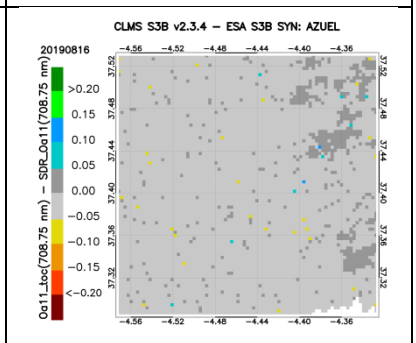
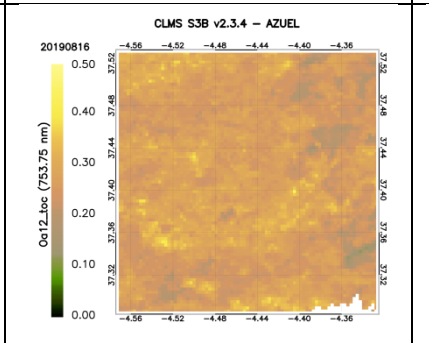
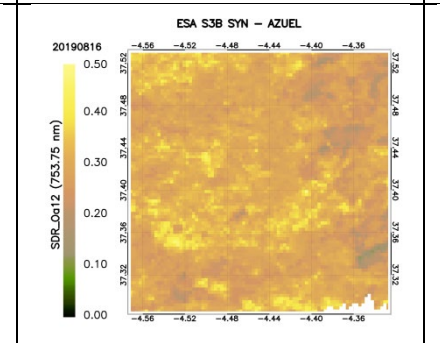
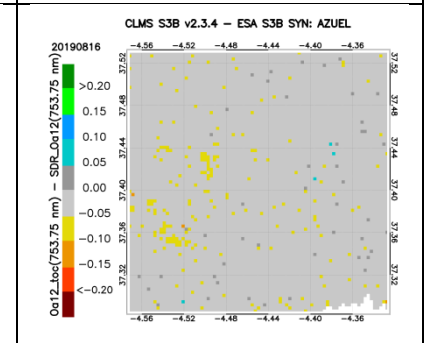
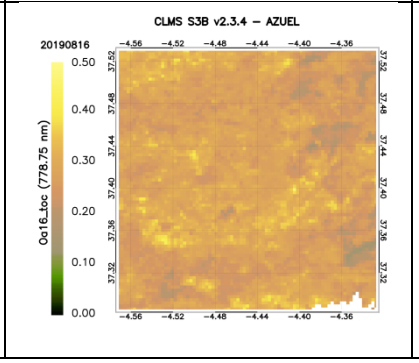
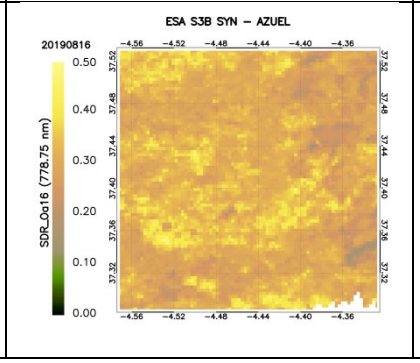
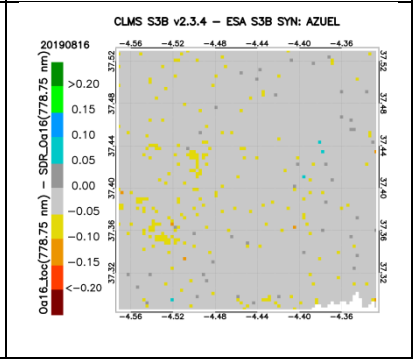


Figure 107: Spatial consistency of Sentinel-3A (2021.07.13) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#3 (Barrax, Spain – Cultivated). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#4 AZUEL (SPAIN) – HERBACEOUS

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

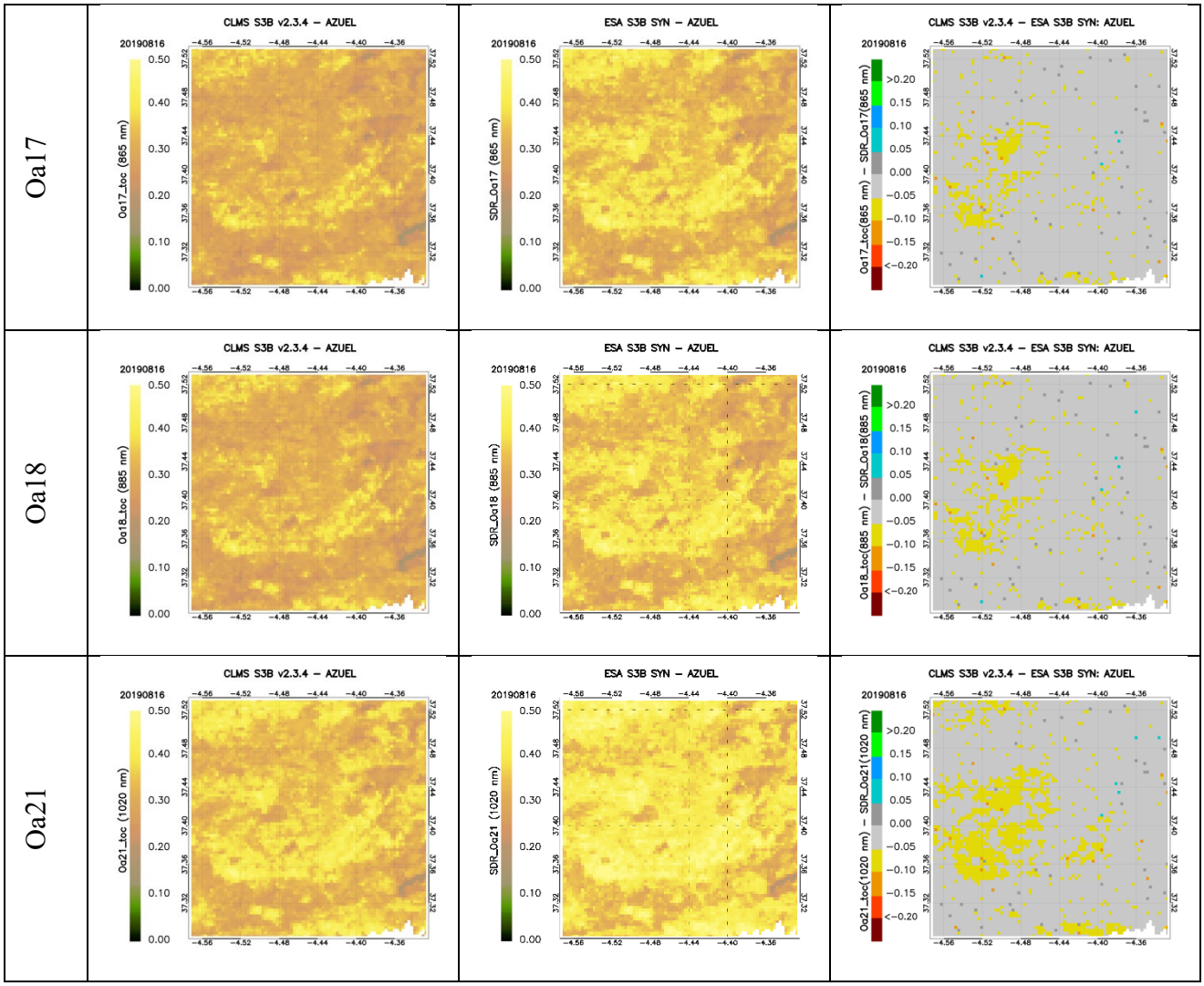
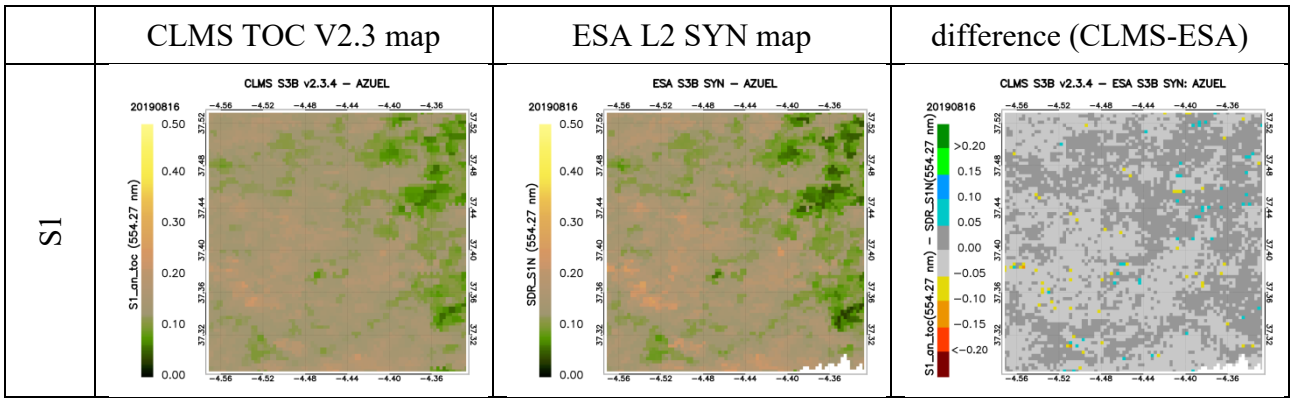


Figure 108: Spatial consistency of Sentinel-3B (2019.08.16) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#4 (Azuel, Spain – Herbaceous). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).



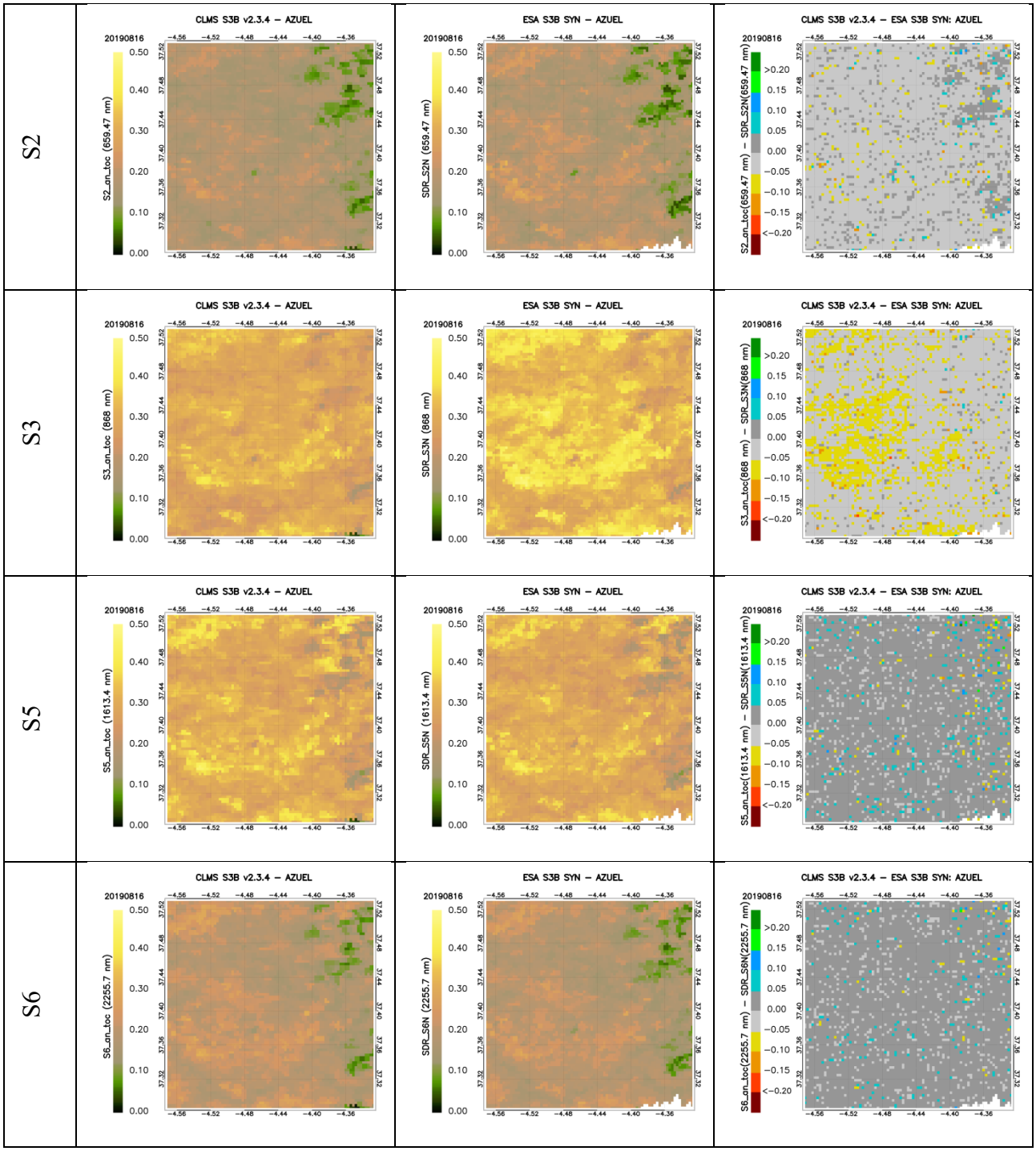
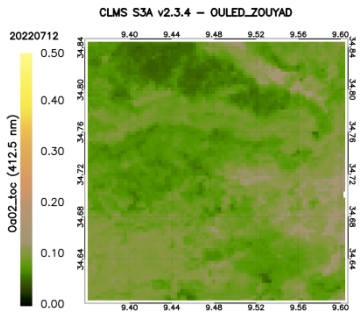
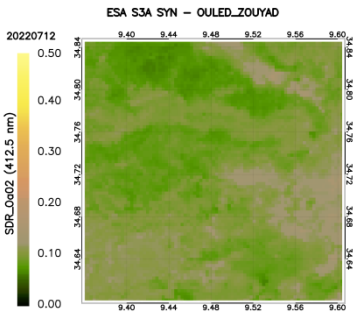
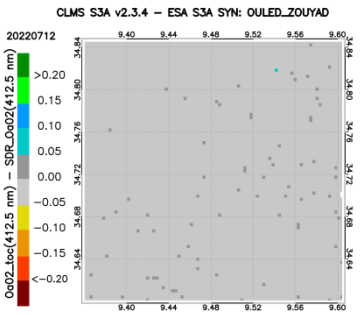
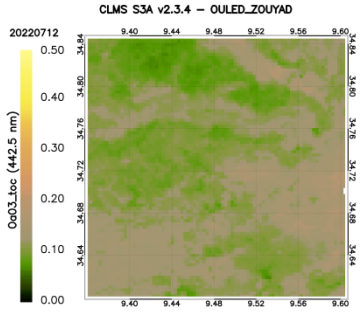
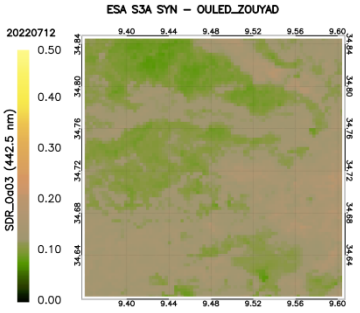
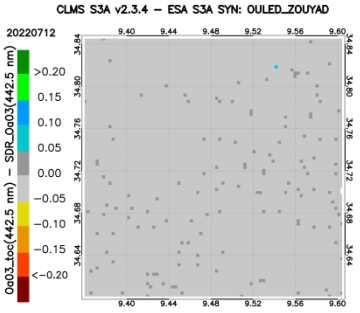
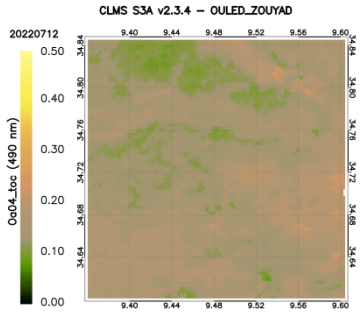
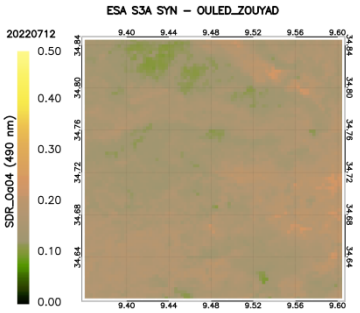
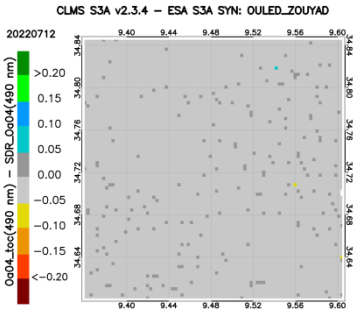
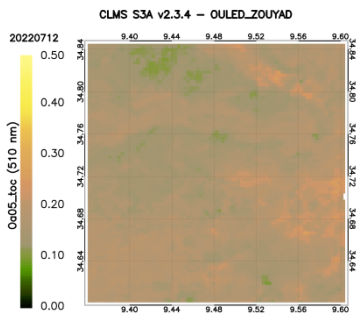
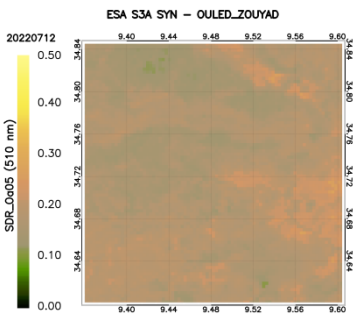
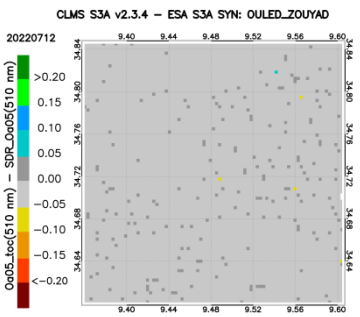
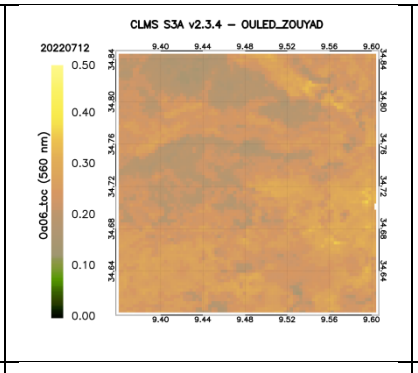
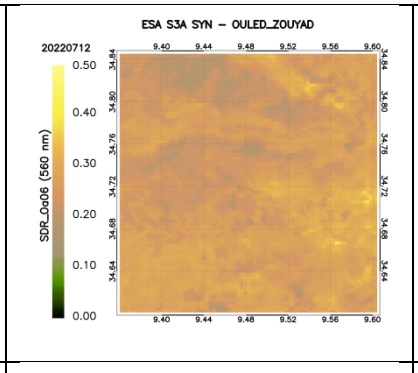
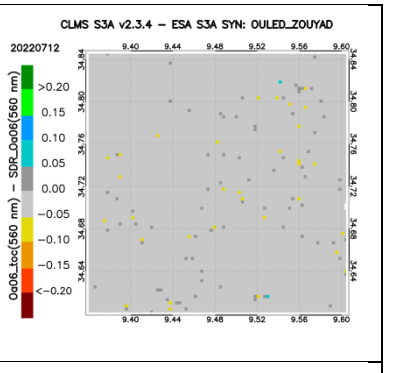
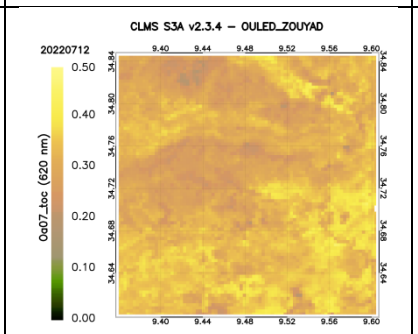
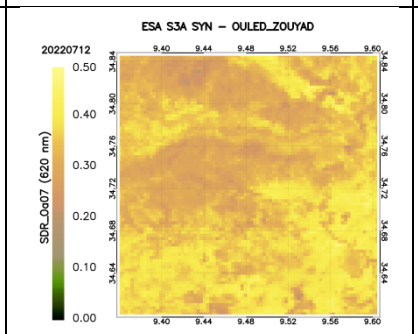
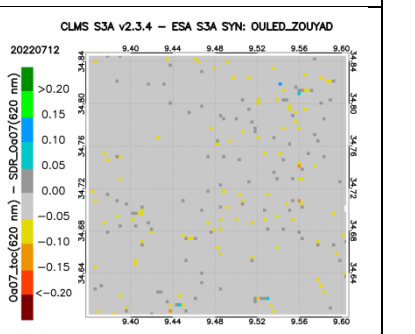
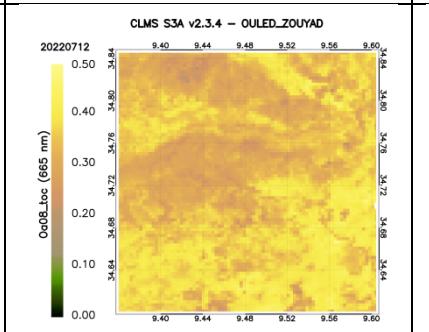
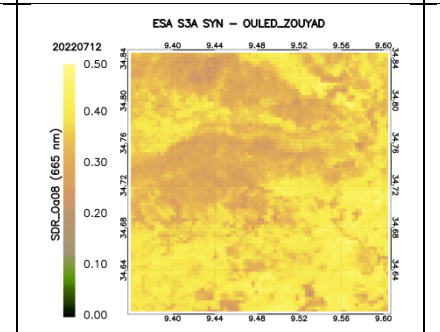
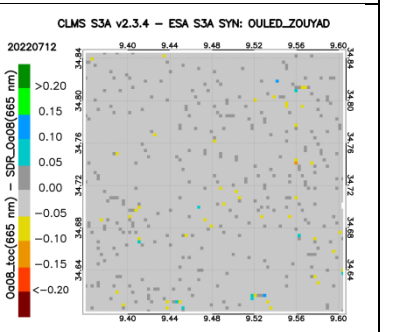
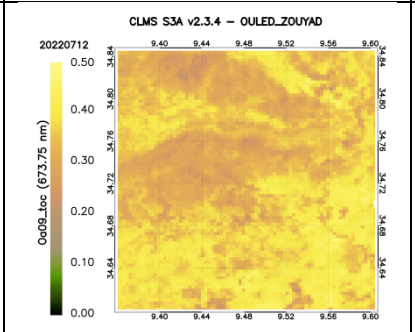
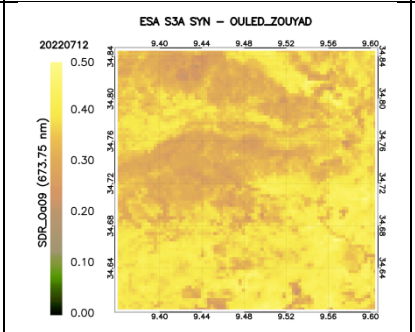
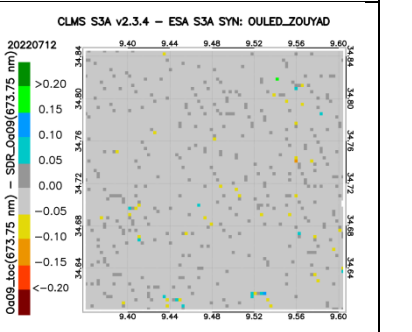
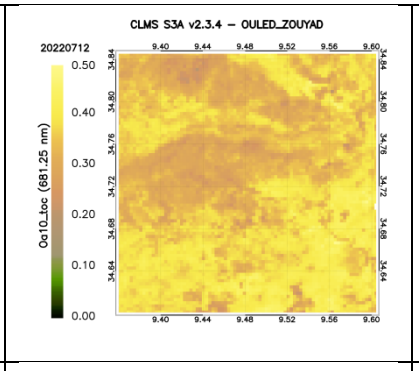
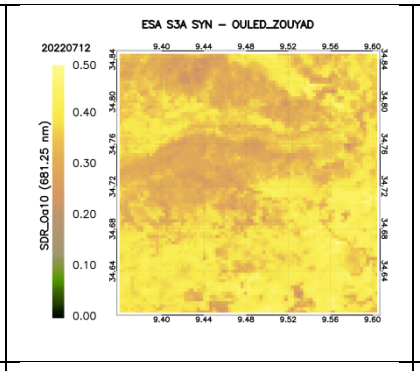
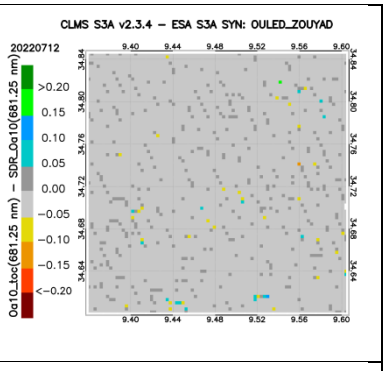
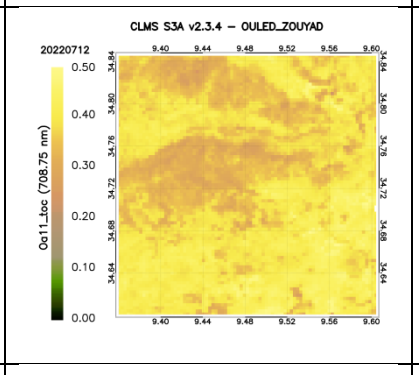
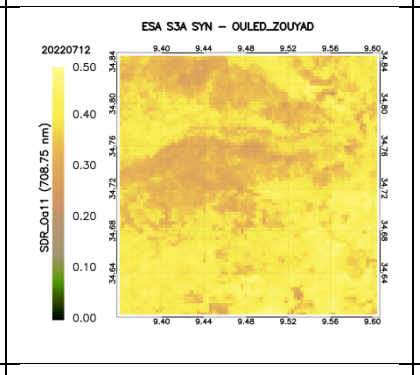
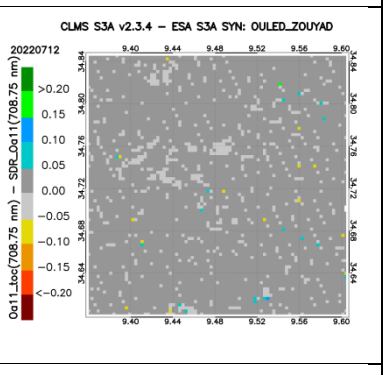
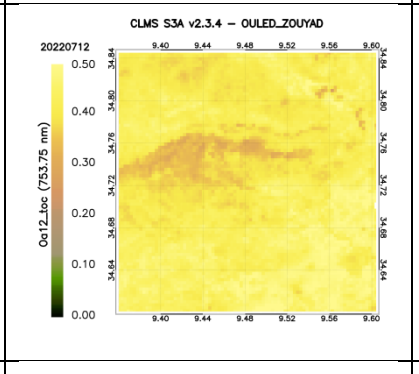
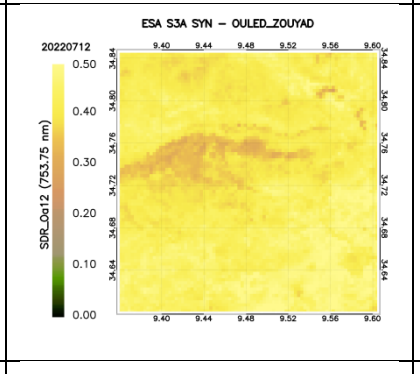
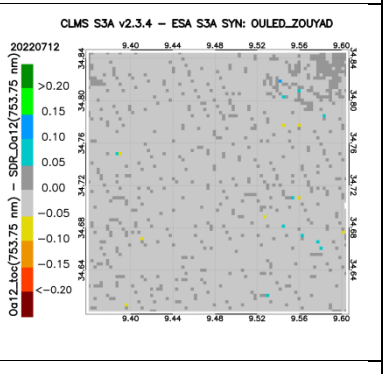
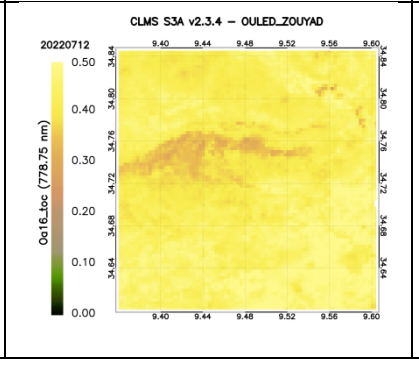
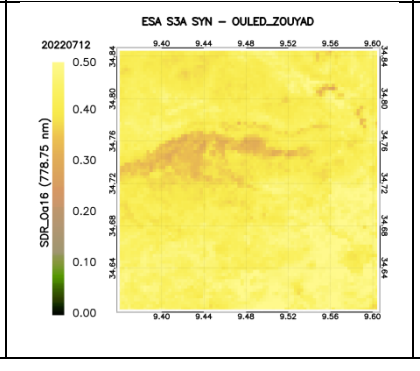
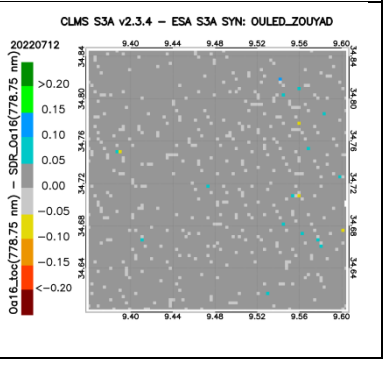


Figure 109: Spatial consistency of Sentinel-3B (2019.08.16) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#4 (Azuel, Spain – Herbaceous). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#5 OULED ZOUYAD (TUNISIA) – SHRUBLANDS

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

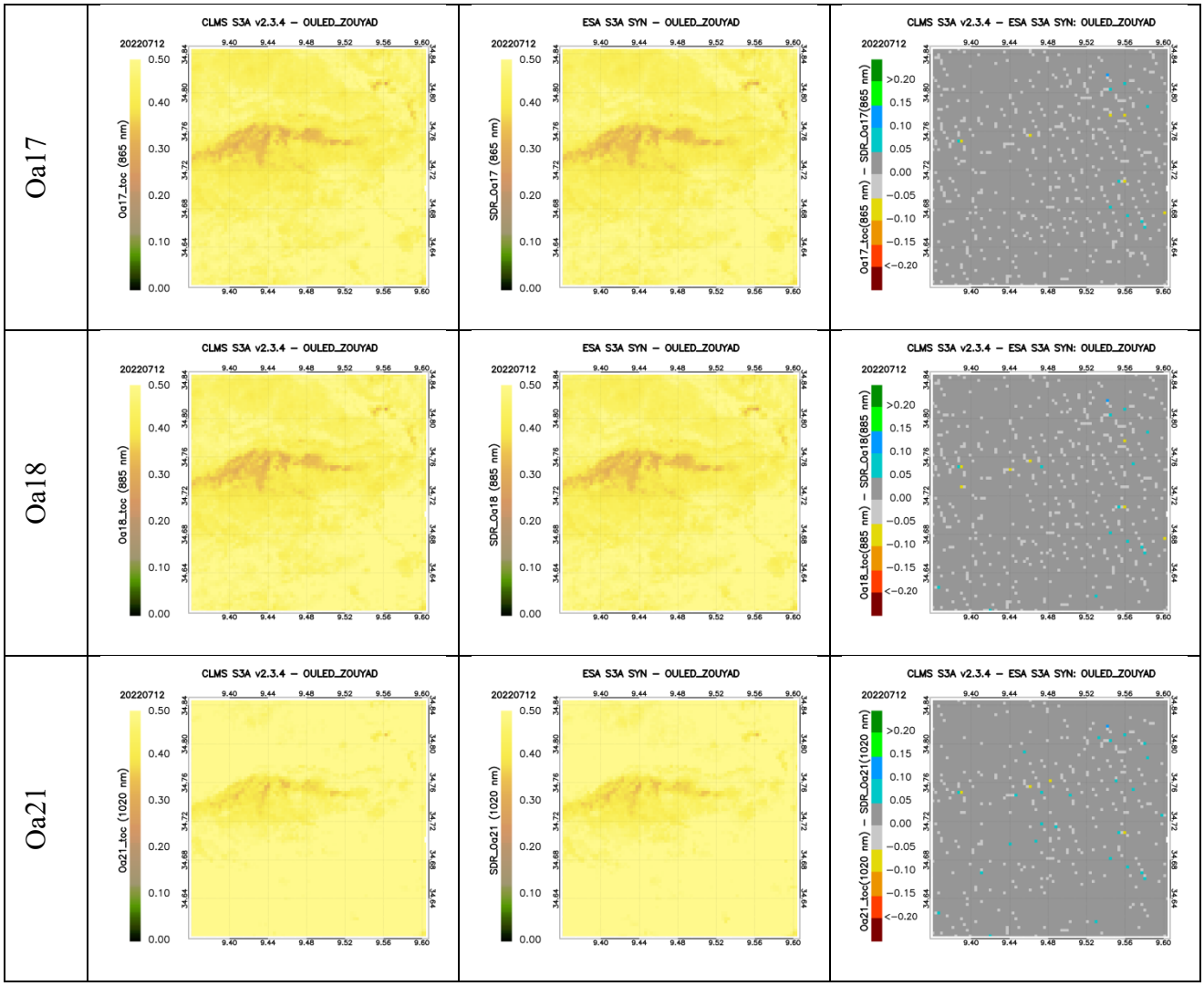
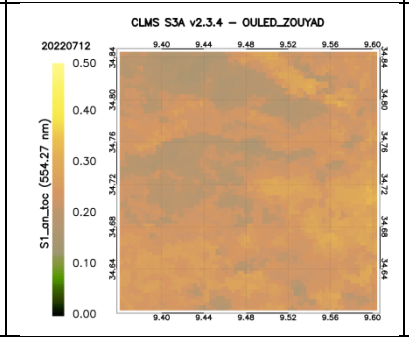
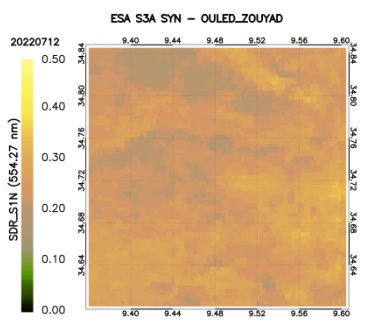
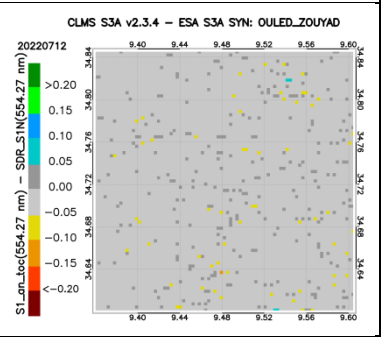


Figure 110: Spatial consistency of Sentinel-3A (2022.07.12) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#5 (Ouled Zouyad, Tunisia – Shrublands). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
S1			

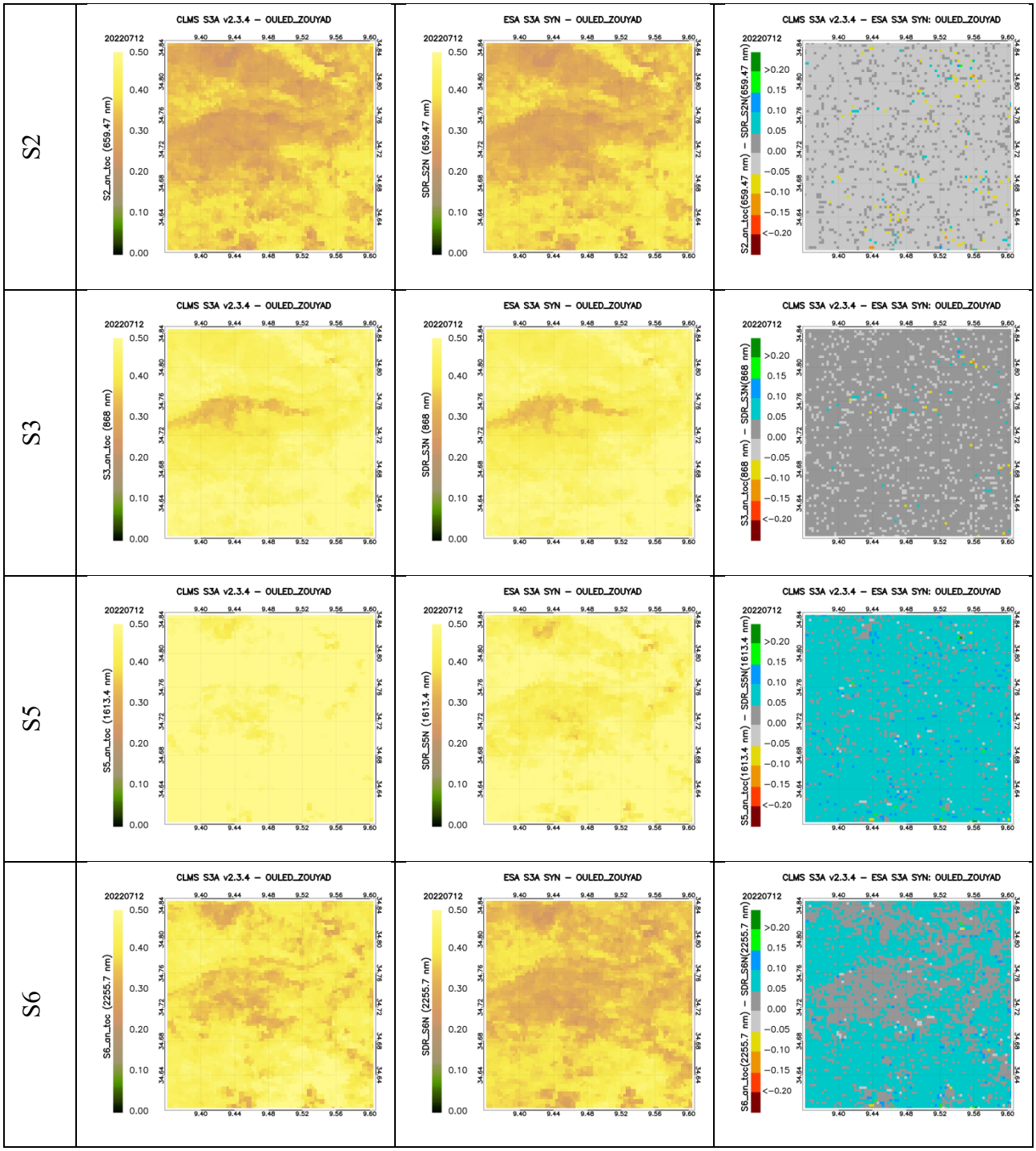
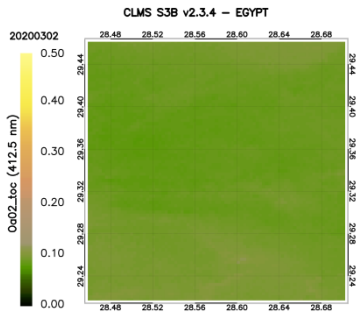
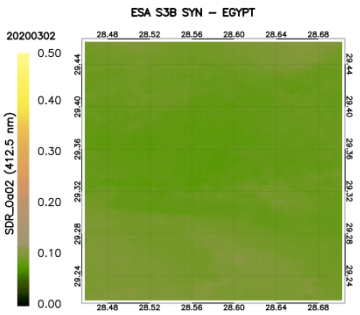
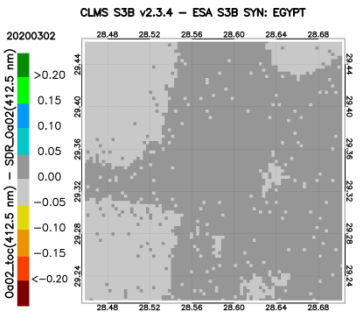
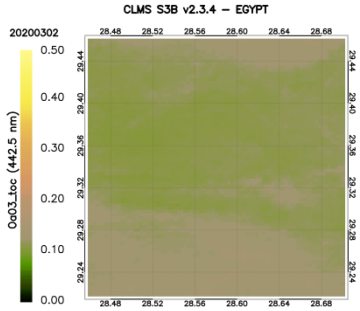
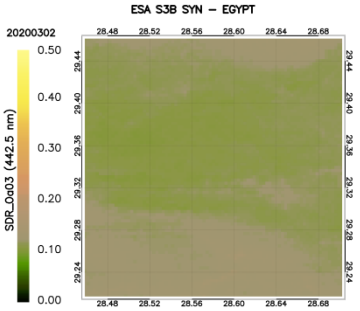
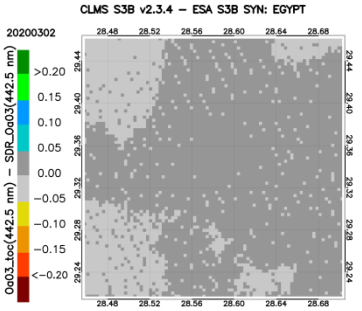
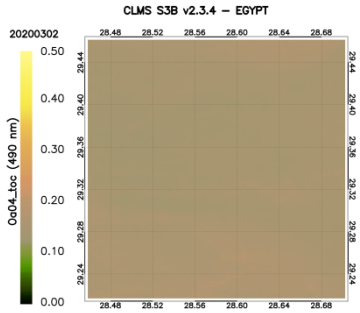
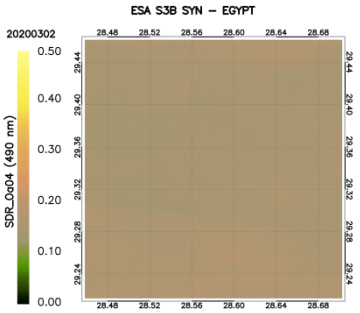
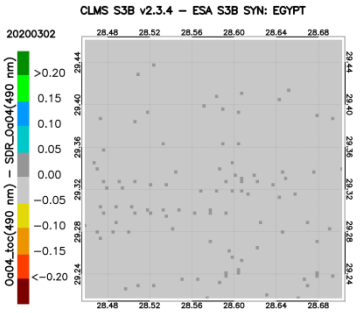
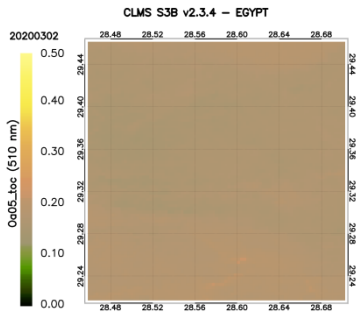
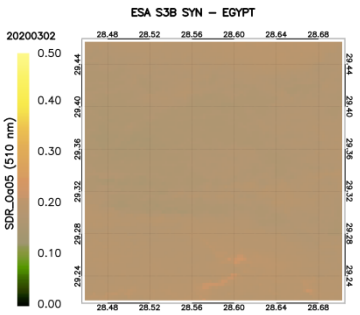
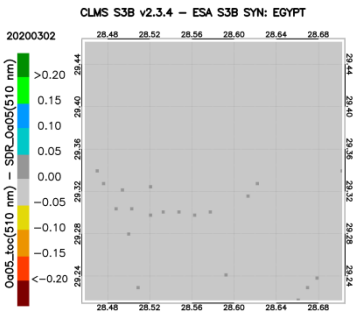
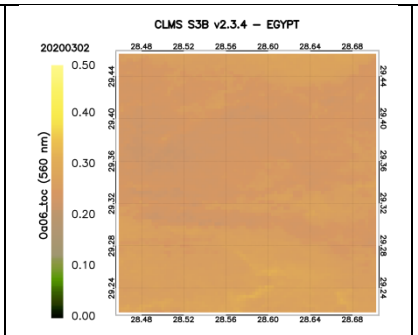
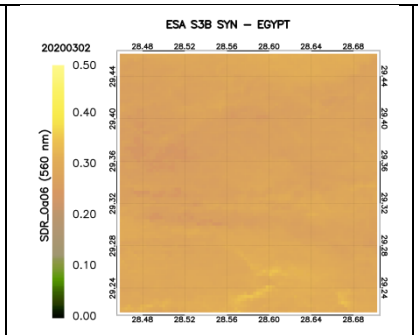
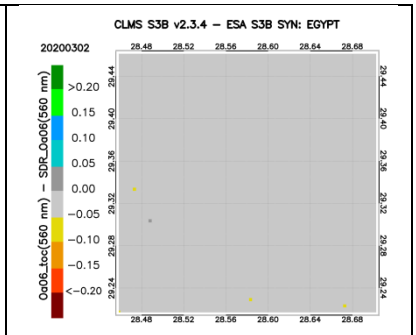
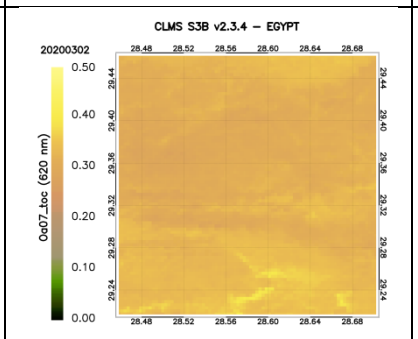
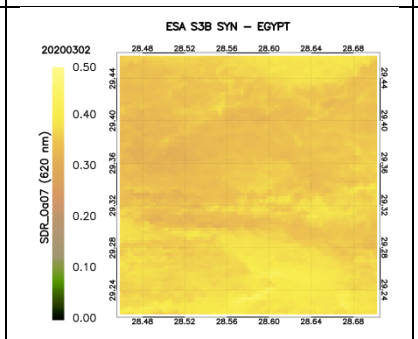
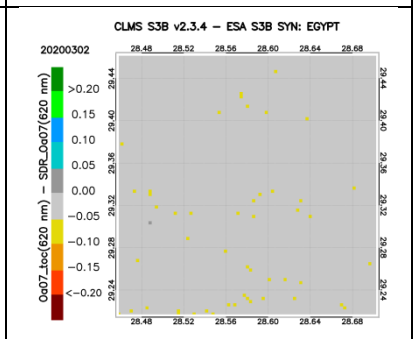
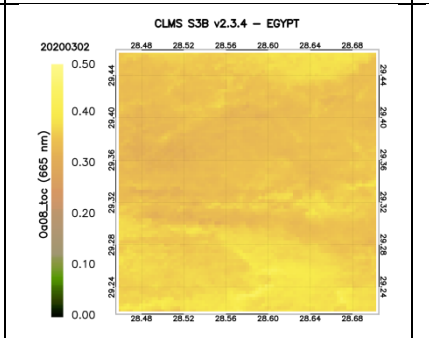
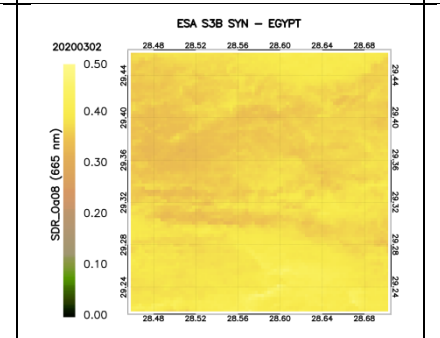
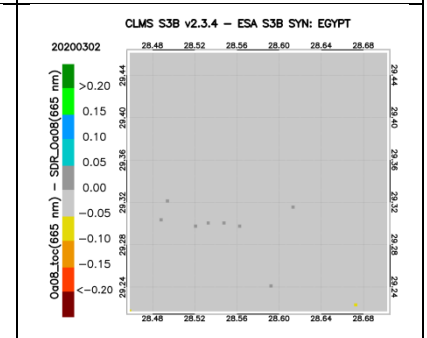
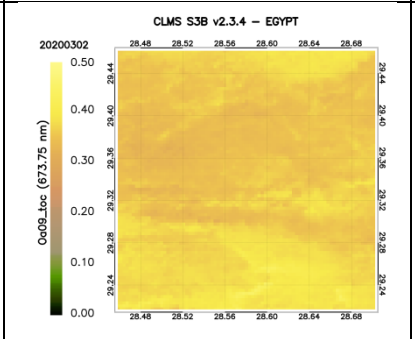
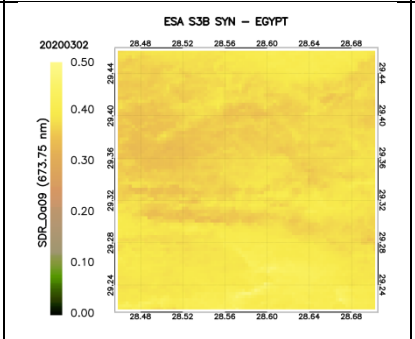
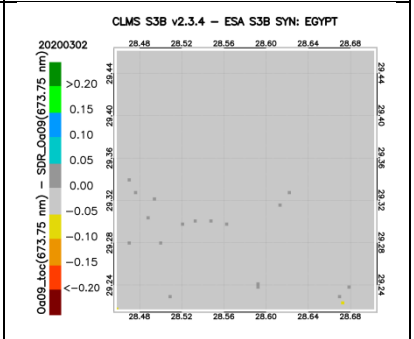
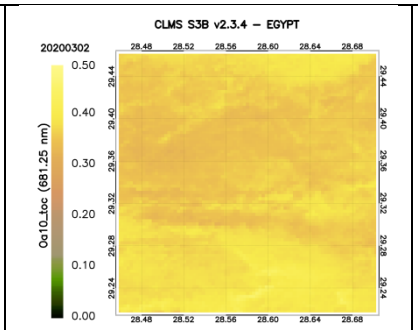
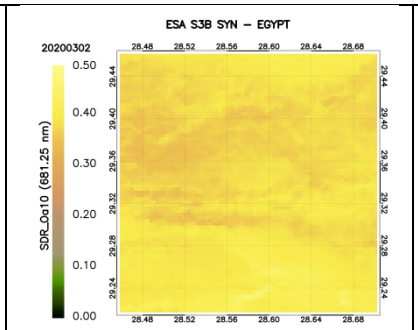
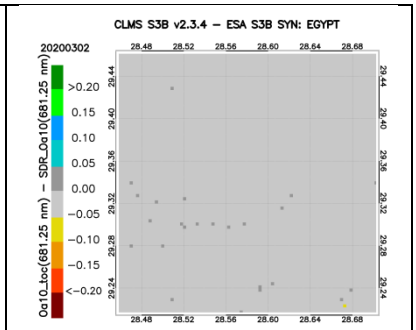
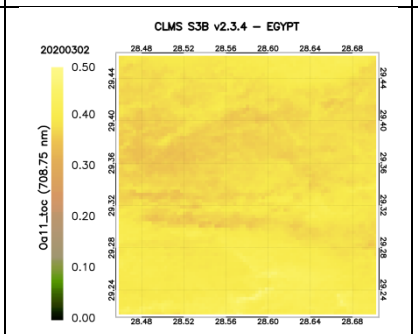
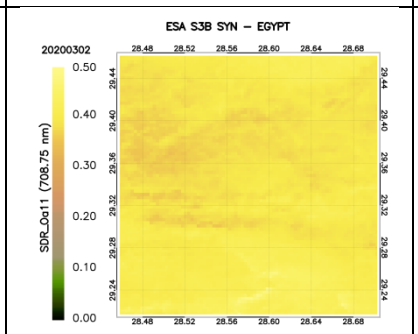
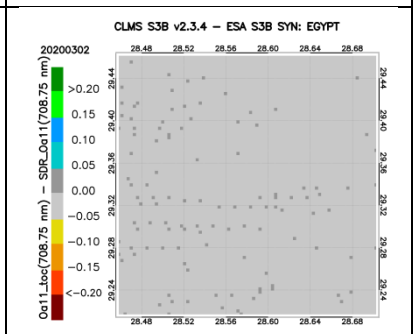
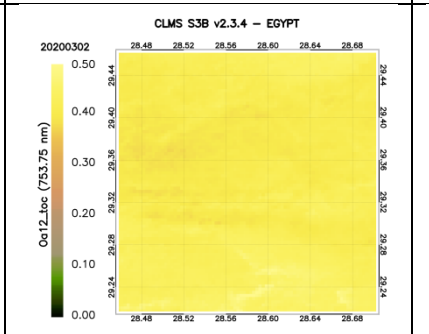
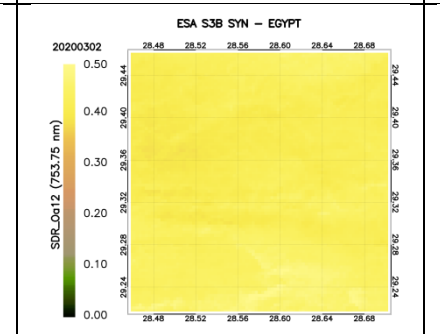
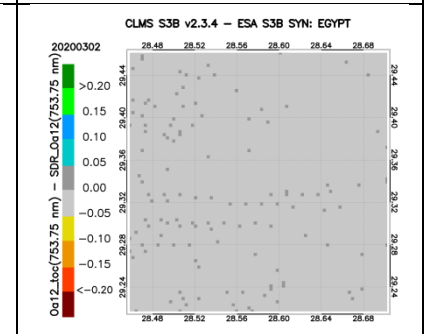
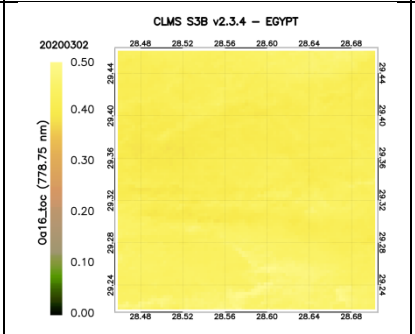
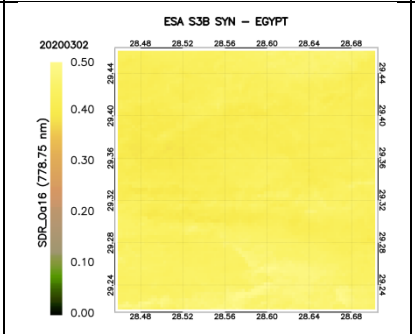
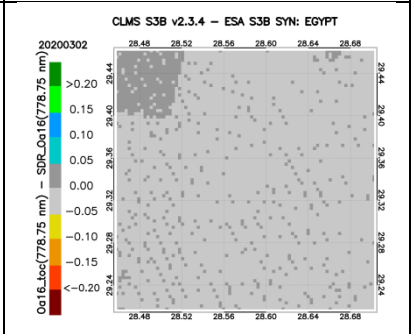


Figure 111: Spatial consistency of Sentinel-3A (2022.07.12) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#5 (Ouled Zouyad, Tunisia – Shrublands). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#6 CALIBRATION SITE (EGYPT) – BARE AREAS

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

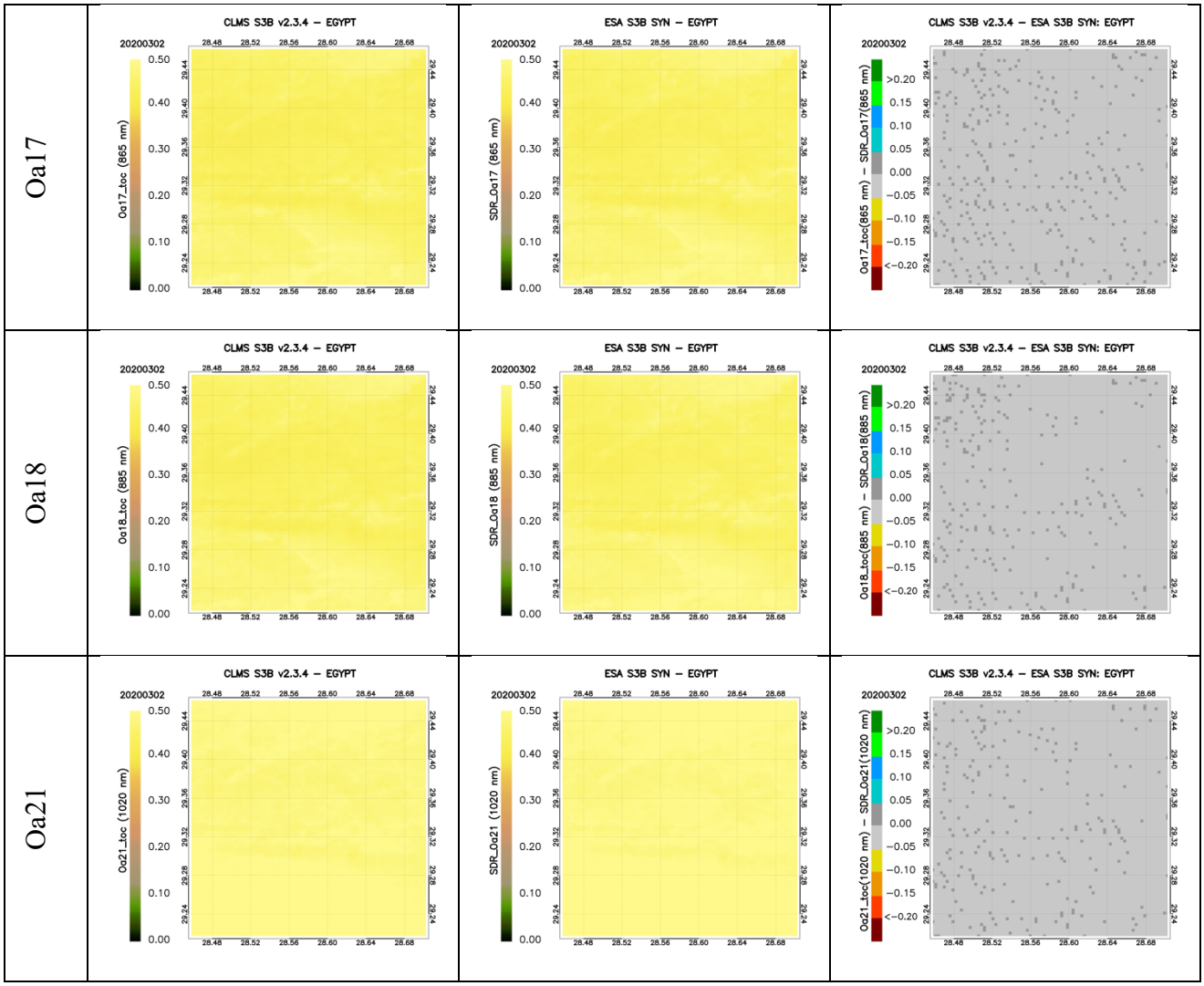
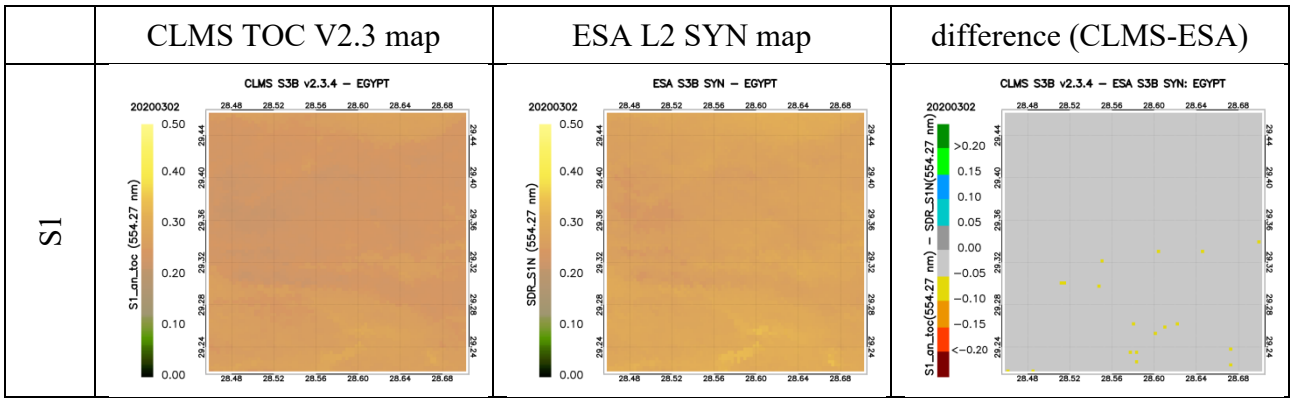


Figure 112: Spatial consistency of Sentinel-3B (2023.03.02) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#6 (Calibration Site, Egypt – Bare Areas). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).



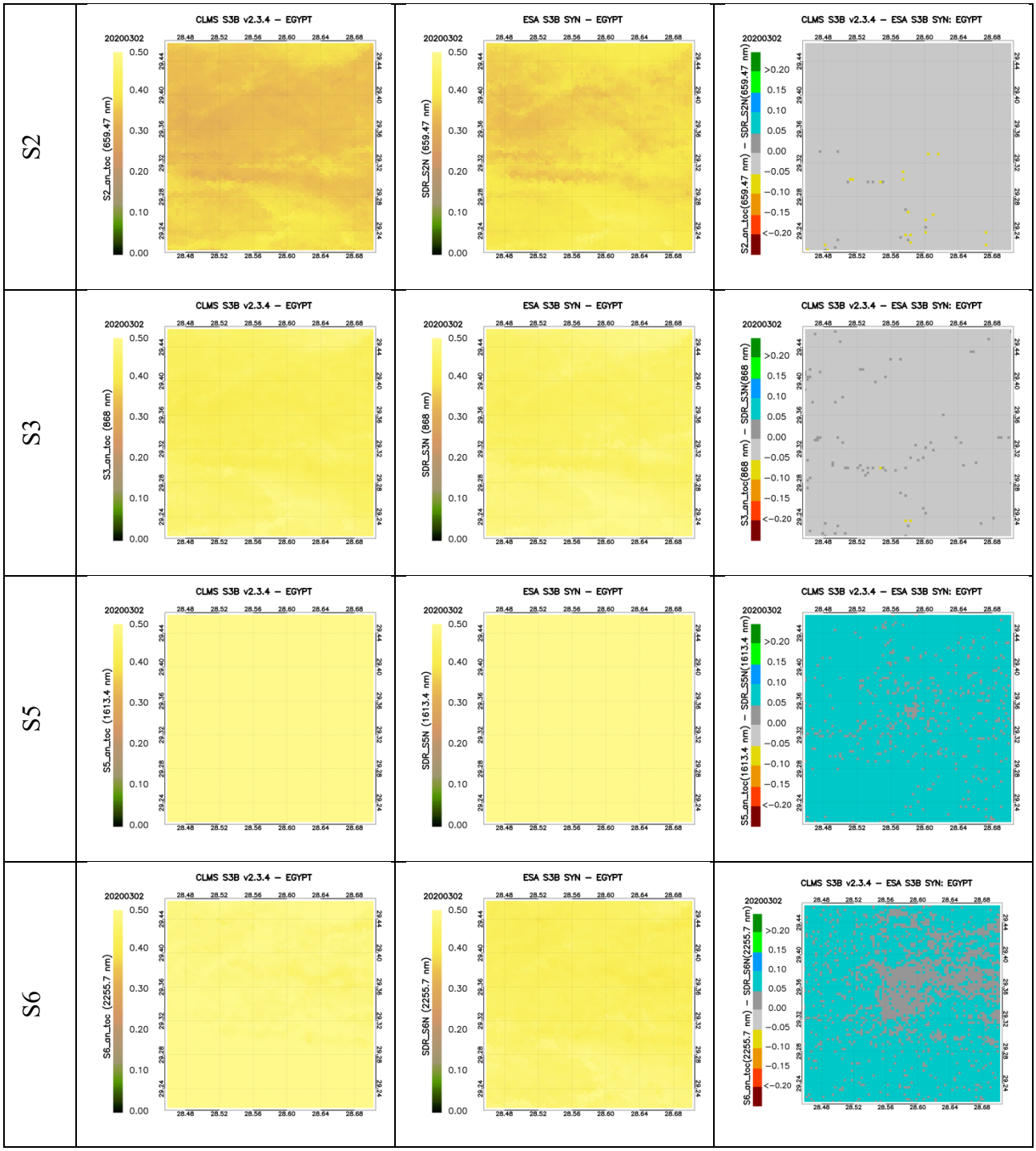
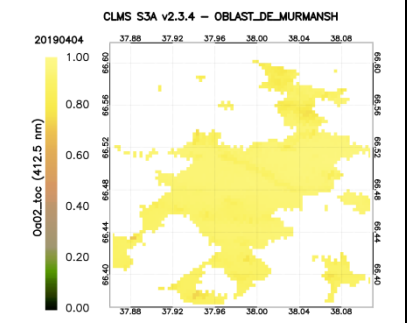
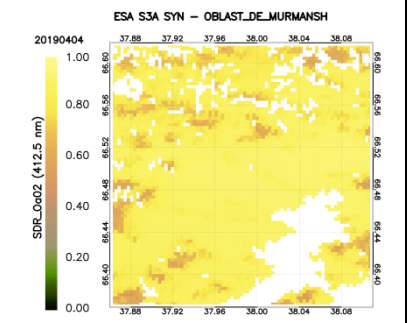
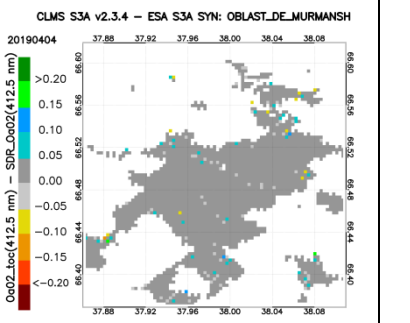
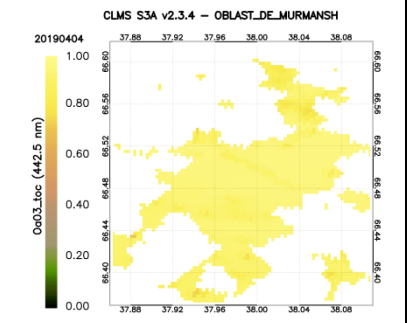
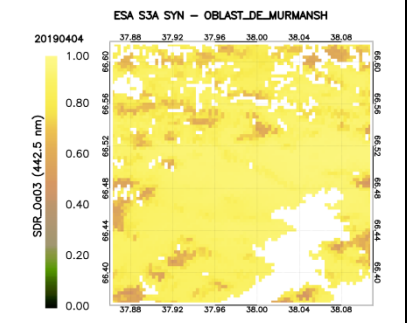
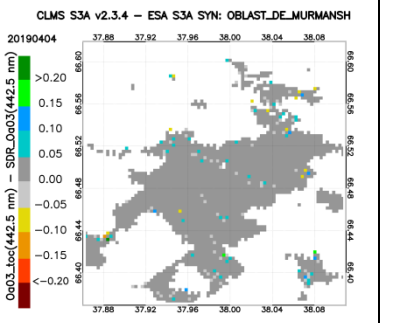
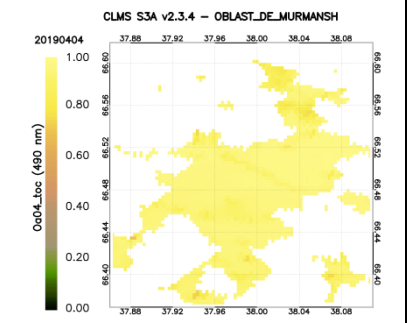
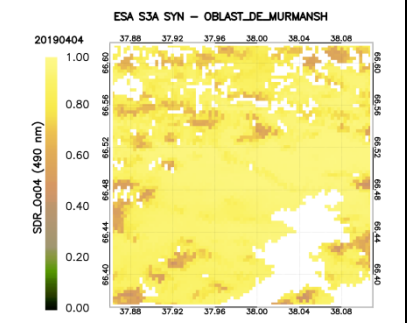
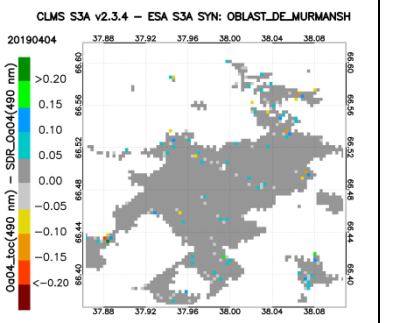
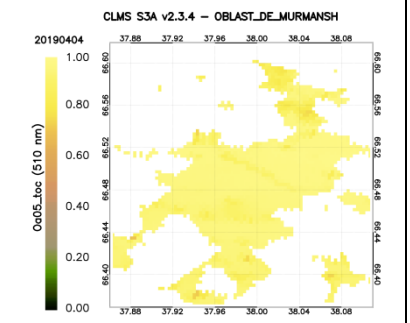
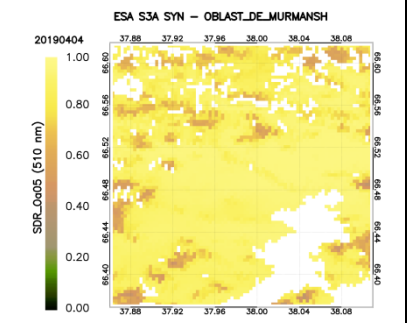
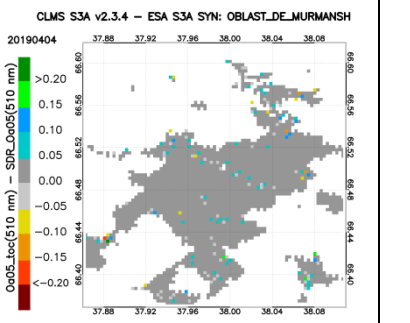
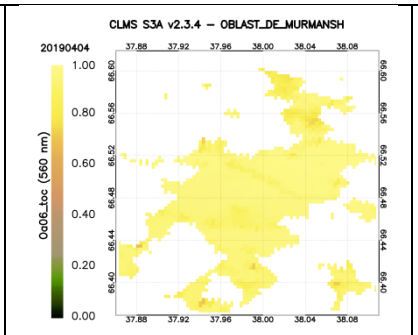
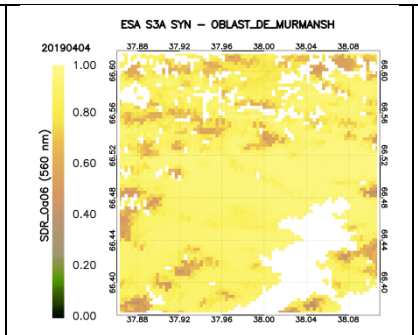
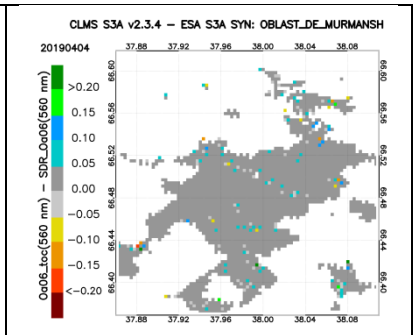
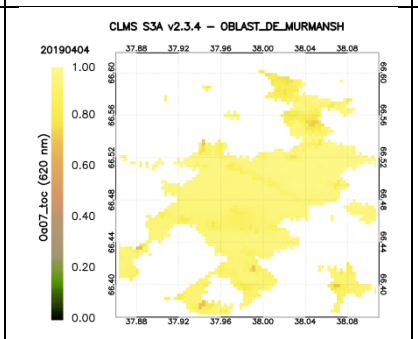
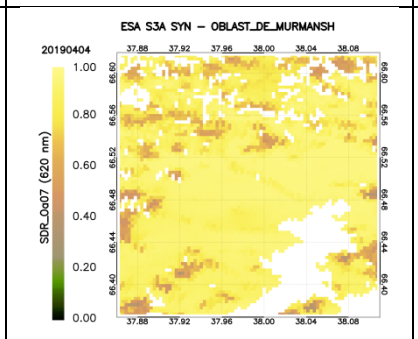
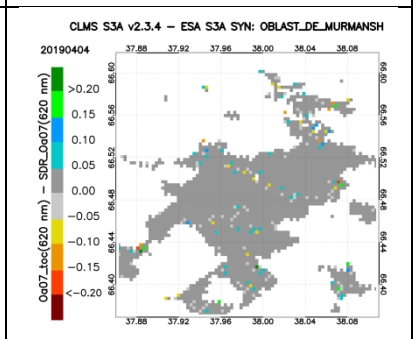
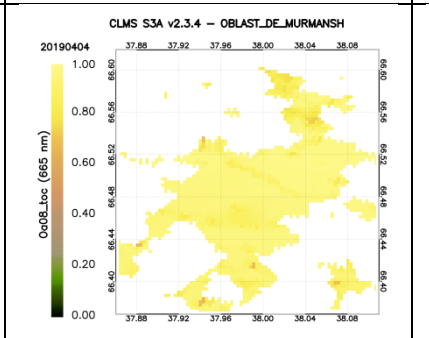
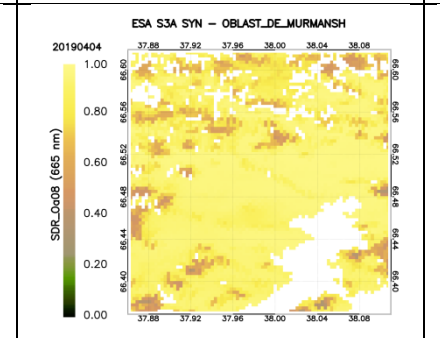
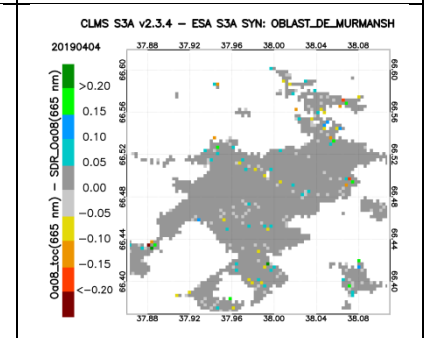
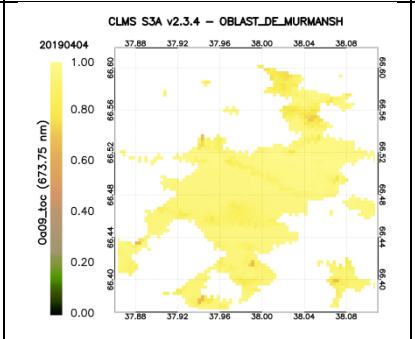
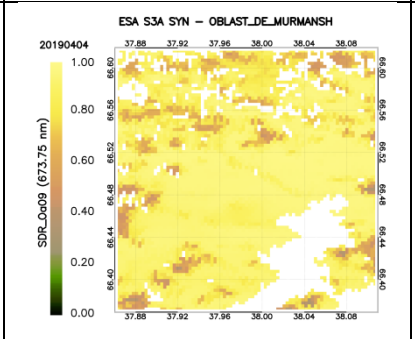
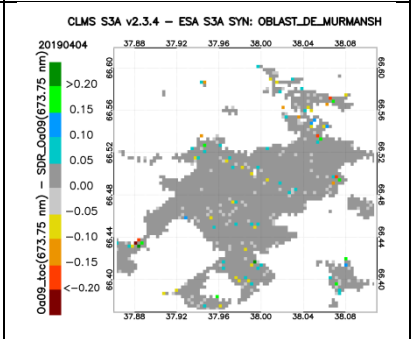
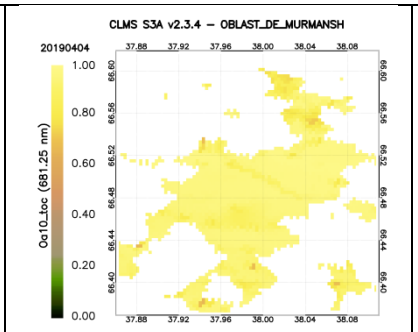
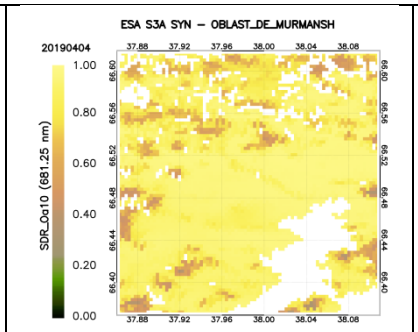
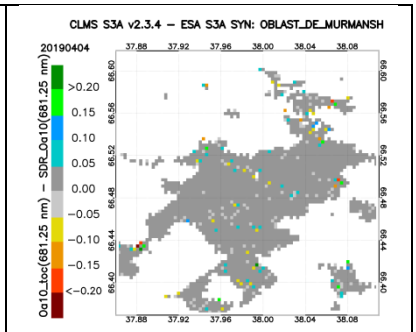
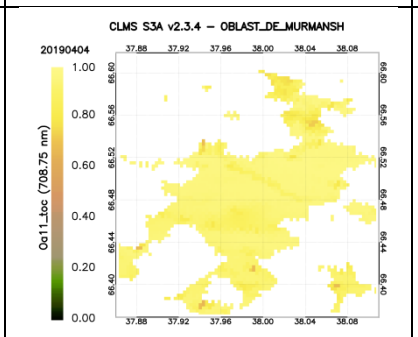
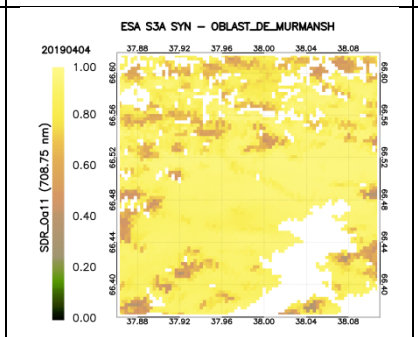
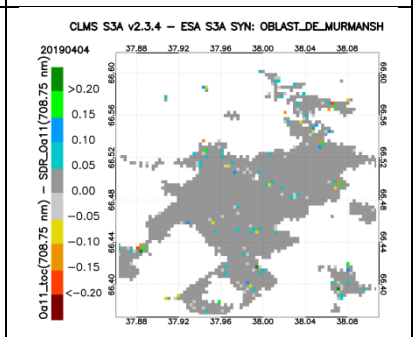
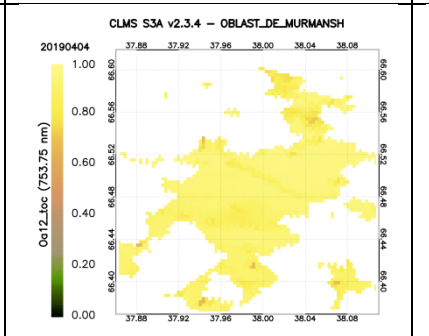
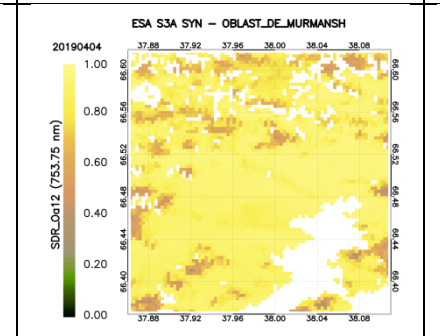
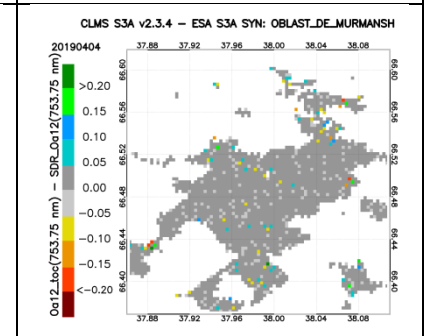
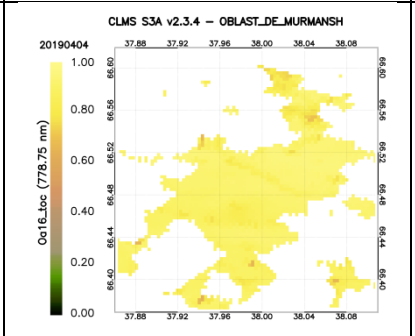
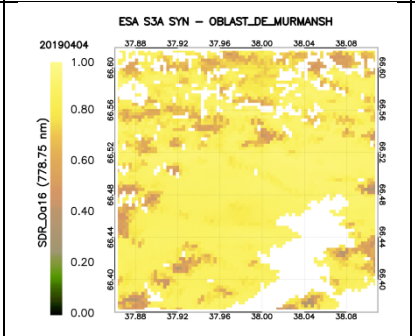
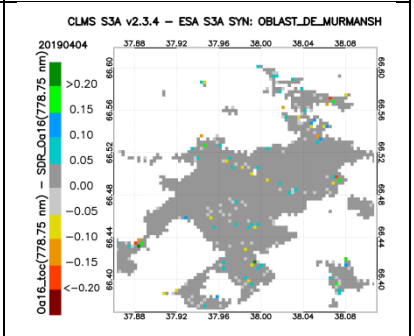


Figure 113: Spatial consistency of Sentinel-3B (2023.03.02) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#6 (Calibration Site, Egypt – Bare Areas). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#7 OBLAST DE MURMANSH (RUSSIA) – SNOW & ICE

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

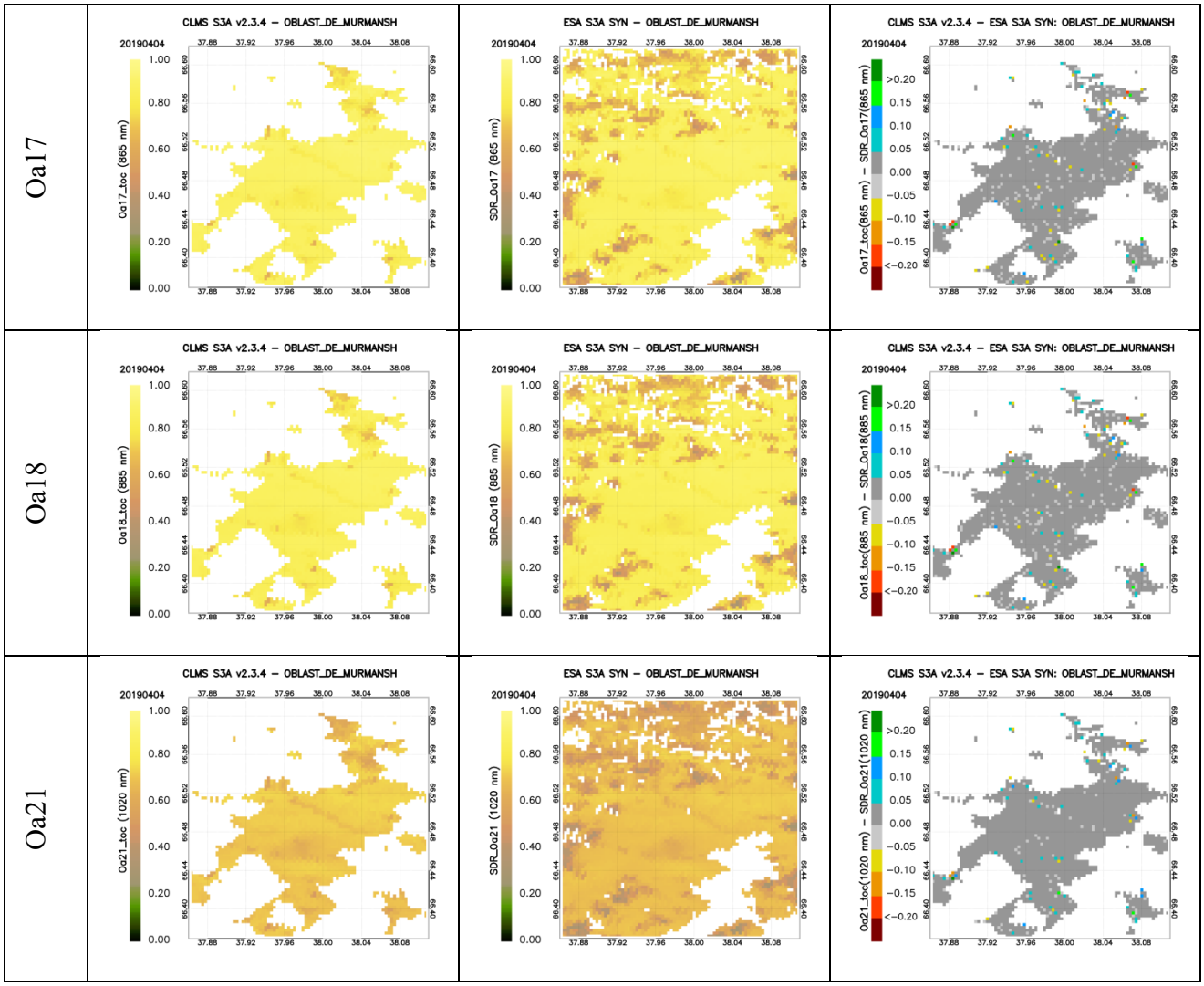
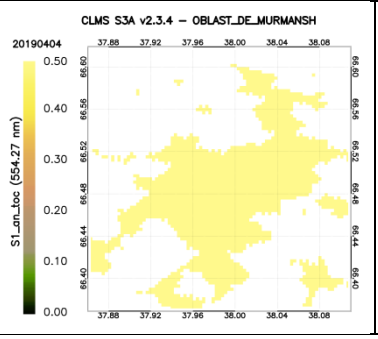
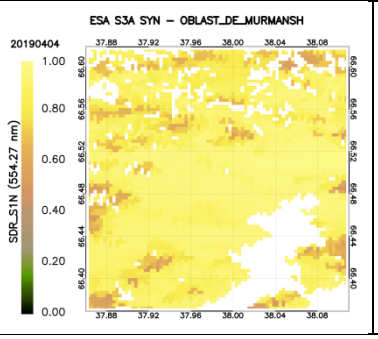
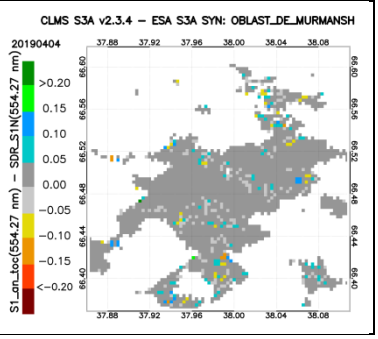


Figure 114: Spatial consistency of Sentinel-3A (2019.04.04) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
S1			

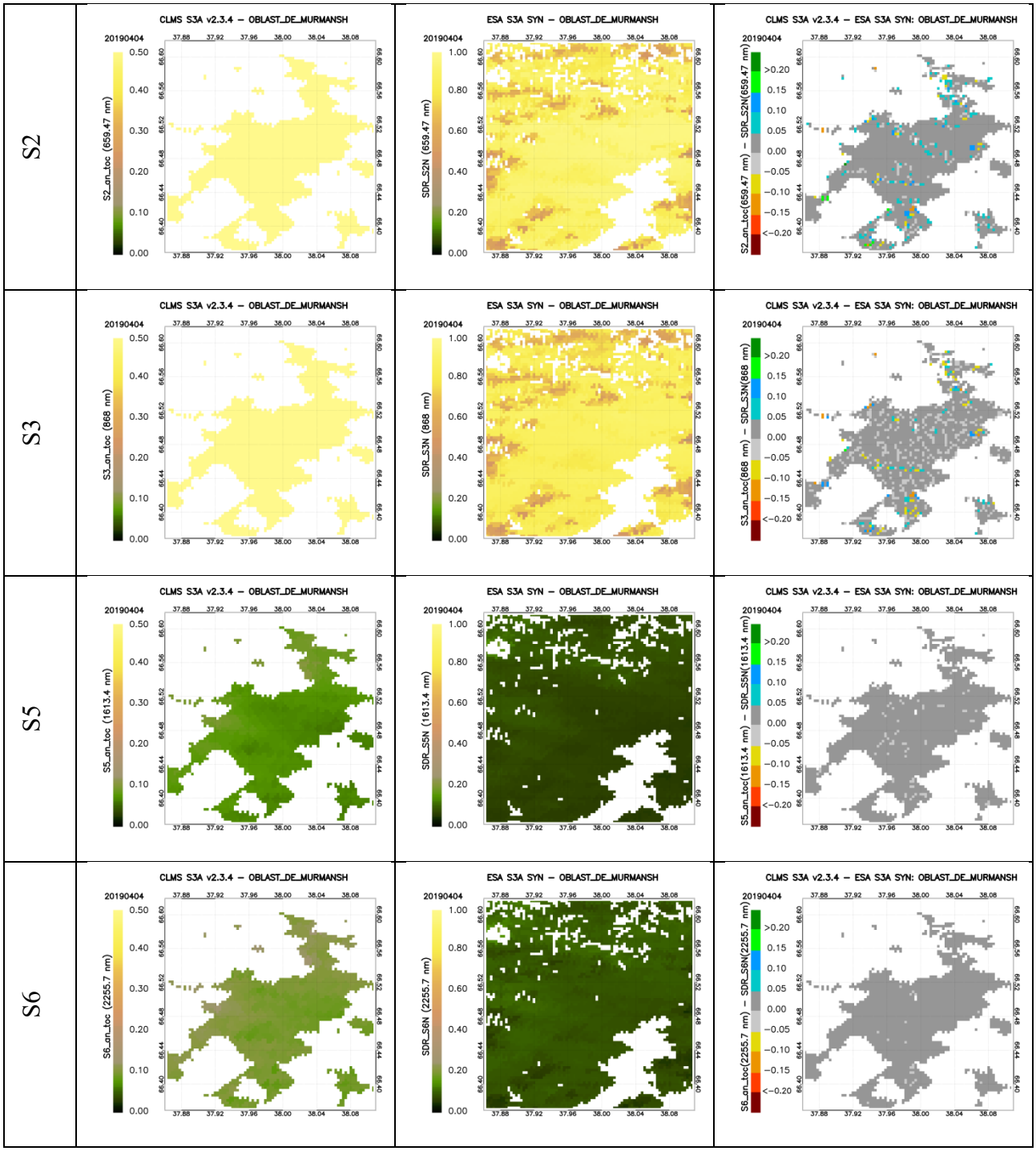
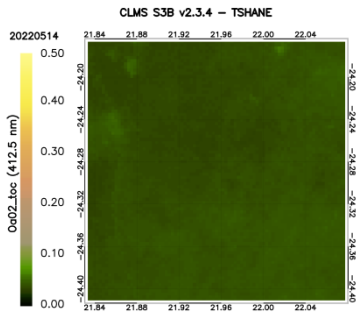
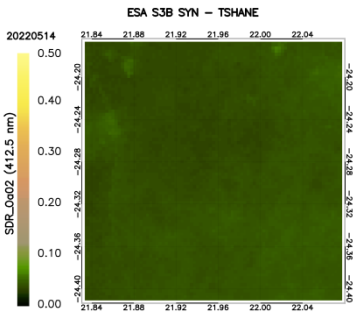
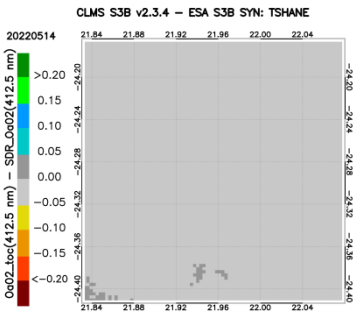
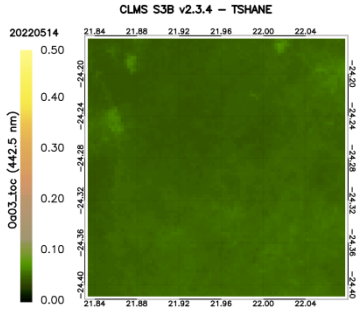
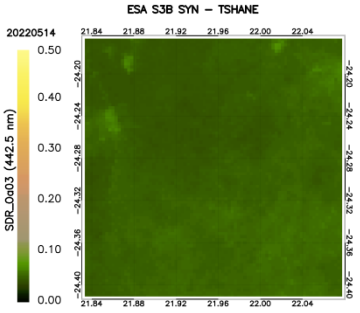
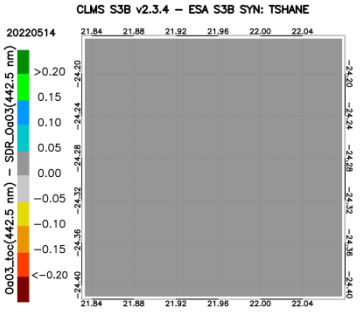
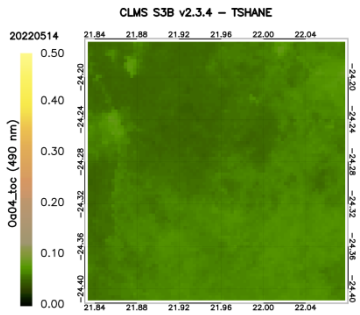
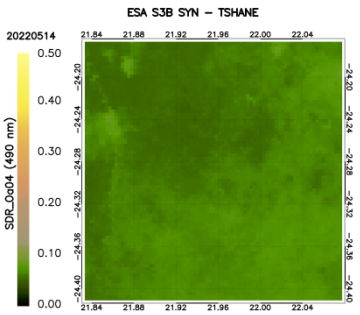
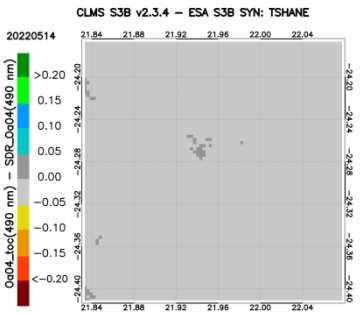
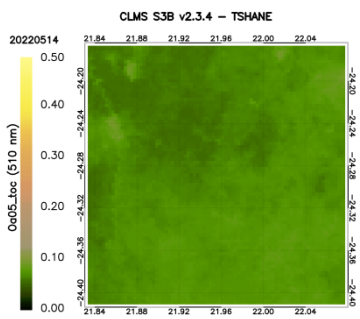
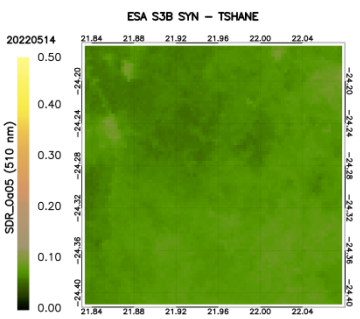
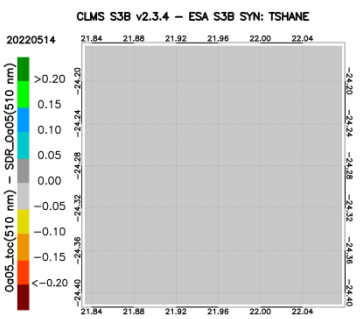
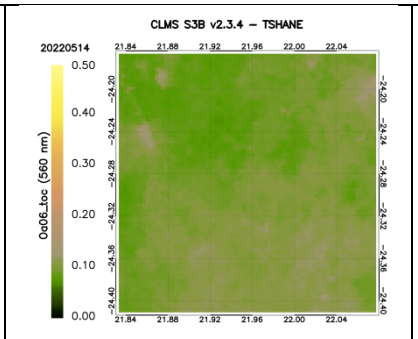
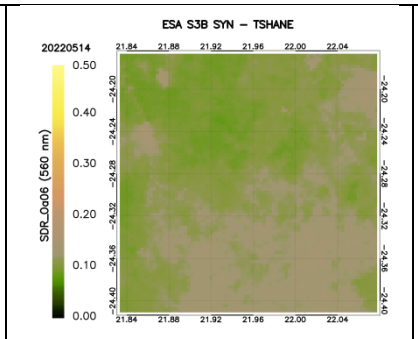
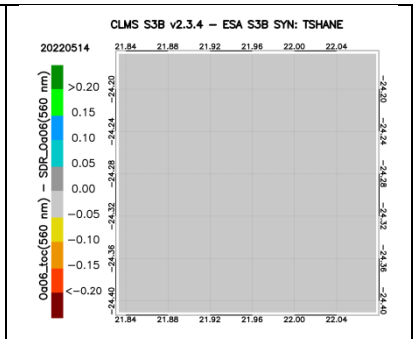
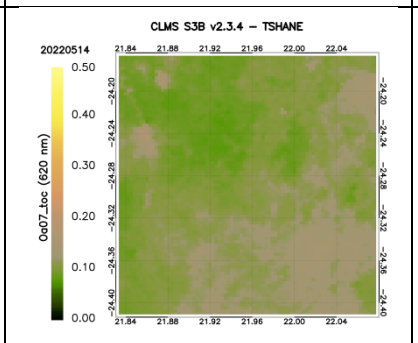
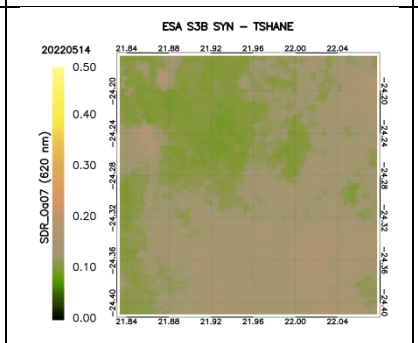
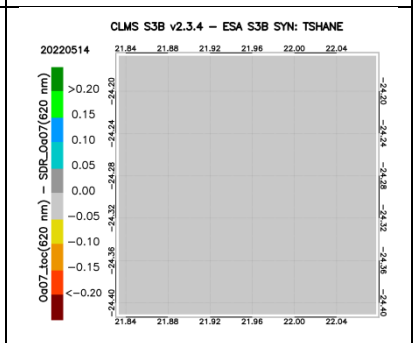
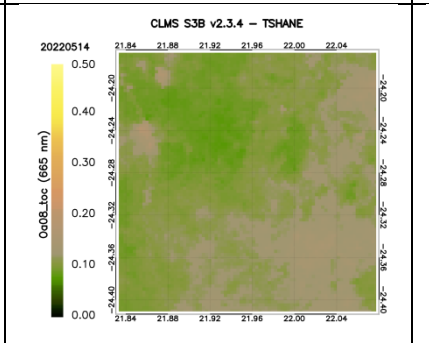
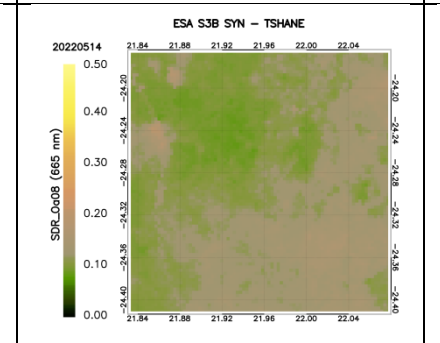
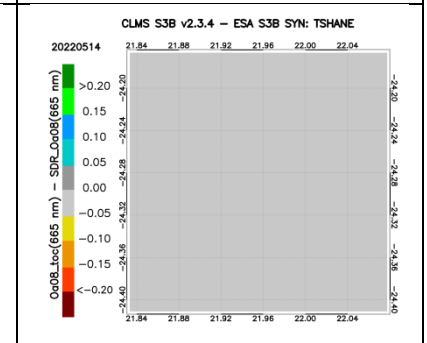
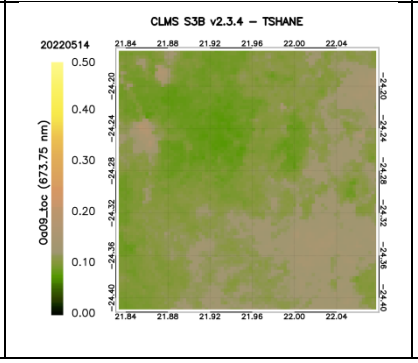
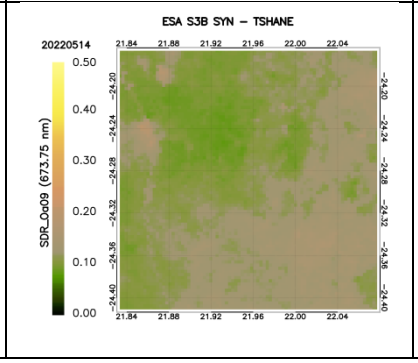
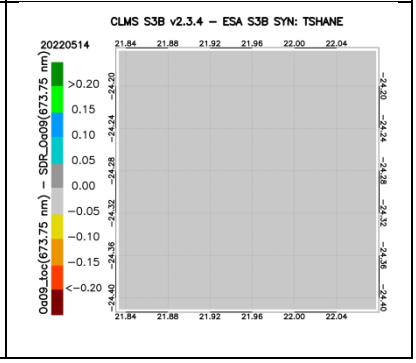
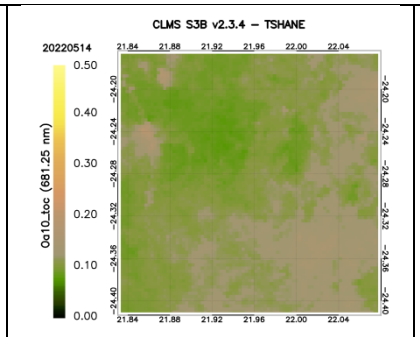
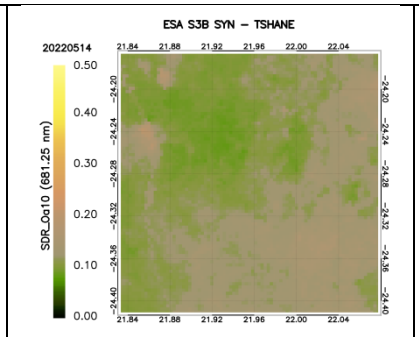
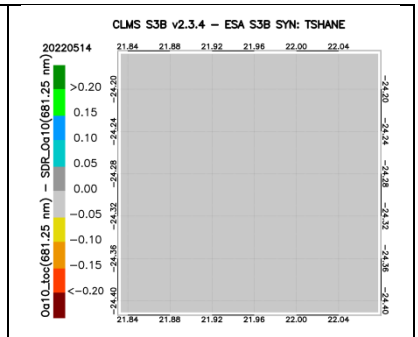
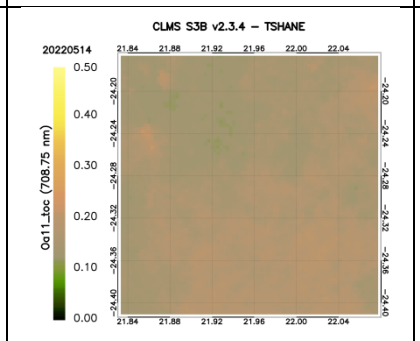
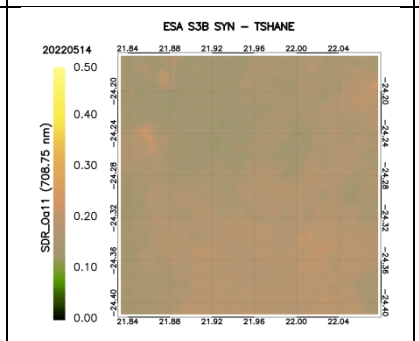
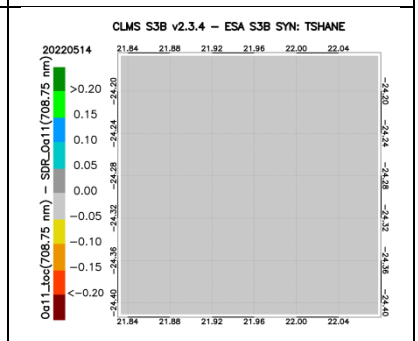
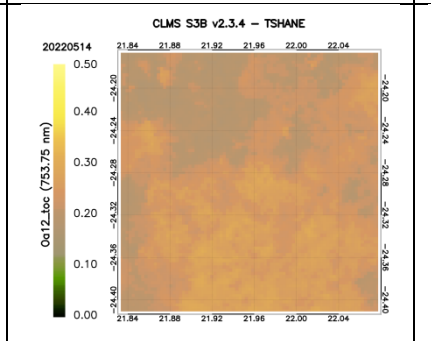
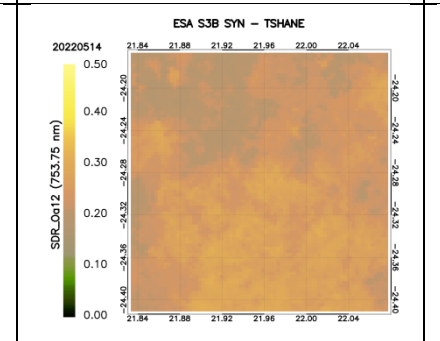
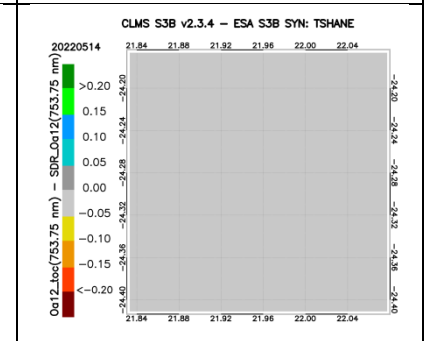
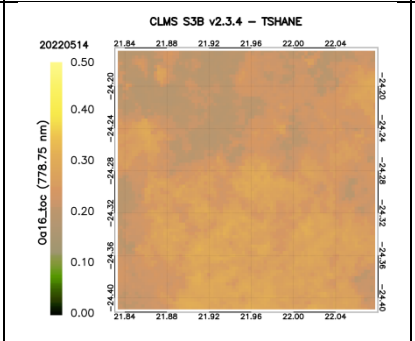
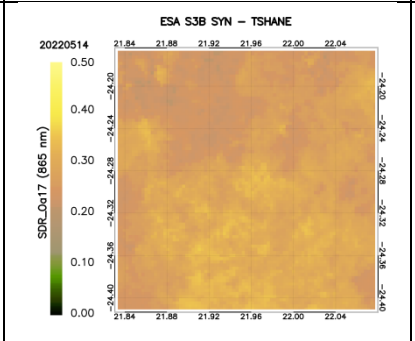
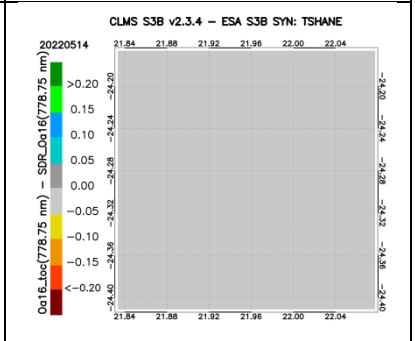


Figure 115: Spatial consistency of Sentinel-3A (2019.04.04) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#8 TSHANE (BOTSWANA) – SAVANNAH

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

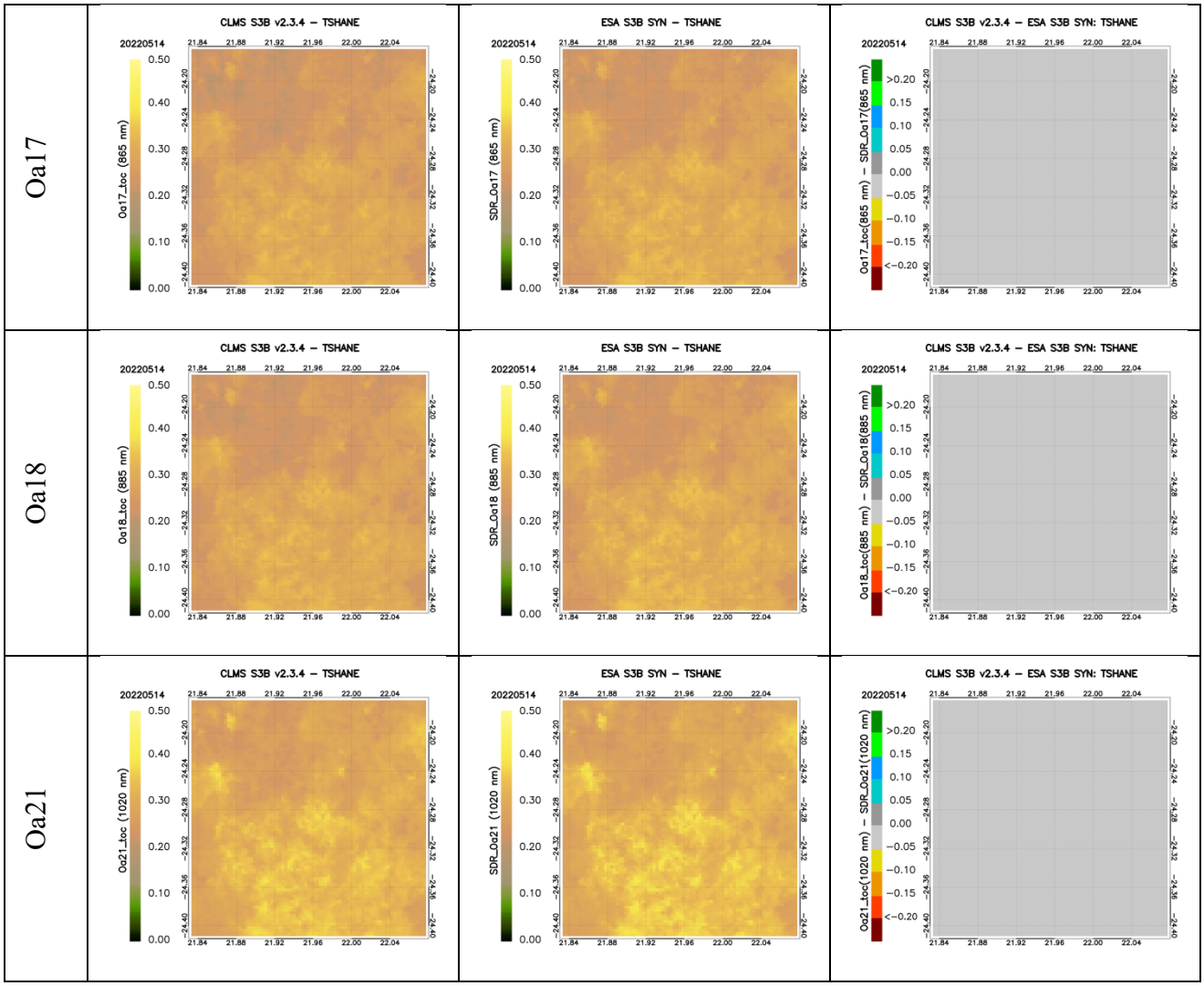
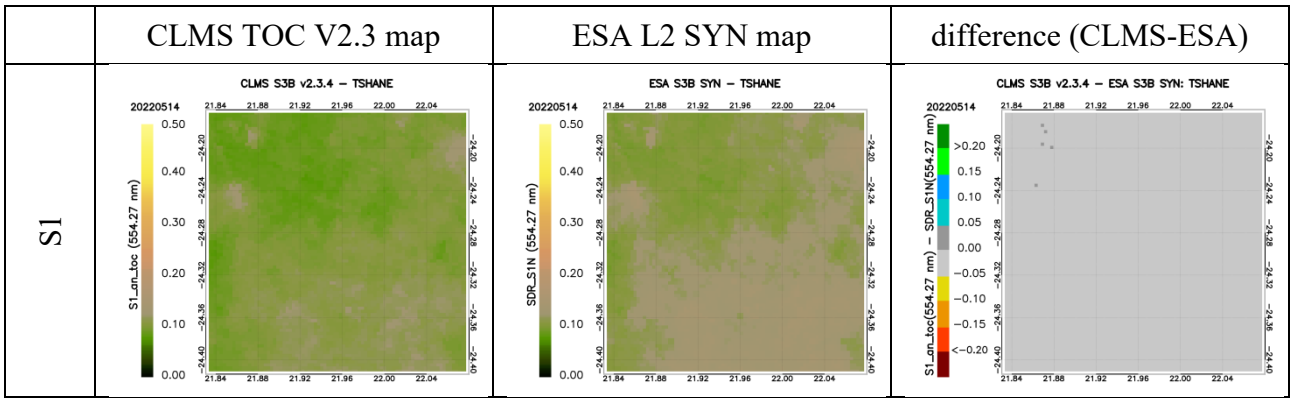


Figure 116: Spatial consistency of Sentinel-3B (2022.05.14) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#8 (Tshane, Botswana – Savannah). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).



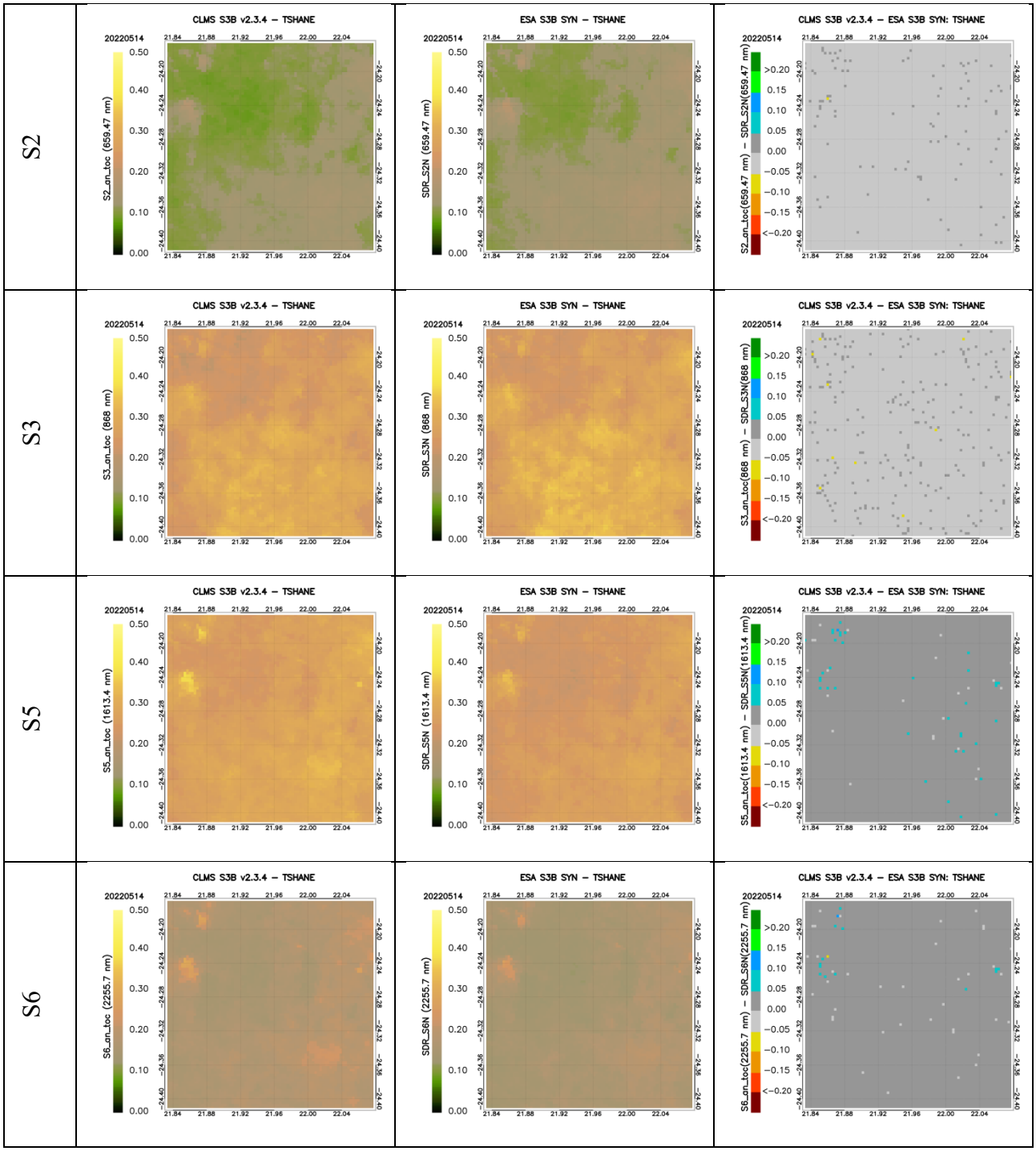
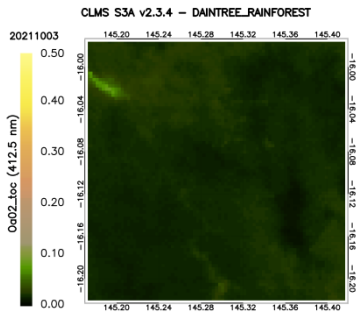
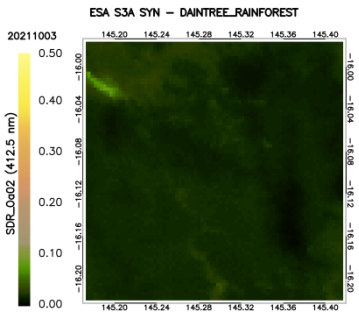
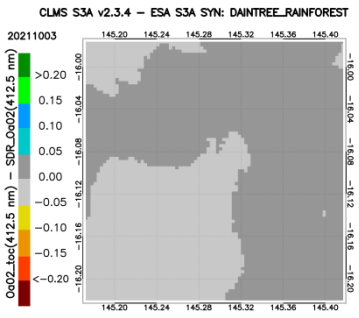
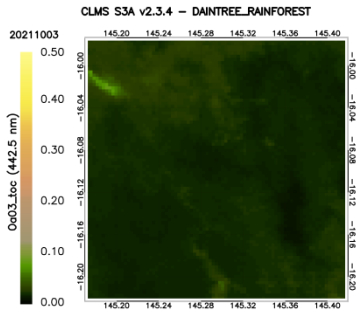
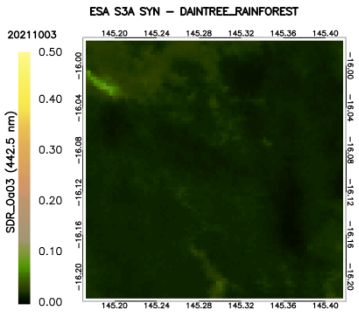
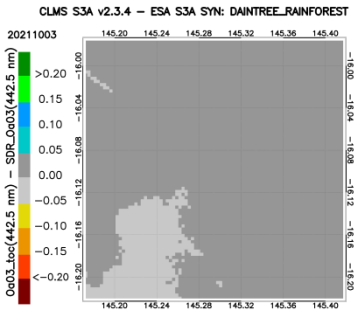
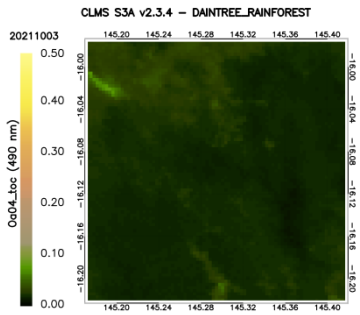
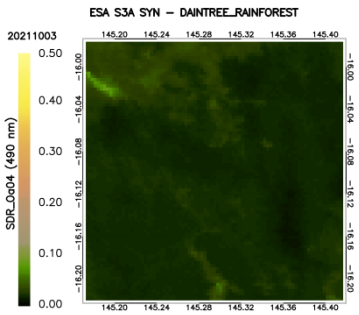
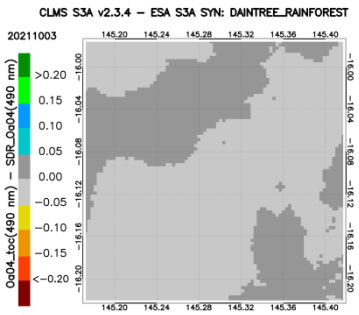
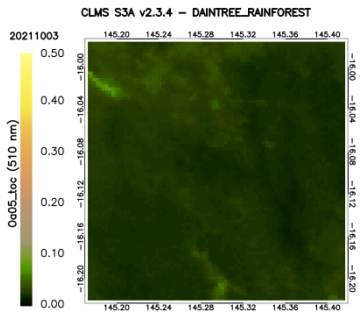
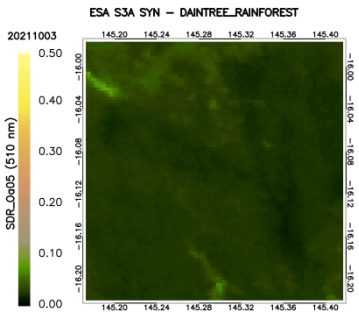
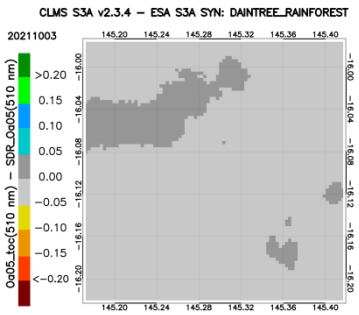
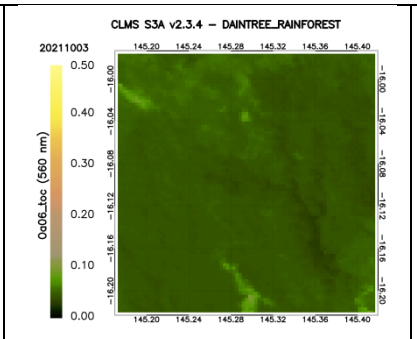
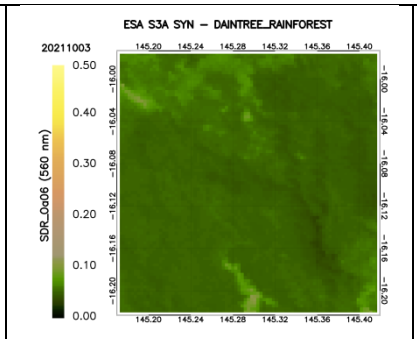
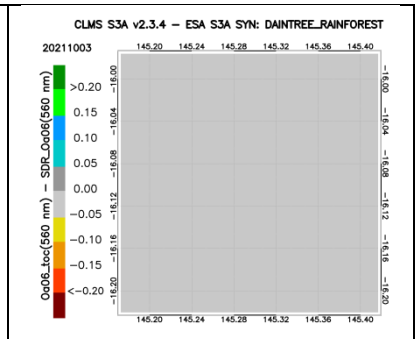
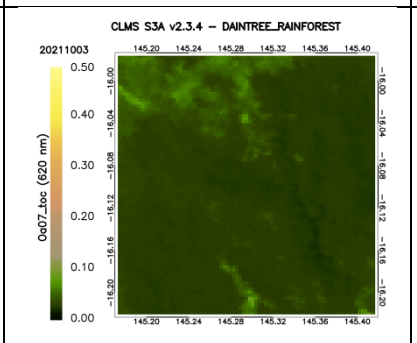
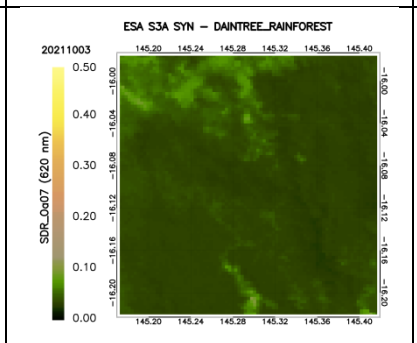
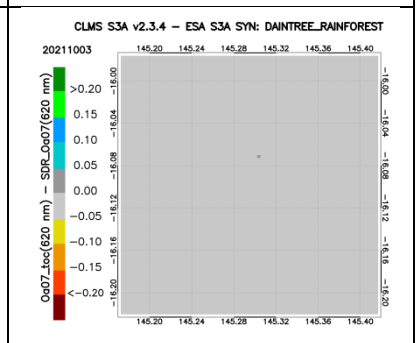
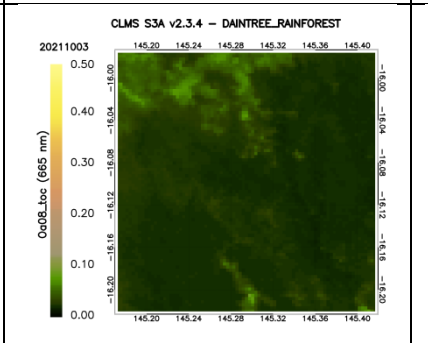
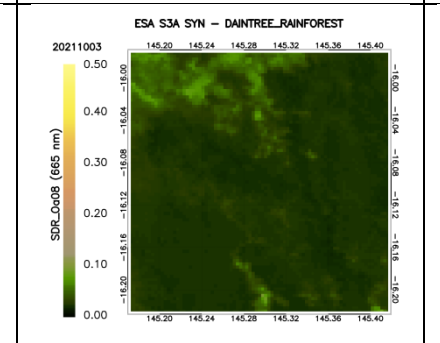
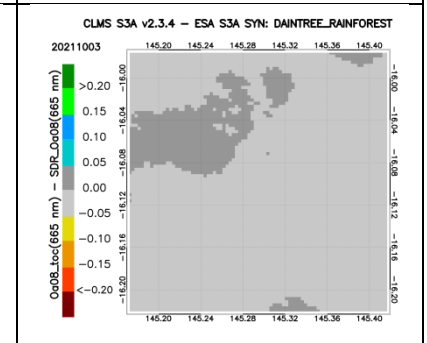
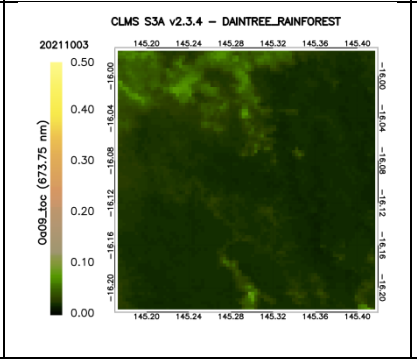
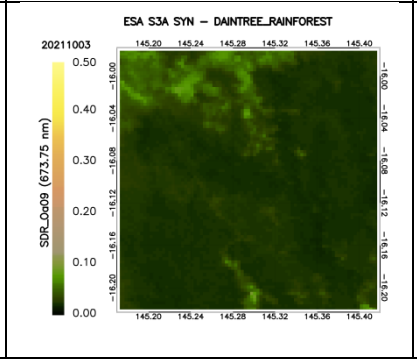
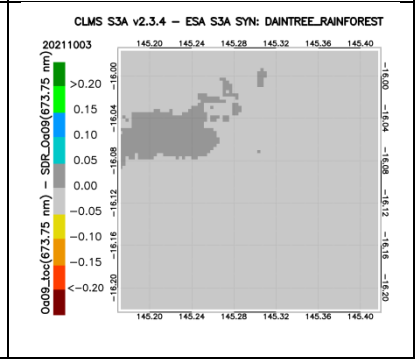
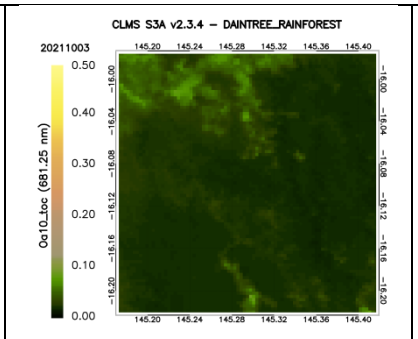
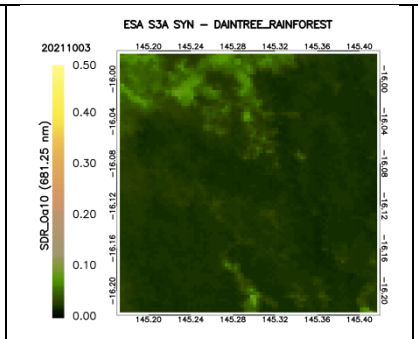
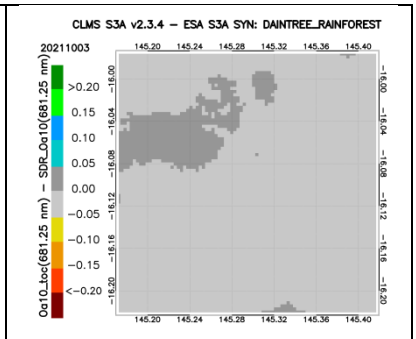
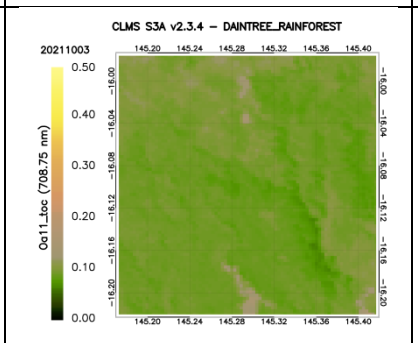
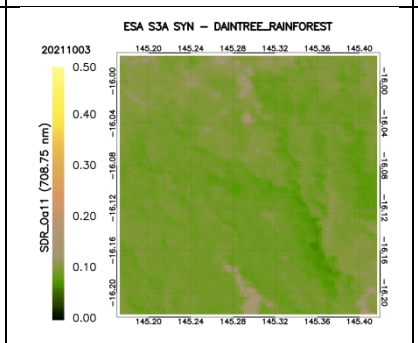
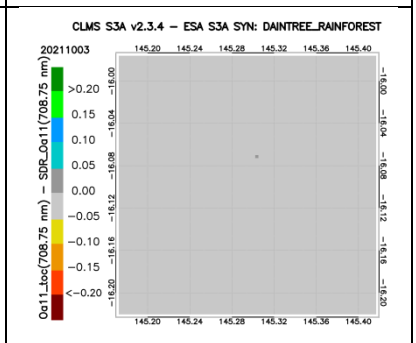
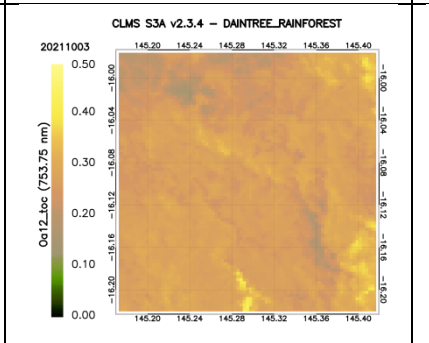
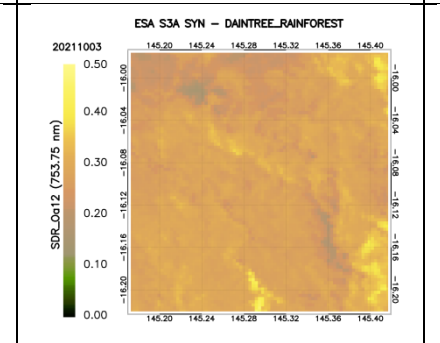
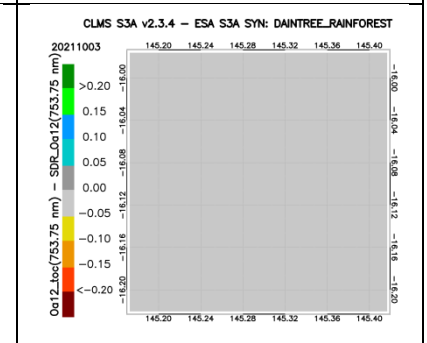
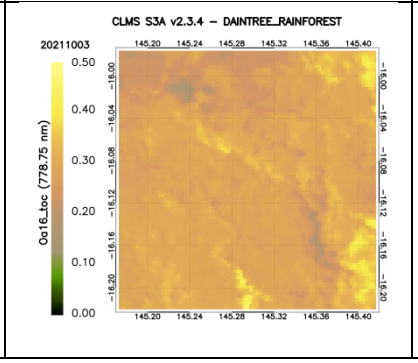
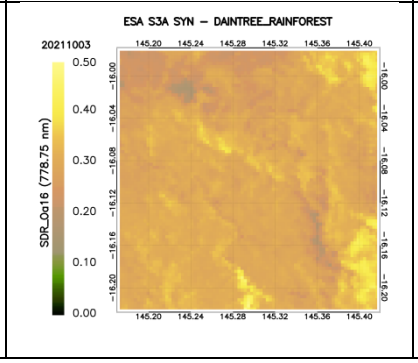
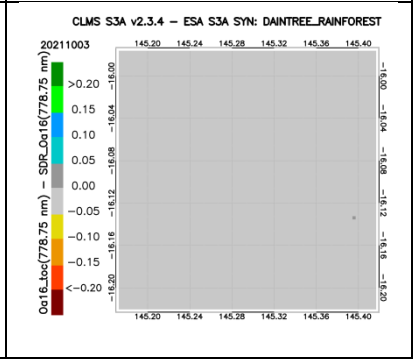


Figure 117: Spatial consistency of Sentinel-3B (2022.05.14) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#8 (Tshane, Botswana – Savannah). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#9 DAINTREE RAINFOREST (AUSTRALIA) - EBF

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

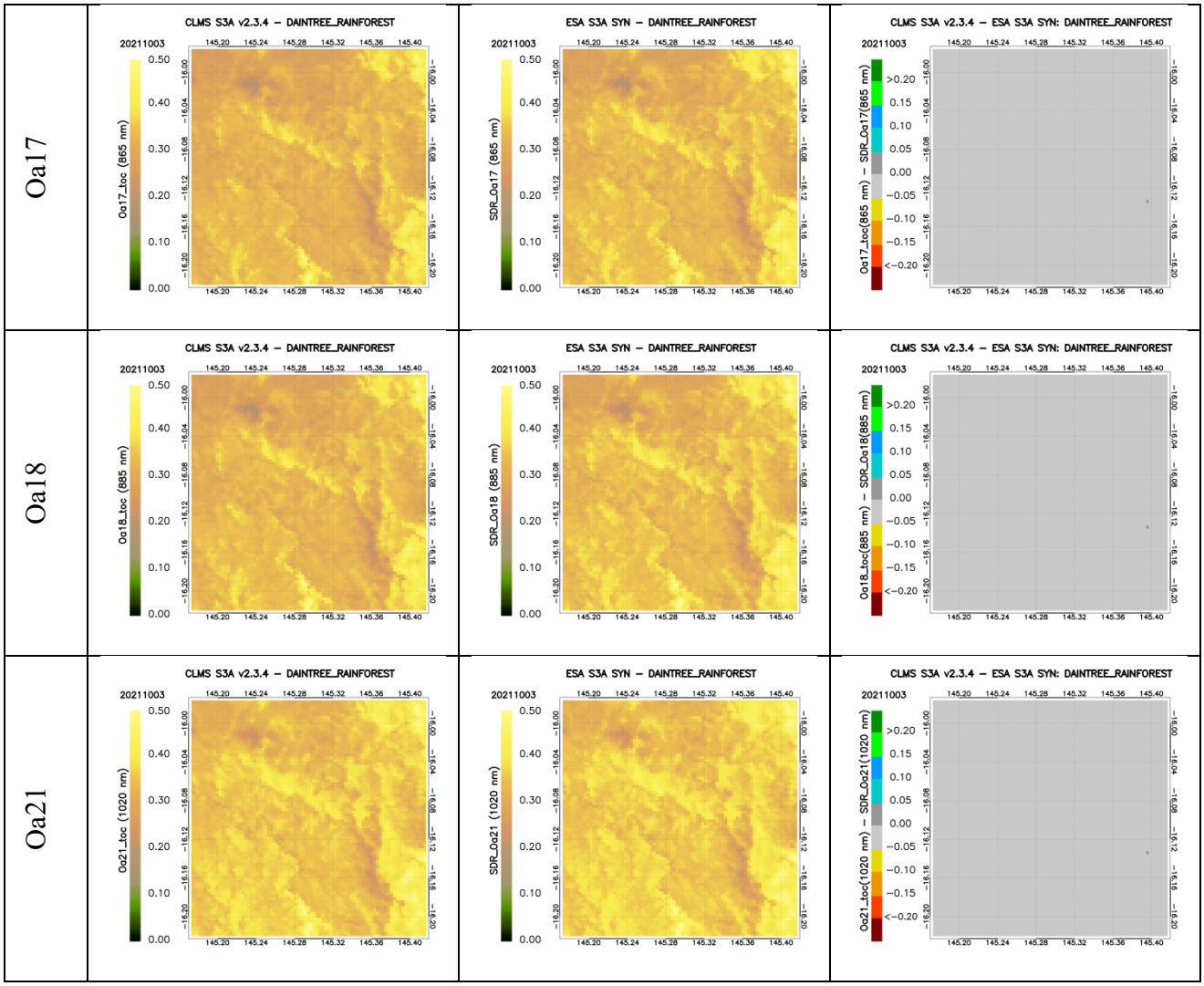
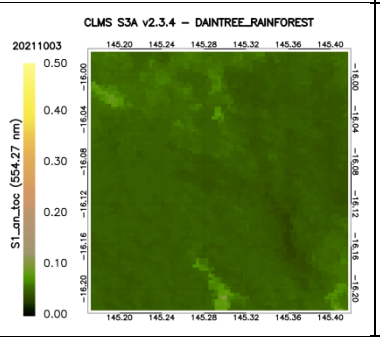
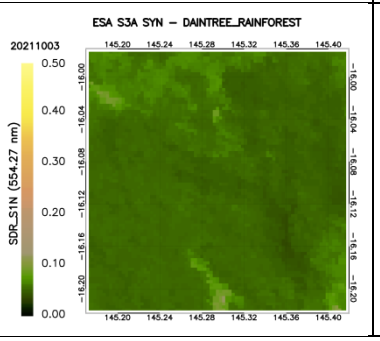
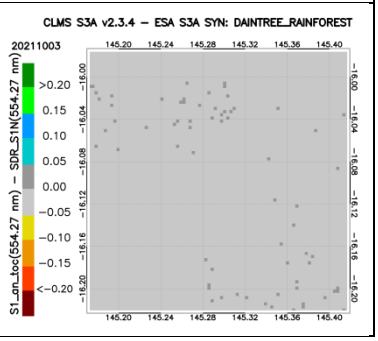


Figure 118: Spatial consistency of Sentinel-3A (2021.10.03) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#9 (Daintree Rainforest, Australia - EBF). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
S1			

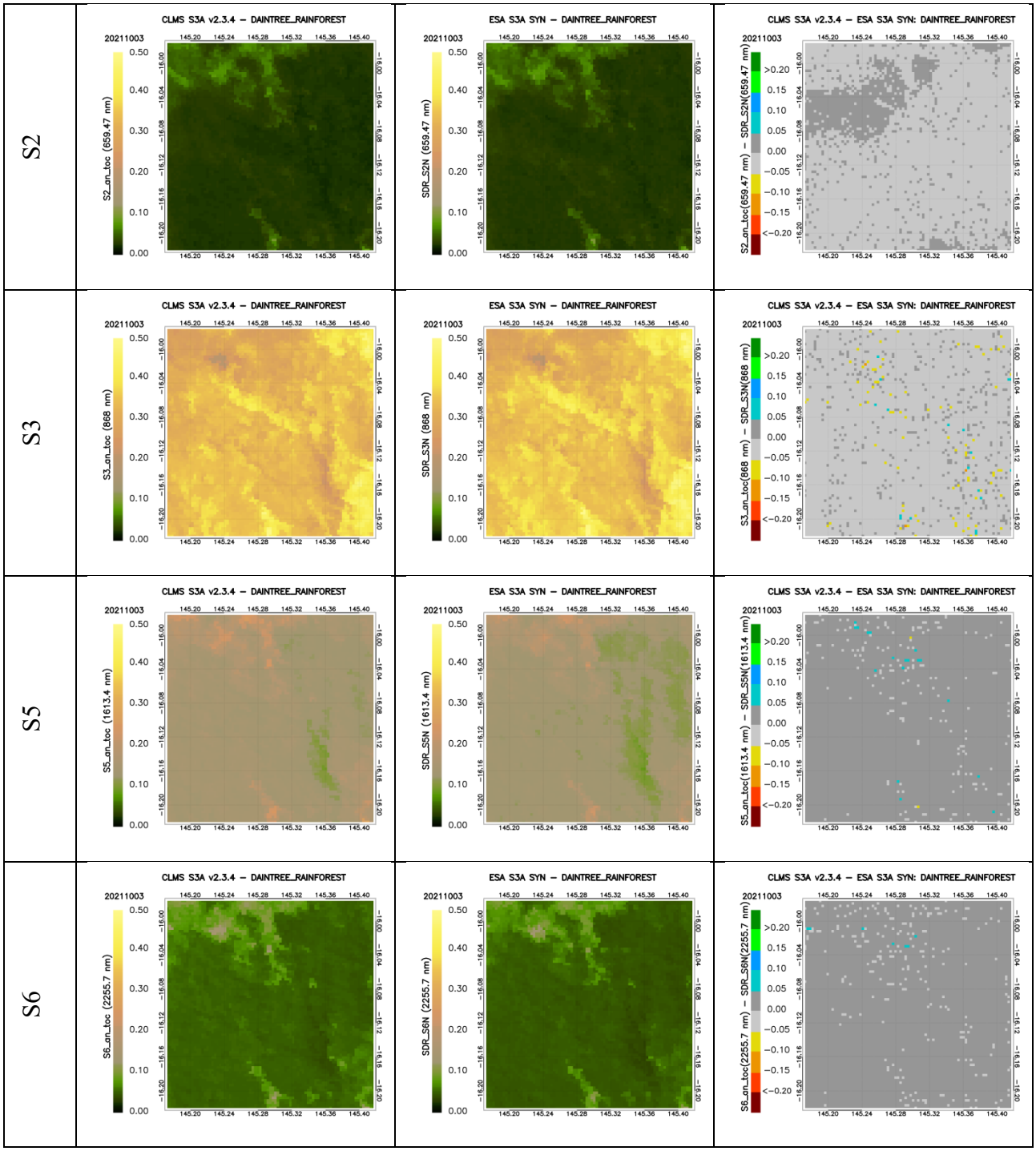
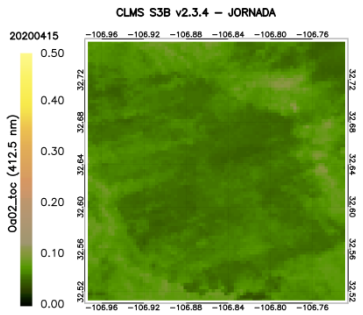
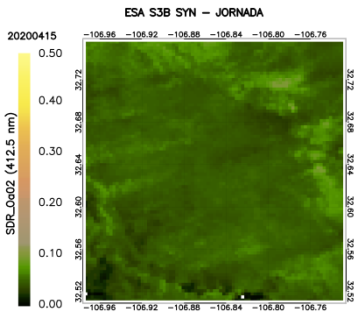
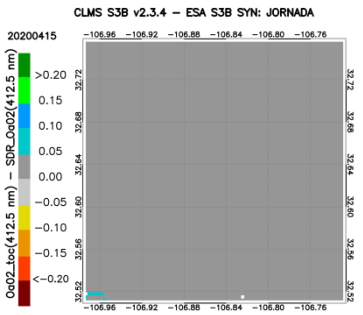
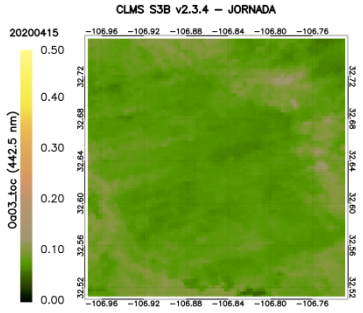
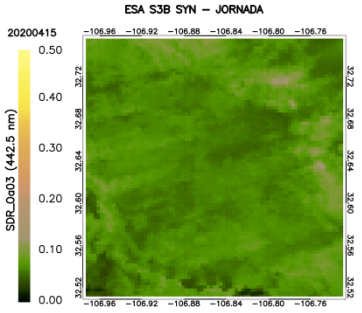
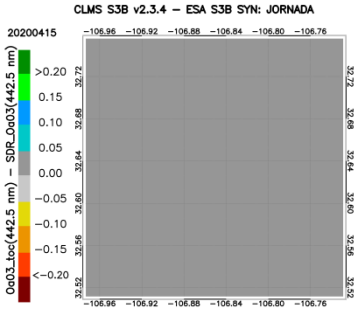
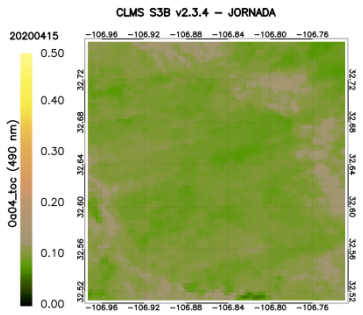
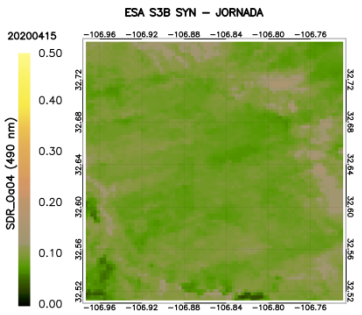
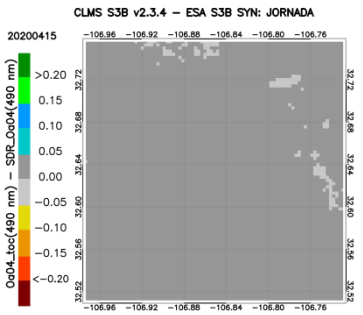
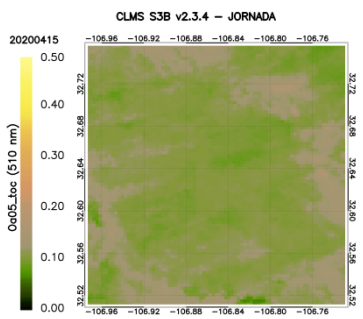
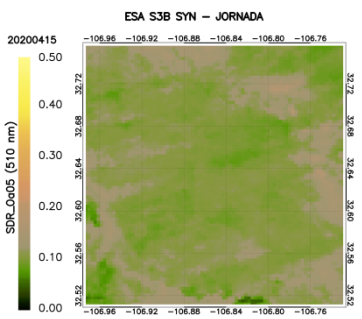
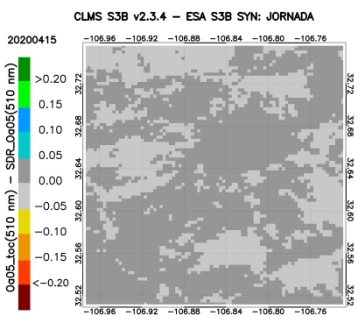
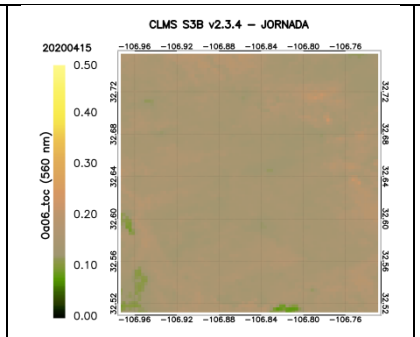
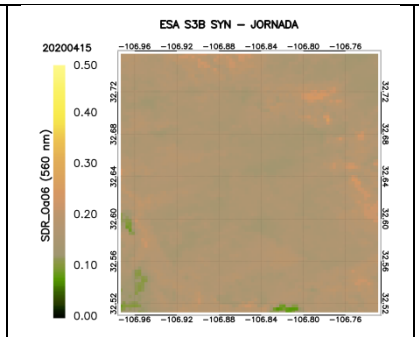
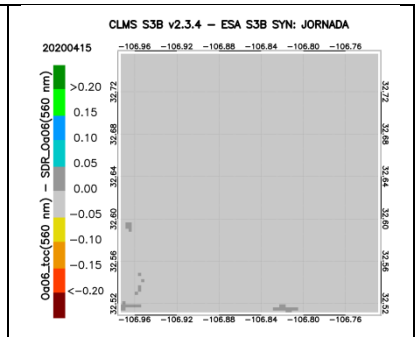
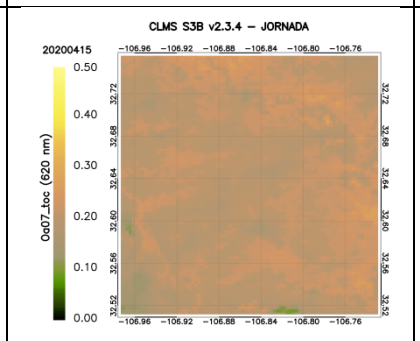
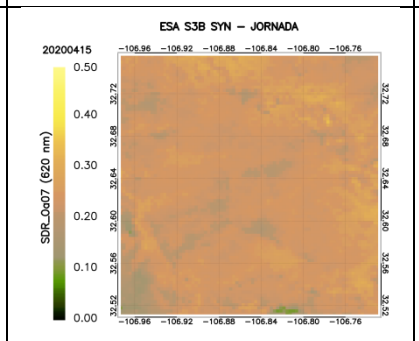
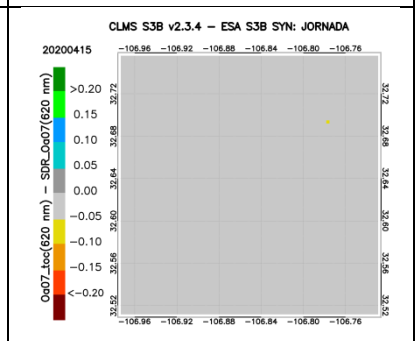
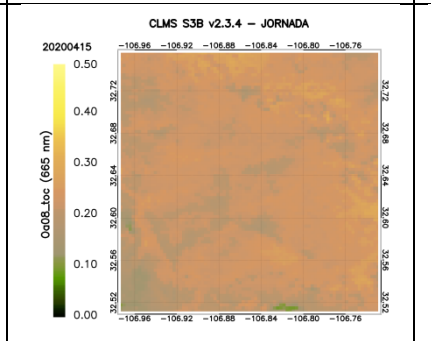
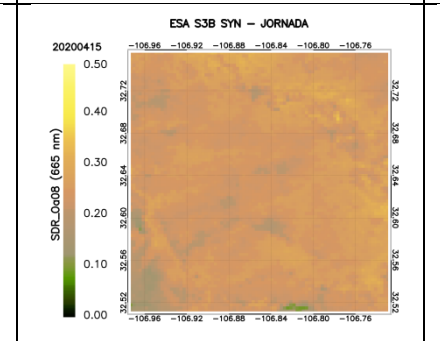
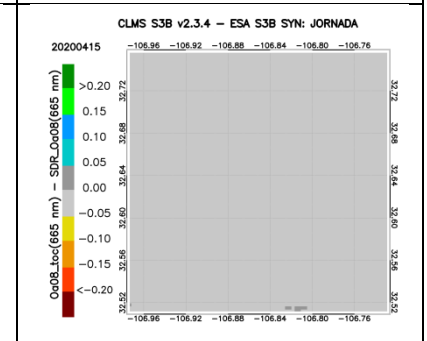
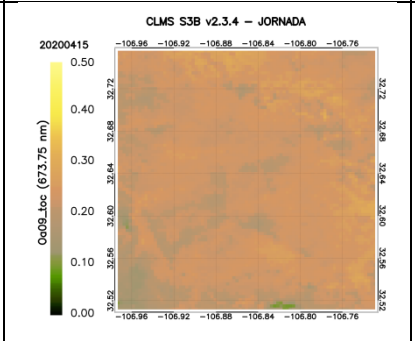
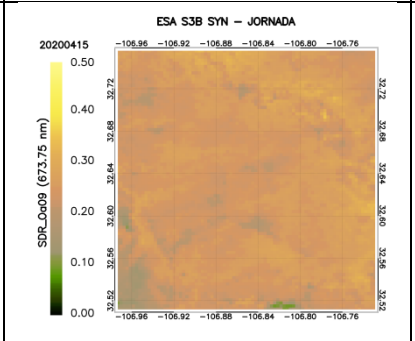
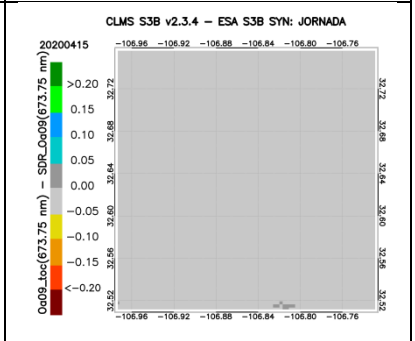
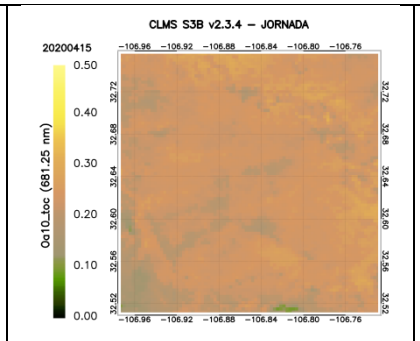
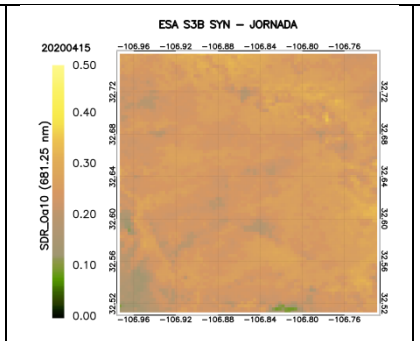
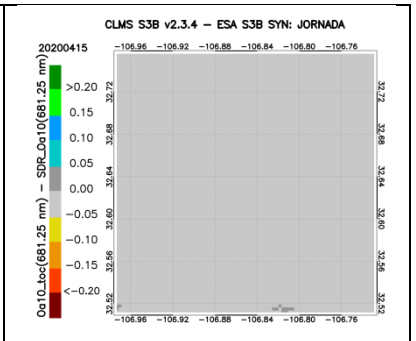
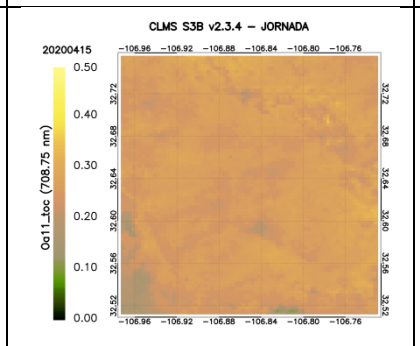
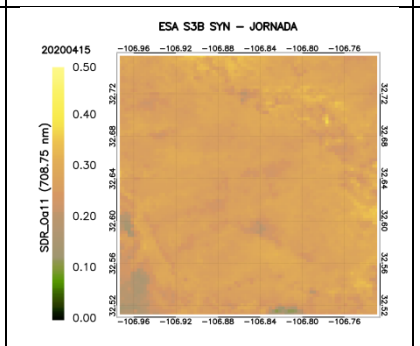
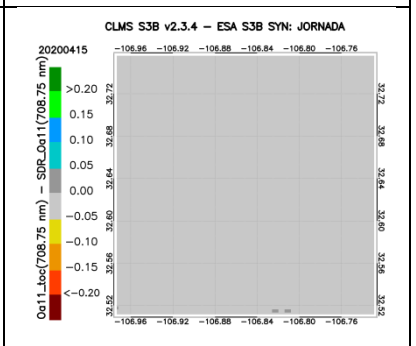
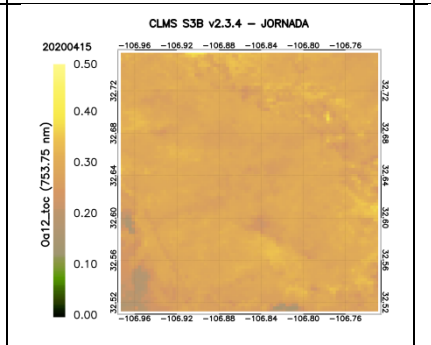
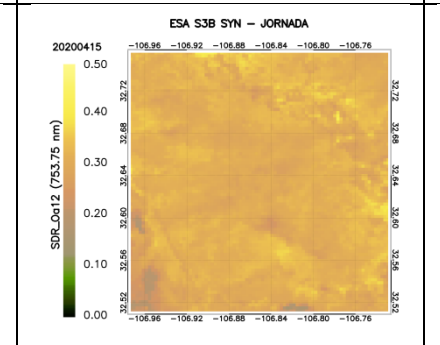
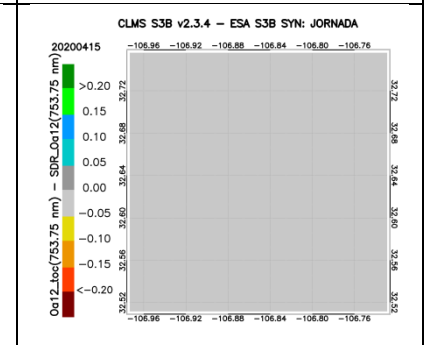
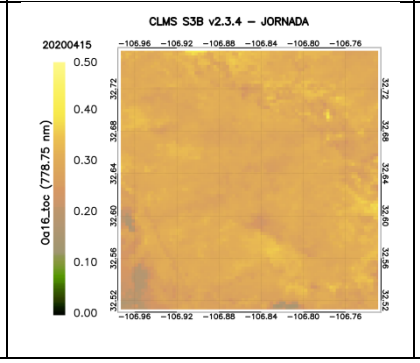
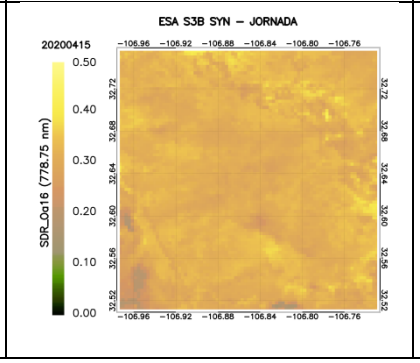
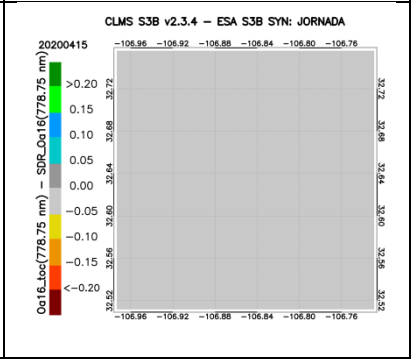


Figure 119: Spatial consistency of Sentinel-3A (2021.10.03) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#9 (Daintree Rainforest, Australia - EBF). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

AOI#10 JORNADA (NEW MEXICO) - SHRUBLANDS

	CLMS TOC V2.3 map	ESA L2 SYN map	difference (CLMS-ESA)
Oa02			
Oa03			
Oa04			
Oa05			

Oa06			
Oa07			
Oa08			
Oa09			

Oa10			
Oa11			
Oa12			
Oa16			

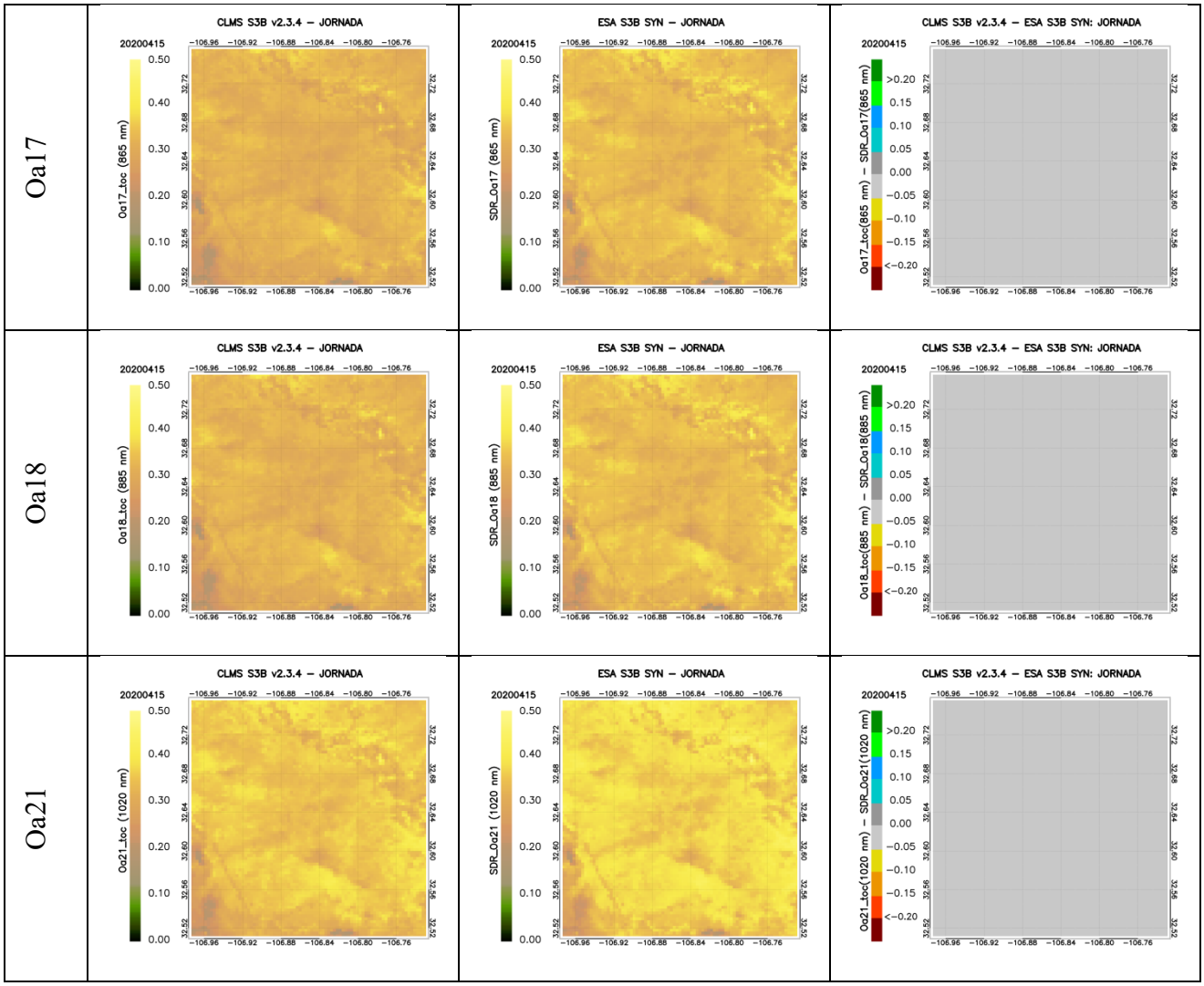
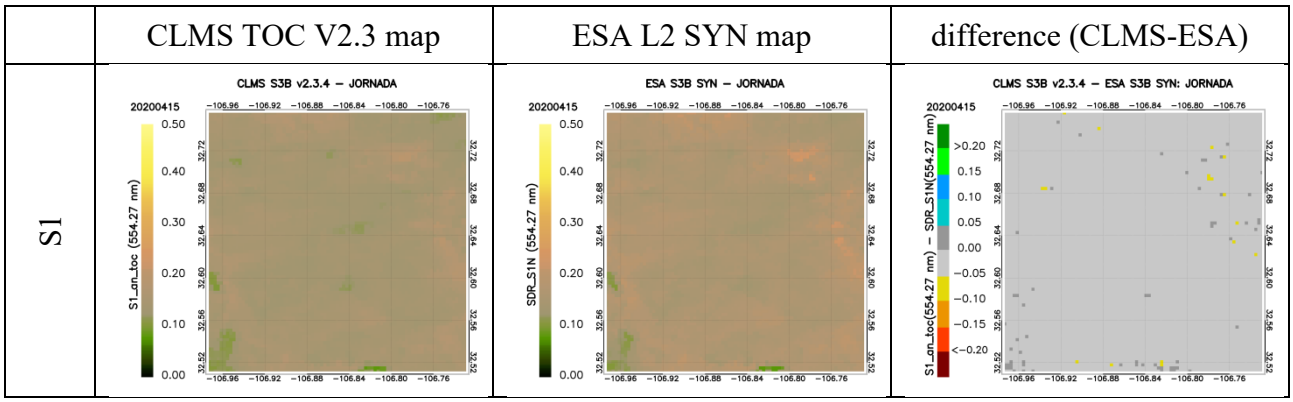


Figure 120: Spatial consistency of Sentinel-3B (2020.04.15) OLCI bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#10 (Jornada, New Mexico - Shrublands). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).



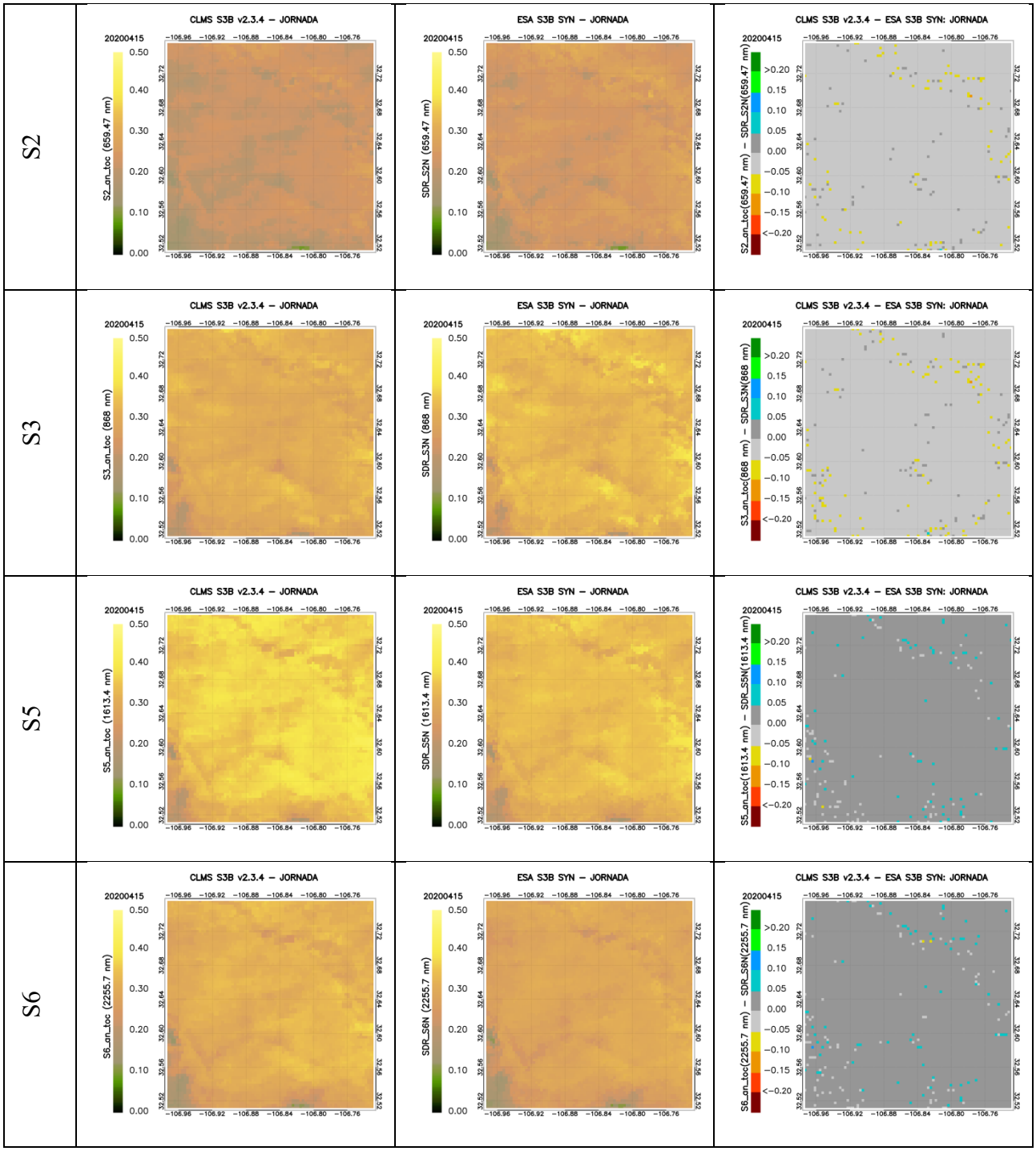


Figure 121: Spatial consistency of Sentinel-3B (2020.04.15) SLSTR bands between CLMS TOC V2.3 and ESA L2 SYN products for the AOI#10 (Jornada, New Mexico - Shrublands). From left to right: maps of CLMS and ESA products and difference map (CLMS – ESA).

ANNEX VIII: CONSISTENCY BETWEEN SENTINEL-3 TOC REFLECTANCE V1 OLCI AND SLSTR OVER AOIS

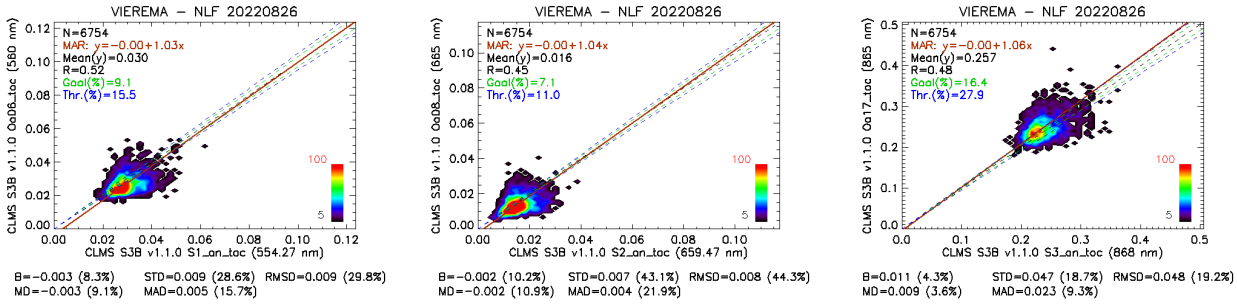


Figure 122: Scatter-plots of Sentinel-3 TOC reflectance V1 (2022.08.26) OLCI versus SLSTR equivalent bands over the AOI#1 (Vieremä, Finland – NLF).

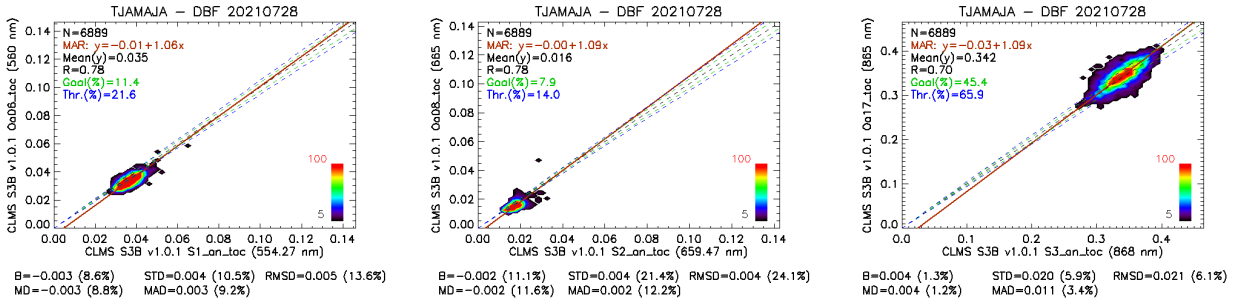


Figure 123: Scatter-plots of Sentinel-3 TOC reflectance V1 (2018.10.12) OLCI versus SLSTR equivalent bands over the AOI#2 (Tjamaja, Russia – DBF).

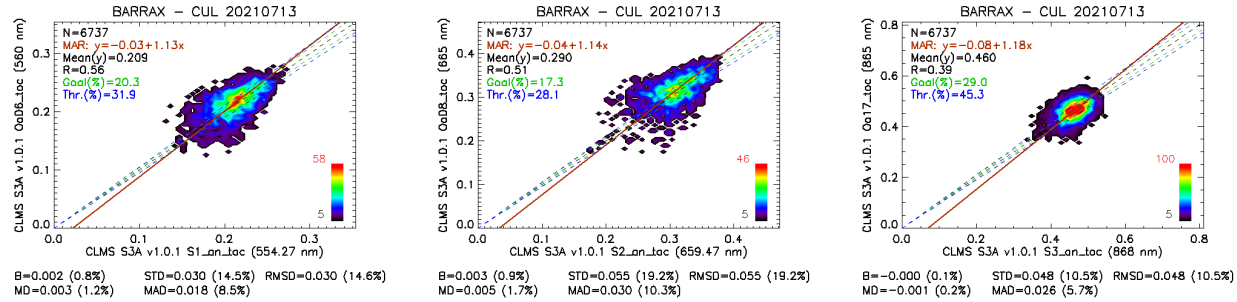


Figure 124: Scatter-plots of Sentinel-3 TOC reflectance V1 (2021.07.13) OLCI versus SLSTR equivalent bands over the AOI#3 (Barrax, Spain – Cultivated).

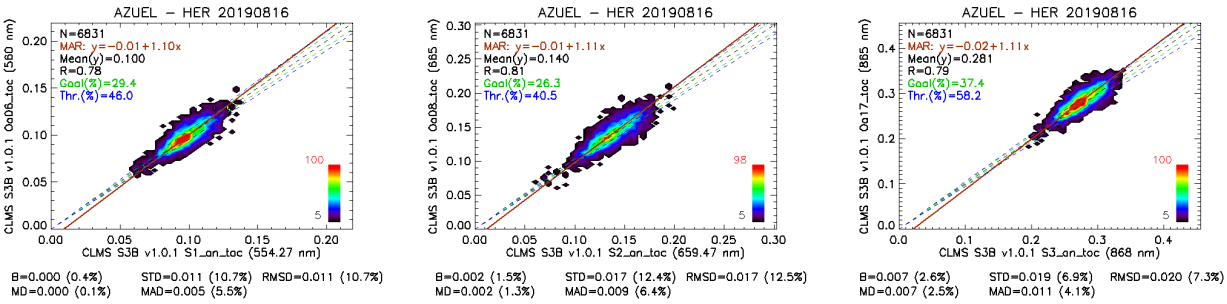


Figure 125: Scatter-plots of Sentinel-3 TOC reflectance V1 (2019.08.16) OLCI versus SLSTR equivalent bands over the AOI#4 (Azuel, Spain – Herbaceous).

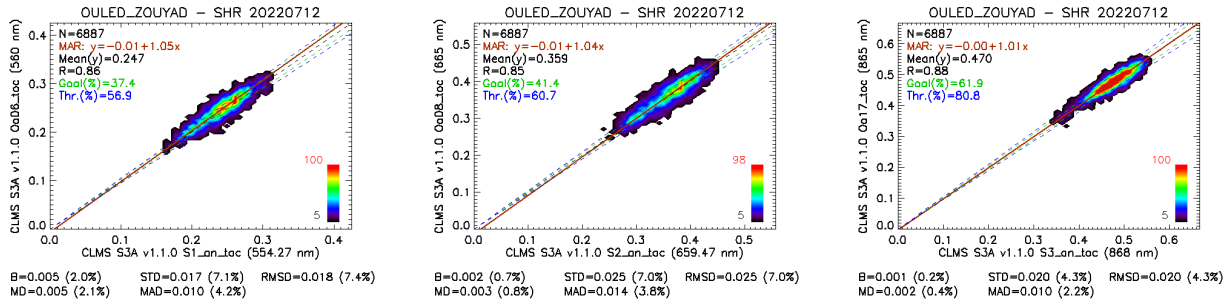


Figure 126: Scatter-plots of Sentinel-3 TOC reflectance V1 (2022.07.12) OLCI versus SLSTR equivalent bands over the AOI#5 (Ouled Zouyad, Tunisia – Shrublands).

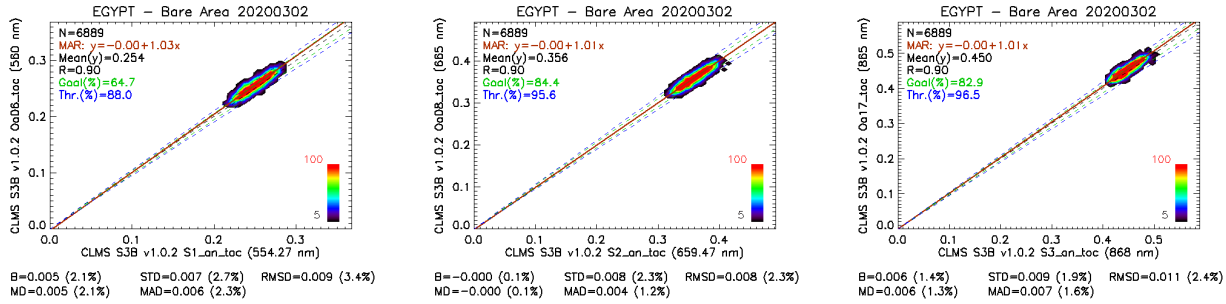


Figure 127: Scatter-plots of Sentinel-3 TOC reflectance V1 (2020.03.02) OLCI versus SLSTR equivalent bands over the AOI#6 (Calibration Site, Egypt – Bare Areas).

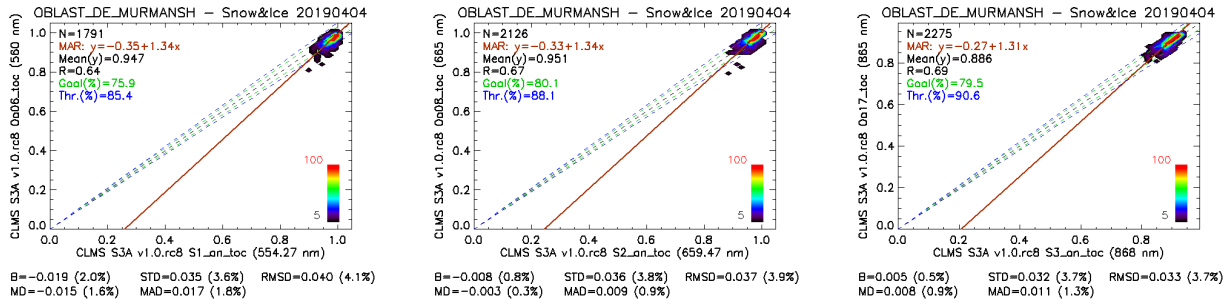


Figure 128: Scatter-plots of Sentinel-3 TOC reflectance V1 (2019.04.04) OLCI versus SLSTR equivalent bands over the AOI#7 (Oblast de Murmansh, Russia – Snow & Ice).

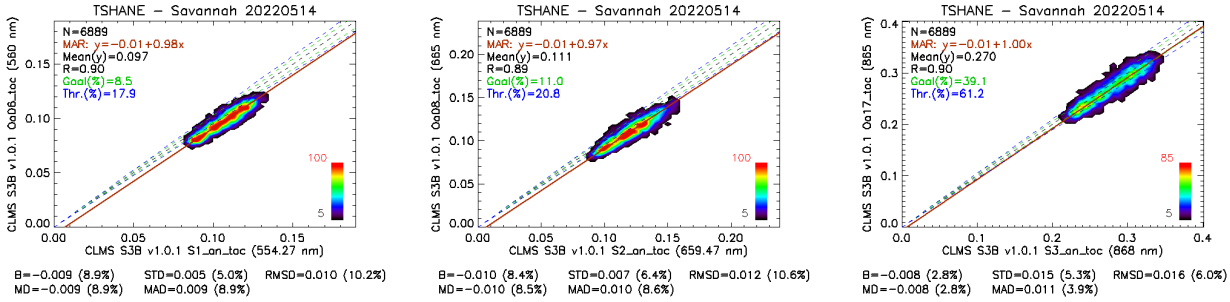


Figure 129: Scatter-plots of Sentinel-3 TOC reflectance V1 (2022.05.14) OLCI versus SLSTR equivalent bands over the AOI#8 (Tshane, Botswana – Savannah).

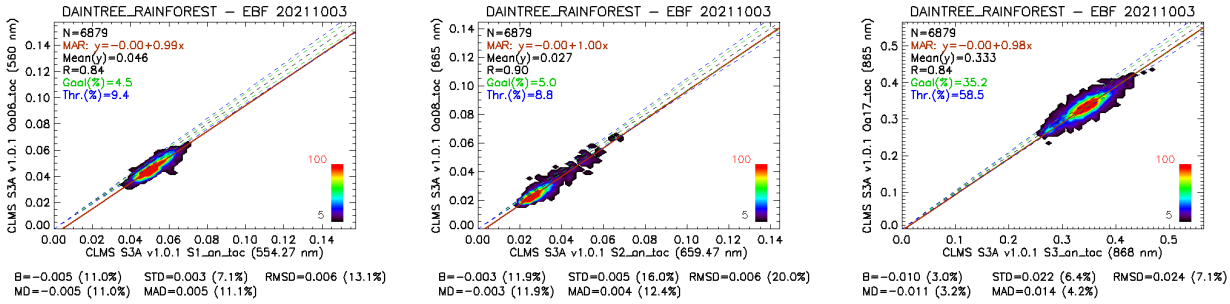


Figure 130: Scatter-plots of Sentinel-3 TOC reflectance V1 (2021.10.03) OLCI versus SLSTR equivalent bands over the AOI#9 (Daintree Rainforest, Australia - EBF).

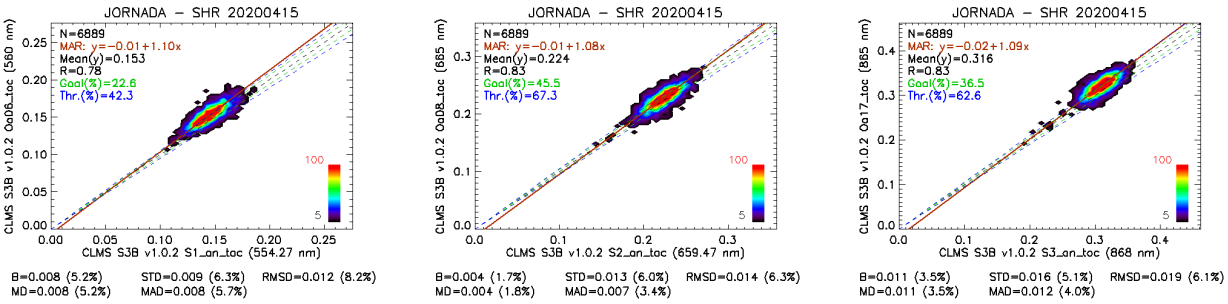


Figure 131: Scatter-plots of Sentinel-3 TOC reflectance V1 (2020.04.15) OLCI versus SLSTR equivalent bands over the AOI#10 (Jornada, New Mexico - Shrublands).